

BATTERY ENERGY STORAGE SYSTEM MARKET WITH COVID-19 IMPACT

GLOBAL FORECAST TO 2027

BY STORAGE SYSTEM, ELEMENT, BATTERY TYPE (LITHIUM-ION, FLOW BATTERIES),
CONNECTION TYPE (ON-GRID AND OFF-GRID), OWNERSHIP, ENERGY CAPACITY,
APPLICATION, AND GEOGRAPHY

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AC	Alternating Current
ALABC	Advanced Lead-Acid Battery Consortium
APAC	Asia Pacific
ARE	Alliance of Rural Electrification
ASME TES	American Society of Mechanical Engineers Thermal Energy Storage
BEIS	Business, Energy, and Industrial Strategy
BESS	Battery Energy Storage System
BHEL	Bharat Heavy Electricals Limited
BMS	Battery Management System
BNEF	Bloomberg New Energy Finance
BOP	Balance-of-point
BPU	Battery Protection Unit
BTM	Behind-the-meter
C&C	Commissioning Costs
C&I	Commercial and Industrial
CAES	Compressed-Air Energy Storage
CEC	Clean Energy Council
CEO	Chief Executive Officer
CESA	California Energy Storage Alliance
CG	Compliance Guide
CSIRO	Commonwealth Scientific and Industrial Research Organization
DC	Direct Current
DER	Distributed Energy Resources
DEWA	Dubai Electricity and Water Authority
DoD	Department of Defense
DoD	Depth of Discharge
DOE	US Department of Energy
DSS	Distributed Storage Solution

EIA	Energy Information Administration
EIB	European Investment Bank
EP	Electrification Products
EPC	Engineering, Procurement, and Construction
ESA	Energy Storage Association
ESMAP	Energy Sector Management Assistance Program
ESS	Energy Storage System
EU	European Union
FTM	Front-of-the-meter
GSS	Grid Storage Solution
ICB	Iron-chromium
IEA	International Energy Agency
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IESA	India Energy Storage Alliance
IFC	International Finance Corporation
IPPs	Independent Power Producers
ISO	International Organization for Standardization
ISO	Independent System Operator
JV	Joint Venture
LFP	Lithium-Iron Phosphate Batteries
Li-Ion	Lithium-Ion Batteries
LiPo	Lithium-Polymer Batteries
Li-S	Lithium-Sulfur
LTO	Lithium Titanate Oxide
MoU	Memorandum of Understanding
MW	Megawatt
MWh	Megawatt Hour
NERL	National Energy Renewable Laboratory
NFPA	National Fire Protection Association
NiFe	Nickel-Iron
NTPC	National Thermal Power Corporation

OPG	Ontario Power Generation
PCS	Power Conversion System
PHEV	Plug-in Hybrid Electric Vehicle
PHS	Pumped Hydroelectric Storage
PLC	Peak Load Contribution
PNNL	Pacific Northwest National Laboratory
PV	Photovoltaic
REACT	Renewable Energy Accumulator and Conversion Technology
Redox	Reduction-Oxidation
RoE	Rest of Europe
RoW	Rest of the World
RTO	Regional Transmission Organizations
SCADA	Supervisory Control and Data Acquisition
SNL	Sandia National Laboratories
T&D	Transmission and Distribution
TOU	Time-of-Use
UET	UniEnergy Technologies
UL	Underwriters Laboratories
UPS	Uninterruptible Power Supply
USD	US Dollar
VP	Vice President
VRB	Vanadium Redox Battery
ZNBR	Zinc–Bromine

1 INTRODUCTION

1.1 STUDY OBJECTIVES

- To define, describe, and forecast the battery energy storage system market based on storage system, element, battery type, connection type, ownership, energy capacity, and application
- To describe and forecast the size of the battery energy storage system market based on four regions, namely, North America, Europe, Asia Pacific (APAC), and the Rest of the World (RoW), along with their respective countries
- To provide detailed information regarding factors (drivers, restraints, opportunities, and challenges) influencing the market growth
- To provide a detailed overview of the process flow of the battery energy storage system market
- To analyze opportunities for stakeholders by identifying high-growth segments of the battery energy storage system market
- To study the complete value chain of the battery energy storage system market
- To provide a detailed analysis of the impact of the COVID-19 crisis on the battery energy storage system market, its segments, and market players
- To strategically analyze micromarkets¹ with respect to individual growth trends, prospects, and contributions to the overall market size
- To strategically profile the key players and comprehensively analyze their market positions in terms of their ranking and core competencies², along with detailing the competitive landscape for market leaders
- To analyze competitive developments, such as product launches, acquisitions, collaborations, agreements, and partnerships, in the battery energy storage system market

1.2 MARKET DEFINITION

A battery energy storage system is embedded with a rechargeable battery in the form of battery packs and modules. The system stores energy from solar arrays, wind, or the electric grid and then discharges that energy later to provide electricity or other grid services when needed. Rechargeable batteries are usually expensive and have less environmental impact than non-rechargeable (disposable) batteries. However, rechargeable batteries incur a high upfront cost as they can be charged several times after the initial purchase. Battery energy storage systems are used in low-peak periods to store surplus energy and provide this energy to the end users during the time of high peak demand for electricity.

¹Micromarkets are defined as the segments and subsegments of the battery energy storage system market included in the scope of the report.

²Core competencies of companies are captured in terms of their key developments and strategies adopted to sustain their positions in the market.

1.2.1 INCLUSIONS AND EXCLUSIONS

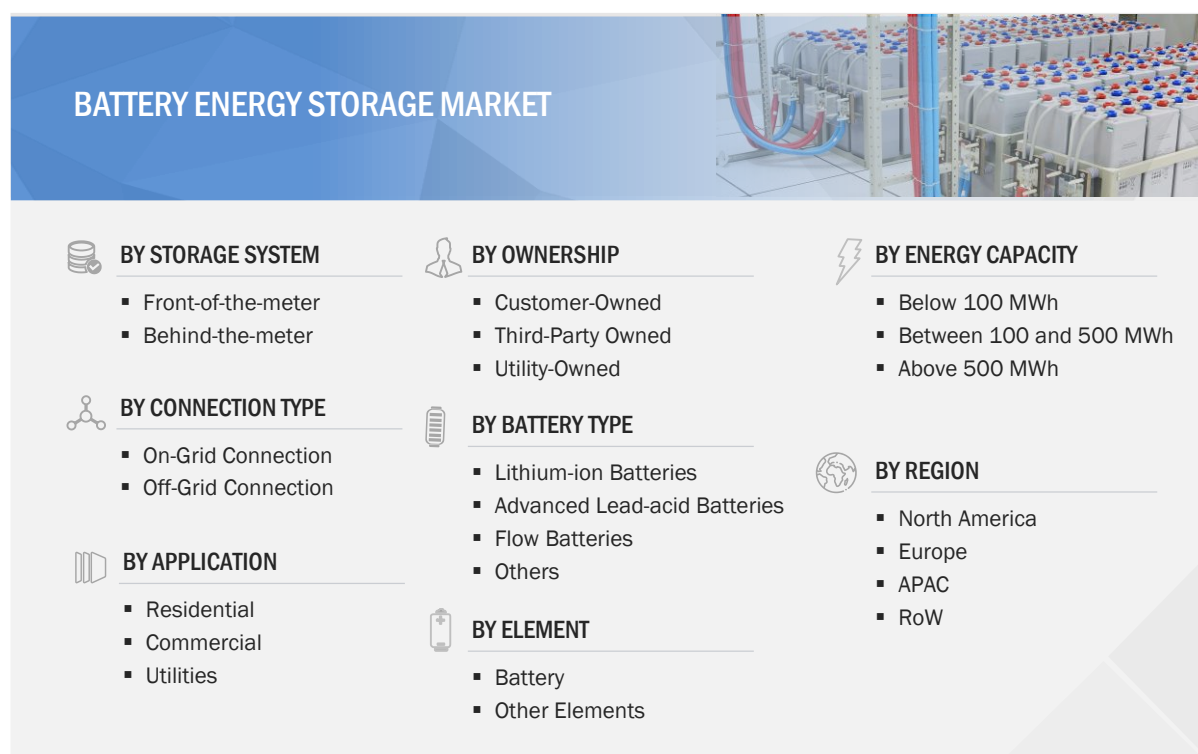
PARTICULARS	INCLUSION	EXCLUSION
Storage System	Storage systems include front-of-the-meter (FTM) systems and behind-the-meter (BTM) systems	NA
Element	Elements include battery and other elements. Other elements include hardware and others (engineering, procurement, and construction (EPC), and development services).	NA
Battery Type	Battery types include lithium-ion batteries, advanced lead-acid batteries, flow batteries, and others. Other batteries include sodium-sulfur batteries, nickel-cadmium batteries, nickel-metal hydride batteries, nickel-iron batteries, and flywheel batteries	Any other battery type is not considered in the study
Connection Type	Connection type includes on-grid connection and off-grid connection	NA
Ownership	The ownership segment includes customer-owned, third-party owned, and utility-owned	NA
Energy Capacity	The energy segment includes below 100 MWh, between 100 and 500 MWh, and Above 500 MWh	NA
Application	The application segment includes residential, commercial, and utilities	NA
Region	North America, Europe, APAC, and RoW (the Middle East & Africa and South America)	NA

1.3 STUDY SCOPE

1.3.1 MARKETS COVERED

The battery energy storage system market has been categorized based on storage system, element, battery type, connection type, ownership, energy capacity, application, and region. The following figure represents the segmentation.

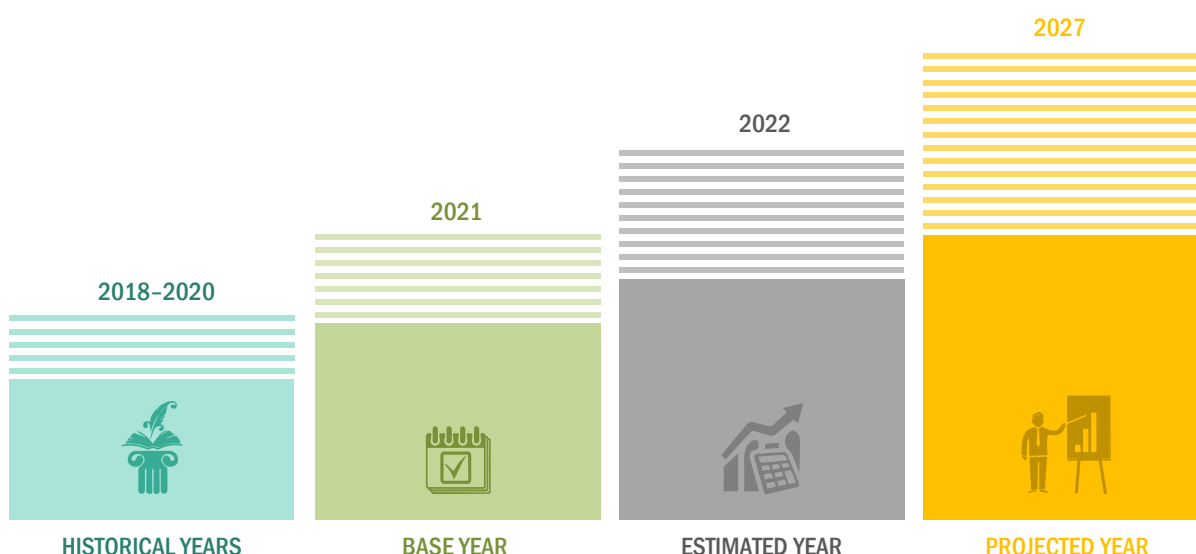
FIGURE 1 BATTERY ENERGY STORAGE SYSTEM MARKET: SEGMENTATION



Note: Other elements include hardware and others (engineering, procurement and construction (EPC), and development services). Other battery types include sodium-sulfur batteries, nickel-cadmium batteries, nickel-metal hydride batteries, nickel-iron batteries, and flywheel batteries. The Rest of Europe primarily comprises Italy, Spain, Norway, Sweden, and Switzerland. The Rest of APAC primarily includes New Zealand, Thailand, Singapore, Taiwan, Malaysia, and Indonesia. RoW includes South America and the Middle East & Africa.

Source: MarketsandMarkets Analysis

1.3.2 YEARS CONSIDERED



Note: The forecast period considered is from 2022 to 2027. The base year considered for the company profiles is 2021. Wherever the information for the base year was unavailable, information of the previous year has been considered.

1.4 CURRENCY

The currency used in the report is the US dollar (USD), with market size indicated in USD million and billion.

- Company revenues have been sourced from the annual reports of respective companies.
- For companies reporting their revenues in currency other than USD, the average annual currency exchange rate for that particular year has been used (website of Oanda Corporation) to convert the values to USD.

1.5 LIMITATIONS

- Industry players have kept quantitative information for a few of the market segments confidential. For such segments, the qualitative insights gathered during the study have been used to arrive at the market size in terms of value.
- The quantitative information of the storage system segment has not been given in this study.
- The size of the battery energy storage system market in RoW is derived from the size of markets in the Middle East & Africa and South America. However, the country-wise market sizes for Africa and South America regions have not been included in the study owing to data granularity.

1.6 STAKEHOLDERS

- Government bodies and policymakers
- Industry organizations, forums, alliances, and associations
- Market research and consulting firms
- Raw material suppliers and distributors
- Research institutes and organizations
- Original equipment manufacturers (OEMs) of battery energy storage systems
- Traders and suppliers

- Battery energy storage system manufacturers
- Energy storage solutions and service providers
- Energy storage research institutions
- Government and research organizations
- Grid operators
- Retail service providers
- Lithium-ion battery, electrochemical capacitor, and other energy storage products manufacturers
- Utility providers
- Power distribution and IT consultants
- Research organizations and consulting companies
- Technology investors
- Technology standard organizations, forums, alliances, and associations
- Associations and industrial bodies
- End-user industries
- Power bank manufacturers
- Power bank traders and suppliers

1.7 SUMMARY OF CHANGES

- In this version of the report, the market segmentation based on storage system (front-of-the-meter (FTM) behind-the-meter (BTM)) has been added. Qualitative information has been provided for this segment.
- An updated company evaluation quadrant for 25 players has been included in this version, along with updated market dynamics.
- Updated financial information of 25 companies has been included in the company profiles section in the graphical representation form until 2021/2020 for each listed company, along with the updated product portfolios of market players. This would help to analyze the present status of the profiled companies in terms of their financial strength, as well as their key revenue-generating region/country/business segment.
- This version of the report includes strategies adopted by the top five players in the global battery energy storage system market under the “right-to-win” section.
- This version includes recent market developments of the profiled players from February 2020 to January 2022.
- The impact of the COVID-19 pandemic has been considered while estimating and forecasting the global market size (in terms of value) and is reflected throughout the market segments in the report.
- This latest version of the report also includes the analysis of the COVID-19 impact on each region.
- This report includes pricing analysis, trade analysis, technology analysis, patent analysis, case study analysis, as well as a few recent changes in the regulatory framework, which might impact the growth of the battery energy storage system market.

- The research study consists of the market share analysis of the top players in the battery energy storage system market.
- Company Evaluation Quadrant: This new section is an output of a comprehensive study of the key vendors offering battery energy storage systems. The top 25 vendors were shortlisted from a list of 50+ vendors and were evaluated based on market share/ranking and product footprint. They have been rated and positioned on a 2x2 quadrant, referred to as Company Evaluation Quadrant, categorizing them as “Star,” “Emerging Leader,” “Pervasive,” and “Participant” companies. The section also includes the small and medium enterprise (SME) evaluation quadrant, categorizing them as “Progressive Company,” “Responsive Company,” “Dynamic Company,” and “Starting Block.”

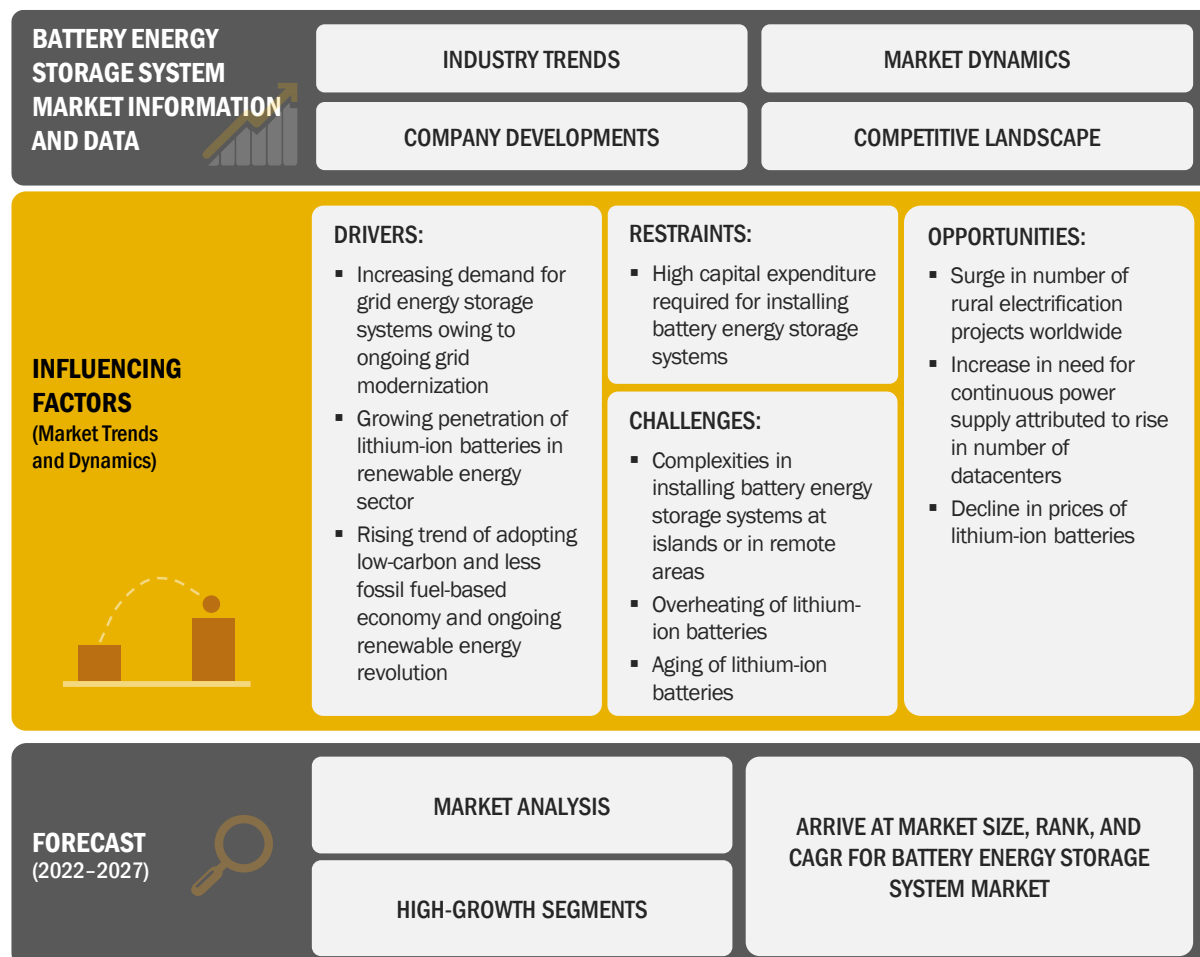
2 RESEARCH METHODOLOGY

2.1 RESEARCH DATA

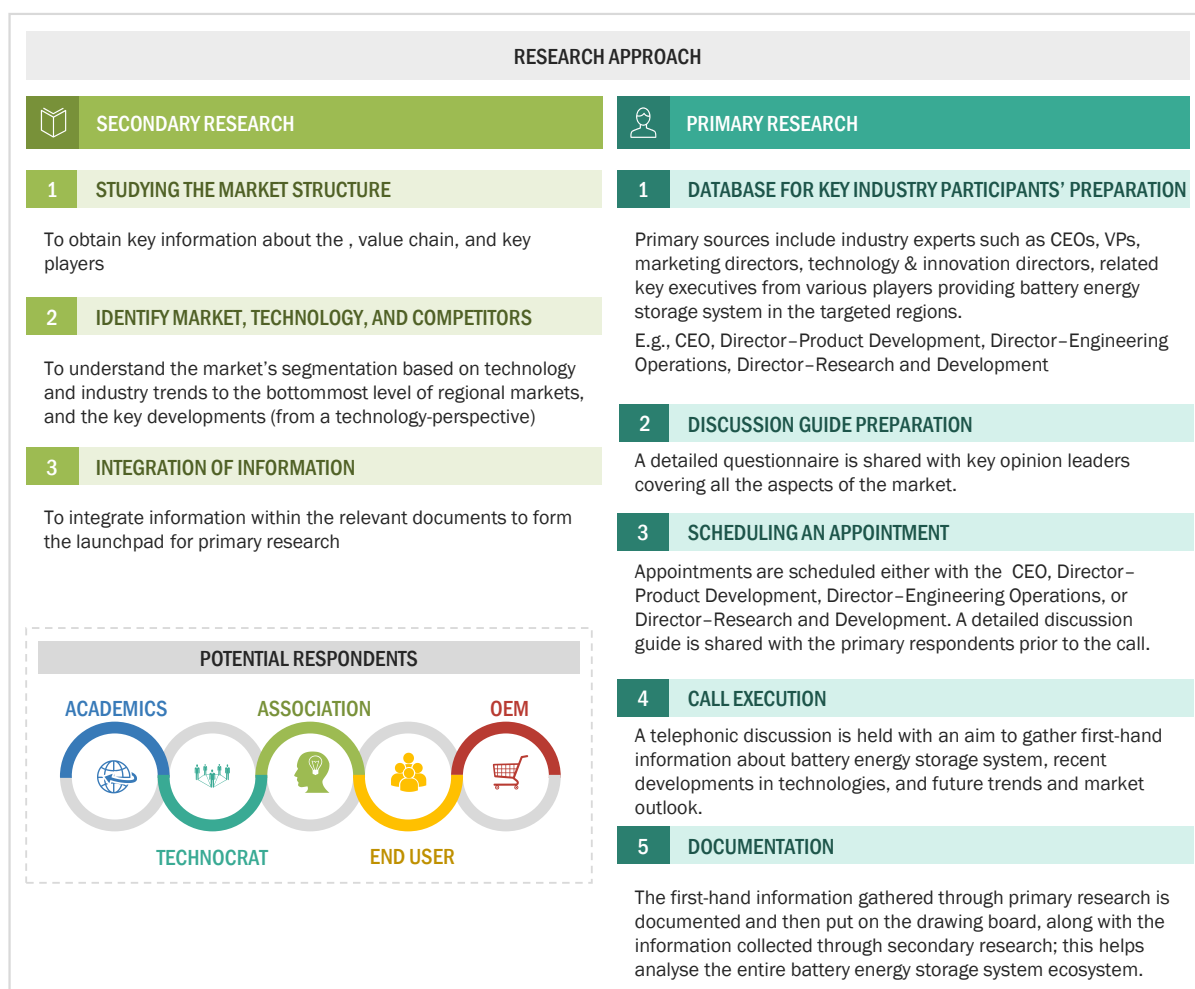
This research study involved the extensive use of secondary sources, directories, and databases (Factiva, Oanda, and OneSource) for identifying and collecting information useful for the technical and commercial study of the battery energy storage system market. In-depth interviews have been conducted with various primary respondents, including experts from core and related industries and preferred manufacturers, to obtain and verify critical qualitative and quantitative information, as well as to assess growth prospects. They have been interviewed to understand the impact of COVID-19 on the ecosystem of battery energy storage systems. Key players in the battery energy storage system market have been identified through secondary research, and their market rankings have been determined through primary and secondary research. This research included studying annual reports of top players and interviewing key industry experts such as CEOs, directors, and marketing executives.

In the primary research process, various key correspondents from the supply and demand sides have been interviewed to obtain the qualitative and quantitative information relevant to this report. The impact and implication of COVID-19 on manufacturing industries, as well as on battery energy storage systems, have also been discussed. Primary sources from the supply side include experts such as CEOs, vice presidents, marketing directors, business development executives, application users, and related executives from various key companies and organizations operating in the ecosystem of the battery energy storage system market.

FIGURE 2 BATTERY ENERGY STORAGE SYSTEM MARKET: RESEARCH DESIGN



2.1.1 SECONDARY AND PRIMARY RESEARCH



Source: MarketsandMarkets Analysis

2.1.2 SECONDARY DATA

The secondary sources referred to for this research study included government sources; corporate filings such as annual reports, investor presentations, and financial statements; and trade, business, and professional associations. Secondary data was collected and analyzed to arrive at the overall market size, which was further validated through primary research.

2.1.2.1 List of major secondary sources

SOURCE	WEB LINK
Energy Storage Association	https://energystorage.org/
International Battery and Energy Storage Alliance (IBESA)	https://www.ibesalliance.org/
California Energy Storage Alliance	https://www.storagealliance.org/
U.S. Energy Information Administration	https://www.eia.gov/

2.1.2.2 Key data from secondary sources

PARAMETER	SOURCE
 MARKET SIZE (In Terms of Value)	▪ Company Financials ▪ Magazines ▪ Journals ▪ Press Releases ▪ Paid Databases ▪ MarketsandMarkets Data Repository
 REVENUE OF COMPANIES	▪ Annual Reports ▪ Company Websites ▪ Public Databases ▪ MarketsandMarkets Data Repository
 QUALITATIVE INFORMATION (Market Dynamics)	▪ Company Websites ▪ Annual Reports ▪ Press Releases ▪ MarketsandMarkets Data Repository
 COVID-19 RELATED INFORMATION	▪ Company Websites ▪ Press Releases

2.1.3 PRIMARY DATA

In the primary research process, extensive primary research has been conducted after understanding and analyzing the current scenario of the battery energy storage system market through secondary research. Several primary interviews have been conducted with the market experts from both demand and supply sides across four major regions, namely, North America, Europe, APAC, and RoW. The primary sources from the supply side included various industry experts, such as chief experience officers (CXOs), vice presidents (VPs), directors from business development, marketing, product development/innovation teams, and related key executives from companies offering battery energy storage system such as LG Energy Solutions Co., Ltd. (South Korea), Panasonic Corporation (Japan), SAMSUNG SDI Co., Ltd. (South Korea), and BYD Company Limited (China); industry associations; independent consultants and importers; distributors; and key opinion leaders.

Primary interviews were conducted to gather insights such as market statistics, revenue generated from products and services, market breakdowns, market size estimations, market forecasting, and data triangulation. Primary research also helped in understanding the various trends related to technology, application, vertical, and region. Stakeholders from the demand side, such as CIOs, CTOs, and CSOs and installation teams of customers/end users were interviewed to understand the buyer's perspective on the suppliers, products, service providers, and their current use of battery energy storage systems and the outlook of their businesses.

2.1.3.1 Primary interviews with experts

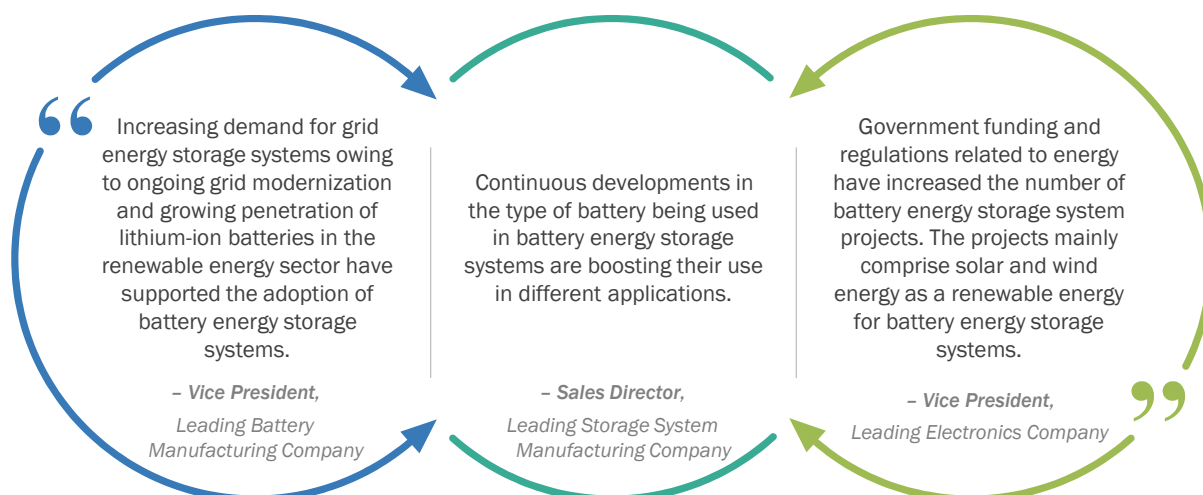
PARTICULARS	INTENDED PARTICIPANTS AND KEY OPINION LEADERS
Primary Interview Breakup	- Manufacturers and distributors of battery energy storage systems - Battery energy storage system applications, based on region - Industry consultants and experts - Government and regulatory bodies

2.1.3.2 Key data from primary sources

TYPE	PARAMETER	KEY DATA
GEOGRAPHIC SPLIT	- Size of the overall market and its sub-segments, 2021 - CAGR of the market in each region during the forecast period	Global battery storage market, by region–North America, Europe, APAC, and RoW
GLOBAL MARKET SIZE	- Global market size in 2021 - CAGR during the forecast period	- Global battery energy storage market - Battery energy storage system market, by storage system, element, battery type, connection type, ownership, energy type, application, and region
MARKET SPLIT	- By Storage System - By Element - By Battery Type - By Connection Type - By Ownership - By Energy Capacity - By Application	- Storage includes front-of-the-meter (FTM) and behind-the-meter (BTM) - Element includes battery and other elements. Other elements include hardware and others (Engineering, Procurement and Construction (EPC), and Development Services) - Battery type includes lithium-ion Batteries, advanced lead-acid batteries, flow batteries, and others. Others include sodium-sulfur batteries, nickel cadmium batteries, nickel metal hydride batteries, nickel iron batteries, and flywheel batteries - Connection type includes on-grid connection, and off-grid connection - Ownership segment includes customer-owned, third-party owned, and utility-owned - Energy Capacity segment includes below 100 MWh, between 100 and 500 MWh, and above 500 MWh - Application includes residential, commercial, and utilities

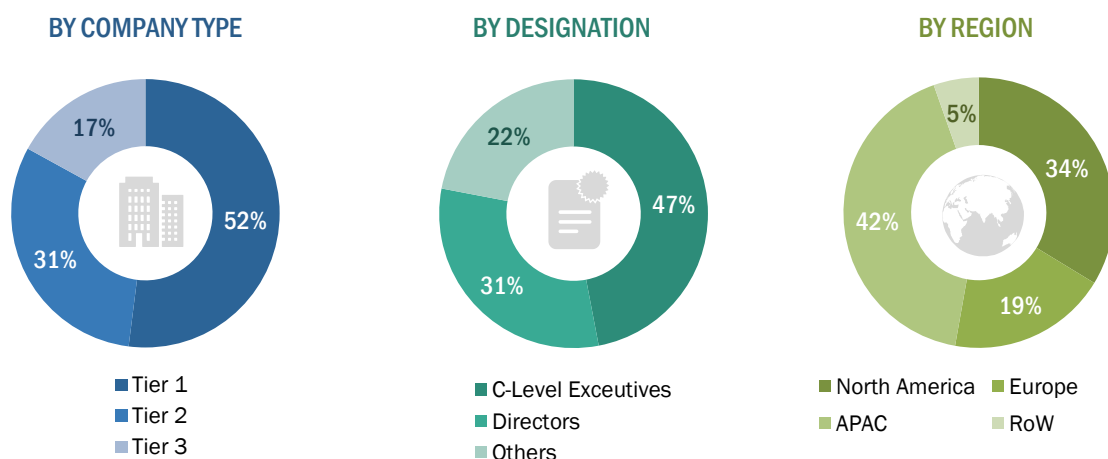
Source: Industry Experts and MarketsandMarkets Analysis

2.1.3.3 Key industry insights



Source: Industry Experts and MarketsandMarkets Analysis

2.1.3.4 Breakdown of primaries

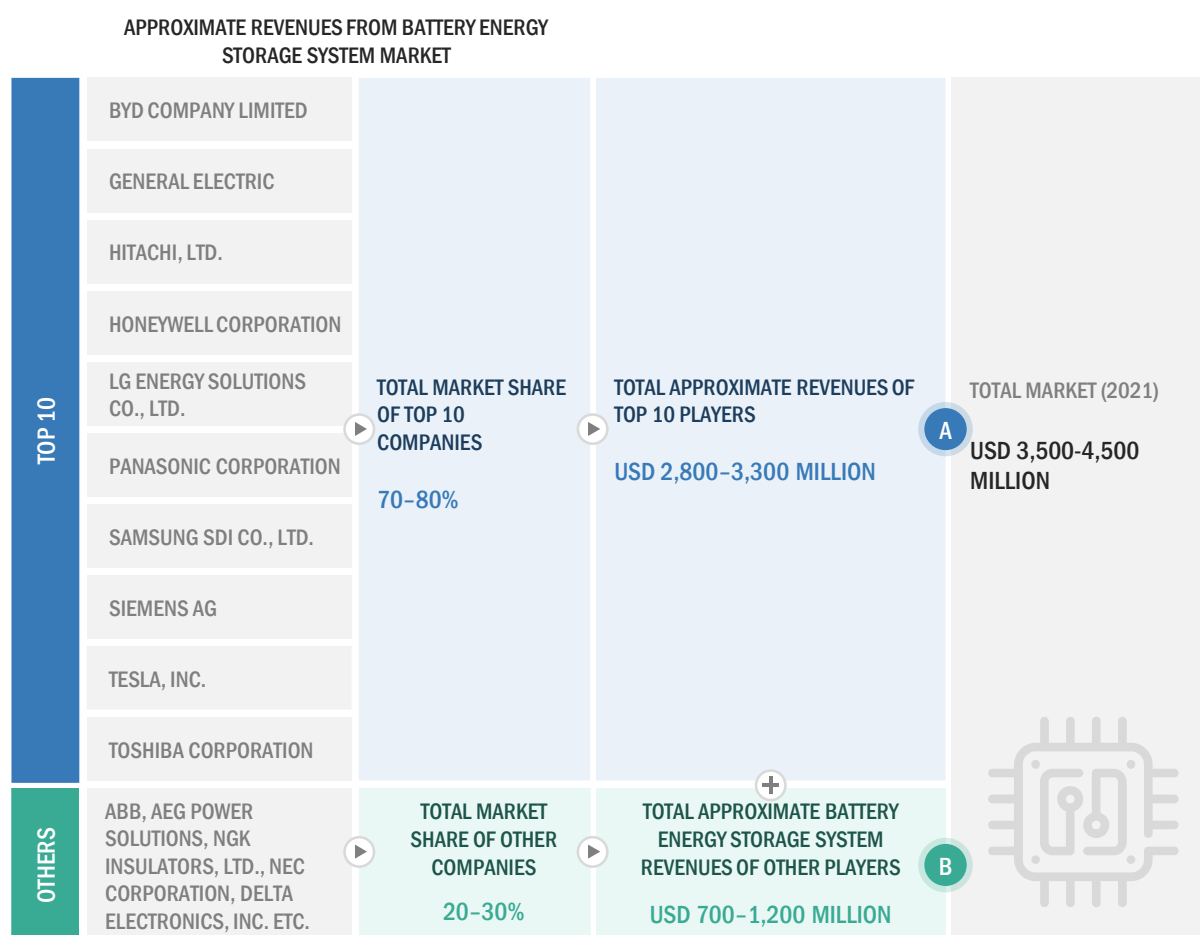


Note: RoW includes the Middle East & Africa and South America; Other designations include product managers, sales managers, and marketing managers. Three tiers of companies have been defined based on their total revenue as of 2021; tier 3: revenue lesser than USD 500 million; tier 2: revenue between USD 1 billion and USD 3 billion; and tier 1: revenue more than USD 3 billion.

2.2 MARKET SIZE ESTIMATION

Multiple approaches were adopted for the estimation and forecasting of the battery energy storage system market. The first approach involved the estimation of the market size by the summation of the company's revenues generated through the sales and services of battery energy storage systems.

FIGURE 3 MARKET SIZE ESTIMATION METHODOLOGY: APPROACH (SUPPLY-SIDE):
REVENUE OF PRODUCTS/SOLUTIONS/SERVICES OF BATTERY ENERGY STORAGE
MARKET SIZE



In this approach for market estimation, we identified the key companies, such as BYD Company Limited (China), General Electric (US), Hitachi, Ltd. (Japan), Honeywell Corporation (US), LG Energy Solutions CO., Ltd. (South Korea), Panasonic Corporation (Japan), SAMSUNG SDI Co., Ltd. (South Korea), Siemens AG (Germany), Tesla, Inc. (US), Toshiba Corporation (Japan), ABB (US), among others offering battery energy storage systems and contributing to almost 70 to 80% of the global market. After confirming these companies through primary interviews with industry experts, we estimated their total revenue through annual reports, SEC filings, and paid databases. These companies' revenues pertaining to the business units (BUs) that offer battery energy storage systems were identified through similar sources. Then, through primaries, we collected data regarding the revenue generated through specific battery energy storage systems. The collective revenue of key companies that offer battery energy storage systems comprised 20 to 30% of the market, which was again confirmed through primary interviews with industry experts. With the assumption that the rest of the market is contributed by smaller players (part of the unorganized market), the sizes of the organized market (70 to 80%) and the unorganized market (20 to 30%) collectively were assumed to be the market size of the global battery energy storage system market for FY 2021.

We identified the BUs catering to battery energy storage systems and segregated the product and services that contribute to battery energy storage systems. We estimated the revenues of products contributing to the battery energy storage system market and validated them with company executives through primary research. We performed similar analyses for all companies in the battery energy storage system market.

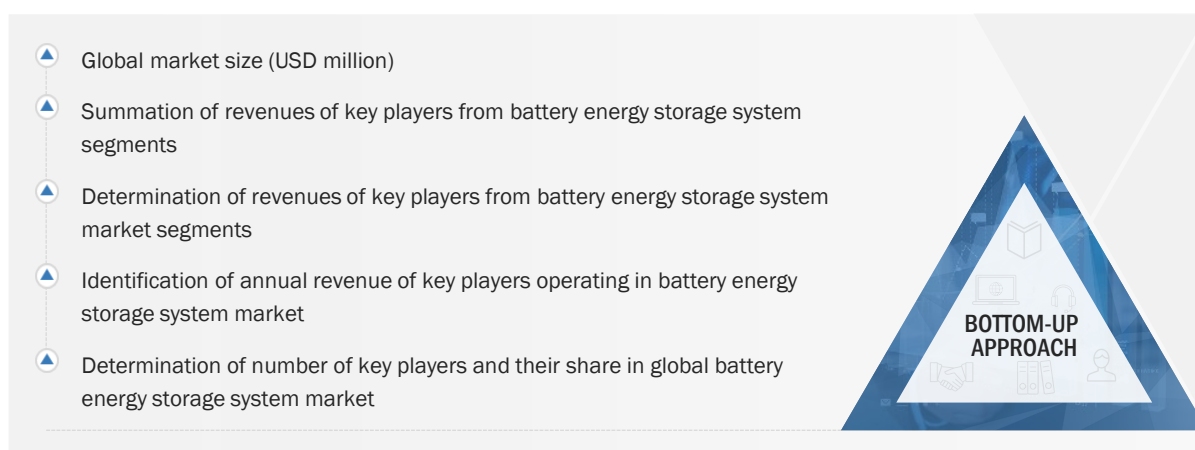
Further both top-down and bottom-up approaches have been used to estimate and validate the size of the battery energy storage system market and other dependent submarkets. The leading players in the market have been identified through secondary research, and their market shares in the key regions have been determined through primary and secondary research. This entire procedure includes the study of the annual and financial reports of the top players and extensive interviews with the industry experts such as chief executive officers, vice presidents, directors, and marketing executives for the key insights. All percentage shares, splits, and breakdowns have been determined using secondary sources and verified through primary sources. All the possible parameters that can affect the markets covered in this study have been accounted for, viewed in detail, verified through primary research, and analyzed to get the final quantitative and qualitative data. This data has been consolidated and supplemented with detailed inputs and analysis from MarketsandMarkets and presented in this report. The three figures below represent the overall market size estimation process employed for this study.

2.2.1 BOTTOM-UP APPROACH

The bottom-up approach has been used to arrive at the overall size of the battery energy storage system market from the revenues of the key players and their share in the market. Calculations based on revenues of the key companies identified in the market led to the estimation of their overall market size.

2.2.1.1 Approach for arriving at market size by bottom-up analysis (demand side)

- Identifying entities in the battery energy storage system market that influence the entire market.
- Analyzing major manufacturers of battery energy storage systems and studying their portfolios and understanding different types of battery energy storage systems.
- Analyzing trends pertaining to the use of different types of battery energy storage systems for different kinds of applications.
- Tracking the ongoing and upcoming developments in the market, such as investments made, R&D activities, product launches, agreements, and partnerships, and forecasting the market based on these developments and other critical parameters.
- Carrying out multiple discussions with key opinion leaders to understand different types of battery energy storage systems, applications, and recent trends in the market, thereby analyzing the breakup of the scope of work carried out by major companies.
- Arriving at the market estimates by analyzing revenues of companies generated and then combining the same to get the market estimate.
- Segmenting the overall market into various other market segments.
- Verifying and cross-checking the estimate at every level from discussion with key opinion leaders such as CXOs, directors, and operations managers, and finally with the domain experts at MarketsandMarkets.
- Studying various paid and unpaid sources of information such as annual reports, press releases, white papers, and databases.

FIGURE 4 BATTERY ENERGY STORAGE SYSTEM MARKET: BOTTOM-UP APPROACH

2.2.2 TOP-DOWN APPROACH

In the top-down approach, the overall market size has been used to estimate the size of the individual markets (mentioned in the market segmentation) through percentage splits from secondary and primary research. For the calculation of the size of specific market segments, the size of the most appropriate immediate parent market has been used to implement the top-down approach. The bottom-up approach has also been implemented for the data obtained from secondary research to validate the market size of various segments. The market share for each company has been estimated to verify revenues used earlier in the bottom-up approach. With the data triangulation procedure and validation of the data through primaries, the sizes of the parent market and each market have been determined and confirmed. The data triangulation procedure used for this study has been explained in the next section.

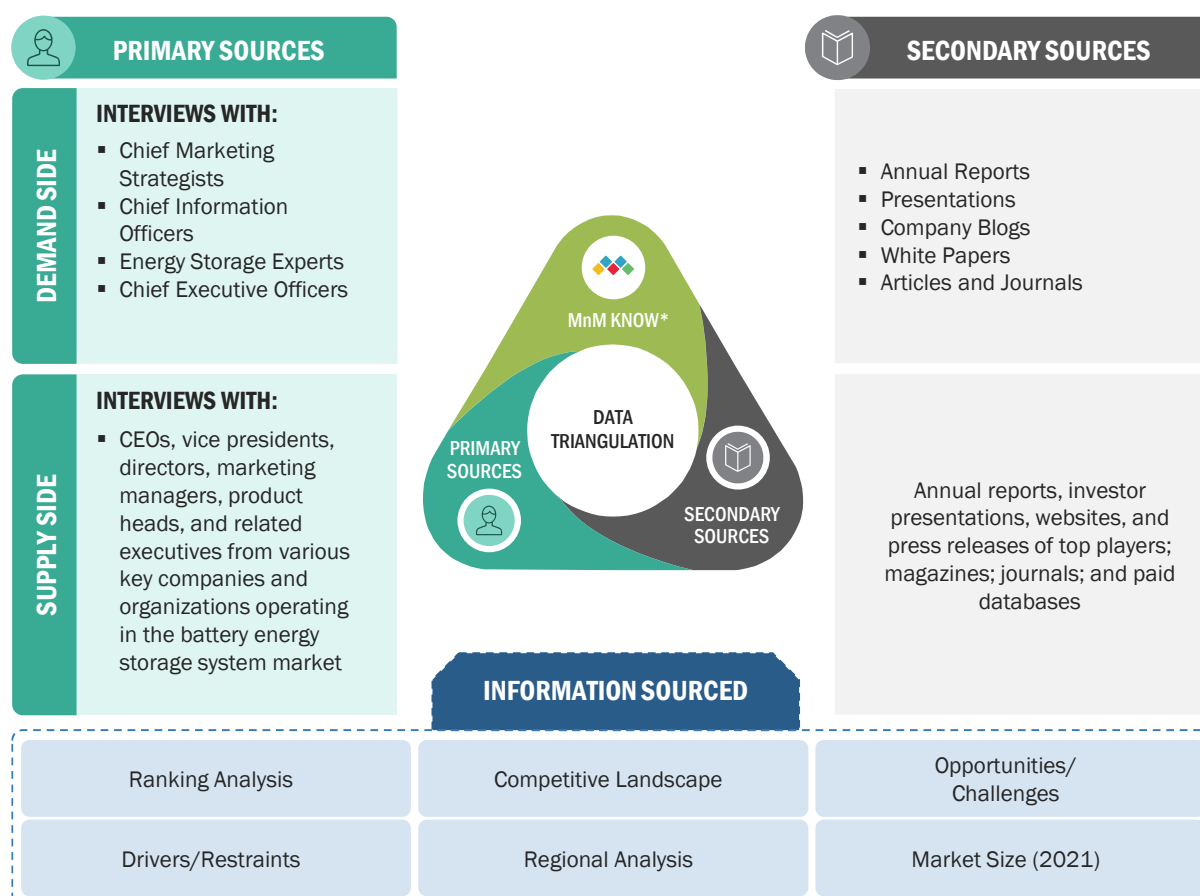
2.2.2.1 Approach for capturing market size by using top-down analysis (supply side)

- Focusing on top-line investments and expenditures made in the battery energy storage system ecosystem.
- Building and developing information related to the revenue generated through offering key types of battery energy storage systems.
- Conducting multiple on-field discussions with key opinion leaders from the major companies involved in the development of battery energy storage systems.
- Calculating the market size considering revenues generated by major players through the sale of battery energy storage systems.
- Further segmenting is based on R&D activities and key developments in major market areas.
- Further segmenting is based on mapping the use of battery energy storage systems for different applications.
- Estimating the geographic split using secondary sources based on various factors, such as the number of players in a specific country or region and the battery energy storage system offerings of these players.
- The impact of COVID-19 on the steps mentioned above has also been considered.

FIGURE 5 BATTERY ENERGY STORAGE SYSTEM MARKET: TOP-DOWN APPROACH

2.3 MARKET BREAKDOWN AND DATA TRIANGULATION

After arriving at the overall size of the battery energy storage system market from the market size estimation processes explained above, the total market has been split into several segments and subsegments. The market breakdown and data triangulation procedures have been employed, wherever applicable, to complete the overall market engineering process and arrive at the exact statistics for all segments and subsegments. The data have been triangulated by studying various factors and trends. Along with this, the market size has been validated using top-down and bottom-up approaches.

FIGURE 6 DATA TRIANGULATION

MnM KNOW* stands for MarketsandMarkets' Knowledge Asset Management framework. In this context, it stands for the existing market research knowledge repository of over 5,000 granular markets, our flagship competitive intelligence and market research platform "RT," subject matter experts, and independent consultants. MnM KNOW acts as an independent source that helps us validate the information gathered from primary and secondary sources.

2.4 RESEARCH ASSUMPTIONS

The following assumptions have been considered to complete the overall market engineering of the battery energy storage system market.

PARAMETER	DESCRIPTION
GLOBAL ECONOMIC OUTLOOK	The global economy has a direct impact on any market. Regarding the global economic landscape, the impact of COVID-19 has been taken into consideration while forecasting the battery energy storage system market size in this study.
POLITICAL OUTLOOK	For the battery energy storage system market, the global political environment has been assumed to remain stable during the forecast period.
CURRENCY VALUE	Forecasts have been done with values calculated under the standard assumption that the value of USD would remain constant during the forecast period.
DATA AUTHENTICITY	Segment-specific information and company revenues have been derived from the annual reports of the respective companies. The information provided in the annual reports has been assumed to be authentic.
GLOBAL ECONOMIC SLOWDOWN DUE TO COVID-19 PANDEMIC	In the study, the impact of the COVID-19 pandemic has been considered while estimating the market size. According to our primary respondents, the impact of COVID-19 is moderate. We have not factored the extension of lockdowns and economic downturns caused by the second or third waves of the pandemic in various countries worldwide.
CLASSIFICATION OF SEGMENTS	All market segments and sub-segments listed in the report through various forms of classification have been considered to be mutually exclusive of each other.

2.5 RISK ASSESSMENT

TABLE 1 LIMITATIONS AND ASSOCIATED RISKS

PARTICULARS	COVERED	RISK
Element	The report covers batteries and other elements. Other elements include hardware and others (Engineering, Procurement and Construction (EPC), and Development Services)	NA
Battery Type	The report covers lithium-ion batteries, advanced lead-acid batteries, flow batteries, and others. Others include sodium-sulfur batteries, nickel-cadmium batteries, nickel-metal hydride batteries, nickel-iron batteries, and flywheel batteries.	NA
Connection Type	The report covers on-grid connection and off-grid connection	NA
Ownership	The report covers customer-owned, third-party owned, and utility-owned	NA
Energy Capacity	The report covers below 100 MWh, between 100 and 500 MWh, and above 500 MWh	NA
Application	The report covers residential, commercial, and utilities	High risk due to declining growth of the utility sector, which may affect the total growth rate of the battery energy storage system market. However, the high demand for battery energy storage systems for residential applications is expected to boost the market growth during and after the COVID-19 pandemic.
Region	The scope of the report includes North America, Europe, APAC, and ROW.	NA
Global Market Size	The global market size is based on the adoption of battery energy storage systems in various industries.	NA
Country-wise Market Size	The report does not cover the country-level quantitative information for the Middle East and South America due to the limited availability of information from primary and secondary sources.	NA

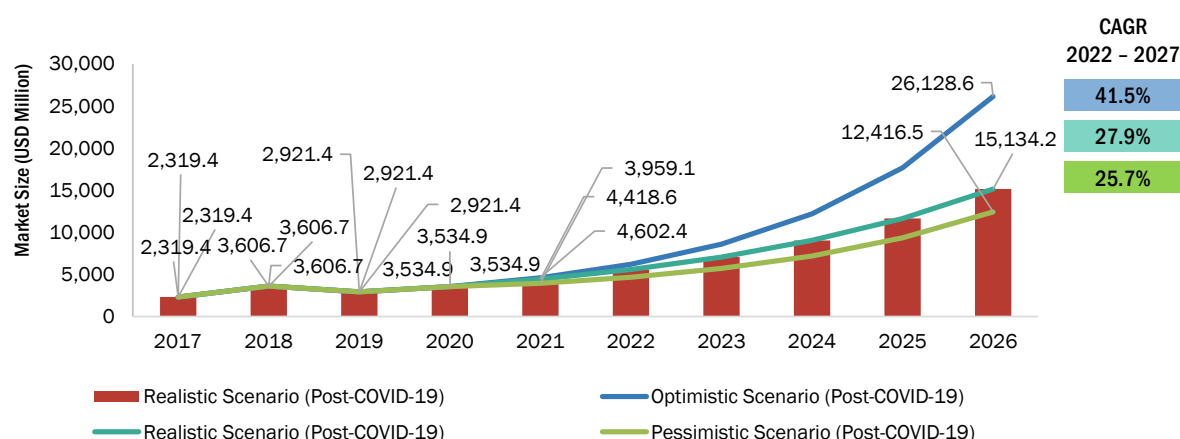
3 EXECUTIVE SUMMARY

A battery energy storage system (BESS) comprises an energy storage medium and other ancillary equipment to protect and control the system. The system quickly injects and absorbs the active power to and from the electrical grid. It is primarily used to reduce peak demand charges, integrate renewable sources, regulate voltage and frequency, and provide a backup power supply. The demand for battery energy storage systems is increasing owing to the growing need for grid-connected solutions and declining prices of lithium-ion batteries. Based on battery type, the battery energy storage system market has been classified into lithium-ion batteries, advanced lead-acid batteries, flow batteries, and other batteries, which include sodium-based batteries, nickel-cadmium batteries, nickel-metal hydride, nickel-iron batteries, and flywheel batteries.

The battery energy storage system market was valued at USD 3,534.9 million in 2021 and is projected to reach USD 15,134.2 million by 2027, growing at a CAGR of 27.9% between 2022 and 2027. The major opportunities for the market players are the increasing number of rural electrification projects worldwide, the growing need for continuous power supply in critical data centers, and the declining prices of lithium-ion batteries. However, high complexity due to the installation of battery energy storage systems in islands and remote areas, overheating and aging of lithium-ion batteries, and aging of lithium-ion batteries are expected to be major challenges for the players operating in the market. Major players operating in the market include SAMSUNG SDI Co., Ltd. (South Korea), BYD Company Limited (China), Panasonic Corporation (Japan), LG Energy Solutions Co., Ltd. (South Korea), Siemens AG (Germany), Tesla, Inc. (US), among others. The battery energy storage system market has been segmented based on storage system, battery type, element, ownership, connection type, energy capacity, application, and region.

The COVID-19 pandemic has affected the battery energy storage system market during the initial phases of 2020. The rapid spread of the virus has canceled, postponed, or suspended the operational projects as well as halted the export of batteries and other components, which has impacted the production of battery energy storage systems during the first two quarters of 2020.

FIGURE 7 COVID-19 IMPACT ANALYSIS ON BATTERY ENERGY STORAGE SYSTEM MARKET



Source: Investor Presentations, Expert Interviews, Industry Journals, Company Websites, Annual Reports, and MarketsandMarkets Analysis

3.1 REALISTIC SCENARIO

In the realistic scenario, the market is projected to reach USD 4,418.6 million in 2022. The COVID-19 pandemic has significantly impacted the value chain of the battery energy storage system market. It has also lowered the installation of battery energy storage systems in countries, which are highly impacted by the spread of the pandemic. The largest battery energy storage system projects were halted or postponed in various countries owing to the adverse effects of the pandemic, which has impacted the growth of the market. Various applications are witnessing low demand, which is expected to continue for the short term with the global slowdown and declining purchasing power. Countries such as the US, Germany, and South Korea accounted for a significant share in the global battery energy storage system market; however, the year-on-year growth is estimated to decrease in 2020 owing to the negative impact of the pandemic. The utilities and commercial applications are witnessing the increasing deployment of battery energy storage systems during the post-COVID-19 period, which is expected to drive market growth.

3.2 PESSIMISTIC SCENARIO

In the pessimistic scenario, the battery energy storage system market may decline further, and its market size may account for USD 3,959.1 million in 2022. The COVID-19 pandemic has led to operational disruptions for the companies operating in hard-hit countries such as the US, France, China, and India, which has delayed their ability to finalize financial statements. Besides, the pandemic has caused disruptions in the supply chain, which has also led to revenue losses for several companies. Further, in the pre-COVID-19 period, governments were supporting companies in the battery energy storage system ecosystem, either financially or through favorable policies. However, governments are now concentrating and spending on healthcare services to contain the pandemic. This may lead to less government support, and the market for battery energy storage systems may decline to the worst state and grow at a low CAGR of 25.7% from 2022 to 2027.

3.3 OPTIMISTIC SCENARIO

In the optimistic scenario, the battery energy storage system market size is estimated to reach USD 26,128.6 million in 2027, from the market size of USD 4,602.4 million in 2022. According to the estimates, applications such as utilities, commercial, and residential may rebound once the lockdown across countries is lifted. The demand for battery energy storage systems has significantly reduced owing to the reduction in demand for electricity, due to the shutdown of commercial facilities and manufacturing plants across the world in the first two quarters of 2020. However, the resumption of production activities and ease in the COVID norms have increased the electricity demand and thus the demand for the installation of battery energy storage systems for various operational and upcoming projects post the pandemic period. The demand rose during the second half of 2020, making up for the revenue loss in the first half. Thus, the battery energy storage system market may grow at a CAGR of 41.5% from 2022 to 2027. However, the market value may not reach that of the pre-COVID-19 value until 2027.

TABLE 2 BATTERY ENERGY STORAGE SYSTEM MARKET, IN TERMS OF VALUE, 2018–2021

Particulars	2018	2019	2020	2021
Total Market Size (USD Million)	2,319.4	3,606.7	2,921.4	3,534.9
Total Market Size (GW Units)	3.3	3.3	5.2	6.3

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 3 BATTERY ENERGY STORAGE SYSTEM MARKET, IN TERMS OF VALUE, 2022–2027

Particulars	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Total Market Size (USD Million)	4,418.6	5,564.3	7,060.6	9,029.8	11,641.7	15,134.2	27.9%
Total Market Size (GW Units)	8.0	10.2	13.2	17.1	22.4	29.6	29.9%

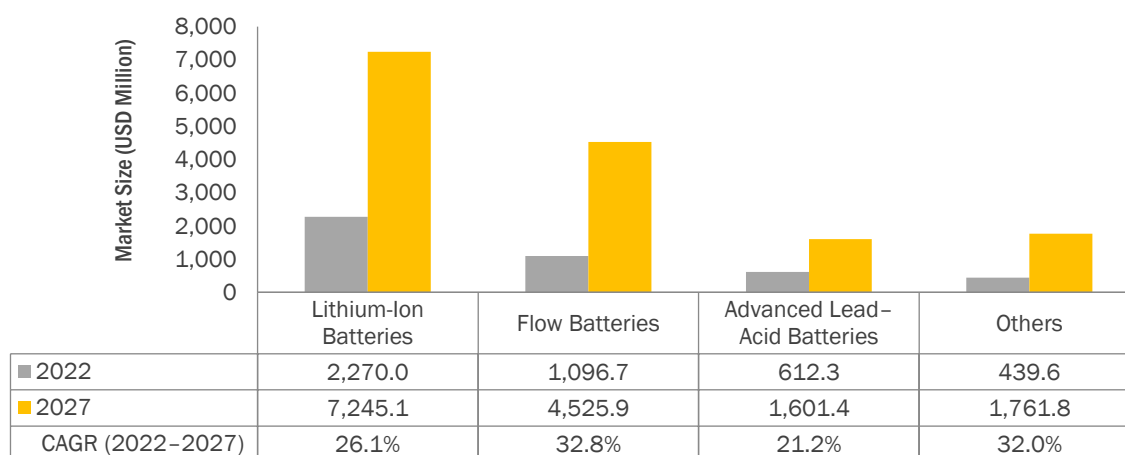
Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The battery energy storage system market was valued at USD 3,534.9 million in 2021 and is projected to reach USD 15,134.2 million by 2027; it is expected to register a CAGR of 27.9% from 2022 to 2027. The battery energy storage system market registered a shipment of 6.3 GW units in 2021 and is projected to reach 29.6 GW units by 2027, at a CAGR of 29.9% from 2022 to 2027. The key factors driving the battery energy storage system market growth include the increasing demand for grid energy storage systems owing to ongoing grid modernization, growing penetration of lithium-ion batteries in the renewable energy sector, and rising trend of adopting low-carbon and less fossil fuel-based economy and ongoing renewable energy revolution. However, the high capital expenditure required for installing battery energy storage systems may hamper the growth of the market.

APAC held a major share of the battery energy storage system market in 2021 and is expected to register an impressive CAGR during the forecast period. The supply chain of the energy industry was highly impacted due to COVID-19, particularly renewable energy technologies and battery energy storage systems. The key components of battery energy storage systems are mainly manufactured in Asia Pacific. The COVID-19 situation has exposed the overdependency on the region for key raw materials, especially China. The businesses of the energy storage system manufacturers in the US, Germany, and Australia have been adversely affected by the shutdown of operations in China and other countries of Asia Pacific. It has also resulted in limited production leading to a serious decline in business inputs. The recovery depends on government assistance as well as the level of corporate debt and how the companies and markets cope with the low demand. It is expected that the demand would increase in the coming years, and the market would recover by 2022. The deployment of battery energy storage in upcoming projects in the utility sector will support the recovery of the market. After the recovery phase, the market is expected to grow at a moderate rate until 2025.

FIGURE 8 LITHIUM-ION BATTERIES TO HOLD LARGEST SIZE OF BATTERY ENERGY STORAGE SYSTEM MARKET FROM 2022 TO 2027

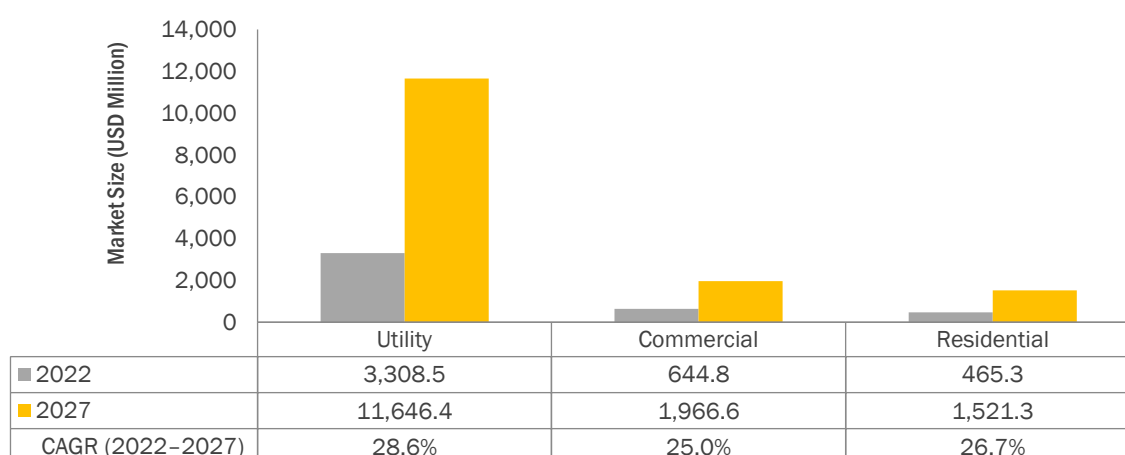


Note: Other battery types include sodium-based batteries, nickel-cadmium batteries, nickel-metal hydride, nickel-iron batteries, and flywheel batteries.

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

The lithium-ion battery energy storage system market was valued at USD 2,270.0 million in 2022 and is projected to reach USD 7,245.1 million by 2027; it is expected to grow at a CAGR of 26.1% from 2022 to 2027. The lithium-ion battery exhibits high energy and power density, which leads to low standby losses and high life expectancy. Due to these characteristics, lithium-ion batteries are likely to account for a major share of the battery energy storage system market by 2027. The lithium-ion-based battery energy storage systems are used in the utility application to address the demand during peak time as these systems offer high response time and can store excess energy. Also, lithium-ion-based battery energy storage systems support frequency regulation, reduction in demand charge, grid-buffering, and renewable integration, which are suitable for on-grid connections. The COVID-19 pandemic has impacted the supply of lithium-ion batteries and raw materials from China from the start of January 2020. Major battery manufacturers, including General Electric (US), Panasonic Corporation (Japan), and Samsung SDI Co., Ltd. (South Korea), have shifted their focus on making healthcare equipment such as masks and sanitizers to combat the spread of COVID-19.

The market for the flow battery type was valued at USD 1,096.7 million in 2022 and is projected to reach USD 4,525.9 million by 2027; it is expected to grow at the highest CAGR of 32.8% from 2022 to 2027. The most popular flow batteries include vanadium redox flow batteries, iron-chromium (ICB) flow batteries, and zinc-bromine (ZNBR) flow batteries. Compared with lithium-ion batteries, vanadium redox flow batteries (VRB) are non-flammable and environmentally friendly, have a long lifespan, and maintain 90% of their capacity over 20 years, resulting in the reduction of the total cost of ownership. A new generation of redox (reduction-oxidation) flow batteries in battery energy storage systems are much better suited for medium- and large-scale applications where heavy use, long-term cycling, durability, and safety are paramount. Their application areas include campuses, community service centers, commercial and industrial sites, and utility grids. The utility sector holds the highest number of operational flow battery projects to date worldwide.

FIGURE 9 UTILITY APPLICATION TO HOLD LARGEST MARKET SHARE FROM 2022 TO 2027

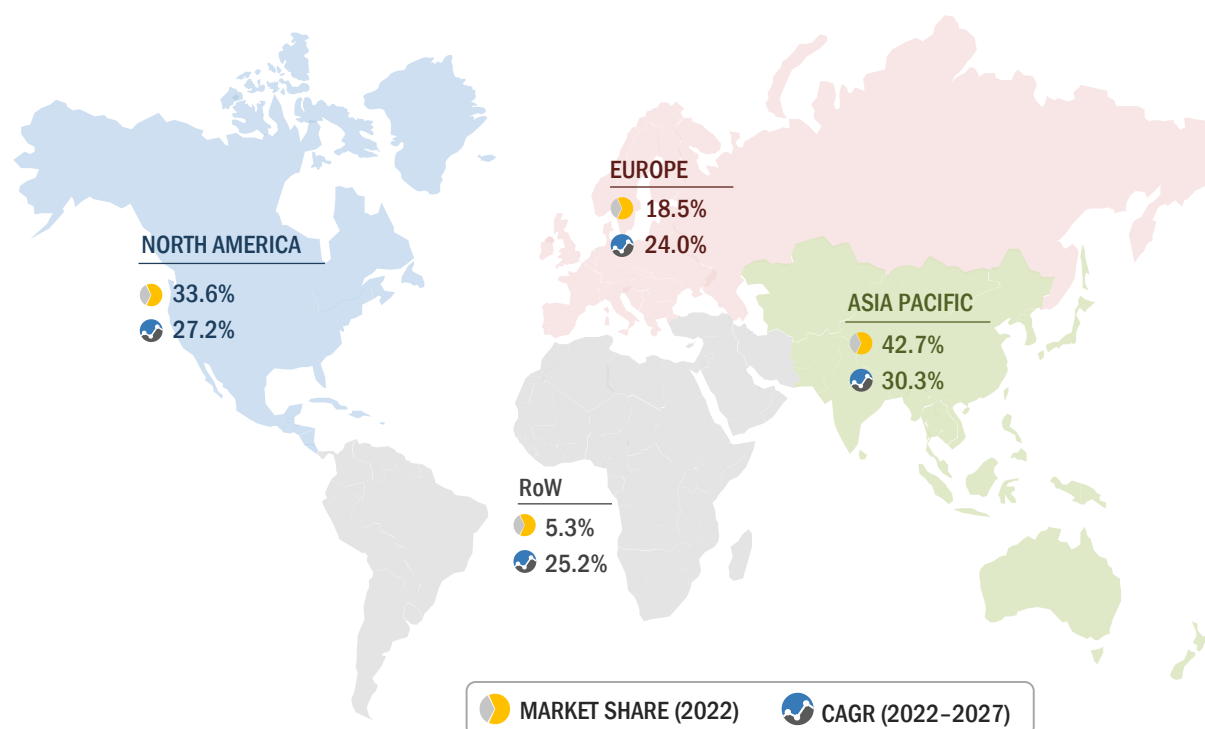
Note: The percentages in the figure represent the CAGR values during the forecast period.

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

The battery energy storage system market for utilities was valued at USD 3,308.5 million in 2022 and is projected to reach USD 11,646.4 million by 2027; it is expected to register the highest CAGR of 28.6% during the forecast period. The utility segment held the largest size of the battery energy storage system market in 2021 and is expected to register the highest growth during the forecast period. The need to fulfill the requirement of peak electricity demand is the major factor contributing to the growth of the utility segment. The flexibility offered by these systems to the grid energy storage to deal with the fluctuations in renewable energy, such as solar and wind energy, is increasing the adoption of battery energy storage systems. They provide frequency and voltage control services in case of sudden fluctuations in the energy supply. The battery energy storage systems enable grid operators to save electricity when there is a surplus of renewable energy.

The commercial application segment accounted for the second-largest share of the battery energy storage system market in 2021. Commercial & industrial customers pay a major portion of their electricity bills based on demand charges, and battery energy storage systems help them save on these bills through peak demand reduction. This is expected to drive the growth of the on-grid connection battery energy storage system market for commercial applications. The battery energy storage system market for commercial applications is expected to grow at a CAGR of 25.0% from 2022 to 2027.

FIGURE 10 BATTERY ENERGY STORAGE SYSTEM MARKET IN APAC TO GROW AT HIGHEST CAGR DURING FORECAST PERIOD



Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

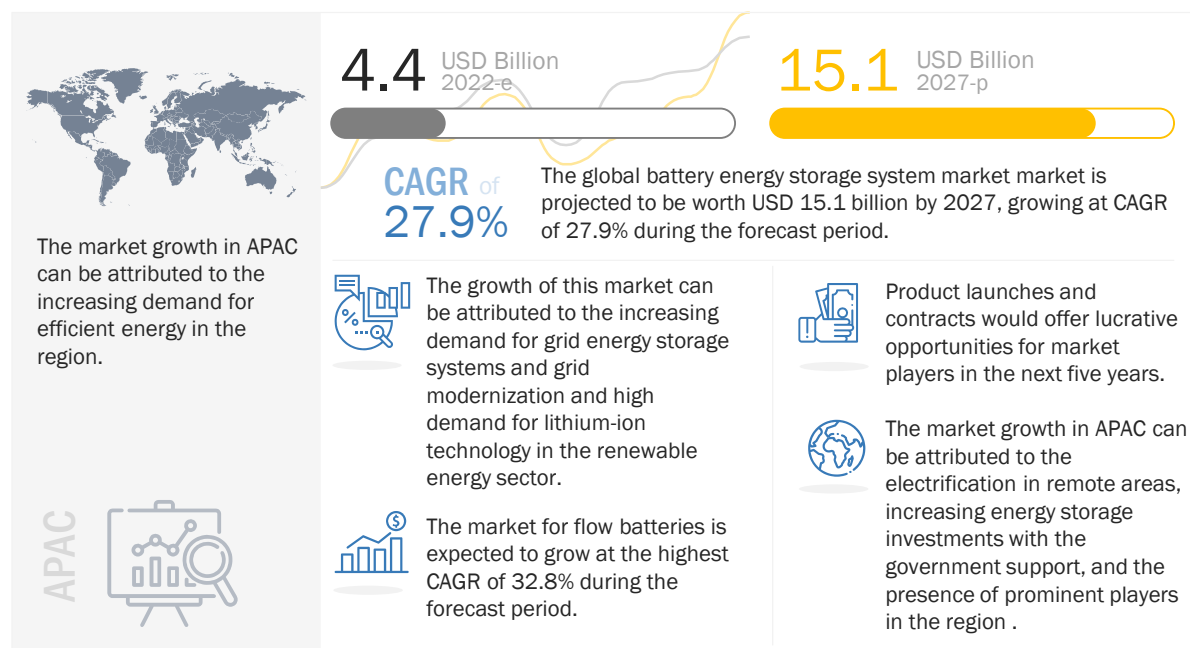
APAC accounted for the largest share of 42.7% of the battery energy storage system market in 2022. The market in APAC is valued at USD 1,884.8 million in 2022 and is projected to reach USD 7,090.6 million by 2027; it is expected to grow at the highest CAGR of 30.3% from 2022 to 2027. APAC is a hub for the battery energy storage system industry. The region accounted for the largest share of the battery energy storage system market in 2021 and is expected to register the highest CAGR during the forecast period. South Korea and China accounted for the highest battery energy storage installations globally. APAC is expected to witness plans for electrification in remote areas, which are mostly off-grid in several countries. The use of residential energy storage technology will enable to increase access to electricity for these remote and island communities in the region. Energy storage investments are expected to increase substantially within the region as the governments in growing economies are making new policies to improve the reliability and quality of power distribution facilities for residential customers. The governments of various countries in this region are focusing on minimizing the adverse effects of the energy sector on the environment. These factors are expected to increase the deployment of battery energy storage systems in residential and public utility applications.

The market in North America was valued at USD 1,482.5 million in 2022 and is projected to register a CAGR of 27.2% from 2022 to 2027. The growth of the North American market is dependent on the increasing demand for renewable energy storage systems in the residential, non-residential, and utility sectors. Companies such as Tesla, Inc., ABB, General Electric, and Honeywell are the key players in the region, which are focusing on installing and providing battery energy storage systems. According to the US Energy Information Administration, large-scale battery storage systems are increasingly being adopted across the power grids in the US. Over 90% of large-scale battery storage power capacity in the US was provided by batteries based on lithium-ion technology. Besides, North America held the largest share of the flow battery market worldwide. The region is expected to hold a significant market size in the near future as it is home to several active players as well as has the highest number of flow battery projects for various applications such as utilities, microgrid, industrial, commercial, and military.

4 PREMIUM INSIGHTS

4.1 ATTRACTIVE GROWTH OPPORTUNITIES FOR BATTERY ENERGY STORAGE SYSTEM MARKET

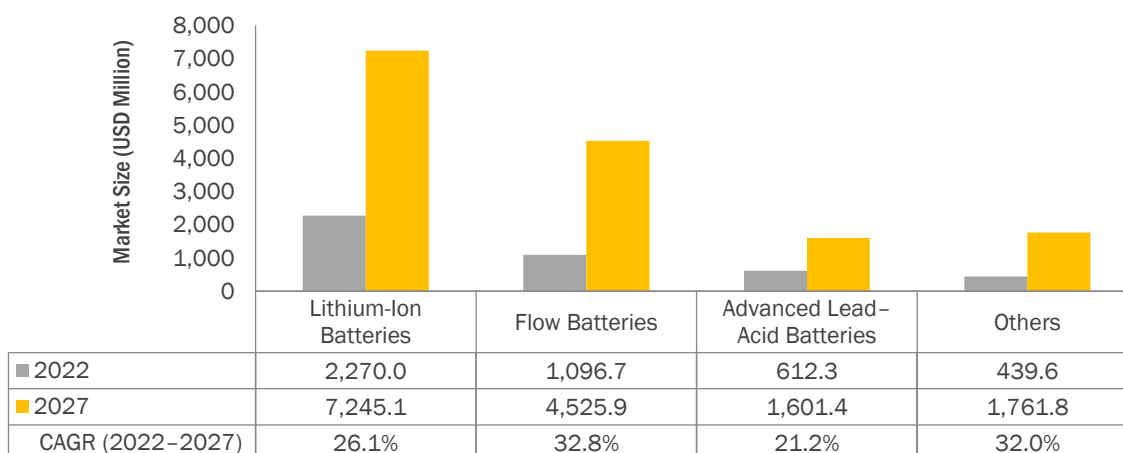
FIGURE 11 INCREASING NUMBER OF PROJECTS, WHICH REQUIRE BATTERY ENERGY STORAGE SYSTEMS IS LIKELY TO CREATE LUCRATIVE OPPORTUNITIES FOR BATTERY ENERGY STORAGE SYSTEM MARKET IN APAC



Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

4.2 BATTERY ENERGY STORAGE SYSTEM MARKET, BY BATTERY TYPE

FIGURE 12 BATTERY ENERGY STORAGE SYSTEM MARKET FOR LITHIUM-ION BATTERIES TO HOLD LARGEST SIZE IN 2027

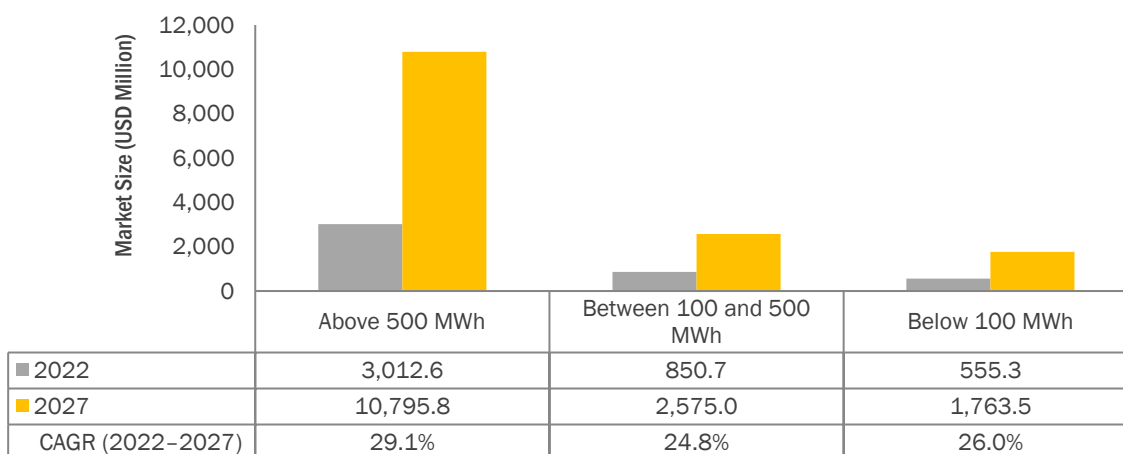


Note: Other battery types include sodium-based batteries, nickel-cadmium batteries, nickel-metal hydride, nickel-iron batteries, and flywheel batteries.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

4.3 BATTERY ENERGY STORAGE SYSTEM MARKET, BY ENERGY CAPACITY

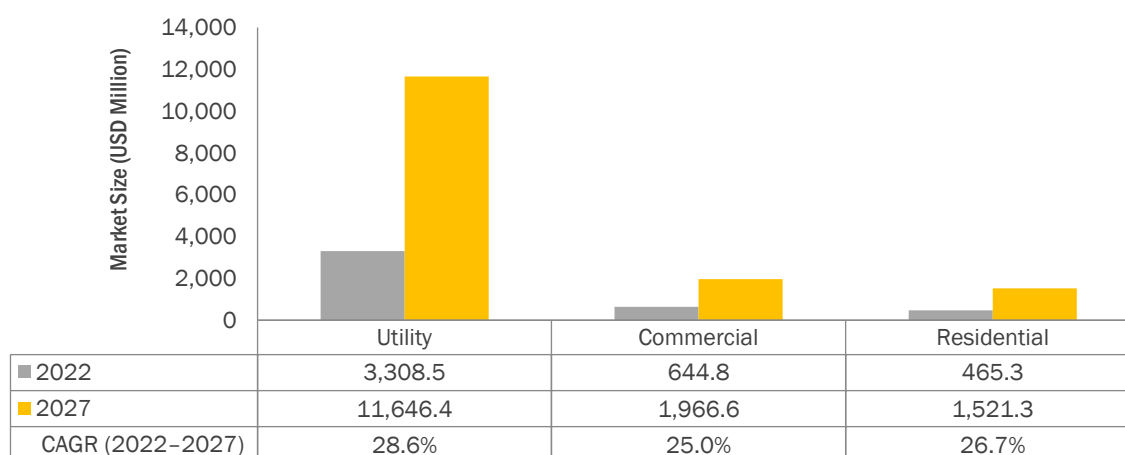
FIGURE 13 ABOVE 500 MWH ENERGY CAPACITY SEGMENT TO DOMINATE BATTERY ENERGY STORAGE SYSTEM MARKET DURING FORECAST PERIOD



Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

4.4 BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION

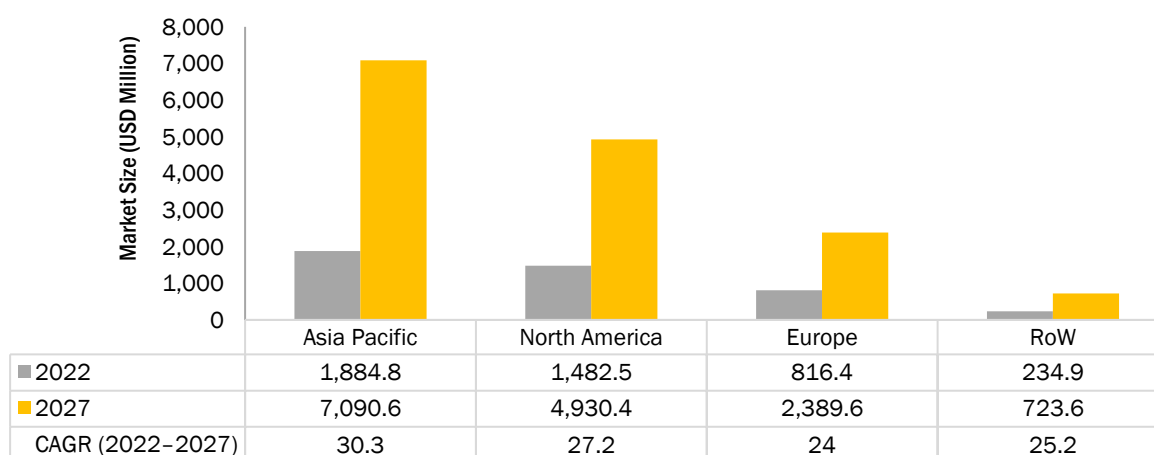
FIGURE 14 UTILITY APPLICATION OF BATTERY ENERGY STORAGE SYSTEM MARKET TO WITNESS HIGHEST RATE DURING FORECAST PERIOD



Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

4.5 BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION

FIGURE 15 APAC TO HOLD LARGEST SIZE OF BATTERY ENERGY STORAGE SYSTEM MARKET IN 2027



Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

5 MARKET OVERVIEW

5.1 INTRODUCTION

Battery energy storage systems (BESS) are rechargeable batteries, wherein the energy is stored from the electric grid or solar arrays. They are used to enhance the efficiency of the energy chain, right from the generation of electricity to its transmission and distribution. They are developed for storing the electric charge by utilizing specially developed batteries such as conventional batteries and flow batteries. These systems are not restricted to any single battery technology and hence, can be easily connected to all available battery types, such as lithium-ion batteries, flow batteries, advanced lead-acid batteries, sodium-sulfur batteries, nickel-cadmium batteries, nickel-metal hydride batteries, nickel-iron batteries, among others. Battery energy storage systems improve the quality of power by ensuring improved voltage and frequency regulation with minimum interruption. Besides, these systems provide backup power during disruption, save the operational cost in powering the grid, and create capabilities and efficiency of the grid. The key players operating in the battery energy storage system market include Samsung SDI Co., Ltd. (South Korea), LG Energy Solutions Co., Ltd. (South Korea), Panasonic Corporation (Japan), BYD Company Limited (China), and Siemens AG (Germany).

5.2 MARKET DYNAMICS

In this section, we have analyzed the dynamics (drivers, restraints, opportunities, and challenges) and value chain of the battery energy storage system market in detail. Market players can analyze these factors to strategize and decide their action plans to boost their position in the market and gain maximum revenue.

FIGURE 16 GLOBAL BATTERY ENERGY STORAGE SYSTEM MARKET DYNAMICS

 DRIVERS	▪ Increasing demand for grid energy storage systems owing to ongoing grid modernization ▪ Growing penetration of lithium-ion batteries in renewable energy sector ▪ Rising trend of adopting low-carbon and less fossil fuel-based economy and ongoing renewable energy revolution
 RESTRAINTS	▪ High capital expenditure required for installing battery energy storage systems
 OPPORTUNITIES	▪ Surge in number of rural electrification projects worldwide ▪ Increase in need for continuous power supply attributed to rise in number of datacenters ▪ Decline in prices of lithium-ion batteries
 CHALLENGES	▪ Complexity in installing battery energy storage systems at islands or in remote areas ▪ Overheating of lithium-ion batteries ▪ Aging of lithium-ion batteries

Source: Industry Journals, Expert Interviews, and MarketsandMarkets Analysis

5.2.1 DRIVERS

5.2.1.1 Increasing demand for grid energy storage systems owing to ongoing grid modernization

Solar and wind energy are the most prominent renewable energy types that are stored in grids. However, the concealing of the sun by clouds or fluctuating wind currents leads to variations in the process of energy generation. Such fluctuations create the requirement for flexible grid systems used to store energy. Battery energy storage systems are becoming an integral part of grid modernization. These systems help grid operators to save electricity when the electricity generated exceeds the electricity demand. The adoption of these systems improves the reliability and flexibility of electricity supply systems with respect to the generation, transmission, and distribution of electric power. During the transmission and distribution of electricity, battery energy storage systems can handle asset deferral, ensure frequency regulation, carry out harmonic suppression, offer voltage support, and ensure power quality. Electric energy time-shifting, baseload leveling and peak shaving, renewable energy time-shifting, and renewable capacity firming are other benefits offered by battery energy storage systems for grid-connected solutions.

Companies are highly involved in modernizing the electricity grid with the help of battery energy storage systems. For instance, The AES Corporation is modernizing India's electricity grid by building the country's first grid-scale battery energy storage facility. The company has entered into a partnership agreement with Mitsubishi Corporation to provide India's first grid-scale BESS project using Advancion Energy Storage Technology Platform of Fluence. In addition, Siemens offers SIESTORAGE, an intelligent grid solution capable of overcoming the challenges in power supply in terms of the optimization of grid connections, the flexibility of energy for modern grids, and support to large energy consumers. Broad Reach Power (US), an independent power generator, is planning to develop 15 small-scale battery storage plant sites in areas near Houston and Odessa in Texas (US) in 2020 and two large systems by 2021. Each site is expected to contain battery systems capable of storing and distributing up to 10 megawatts of power. The small systems developed by the company are expected to provide energy stabilization services, such as mitigating the impact of price spikes and dips that expose generators, utilities, and retail electric providers to uncertainty in the supply-demand matching that occurs in power grids.

The demand for battery energy storage systems goes hand-in-hand with grid modernization efforts, including the transition from traditional systems to new technologies such as smart grids. Batteries help unlock the full potential of smart technologies. All these factors drive the adoption of these battery energy storage systems, thereby fueling the growth of the market.

5.2.1.2 Growing penetration of lithium-ion batteries in renewable energy sector

Lithium-ion batteries have better electrochemical performance parameters than lead-acid batteries, nickel-cadmium batteries, and nickel-metal hydride batteries. These batteries have a long life and offer increased power, excellent efficiency, and low self-discharge. As such, they are commonly used in energy storage systems, which store energy generated from renewable energy sources such as solar and wind. According to the recent study conducted by Energy Information Administration (EIA), more than 80% of the large-scale battery energy storage systems are powered by lithium-ion batteries across the world. Various players in the market are working toward improving the performance of lithium-ion batteries making them an attractive option for stationary energy storage applications. Cells or batteries embedded in battery energy storage systems play an important role in deciding the service life of these systems.

LG Energy Solutions Co., Ltd. (South Korea), Panasonic Corporation (Japan), and Samsung SDI Co., Ltd. (South Korea) are a few companies providing lithium-ion batteries and using them in their battery energy storage systems. These companies have expertise in lithium-ion technology, thus supporting the growth of the battery energy storage system market. A few key companies are also receiving investments for accelerating the development of their lithium-ion cells and batteries. For instance, in April 2020, LG Energy Solutions Co., Ltd. (South Korea) received a USD 526, 6 million (Euro 480 million) loan from the European Investment Bank (EIB) to support USD 1.6 billion (Euro 1.5 billion) project of LG Energy Solutions Co., Ltd. that includes the construction of its battery production facilities in Europe. These developments support the increased demand for lithium-ion battery energy storage systems.

Lithium-ion batteries are lightweight and compact, require low maintenance, have a long lifetime, and have high energy density and capacity, which makes them ideal for various applications. Lithium-ion energy storage systems allow increased energy storage, thereby enabling low-cost wind and solar energy to enter grids. This, in turn, reduces carbon dioxide emissions and lowers the overall power costs. According to a report—Energy Storage Trends and Opportunities in Emerging Markets—by the International Finance Corporation (IFC) and the Energy Sector Management Assistance Program (ESMAP) of the World Bank Group, the use of lithium-ion technology in utilities leads to 90% efficiency of batteries. The adoption of advanced lead-acid batteries and flow batteries results in 80% and 70% efficiencies, respectively, in public utility applications. This factor further leads to the increasing requirement for lithium-ion batteries in battery energy storage systems.

5.2.1.3 Rising trend of adopting low-carbon and less fossil fuel-based economy and ongoing renewable energy revolution

Battery energy storage systems have unlocked a wide range of solutions for efficient and sustainable energy use that deliver economic and environmental benefits. The prevailing trend of adopting a low-carbon and less fossil fuel-based economy dependent on low-carbon power sources that have minimal greenhouse gas emissions is leading to the increased use of battery energy storage systems. The deep carbonization of electricity production is a societal challenge that can be achieved with a rise in the adoption of variable renewable energy. Battery energy storage systems aid in the low-carbon energy transition by enabling the use of renewable sources of energy and achieving cost savings. These storage systems play an important role in decarbonizing the electricity sector by offering a new, carbon-free source with operational flexibility that improves the utilization of electricity generation assets.

It also facilitates the integration of variable renewable energy sources such as wind, solar, and thermal energy with battery energy storage systems. Battery storage can enhance power system flexibility and allows the integration of renewable energy. Interconnected power systems can reliably and safely integrate renewable energy from variable renewable energy (VRE) sources with no new energy storage resources. Using renewable energy for energy storage in large quantities can lower the prices of natural gas and coal and the demand for them by increasing the competition among energy firms and diversifying energy supplies. Increased dependence on renewable energy can help protect consumers when the prices of fossil fuels spike. Thus, the economic benefits offered by renewable energy systems are expected to fuel the growth of the battery energy storage system market.

FIGURE 17 DRIVERS FOR BATTERY ENERGY STORAGE SYSTEM MARKET AND THEIR IMPACT

DRIVERS	2 YEARS	5 YEARS	REASON
Increasing demand for grid energy storage systems owing to ongoing grid modernization			Solar and wind energy are the most prominent renewable energy types that are stored in grids. However, the concealing of the sun by clouds or fluctuating wind currents lead to variations in the process of energy generation. Such fluctuations create the requirement for flexible grid systems used to store energy. Battery energy storage systems are becoming an integral part of grid modernization.
Growing penetration of lithium-ion batteries in renewable energy sector			Lithium-ion batteries have better electrochemical performance parameters than lead-acid batteries, nickel-cadmium batteries, and nickel-metal hydride batteries. These batteries have a long life and offer increased power, excellent efficiency, and low self-discharge. As such, they are commonly used in energy storage systems, which store energy generated from renewable energy sources such as solar and wind. According to the recent study conducted by Energy Information Administration (EIA), more than 80% of the large-scale battery energy storage systems are powered by lithium-ion batteries across the world.
Rising trend of adopting low-carbon and less fossil fuel-based economy and ongoing renewable energy revolution			Battery energy storage systems have unlocked a wide range of solutions for efficient and sustainable energy use that deliver economic and environmental benefits. The prevailing trend of adopting low-carbon and less fossil fuel-based economy dependent on low-carbon power sources that have minimal greenhouse gas emissions is leading to the increased use of battery energy storage systems.
IMPACT LEVEL:  HIGH  MEDIUM  LOW  VERY LOW			

Source: Industry Journals, Expert Interviews, and MarketsandMarkets Analysis

5.2.2 RESTRAINTS

5.2.2.1 High capital expenditure required for installing battery energy storage systems

Battery energy storage technologies, including lithium-ion batteries, flow batteries, and lead-acid batteries, require increased installation investments owing to the high energy density and improved performance offered by them. Lithium-ion batteries are costly as they offer high energy density, have a low rate of self-discharge, and require less maintenance. However, in the future, the costs of lithium-ion batteries are expected to decline. These batteries are also used in electric vehicles (EVs) as they are lightweight and compact, as well as have a large capacity. Besides, the high initial investment cost required for the manufacturing of flow batteries can also act as a major restraint for the growth of the market. The overall cost associated with the flow battery includes capital costs, component costs, material costs, installation costs, and repair & maintenance costs, which is an extensive investment for small- and medium-sized enterprises.

The costs of battery energy storage systems depend on the type and the number of batteries used in them. They also vary according to the intended application (residential, commercial, and utilities) of these systems. The integration of different batteries (in the form of packs, modules, and racks), inverters, battery management systems (BMS), and wiring in battery energy storage systems results in their increased costs. During the installation of battery energy storage systems, the required capital cost is dependent on the footprint and space requirements (energy and power density) of storage systems; components (batteries, power conversion systems, and BMS) used in them; and facility infrastructures (communications and control, environmental control, and grid interconnection) wherein they are deployed. Additionally, the capital cost is also dependent on the system control cost, the electric balance of point cost (BOP), and construction and commissioning costs (C&C).

Moreover, the total operating cost of battery energy storage systems is a combination of a number of factors, including costs of charging batteries, labor associated with their installation; the maintenance of plants wherein they are installed; replacement and repair costs of these systems; and costs for their decommissioning and disposal. Thus, the installation of battery energy storage systems in a plant or any required place needs high capital investments that act as a restraint for the growth of the market.

FIGURE 18 RESTRAINTS FOR BATTERY ENERGY STORAGE SYSTEM MARKET AND THEIR IMPACT

RESTRAINTS	2 YEARS	5 YEARS	REASON
High capital expenditure required for installing of battery energy storage systems			The overall cost associated with the flow battery include capital cost, component cost, material cost, installation cost, and repair & maintenance cost, which is an extensive investment for small and medium-sized enterprises. Flow batteries require high capital for production as compared with other technologies due to their benefits of high energy density and improved performance
IMPACT LEVEL:  HIGH  MEDIUM  LOW  VERY LOW			

Source: Investor Presentations, Expert Interviews, Industry Journals, Magazines, and MarketsandMarkets Analysis

5.2.3 OPPORTUNITIES

5.2.3.1 Surge in number of rural electrification projects worldwide

Rural electrification is the process of making electrical power accessible in rural or remote areas. Battery energy storage systems are effective for rural electrification. Various countries in the world are trying to develop their remote areas such as isolated rural areas and peri-urban areas, especially in developing countries; and small islands separated from the national grids. As such, they require to have a reliable and uninterrupted power supply from renewable and conventional sources of energy that increases the overall system efficiency and ensures economic savings across the system life cycle. The installation of battery energy storage systems in rural areas would act as a solution to serve the businesses and local communities with a reliable and safe power supply.

The requirement for an uninterrupted and reliable power supply acts as a long-term opportunity for the manufacturers of battery energy storage systems post-COVID-19. For instance, stand-alone off-grid rural electrification systems are designed to harness the primary energy from local renewable energy sources such as wind, solar, and biomass rather than conventional fuels. These renewable energy sources are intermittent, thereby leading to the requirement for storage technologies for short-term power balancing or long-term energy management to ensure that the electricity is generated even when the energy supply is low. Moreover, battery energy storage systems enable the self-reliance of residential power consumers so that they are less affected by grid blackouts and harsh weather conditions. There is a surge in the number of rural electrification projects owing to power outages or for emergency use in rural areas across the world. For instance, in June 2019, a 500 KW battery energy storage system was deployed in rural Alaska (US) to ensure energy stability in the area.

According to the International Energy Agency (IEA), battery energy storage systems are the most sustainable and effective mode to electrify isolated rural areas. The study conducted by IEA has estimated that electricity generation of 379 TWh from on-grid, 171 TWh from off-grid, and 399 TWh from mini-grid systems would be required to achieve universal energy access by 2030. On the other hand, the Alliance for Rural Electrification (ARE) is an international association that works for the integration of renewable sources of energy with rural electrification projects in emerging economies. The increase in entrepreneurial seed capital, utility corporate social responsibility funds, and corporate and social investor capital is supporting investments for rural electrification projects. Thus, the increase in rural electrification projects in the emerging economies, as well as developed countries, is expected to create huge opportunities for the growth of the battery energy storage system market.

5.2.3.2 Increase in need for continuous power supply attributed to rising number of datacenters

The data center is one of the major clients of power utilities and plays a vital role in peak power draw. According to the data from the US Department of Energy, the annual energy consumption by data centers in 2020 covered a range between 200 and 1,000 TWh, with an assumption that data centers would need to meet the average load of 600 TWh for up to 20 minutes. Power consumption in data centers has raised cost, environmental, and scalability concerns. They need to maintain the uptime of the site by deploying cost-effective and energy-efficient alternatives. Datacenters use an uninterrupted power supply (UPS) to provide power to loads in emergencies when the main power supply fails. In recent years, energy storage systems in data centers are primarily used as a backup to handle power outages when the diesel power generators stop working. Battery energy storage systems play an important role in eliminating electricity fluctuations in data centers. The combination of battery energy storage systems and UPS ensures continuous power supply to designated critical loads when utility disturbances or outages are detected. This leads to a rise in demand for UPS from large data centers. The growing demand for UPS from large data centers and service providers to build lighter physical infrastructure at low capital and operational costs is expected to create opportunities for battery storage technology. This, in turn, is anticipated to fuel the demand for energy storage systems based on lithium-ion and lead-acid batteries.

Companies such as Microsoft Corporation (US) and Google Inc. (US) extensively use battery storage systems in their data centers as failover to onsite local generation. In 2020, Google Inc. revealed the plan to install the first backup battery by replacing the diesel generators at its data center located in Belgium. The 3MW-2H duration battery will be installed at the data center. This installation aims to meet the need for round-the-clock carbon-free electricity at its data center by 2030.

5.2.3.3 Decline in prices of lithium-ion batteries

The price of lithium-ion batteries has been a major factor hindering their adoption since their introduction in the 1990s. A lithium-ion battery consists of several components. The primary component of any lithium-ion battery is the cell, which accounts for almost 50% of its cost, electronics, assembly, and packaging account for the rest. However, recent developments and claims made by lithium-ion battery manufacturing companies and automobile companies suggest that the price of these batteries is expected to decline substantially. According to General Motors Company (US), the cost of a cell is expected to drop to USD 100 per kWh by 2021. Tesla Motors Inc. (US) has begun the construction of its Gigafactory 1, a lithium-ion battery factory, which is expected to help bring down the cost of lithium-ion batteries.

Developments such as manufacturing on a large scale, declining prices of components, and the adoption of advanced technologies to boost battery capacity are a few of the factors leading to a decline in price. A reduction in the price would catalyze the adoption of these batteries across various new applications. Renewable energy storage and other grid storage markets, in particular, are expected to opt for lithium-ion batteries over other rechargeable batteries.

FIGURE 19 OPPORTUNITIES FOR BATTERY ENERGY STORAGE SYSTEM MARKET AND THEIR IMPACT

OPPORTUNITIES	2 YEARS	5 YEARS	REASON
Surge in number of rural electrification projects worldwide			Rural electrification is the process of making electrical power accessible in rural or remote areas. Battery energy storage systems are effective for rural electrification. Various countries in the world are trying to develop their remote areas such as isolated rural areas and peri-urban areas, especially in developing countries; and small islands separated from the national grids.
Increase in need for continuous power supply attributed to rise in number of datacenters			The data center is one of the major clients of power utilities and plays a vital role in peak power draw. According to the data from the US Department of Energy, the annual energy consumption by data centers in 2020 covered a range between 200 and 1,000 TWh, with an assumption that data centers would need to meet the average load of 600 TWh for up to 20 minutes.
Decline in prices of lithium-ion batteries			The price of lithium-ion batteries has been a major factor hindering their adoption since their introduction in the 1990s. A lithium-ion battery consists of several components. The primary component of any lithium-ion battery is the cell, which accounts for almost 50% of its cost, electronics, assembly, and packaging account for the rest.
IMPACT LEVEL:  HIGH  MEDIUM  LOW  VERY LOW			

Source: Investor Presentations, Expert Interviews, Industry Journals, Magazines, and MarketsandMarkets Analysis

5.2.4 CHALLENGES

5.2.4.1 Complexities in installing battery energy storage systems at islands or in remote areas

Battery energy storage systems are mainly deployed on islands or in remote areas. The installation of these systems in remote locations is difficult as they are difficult to reach. The remote locations usually include islands and off-grid remote locations, which face various challenges owing to variable generation and supply of power from renewable energy sources. Challenges are related to ambient conditions such as temperature variation during day and night, costly maintenance owing to commuting difficulties to reach these locations, and the lack of equipment installation infrastructures.

The off-grid regions exceed the local load during the sunniest hours of the day, and battery energy storage systems store excess solar energy and compensate for the photovoltaic (PV) power fluctuations. Reaching remote areas frequently is not possible for operators; hence, they also use diesel generators, which are activated only when the status of batteries falls below a minimum set point. These generators are expensive and have high carbon dioxide emissions. Moreover, the shipping of battery energy storage systems and diesel generators to remote areas becomes difficult owing to restrictive government regulations. All these factors act as a challenge for the growth of the battery energy storage system market in remote islands and off-grid regions.

5.2.4.2 Overheating of lithium-ion batteries

Lithium-ion batteries are predominantly used in energy storage systems apart from applications such as consumer electronics, automobiles, electrical systems, aviation components, automated guided vehicles (AGVs), forklifts, pallet trucks, and material-handling equipment. While lithium-ion batteries can store a relatively large amount of energy in a small space, they tend to heat up quickly and may catch fire in case of any malfunctioning.

The overheating of lithium-ion batteries takes place during abnormal discharges and normal high-rate discharges such as short circuits. In abnormal discharges, the temperature of the batteries might reach the threshold for exothermic reactions and trigger thermal runaway, whereas, in the normal high-rate discharges, the parasitic heat generation of these batteries can hasten the capacity fading. The overheating of lithium-ion batteries is caused by a number of reasons. Lithium is a highly unstable substance. When separators are placed between cathodes and anodes in batteries get disrupted, a short circuit occurs. The physical damage inflicted upon separators causes their overheating.

In the event of leakages of liquid chemicals, they react with surrounding components, causing overheating. Software issues in devices, where automated instructions to turn off charging ports are missed while batteries are charging, can also cause overheating. This problem can lead to the overcharging and swelling up of batteries, thereby resulting in disruptions in the functioning of battery energy storage systems.

5.2.4.3 Aging of lithium-ion batteries

Despite remarkable recent developments and improvements, lithium-ion battery technology has a significant flaw—aging. Lithium-ion cells age with time and usage, reducing energy and storage capability. Battery cell deterioration is noticeable in mobile phones and laptops. While such deterioration is an inconvenience in portable devices, it is a serious issue for electric vehicles. Cell aging directly reduces the available range in electric vehicles. Lithium-ion cells will always age due to innate chemical reactions between the negative electrode and electrolyte. The rate at which aging occurs is a complex relationship between cell usage and cell design. Certain factors increase the rate of aging, such as storing at full charge for long periods, applying high charge currents, charging at low temperatures, storing or using cells at high charge currents, and cycling cells in the extremes of their operating voltage range. Currently, there is no solution to ensure an acceptable battery life in all conditions; this can only be achieved through further improvements in cell design, battery pack construction, and usage optimization, in conjunction with gaining a more thorough understanding of the fundamental aging mechanisms of emerging cell chemistries.

FIGURE 20 CHALLENGES FOR BATTERY ENERGY STORAGE SYSTEM MARKET AND THEIR IMPACT

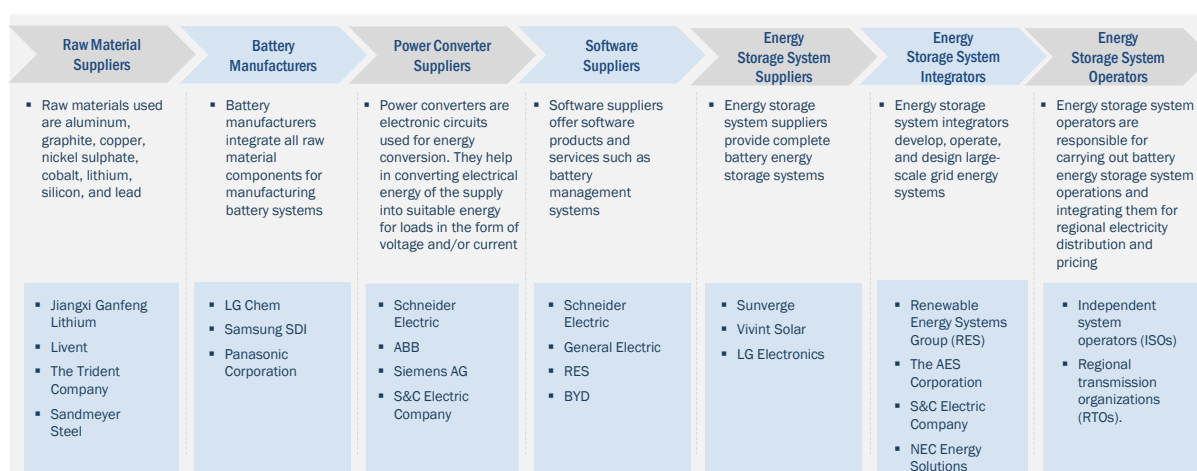
CHALLENGES	2 YEARS	5 YEARS	REASON
Complexity in installing battery energy storage systems at islands or in remote areas			Battery energy storage systems are mainly deployed at islands or in remote areas. The installation of these systems in remote locations is difficult as they are difficult to reach. The remote locations usually include islands and off-grid remote locations, which face various challenges owing to variable generation and supply of power from renewable energy sources
Overheating of lithium-ion batteries			Lithium-ion batteries are predominantly used in energy storage systems apart from applications such as consumer electronics, automobiles, electrical systems, aviation components, automated guided vehicles (AGVs), forklifts, pallet trucks, and material-handling equipment. While lithium-ion batteries can store a relatively large amount of energy in a small space, they tend to heat up quickly and may catch fire in case of any malfunctioning.
Aging of lithium-ion batteries			Despite remarkable recent developments and improvements, the lithium-ion battery technology has a significant flaw—aging. Lithium-ion cells age with time and usage, reducing energy and storage capability. Battery cell deterioration is noticeable in mobile phones and laptops. While such deterioration is an inconvenience in portable devices, it is a serious issue for electric vehicles. Cell aging directly reduces the available range in electric vehicles.
IMPACT LEVEL:  HIGH  MEDIUM  LOW  VERY LOW			

Source: Industry Journals, Expert Interviews, and MarketsandMarkets Analysis

5.3 VALUE CHAIN ANALYSIS

Battery energy storage systems are rechargeable battery systems that store energy from renewable energy sources and/or electric grids. These systems are deployed across multiple levels of the electricity value chain right from the generation, transmission, and distribution of electricity to its consumption. Battery energy storage systems are implemented at two levels, grid-scale (in front of meters) and behind-the-meter (BTM) (in residential, commercial, and industrial facilities). Grid-scale batteries are integrated with electricity generation assets and are then connected to transmission or distribution networks. Grid-scale battery systems are used for integrating excess renewable energy generated with enhanced load management solutions. BTM batteries are connected through electricity meters and are increasingly used for carrying out emergency power supplying, peak shaving, and load shifting operations owing to their superior technologies as compared with basic regular batteries. They are predominantly used in industrial, commercial, and residential applications. Battery energy storage systems consist of several components, which include modules, control systems, onboard sensors, and inverters.

The value chain of battery energy storage systems comprises raw material suppliers, battery manufacturers, power converter suppliers, software suppliers, energy storage system suppliers, energy storage system integrators, and energy storage system operators. Battery manufacturers purchase raw materials for manufacturing the entire battery system. A value chain is a systematic approach for examining the activities taking place during the manufacturing process. The model works by segregating a firm into its strategically relevant activities to understand the cost behavior and to identify the sources of differentiation. For a firm to gain a competitive advantage, it needs to achieve cost reduction in these strategically important activities. The battery energy storage system value chain model has been illustrated in the figure given below.

FIGURE 21 BATTERY ENERGY STORAGE SYSTEM: VALUE CHAIN

Source: Secondary Research, Industry Journals, Expert Interviews, and MarketsandMarkets Analysis

Raw material suppliers

This phase predominantly consists of raw material suppliers that contribute significant value to the overall price of the system and thus to the value chain of the battery energy storage system market. They supply raw materials that are used by battery manufacturers to assemble the entire battery system. Raw materials used in lithium-ion and lead-acid batteries are aluminum, graphite, copper, nickel sulfate, cobalt, lithium, silicon, and lead. Raw materials utilized in the manufacturing of flow batteries are vanadium, iron, and zinc-bromine. The key raw material suppliers of lithium-ion and lead-acid batteries are Ganfeng Lithium Co., Ltd. (China), Livent (US), Sandmeyer Steel Company (US), Glencore (Switzerland), China Molybdenum (China), The Trident Company (US), and Tianqi Lithium (China) in regions such as Europe, APAC, and North America. The top countries for supplying raw materials are Australia, the US, China, Chile, the Democratic Republic of Congo, and Russia.

Battery manufacturers

This stage of the value chain adds a major value along with the raw material suppliers to the overall battery energy storage system market. After procuring all required raw materials from various vendors, the system is in the manufacturing phase. Battery manufacturers integrate all raw material components for manufacturing battery systems. The battery system must be well tested and should deliver accurate test results. The key battery manufacturers are LG Energy Solutions Co., Ltd. (South Korea), Samsung SDI Co., Ltd. (South Korea), and Panasonic Corporation (Japan).

Power converter suppliers

Power converters or power conversion systems are electronic circuits used for energy conversion. They convert electrical energy received from the main supply into suitable energy for loads in the form of voltage and/or current. The key power converter suppliers are Schneider Electric (France), ABB (Switzerland), AEG Power Solutions (the Netherlands), Siemens AG (Germany), and S&C Electric Company (US).

Software suppliers

Software suppliers offer software products and services such as battery management systems. The key software suppliers include Schneider Electric (France), General Electric (US), Renewable Energy Systems Ltd. (UK), and BYD Company Limited (China).

Energy storage system suppliers

Energy storage system suppliers provide complete battery energy storage systems. These battery energy storage systems are utilized in utilities, residential, and commercial applications. The key companies operating in this space include Sunverge Energy Inc. (US), Vivint Solar Developer, LLC. (US), and LG Electronics (South Korea).

Energy storage system integrators

Energy storage system integrators develop, operate, and design large-scale grid energy systems. The integration includes putting together the entire battery energy storage system and testing the same if the prerequisites of storing energy are achieved by the system. The key companies offering energy storage system integration include Renewable Energy Systems Ltd. (UK), The AES Corporation (US), S&C Electric Company (US), and NEC Energy Solutions (US).

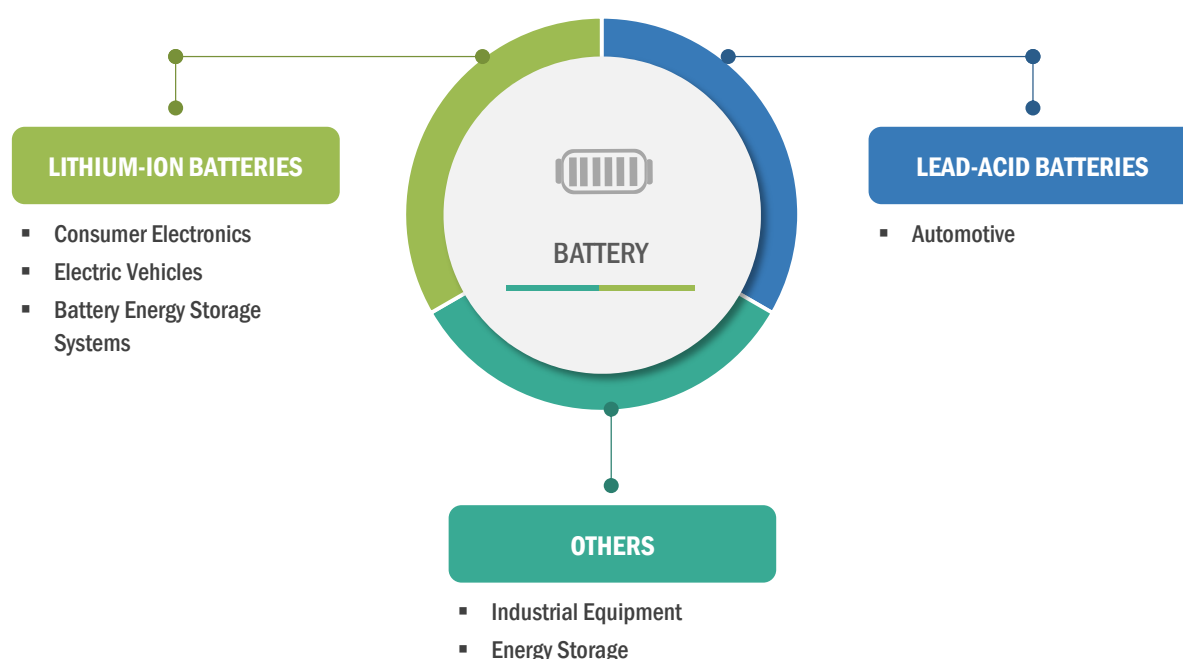
Energy storage system operators

Energy storage system operators are responsible for carrying out battery energy storage system operations for regional electricity distribution and pricing. The key operators include independent system operators (ISOs) and regional transmission organizations (RTOs).

5.4 ECOSYSTEM

The high power and energy density requirements have led to a rise in demand for reliable and safe batteries for use in different industries. The advent of electric vehicles, renewable energy storage systems, and consumer electronics has been driving increased investments in the field of batteries, from the aspects of both incumbents and new players. With the rise in the adoption of electronic devices, the demand for battery energy storage systems has also grown significantly. There are several participants in the ecosystem of batteries, and each participant, right from raw material suppliers to end users, has a specific and important role to play.

FIGURE 22 GLOBAL BATTERY ENERGY STORAGE SYSTEM MARKET: ECOSYSTEM



Source: Secondary Research, Industry Journals, Expert Interviews, and MarketsandMarkets Analysis

The major companies offering battery energy storage systems are Samsung SDI Co., Ltd. (South Korea), LG Energy Solutions Co., Ltd. (South Korea), General Electric (US), BYD Company Limited (China), Panasonic Corporation (Japan), Hitachi, Ltd. (Japan), Tesla, Inc. (US), Toshiba Corporation (Japan), ABB (Switzerland), Siemens AG (Germany), and AEG Power Solutions (Netherlands). The ecosystem comprises battery energy storage system raw material suppliers, battery manufacturers, power converters suppliers, software suppliers, energy storage system suppliers, energy storage system integrators, and energy storage system operators. Suppliers of raw materials offer materials based on battery types to be developed. Batteries are manufactured using a combination of various materials used in cathodes, anodes, electrolytes, and separators. The cathode and anode electrodes generate electricity with the help of electrolyte solutions filled in cells. Batteries have materials immersed in electrolyte solutions that act as separators between cathodes and anodes. Other materials required to manufacture batteries are binders, current collectors, and packaging materials.

TABLE 4 BATTERY ENERGY STORAGE SYSTEM MARKET: ECOSYSTEM

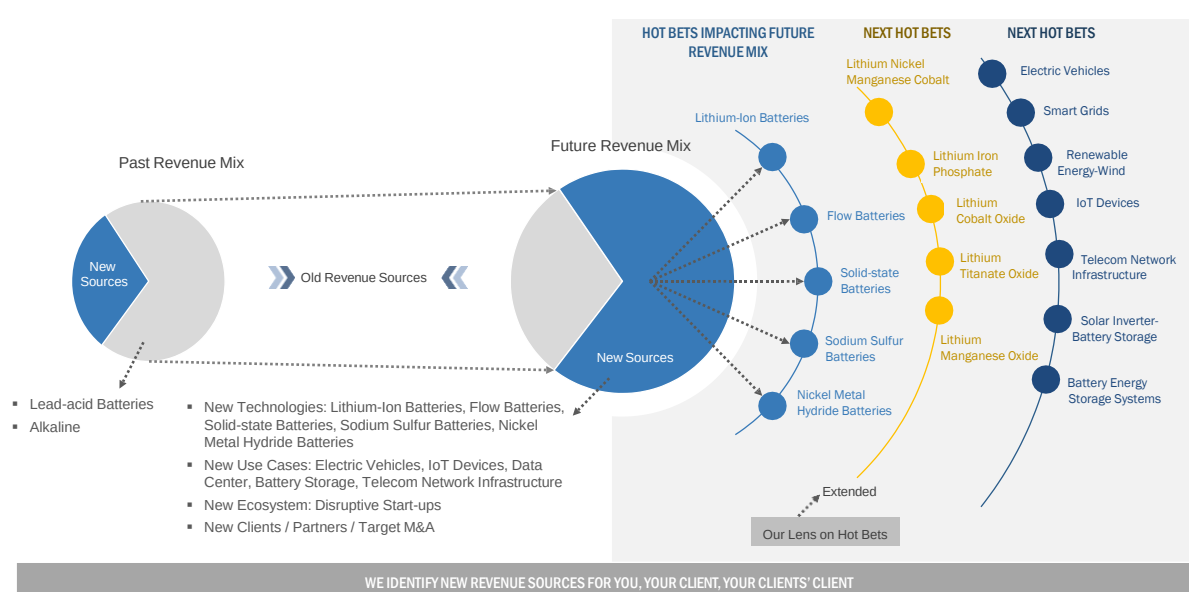
COMPANY	ROLE IN ECOSYSTEM
ABB	Component Provider and Product Manufacturer
LG Energy Solutions Co., Ltd.	Product Manufacturer
Panasonic Corporation	Product Manufacturer
Samsung SDI Co., Ltd.	Product Manufacturer
AEG Power Solutions	Product Manufacturer and Solution Provider
General Electric	Product Manufacturer and Solution Provider
Hitachi, Ltd.	Product Manufacturer
Siemens AG	Product Manufacturer
Tesla, Inc.	Product Manufacturer
Toshiba Corporation	Product Manufacturer
Honeywell International, Inc.	Product Manufacturer and Service Provider
BYD Company Limited	Product Manufacturer
Delta Electronics, Inc.	Product Manufacturer
NEC Corporation	Product Manufacturer and Solution Provider
NGK INSULATOR, LTD.	Product Manufacturer
The AES Corporation	Product Manufacturer
Exergonix	Product Manufacturer and Product Supplier
East Penn Manufacturing Company	Designer and Product Manufacturer
EnerDel, Inc.	Designer and Product Manufacturer
TrinaBESS	Product Manufacturer
Primus Power	Product Manufacturer

NextEra Energy, Inc.	Product Manufacturer
Parker Hannifin Corp	Product Manufacturer and Solution Provider
Johnson Controls	Product Manufacturer and Service Provider
Kokam	Product Manufacturer

5.5 TRENDS/DISRUPTIONS IMPACTING CUSTOMER'S BUSINESS

The MnM engagement model can help battery energy storage system manufacturers prioritize high-growth niche opportunities to achieve the target revenue.

FIGURE 23 TRENDS/DISRUPTIONS IMPACTING CUSTOMER'S BUSINESS



Source: MarketsandMarkets Analysis

Increasing demand for grid energy storage systems owing to ongoing grid modernization, growing penetration of lithium-ion batteries in the renewable energy sector, the prevailing trend of adopting low-carbon and less fossil fuel-based economy and ongoing renewable energy revolution, and continued development of smart devices and other industrial goods are the factors driving the growth of the battery energy storage system market. Battery energy storage systems are highly adopted for utility and commercial operations as they ensure the availability of power when customers need the power most. The ecosystem of battery energy storage systems can generate new revenue capabilities with the introduction of batteries compatible with smart factories and IIoT devices and the introduction of liquid metal batteries. Manufacturers are developing batteries by estimating the production cost and performance of the batteries that can meet the increased requirements of various applications. Besides, technological innovations with improved capabilities of batteries offer high-growth opportunities for battery developers to grow in emerging markets. The companies are heavily investing in research and development activities to develop innovative and improved batteries to enhance their product offerings and meet the requirements of end users. The developments are estimated to create new revenue streams for companies operating in the battery energy storage system market.

5.6 PORTER'S FIVE FORCES ANALYSIS

The competitive environment prevailing in the battery energy storage system industry can be analyzed through Porter's five forces analytical framework. Porter's five forces model describes the current scenario of the battery energy storage system market. The model provides information about profitability and helps companies in decision-making about their entry and exit strategies in the market. Changes in any of the forces can have a significant impact on the battery energy storage system market. Porter's five forces model mainly analyzes the competition in the battery energy storage system market from five different perspectives, namely, the degree of competition, the bargaining power of suppliers, the bargaining power of buyers, the threat of substitutes, and the threat of new entrants. These forces determine the structure of a market/industry and the level of competition in it.

TABLE 5 **IMPACT OF PORTER'S 5 FORCES ON BATTERY ENERGY STORAGE SYSTEM MARKET**

FORCE	IMPACT (2021)
Degree of Competition	High
Bargaining Power of Suppliers	High
Bargaining Power of Buyers	Medium
Threat of Substitutes	Medium
Threat of New Entrants	Low

Source: Company Websites, Expert Interviews, White Papers, and MarketsandMarkets Analysis

5.6.1 THREAT OF NEW ENTRANTS

The threat of new entrants refers to the force of new potential competitors in the market who can capture the market share of existing companies. The threat of new entrants in the battery energy storage system market is low, as several global and domestic players have an established base in the market. Siemens AG (Germany), Samsung SDI Co., Ltd. (South Korea), LG Energy Solutions Co., Ltd. (South Korea), General Electric (US), BYD Company Limited (South Korea), and Panasonic Corporation (Japan) are a few of the established players operating in the battery energy storage system market. These players collectively account for a significant share of the global market. Besides, there are several established players in the US, Japan, Germany, and other developing countries, which poses a challenge for new entrants. Moreover, the capital requirement is a major factor of concern for any new entrant in the battery energy storage system market as there is a high deployment cost in the initial stage of setting up a manufacturing plant and R&D facilities. This has also led to a challenge for the newly emerging organizations.

5.6.2 THREAT OF SUBSTITUTES

The threat of substitutes arises when customers or buyers in the market can achieve the same results using other means than those existing in the market under consideration. Supercapacitors, fuel cells, and flywheels powered with UPS are substitutes for rechargeable batteries. The automotive and stationary applications have started utilizing flywheels and fuel cells for power requirements. In addition, the advancements in supercapacitors have made it well-suited for energy storage applications in the automotive sector, and further rise is expected in its usage. Increasing investments in its research have the potential to change the dynamics of this market in the coming years. Hence, the threat of substitutes is considered moderate in the battery energy storage system market.

5.6.3 BARGAINING POWER OF SUPPLIERS

The bargaining power of suppliers evaluates the ease with which suppliers can drive up the prices of raw materials leveraging the unique proposition of their products or put pressure on buyers by raising their prices, lowering their quality, or reducing the availability of their products in the battery energy storage system market. The number of raw material suppliers in the market is an important factor as it can influence decisions regarding the purchase of energy storage systems. The low concentration of suppliers in the market reduces the switching cost for battery manufacturers; thus, increasing the bargaining power of suppliers. As a result, the bargaining power of suppliers is very high because consumers prefer the suppliers, which provide reliable, effective, and quality products in the market. A few of the major players in this market are General Electric (U.S.), Hitachi, Ltd. (Japan), LG Energy Solutions Co., Ltd. (South Korea), NEC Corporation (Japan), Panasonic Corporation (Japan), and Samsung SDI Co., Ltd. (South Korea), among others.

5.6.4 BARGAINING POWER OF BUYERS

The bargaining power of buyers refers to the pressure that buyers exert on the providers of battery energy storage systems. Buyers can choose the battery energy storage system provided by manufacturers according to their requirements. Buyers from the industrial and commercial sectors have specialized requirements and their buying decision is based on factors such as longevity, durability, usage, and vendor rating. Hence, for industrial buyers, the acceptance of a particular brand of battery storage units is essential. Moreover, for residential users, power requirements are almost similar and have few variations. Thus, for residential buyers, the bargaining power is medium; therefore, the overall impact of this force on the market is medium.

5.6.5 INTENSITY OF COMPETITIVE RIVALRY

The degree of competition evaluates the capabilities of the players in the market and the intensity of rivalry among competitors in the battery energy storage system market. The impact of the intensity of competitive rivalry on the battery energy storage system market has currently been analyzed to be high. The intensity of competitive rivalry is considered to have a high impact owing to various factors such as the number of suppliers, industry growth rate, and product innovations. Thus, huge competition in the battery energy storage system market has pushed companies to adopt powerful competitive strategies to guarantee sustainability. This is intensifying the rivalry among the existing players.

5.7 CASE STUDY ANALYSIS

5.7.1 GENERAL ELECTRIC TO DELIVER BATTERY ENERGY STORAGE SYSTEMS TO CONVERGENT IN CALIFORNIA

General Electric Renewable Energy has been selected by Convergent Energy + Power (Convergent) for the supply of battery energy storage systems with a total capacity of 100 MWh for three projects in California, US. The deployment of the energy storage system in these projects would support two primary goals: firstly, the energy storage system delivers targeted local capacity to improve grid reliability during peak periods. Secondly, these storage systems are fast-acting stabilization devices due to which they can be discharged and charged rapidly to contribute to grid stability and regulate frequency. This helps in facilitating and balancing the rapidly growing penetration of variable renewable energy. With these projects, General Electric Renewable Energy would reach a total capacity of battery energy storage systems of 495 MWh in construction or operations.

5.7.2 ADOPTION OF BATTERY ENERGY STORAGE SYSTEMS BY UTILITY PROVIDER COMPANY

Eskom, a state-owned utility of South Africa, launched a tender in 2019 for 1.4 gigawatt-hours of battery storage to be installed by December 2021. The wide-ranging plan includes energy storage systems to be deployed across all nine provinces of South Africa. The large-scale energy storage system tender is expected to be split into two phases. Phase 1 includes the deployment of 800 MWh battery energy storage at various points along distribution sites operated by Eskom in the Northern Cape, Eastern Cape, Western Cape, and Kwa-Zulu Natal. This phase is scheduled to be completed by December 2020. Phase 2 involves the deployment of 640 MWh of battery energy storage systems in combination with 60 MW of distributed solar PV across all nine provinces of South Africa. It is scheduled to be completed by December 2021.

5.7.3 DEPLOYMENT OF MICROGRID BATTERY ENERGY STORAGE SYSTEMS FOR RURAL ELECTRIFICATION IN NIGERIA

Eos Energy Storage, a leading manufacturer of safe, low-cost, and long-duration zinc battery storage systems, partnered with an EPC firm, Nayo Tropical Technology Ltd., in July 2020 to deploy the Aurora EnergyBlock zinc battery energy storage system (BESS) of the company. The EnergyBlock offered by Eos Energy Storage is a part of the project planned to bring electricity to remote areas and integrate it with rural microgrid systems for electrifying rural communities in Nigeria. The deployment of this BESS is expected to give utilities and energy providers options for the best storage technologies in rural locations and environments.

5.7.4 DEPLOYMENT OF OFF-GRID BATTERY ENERGY STORAGE SYSTEMS FOR RENEWABLE ENERGY PROJECT

A 100 MW battery energy storage system will be built near Wandoan (Australia) by the end of 2020, marking the first stage of the largest grid-scale battery project in Queensland that is projected to create 30 jobs over an estimated 12-month construction period. The first phase is expected to be USD 120 million. It is expected to deploy BESS of Vena Energy that has an initial capacity of 100 MW and can store 150 MWh of energy to power up to 57,000 average homes annually. The first phase is a milestone of the USD 650 million of the Wandoan South Project, which has the potential to supply power to up to 400,000 households with solar energy in Queensland.

5.7.5 ABB TO INSTALL BATTERY ENERGY STORAGE SYSTEMS IN PHILIPPINES

In June 2021, ABB partnered with SMC Global Power Holdings Corp., one of the leading suppliers of power to the national grid in the Philippines. The companies have partnered to install the battery energy storage system to help enhance grid reliability and flexibility in the country. BESS is helping to enhance the grid performance as well as driving wider adoption of renewable energy solutions. ABB has won this contract in 2019 and supports two 20 MW sites and a further 40 MW site, to be commissioned in 2021. ABB has also offered the proprietary software platform ABB Ability Zenon to act as the interface to the BESS, which permits the customers to make real-time decisions based on grid parameters to ensure performance stability.

5.8 TECHNOLOGY ANALYSIS

Battery energy storage systems are rechargeable systems that store energy from solar arrays or electric grids and offer it for use in residential and commercial applications. Since these systems are equipped with advanced technologies that are not present in regular batteries, they can easily perform certain tasks, such as peak shaving and load shifting that used to be difficult or impossible in traditional systems.

BESS offers advantages over other storage technologies as they have a small footprint with no restrictions on geographical locations wherein, they can be deployed. Other storage technologies such as pumped hydro storage (PHS) and compressed air energy storage (CAES) are suitable for a limited number of locations only, considering water and siting-related restrictions and transmission constraints. Energy and power densities of a few technologies are as follows:

TABLE 6 COMPARISON OF ENERGY STORAGE TECHNOLOGIES

TECHNOLOGY	POWER DENSITY (W/L)	ENERGY DENSITY(WH/L)
PHS	0.1–0.2	0.2–2
CAES	0.2–0.6	2–6
Lithium-ion Batteries	1,300–10,000	200–400
Lead-acid Batteries	90–700	50–80
NaS	120–160	150–300

Source: Secondary Research

TABLE 7 CHARACTERISTICS OF BATTERY TECHNOLOGIES IN BATTERY ENERGY STORAGE SYSTEMS

TECHNOLOGY	CAPITAL COSTS USD KWH	LIFE CYCLE (NUMBER OF CHARGE/DISCHARGE CYCLES TO 80% DOD)	PERCENTAGE ROUND TRIP EFFICIENCY AC TO DC	ENVIRONMENTAL IMPACT	ENERGY TO SIZE ENERGY DENSITY WH/L
Lithium-ion	Very High	Medium (2,000–5,000)	Very High (85–95%)	Low	Medium (80–200 Wh/L)
Lead-acid	Medium	High (up to 10,000)	Medium (70–75%)	High	Low (15–20 Wh/L)
Sodium-sulfur	Medium	Medium (3,000–5,000)	High (85–90%)	Medium	High (145–150 Wh/L)
Zinc-bromine	Low	High (above 10,000)	Medium (70–75%)	Low	High (130–150 Wh/L)
Vanadium Redox Flow	Medium	High (2,000–5,000)	Medium (70–75%)	High	Low (15–25 Wh/L)

Source: Secondary Research

Below are a few of the future battery technologies that are expected to boost the battery market over the forecast period. The manufacturers are focusing on the development of enhanced and innovative batteries to meet the requirements of the end users. Companies are heavily investing in research and development activities for the production of improved batteries.

5.8.1 SODIUM-SULFUR BATTERY

A sodium-sulfur battery is a molten-salt battery built from liquid sulfur and sodium. Raw materials such as sodium and sulfur are abundant and inexpensive materials. This battery has a longer life cycle of about 15 years, high efficiency of charging and discharging, and high energy density. Sodium-sulfur batteries operate in temperatures between 300°C and 350°C, and the corrosive nature of sodium polysulfide makes it suitable for stationary energy storage applications.

5.8.2 COBALT-FREE BATTERY

Cobalt material is usually used in the fabrication of rechargeable batteries for stationary energy storage units, consumer electronic devices, and electric vehicles. This material is relatively high in cost, and high exploration is required to meet the increasing demand from consumers, which has forced several companies to develop materials that use iron, which is abundant, as a constituent element instead of cobalt. However, cobalt batteries are highly preferable in electric vehicles owing to their capabilities such as high energy density, modest internal resistance, and long life cycle. Manufacturers using cobalt have a few supplier options because 60% of the world's cobalt is mined in one place, i.e., the Democratic Republic of Congo (DRC).

5.8.3 METAL-AIR BATTERY

Metal-air batteries combine the design features of both conventional batteries and fuel cells. The positive electrode in a metal-air battery is carbon-based, whereas the other electrode is made of aluminum, zinc, lithium, and magnesium metals. They are assembled from a metal anode and an air-breathing cathode in a proper electrolyte, which broadly includes running on oxygen or air feedstock. Metal-air batteries have much higher theoretical energy density and are frequently advocated as the solution toward next-generation electrochemical energy storage for applications, including electric vehicles or grid energy storage.

5.8.4 LIQUID METAL BATTERY

The liquid-metal battery is a concept in reversible chemical in which metal cathode, metal anode, and salt electrolyte are all in the liquid phase. The electrodes and electrolytes are both in the liquid form in the liquid metal batteries, which is one of the promising developments in battery technologies. Besides, the metal cathode, metal anode, and salt electrolyte are all in the liquid phase. The advantage of liquid metal batteries over solid-state batteries is the rapid diffusion of ions into electrodes, which transform into rapid charging and discharging cycles. The bigger the size of the battery, the more power is delivered. These batteries are highly suitable for large-scale electricity storage. These batteries are utilized in electric vehicles and for industrial power backup and grid-energy storage.

5.8.5 POTASSIUM METAL BATTERY

Potassium metal batteries operate as a substitute for lithium-ion batteries owing to their low cost. These batteries utilize potassium ions for the transfer of charge in contrast to lithium ions. The potassium metal batteries self-heal by combining gentle heat with high surface mobility. The mechanism is enabled partly by the heat of the battery itself, which makes the potassium battery design a kind of self-healing closed system. A battery with potassium cathode and anode can compete with a lithium-ion battery as the battery is cheaper, has higher surface mobility, and can heal itself from dendrites.

5.8.6 ZINC MANGANESE BATTERY

Manganese-based batteries have attracted remarkable attention due to their attractive features of low cost, earth abundance, and environmental friendliness. However, the poor stability of the positive electrode due to phase transformation and structural collapse issues hindered their validity as rechargeable batteries. Zinc-manganese batteries once dominated the energy storage market, but their application was limited to being used as primary batteries. A new generation of rechargeable zinc-manganese batteries is poised to compete with Li-ion and Pb-acid batteries on cost, safety, and energy density.

5.8.7 LITHIUM-SULFUR BATTERY

In lithium-sulfur (Li-S) batteries, lithium ions are stored in active materials acting as stable host structures during charge and discharge. In Li-S batteries, there are no host structures. A Li-S battery uses very light active materials, sulfur in the positive electrode, and metallic lithium as the negative electrode. Therefore, its theoretical energy density is extraordinarily high—four times greater than that of Li-ion. This makes it suitable for the aviation and space industries. Saft has selected the most promising Li-S technology based on solid-state electrolytes. This technical path brings very high energy density and long life and overcomes the main drawbacks of the liquid-based Li-S. This technology is supplementary to solid-state Li-ion owing to its superior gravimetric energy density (+30% at stake in Wh/kg). Major technology barriers have already been overcome, and the maturity level is progressing very quickly toward full-scale prototypes. This technology is expected to reach the market just after solid-state Li-ion for applications requiring long battery life.

5.8.8 LITHIUM METAL BATTERY

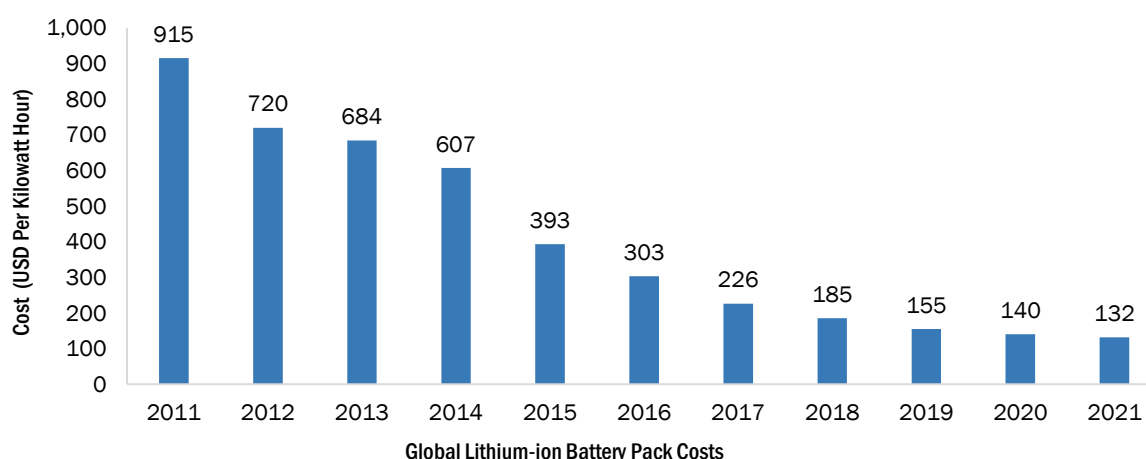
Lithium metal batteries are lithium batteries with an anode from lithium metal. The term 'lithium metal batteries' was coined only after lithium-ion batteries had been invented. Metallic lithium is attractive as a battery material because it easily sheds electrons and positively charged lithium ions. Most lithium metal batteries are non-rechargeable. However, rechargeable lithium metal batteries are also under development. Rechargeable lithium metal batteries are secondary lithium metal batteries. They have metallic lithium as a negative anode. Rechargeable lithium metal batteries can have a long run time due to the high charge density of lithium as an anode material. A rechargeable lithium metal battery was commercialized by Moli Energy (now known as E-One Moli Energy) in the 1980s, but after several cells caught fire, devices using Moli batteries were recalled, and the company went into receivership. However, over the years, the technology has caught the attention of various researchers again, owing to the beneficial characteristics lithium metal batteries offer. Several companies and many academic research groups are currently researching and developing rechargeable lithium metal batteries. Teams of researchers from organizations such as Exxon (US), XNRGI (Exponential Energy) (US), and Harvard (US) are engaged in the development of lithium metal batteries.

5.9 PRICING ANALYSIS

The prime reason behind the adoption of batteries is the benefits provided by them. Lithium-ion, lead-acid, and flow batteries are among the highly adopted batteries in commercial, residential, and utility applications. The standard and customized batteries are manufactured by the market players to meet the rising need of customers. Lithium-ion batteries are one of the most efficient energy storage devices used worldwide. Over recent years, large-scale battery manufacturing and increased capital investments in the battery production process have made lithium-ion battery packs cost-effective and efficient. The price of lithium-ion batteries has been a key hurdle in their adoption since their introduction in the 1990s. A state-of-the-art lithium-ion battery consists of a number of components that are provided at subsidized costs to electric vehicle OEMs. The primary component of any lithium-ion battery is a cell, which accounts for ~50% of its cost, while electronics, assembly, and packaging account for the rest of the cost of the battery.

However, recent developments and claims made by lithium-ion battery manufacturers and automobile companies suggest that the prices of these batteries are expected to decline substantially in the coming years.

FIGURE 24 REDUCING GLOBAL COSTS OF LITHIUM-ION BATTERY PACKS FROM 2011 TO 2021



Source: Bloomberg New Energy Finance (BNEF), Secondary Research, Industry Journals, Expert Interviews, and MarketsandMarkets Analysis

According to Bloomberg New Energy Finance (BNEF), the costs of lithium-ion battery packs dropped to ~140 USD/kWh in 2020, and they are further expected to reduce by ~USD 132 USD/KWh by 2021. Besides, the rising cost of raw materials is estimated to push the average price of the lithium-ion battery pack to 135 USD/kWh by 2022. This is expected to lead to a staggering demand for lithium-ion batteries for use in battery energy storage systems. Large-scale manufacturing of lithium-ion batteries, low prices of components used, and the adoption of technologies to boost the capacity of batteries are a few key factors leading to a decline in the prices of lithium-ion batteries. This, in turn, is expected to contribute to an increased demand for renewable energy storage systems and other grid storage systems.

Apart from lithium-ion batteries, lead-acid batteries also hold a prominent share in the market. They are an ideal option for residential applications, which require off-grid systems and many energy storage systems. Besides, flow batteries are also highly adopted in stationary applications, owing to their benefits, such as large energy storage capacity, uninterrupted power supply, no downtime, and long-life cycle. As the prices of batteries are reducing, they are expected to be used in various industry verticals.

FIGURE 25 BATTERY ENERGY STORAGE SYSTEM: COST BREAKDOWN

				LITHIUM-ION BATTERY - 100 MW/4 HR		LEAD-ACID BATTERY- 100 MW/4 HR		VANADIUM REDOX FLOW BATTERY - 100 MW/4 HR		
PARAMETERS				UNIT	ASP (2020)	ASP (2030)	ASP (2020)	ASP (2030)	ASP (2020)	ASP (2030)
Energy Storage System Installed Cost	Energy Storage System	Storage System	Storage Block	USD/kWh	149-182	79-116	153-172	145-159	235-287	166-238
			Storage BOS	USD/kWh	35-42	23-32	42-47	27-37	47-57	31-43
		Power Equipment	USD/kW	57-69	44-57	108-122	80-104	103-126	80-104	
		Control & Communication	USD/kW	1-2	1-1	1-2	1-1	1-2	1-1	
		System Integration	USD/kWh	33-49	33-40	39-44	31-28	43-53	35-44	
		Engineering, Procurement and Construction (EPC)	USD/kWh	42-64	39-48	43-49	34-42	49-60	40-49	
		Project Development	USD/kWh	49-78	47-58	54-61	43-53	61-75	50-62	
		Grid Integration	USD/kW	18-22	15-18	19-21	15-18	18-22	15-18	
	Total Energy Storage System Installed Cost				USD/kWh	326-438	236-312	355-418	303-359	466-569
USD/kW					1,302-1,752	944-1,249	1419-1672	1211-1436	1,8632-,277	1,388-1,864

Source: 2020 Grid Energy Storage Technology Cost and Performance Assessment, Secondary Research, Battery Journals, and MarketsandMarkets Analysis

As the growth and evolution of the battery storage industry continues, the importance of examining various technologies and comparing their costs and performance on an equitable basis has also increased over the years. The cost of battery energy storage systems depends on various factors, such as project cost, location, battery technology, capacity, and the manufacturer, among others. The figure above demonstrates the cost breakup of lithium-ion batteries, specifically lithium iron phosphate batteries (LFP); lead-acid batteries, and vanadium redox flow batteries, which are the most common batteries adopted as battery energy storage systems.

5.9.1 AVERAGE SELLING PRICES OF KEY PLAYERS, BY BATTERY TYPE

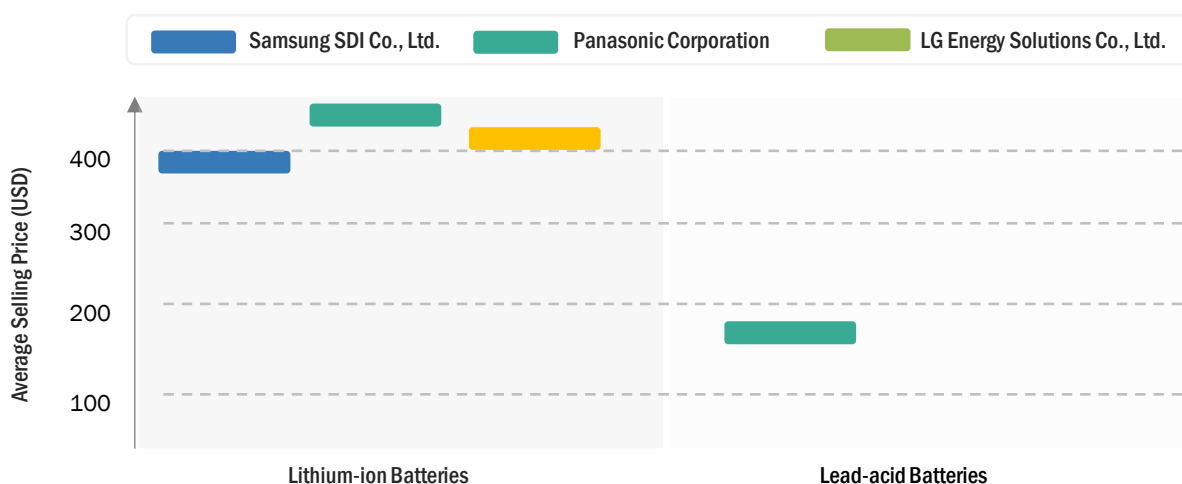
FIGURE 26 AVERAGE SELLING PRICES OF KEY PLAYERS, BY BATTERY TYPE

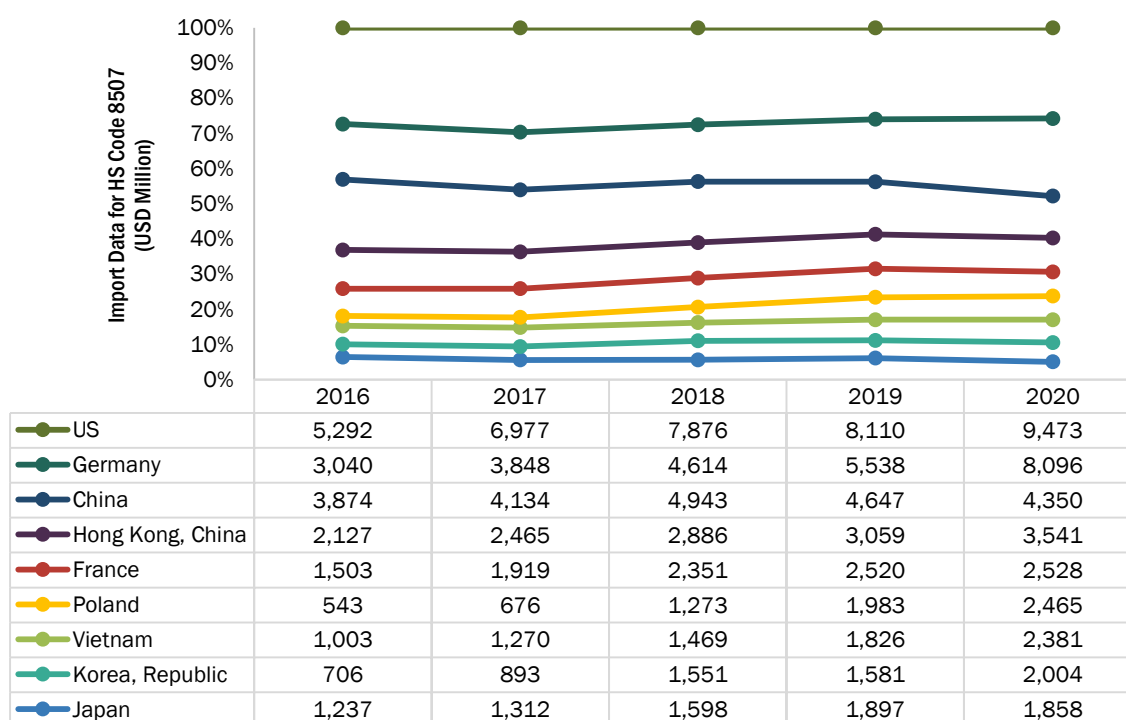
TABLE 8 AVERAGE SELLING PRICES OF KEY PLAYERS, BY BATTERY TYPE (USD)

PRODUCT OFFERING/COMPANY	SAMSUNG SDI CO., LTD.	PANASONIC CORPORATION	LG ENERGY SOLUTIONS CO., LTD.
Lithium-ion Batteries	USD 400	USD 500	USD 430
Lead-acid Batteries	-	USD 190	-

The prices mentioned above for lithium-ion and lead-acid batteries are not fixed. These prices are indicative and are provided based on requirements, capacity, and other properties of batteries used in the battery energy storage system projects. There might be instances where the prices of the batteries may be higher or lower, considering the specifications of batteries and the capacity of the battery energy storage system.

5.10 TRADE ANALYSIS

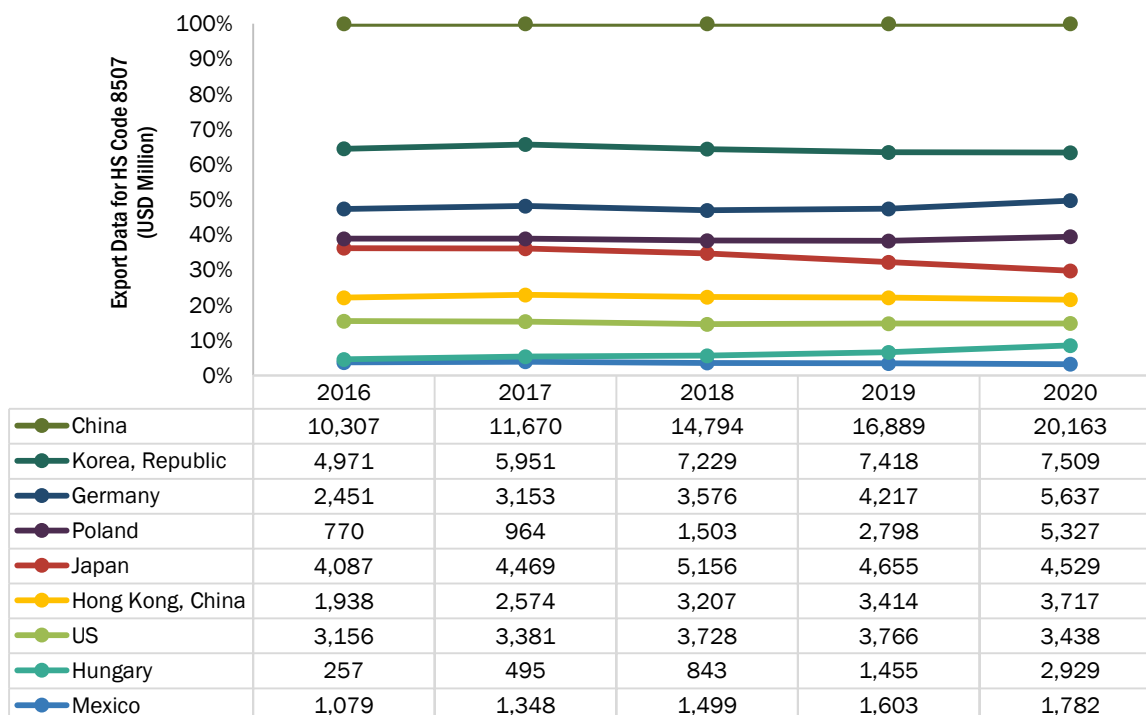
The section describes the trade data of battery energy storage system-related batteries. For battery energy storage systems, the below trade data gives global import and export data analysis for the HS code 8507, which encompasses electric accumulators, incl. separators, therefore, whether or not square or rectangular; parts thereof (excluding spent and those of unhardened rubber or textiles).

FIGURE 27 IMPORT DATA FOR HS CODE 8507, BY KEY COUNTRY, 2016–2020 (USD MILLION)

Source: International Trade Centre Website

In 2020, the US was the largest importer of electric accumulators, incl. separators therefore, whether or not square or rectangular; parts, with an import value of USD 9,473 million. It held a 26% share of the total imports, while most of the countries held a single-digit import share in 2020. Germany had the second-highest import share of USD 8,096 million in 2020, followed by China at USD 4,350 million and France at USD 2,528 million.

FIGURE 28 EXPORT DATA FOR HS CODE 8507, BY KEY COUNTRY, 2016–2020 (USD MILLION)



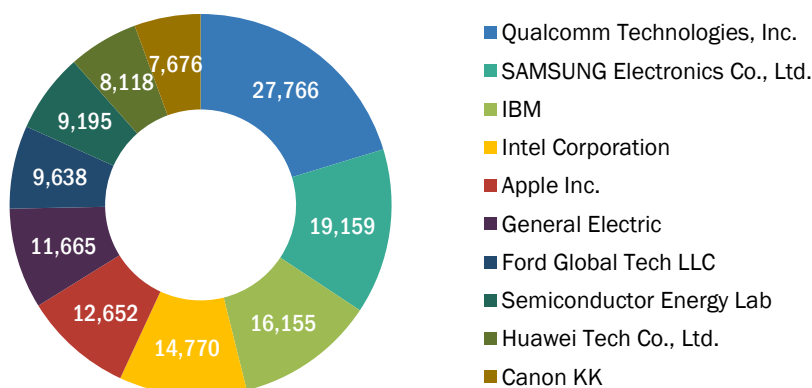
Source: International Trade Centre Website

China was the largest exporter of electric accumulators, incl. separators therefore, whether or not square or rectangular; parts in 2020, with exports value of USD 20,163 million. The country accounted for a 37% share of the overall export in 2020. Korea, Republic was the second-largest exporter, with a value of USD 7,509 million in 2020. The top 3 countries that are exporting these batteries after China and Korea, Republic are Germany, Poland, and Japan, with export values of USD 5,637 million, USD 5,327 million, and USD 4,529 million, respectively, in 2020.

5.11 PATENT ANALYSIS

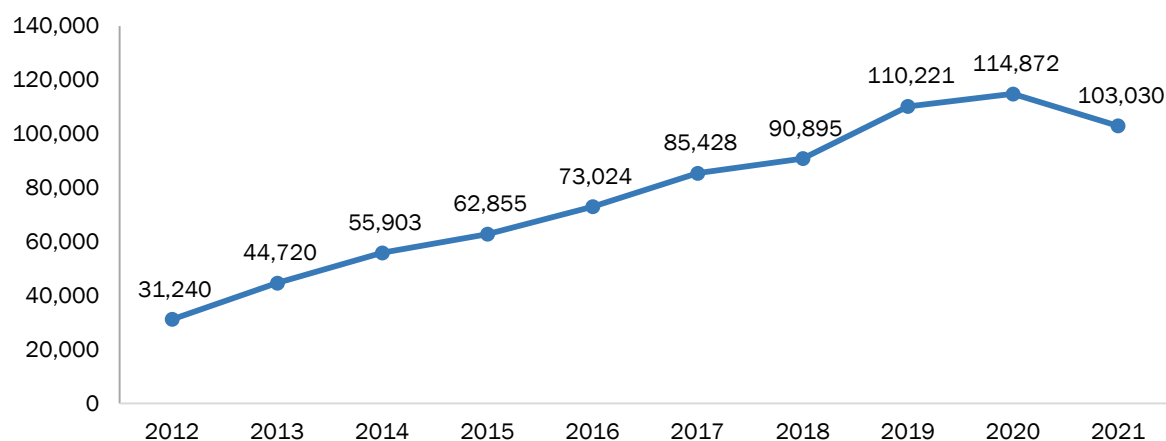
This section provides an indicative analysis of a few of the global patents pertaining to battery energy storage systems.

FIGURE 29 TOP 10 COMPANIES WITH HIGHEST NUMBER OF PATENT APPLICATIONS IN LAST 10 YEARS



Source: Lens.org

FIGURE 30 PATENT ANALYSIS RELATED TO BATTERY ENERGY STORAGE SYSTEM MARKET



Source: Lens.org

TOP 20 PATENT OWNERS IN LAST 10 YEARS	NO. OF PATENTS
Qualcomm Technologies, Inc.	17,094
International Business Machines Corporation	14,919
Samsung Electronics Co LTD	12,414
Intel Corporation	11,268
Ford Global Technologies LLC	9,089
General Electric Company	8,981
Apple Inc.	8,689
Microsoft Technology Licensing LLC	6,429
AT & T Intellectual Property LLP	6,415
Toyota Jidosha Kabushiki Kaisha	5,875
Google LLC	5,749
Semiconductor Energy Laboratory Co LTD	5,539
Canon Kabushiki Kaisha	5,538
Gm Global Technology Operations LLC	5,490
Monsanto Technology LLC	5,463
LG Electronics	4,356
Sony Corporation	4,120
Cisco Technology INC	3,929
Huawei Technologies Co LTD	3,877
Koninklijke Philips Nv	3,829

Source: Lens.org

5.11.1 LIST OF MAJOR PATENTS

The section showcases product innovations through an overview of the patents.

MONTH & YEAR	TITLE	PUBLICATION NUMBER	INVENTOR	DESCRIPTION
June 2021	Building energy storage system with peak load contribution cost optimization	US11036249B2	Mohammad N. EIBsat	An energy storage system for a building includes a battery and an energy storage controller. The battery is configured to store electrical energy purchased from a utility and to discharge stored electrical energy for use in satisfying a building energy load. The energy storage controller is configured to generate a cost function

				including a peak load contribution (PLC) term. The PLC term represents a cost based on electrical energy purchased from the utility during coincidental peak hours in an optimization period.
February 2021	Building energy storage system with multiple demand charge cost optimization	US10909642B2	Mohammad N. EIBsat Michael J. Wenzel Robert D. Turney	An energy storage system includes a battery and an energy storage controller. The battery is configured to store electrical energy purchased from a utility and to discharge the stored electrical energy for use in satisfying a building energy load. The energy storage controller is configured to generate a cost function including multiple demand charges. The controller is configured to modify the cost function by applying a demand charge mask to each of the multiple demand charges.
September 2020	Fe—Cr redox flow battery systems including a balanced arrangement and methods of manufacture and operation	US10777836B1	Kui Wei Liyu Li	A system includes a redox flow battery system that includes an anolyte having chromium ions in a solution, a catholyte having iron ions in a solution, a first half-cell having a first electrode in contact with the anolyte, a second half-cell having a second electrode in contact with the catholyte, and a first separator separating the first half-cell from the second half-cell.
January 2020	Battery energy storage system and control system and applications thereof	US10536007B2	Virgil Lee Beaston Robert Nelson TYRRELL	The patent relates to the embodiments of an electrical energy storage unit. In an embodiment, the electrical energy storage unit (which may also be referred to as a battery energy storage system (BESS)) includes a battery system controller and a plurality of battery packs. Each battery pack of the plurality of battery packs has a plurality of battery cells, a battery pack controller that monitors the plurality of battery cells, a battery pack cell balancer that adjusts an amount of energy stored in each battery cell of the plurality of battery cells, and a battery pack charger.

May 2019	Energy storage system	US10291026B2	Michelle Hunt Elias M. Ayana Bradford K. Palmer	One energy storage system includes an energy storage device and a bidirectional inverter configured to control the flow of power into or out of the energy storage device. The energy storage system further includes a controller configured to control the bidirectional inverter based on one or more signals received from the generator set coupled to the inverter via an AC bus. The controller is configured to, based on the one or more signals, control the bidirectional inverter to store power generated by the generator set in the energy storage device and transmit power from the energy storage device to a load driven by the generator set to maintain the generator set within a range of one or more operating conditions.
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Source: Secondary Research and Google Patents

5.12 REGULATIONS AND STANDARDS

The standards and codes developed by various governments and agencies globally play a vital role in the battery energy storage system market. These codes and standards should be reviewed before the installation of battery energy storage systems. These systems are installed in electricity distribution networks, homes, remote area power supplies, and commercial and industrial installations to improve the transmission and distribution deferral of electricity. Various government organizations, such as the International Standard (IEC), the International Organization for Standardization (ISO), the Underwriters Laboratory (UL), and the American Society of Mechanical Engineers Thermal Energy Storage (ASME TES) have published codes and standards that are related to battery energy storage technologies (for instance, lithium-ion and lead-acid batteries), as well as battery energy storage systems.

The US Department of Energy (DoE) provides an energy storage system guide as a reference to manufacturers to ensure that these systems comply with the safety codes and standards. It provides this guide with the help of partner laboratories such as Sandia National Laboratories (SNL), Oak Ridge National Laboratory, and Pacific Northwest National Laboratory (PNNL). The DoE compliance guide (CG) covers the design and construction of stationary energy storage systems related to batteries and their parts (batteries packs and modules), as well as siting, installation, commissioning, operations, maintenance, and repair/renovation within a built environment. The Australian standards and code of practices for battery storage system installations are currently under development to supplement the requirements of AS/NZS 3000 electrical installations.

Associations such as the Energy Storage Association (ESA), the California Energy Storage Alliance (CESA), the National Fire Protection Association (NFPA), and the Clean Energy Council (CEC) are the non-profit organizations involved in the battery energy storage system market. They work toward producing an innovative ecosystem for battery energy storage systems. Components used in battery energy storage systems and their installations should comply with all statutory regulations. Following are the regulations and standards to be followed by battery energy storage systems.

TABLE 9 REGULATIONS AND STANDARDS FOR BATTERY ENERGY STORAGE SYSTEMS

STATUTORY REGULATIONS AND STANDARDS	DESCRIPTION
AS 1319	Safety signs for the occupational environment
AS 1530.4	Methods for fire tests on building materials, components, and structures—fire-resistance test of elements of construction
AS 3011.2	Electrical installations—secondary batteries installed in buildings—sealed cells
AS 4086.2	Secondary batteries for use with stand-alone power systems—installation and maintenance
AS 4777.1-2005	Grid connection of energy systems via inverters—installation requirements
AS 62040.1.1	Uninterruptible power systems (UPS)—general and safety requirements for UPS used in operator access areas
AS 62040.1.2	Uninterruptible power systems (UPS)—general and safety requirements for UPS used in restricted access locations
ASME TES	Safety standard for molten salt thermal energy storage systems
AS/NZS 4509.1	Stand-alone power systems—safety and installation
AS/NZS 3000	Electrical installations (known as the Australian/New Zealand wiring rules)
AS/NZS 5033	Installation and safety requirements for photovoltaic (PV) arrays
AS/NZS 4777.1	Grid connection of energy systems via inverters—installation requirements
AS/NZS 4777.2	Grid connection of energy systems via inverters—inverter requirements
AS/NZS 60529	Degrees of protection provided by enclosures
AS/NZS 60950.1	Information technology equipment—safety
AS/NZS 5139	Electrical installations—the safety of battery systems for use with power conversion equipment.
AS/NZS 4755.3.5	Instructions and connections for grid-connected electrical energy storage (EES) systems
CWA 50611: 2013	Guidance on the operation, specification, and installation of the flow batteries
EN 50272-2	Safety requirements for secondary batteries and battery installations—part 2: stationary batteries
IEC 62109-1 Ed. 1.0	Safety of power converters for use in photovoltaic power systems—part 1: general requirements
IEC 62109-2 Ed. 1.0	Safety of power converters for use in photovoltaic power systems—part 2: requirements for inverters
IEC 62619:2017	Secondary cells and batteries containing alkaline or other non-acid electrolytes—safety requirements for secondary lithium cells and batteries, for use in industrial applications
IEC 62932-1:2020	Flow battery energy systems for stationary applications - Part 1: Terminology and general aspects

IEC 62932-2-1:2020	Flow battery energy systems for stationary applications - Part 2-1: Performance general requirements and test methods
IEC 62932-2-2:2020	Flow battery energy systems for stationary applications - Part 2-2: Safety requirements
IEC 60622	Secondary cells and batteries containing alkaline or other non-acid electrolytes—sealed nickel-cadmium prismatic rechargeable single cells
IEC 61427-1	Secondary cells and batteries for photovoltaic energy systems (PVES)—general requirements and methods of test
IEC TC 120	Electrical energy storage systems
IEC TC 21	To provide standards for all secondary cells and batteries related to products (dimension and performance), safety (including marking and labeling), testing, and safe application (installation, maintenance, and operations) irrespective of type or application or configuration (hybrid, stand-alone, and module)
IEEE 2030.2.1	Guide for design, operation, and maintenance of battery energy storage systems, both stationary and mobile, and applications integrated with electric power systems
IEEE 1491	Guide for selection and use of BMS in stationary applications
IFC 1206	Electrical energy storage systems
ISO 6469-1:2009	Electrically propelled road vehicles—safety specifications—part 1: on-board rechargeable energy storage systems (RESS)
NFPA 791	Recommended practices and procedures for unlabeled electrical equipment evaluation
NFPA 855	The standard for the Installation of Stationary Energy Storage Systems
PNNL-23618	Inventory of safety-related codes and standards for energy storage systems
UL 1973	Batteries for use in stationary applications
UL 9540	Outlines for investigation for safety for ESS and equipment
UL 3001	Safety for distributed energy generation and storage systems
UL 1642	Safety for lithium batteries
UL 1973	Batteries for use in light electric rail applications and stationary applications
UL 1741	Standards for inverters, converters, controllers, and interconnection system equipment for use with distributed energy resources
UL 9540	Safety of energy storage systems and equipment
UL 9540A	Test method for evaluating thermal runaway fire propagation in battery energy storage systems

Source: Sandia National Laboratories, ISO, IEC, IEEE Publications, and MarketsandMarkets analysis

5.13 KEY CONFERENCES & EVENTS IN 2022–2023

TABLE 10 BATTERY ENERGY STORAGE SYSTEM MARKET: DETAILED LIST OF CONFERENCES & EVENTS

YEAR/QUARTER	NORTH AMERICA	EUROPE	ASIA PACIFIC	REST OF THE WORLD
Q1'22	[March] ICPTST 2022: 16. International Conference on Power Transmission Systems and Transmission	[February] ICAETA 2022: 16. International Conference on Advanced Energy Technologies and Applications [March] ICESSD 2022: 16. International Conference on Energy Storage and Storage Devices	* [February] ICBEST 2022: 16. International Conference on Battery Energy Storage Technology [March] ICAEST 2022: 16. International Conference on Advanced Energy Storage Technologies	[February] ICEEST 2022: 16. International Conference on Electrochemical Energy Storage Technologies [March] ICBESS 2022: 16. International Conference on Battery Energy Storage Systems
	[April] ICESEA 2022: 16. International Conference on Energy Storage Engineering and Applications [April] ICESST 2022: 16. International Conference on Energy Storage Science and Technology [May] ICDEGEE 2022: 16. International Conference on Distributed Energy Generation and Energy Efficiency [June] ICESTFC 2022: 16. International Conference on Energy Storage Technologies and Fuel Cells	* [April] ICSSG 2022: 16. International Conference on Energy Storage in Smart Grids [April] ICSSG 2022: 16. International Conference on Energy Storage in Smart Grids [May] ICTSGDPE 2022: 16. International Conference on Transmission Systems, Generation and Distribution in Power Engineering [June] ICPGTDUE 2022: 16. International Conference on Power Generation, Transmission, Distribution, Utilization, and Engineering	[April] ICPSESTCAES 2022: 16. International Conference on Power System Energy Storage Technology and Compressed-Air Energy Storage [May] ICESEA 2022: 16. International Conference on Electrical Sustainable Energy Systems and Applications [June] ICTSGDPE 2022: 16. International Conference on Transmission Systems, Generation and Distribution in Power Engineering	[April] ICBES 2022: 16. International Conference on Biofuels and Energy Storage [May] ICDEMSG 2022: 16. International Conference on Distributed Energy Management for Smart Grids [June] ICPDEST 2022: 16. International Conference on Power Distribution and Energy Storage Technology
Q3'22	[July] ICAESM 2022: 16. International Conference on Advances in Energy Storage Materials [August] ICBESST 2022: 16. International Conference on Battery Energy Storage Systems and	* [July] ICBES 2022: 16. International Conference on Battery Energy Storage [August] ICBESMS 2022: 16. International Conference on Battery Energy Storage and Management Systems [September] ICECSM 2022: 16. International	[July] ICFEST 2022: 16. International Conference on Future Energy Systems and Technologies [August] ICDGSTA 2022: 16. International Conference on Distributed Generation Systems, Technologies and Applications	[July] ICHESS 2022: 16. International Conference on Hydrides for Energy Storage Systems [August] ICPESTPSH 2022: 16. International Conference on Power System Energy Storage Technologies and Pumped Storage

	Technologies [September] ICESE 2022: 16. International Conference on Energy Storage and Electrochemistry	Conference on Energy Conversion, Storage and Management	[September] ICPSDGT 2022: 16. International Conference on Power Systems, Distribution, Generation and Transmission	Hydropower [September] ICPMESE 2022: 16. International Conference on Power Management and Energy Systems Engineering
Q4'22	* [October] 4th Battery and Energy Storage Conference [October] ICHRESS 2022: 16. International Conference on Hybrid Rechargeable Energy Storage Systems [November] ICESTFC 2022: 16. International Conference on Energy Storage Technologies and Fuel Cells [December] ICGESST 2022: 16. International Conference on Grid Energy Storage Systems and Technologies	[October] ICABTICSBESS 2022: 16. International Conference on Advanced Battery Technologies and Innovative Control Strategies for Battery Energy Storage Systems [November] ICGRSETS 2022: 16. International Conference on Green, Renewable and Sustainable Energy Technologies and Systems [December] ICPESGS 2022: 16. International Conference on Power, Energy and Smart Grid Systems	[October] ICDESDSG 2022: 16. International Conference on Distributed Energy Storage Devices in Smart Grids [November] ICRERA 2022: 16. International Conference on Renewable Energy Resources and Applications	[October] ICPES 2022: 16. International Conference on Power and Energy Systems [November] ICWEEM 2022: 16. International Conference on Wind Energy and Ecosystems Management [December] ICBES 2022: 16. International Conference on Batteries and Energy Storage
Q1'23	[January] ICBEST 2023: 17. International Conference on Battery Energy Storage Technologies	[January] ICBESS 2023: 17. International Conference on Battery Energy Storage Solutions [February] ICHRESS 2023: 17. International Conference on Hybrid Rechargeable Energy Storage Systems [March] ICESSD 2023: 17. International Conference on Energy Storage and Storage Devices	[January] ICBRES 2023: 17. International Conference on Battery Storage for Renewable Energy Systems [February] ICPESTHES 2023: 17. International Conference on Power System Energy Storage Technologies and Hydrogen Energy Storage	[January] ICPESTCAES 2023: 17. International Conference on Power System Energy Storage Technologies and Compressed-Air Energy Storage [February] ICESSTG 2023: 17. International Conference on Energy Storage for Smart Grids [March] ICBESS 2023: 17. International Conference on Battery Energy Storage Systems
Q2'23	[April] ICESST 2023: 17. International Conference on Energy Storage Science and Technology	[April] ICESSTG 2023: 17. International Conference on Energy Storage in Smart Grids [May] ICPSA 2023: 17.	[April] ICPESTCAES 2023: 17. International Conference on Power System Energy Storage Technologies and Compressed-Air Energy	[May] ICESSTA 2023: 17. International Conference on Energy Storage Science, Technologies and Applications

	[June] ICSTFC 2023: 17. International Conference on Energy Storage Technologies and Fuel Cells	International Conference on Power Systems and Applications [June] ICDSOESD 2023: 17. International Conference on Distribution System Optimization and Energy Storage Devices	Storage * [May] ICPSESTEPH 2023: 17. International Conference on Power System Energy Storage Technology and Pumped Storage Hydropower
Q3'23	[August] ICBESST 2023: 17. International Conference on Battery Energy Storage Systems and Technologies	[July] ICBESS 2023: 17. International Conference on Battery Energy Storage Systems [August] ICBESMS 2023: 17. International Conference on Battery Energy Storage and Management Systems [September] ICBMBMS 2024: 18. International Conference on Battery Monitoring and Battery Management Systems	[August] ICECST 2023: 17. International Conference on Energy Conversion Systems and Technologies [August] ICESMD 2023: 17. International Conference on Energy Storage Materials and Devices
Q4'23	[October] ICESSD 2023: 17. International Conference on Energy Storage and Storage Devices [December] ICECSGM 2023: 17. International Conference on Energy Conversion, Storage and Grid Modernization	[December] ICAPSP 2023: 17. International Conference on Advances in Power System Protection	[December] ICMEESEBT 2023: 17. International Conference on Materials for Electrochemical Energy Storage and Emerging Battery Technologies [November] ICWEEM 2023: 17. International Conference on Wind Energy and Ecosystems Management [December] ICBES 2023: 17. International Conference on Batteries and Energy Storage

Note: * indicates conferences and events that may be more important because of wider reach, sponsorship from leading organizations, thought leadership, etc.

5.14 TARIFF AND REGULATORY LANDSCAPE

5.14.1 REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

This section includes a region-wise list of key regulatory bodies, government agencies, industry associations, and non-profit organizations relevant to the market.

TABLE 11 NORTH AMERICA: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	SHORT DESCRIPTION
US	US Department of Energy	Government Regulatory	It is a cabinet-level department of the US government concerned with its policies on energy and safety.
US	National Fire Protection Association	Non-profit Organization	It is a global self-funded non-profit organization of the US, concerned with the death, economic loss, and injury due to fire and electrical hazards.
US	The American Society of Mechanical Engineers	Non-profit Organization	It is a non-profit organization that guides the engineering community by developing codes and standards and conducting research, conferences, certifications, etc.
US	Energy Storage Association	Association	It is the national trade association dedicated to energy storage and working toward a sustainable, efficient, resilient, and affordable electricity grid.
Canada	Energy Storage Canada	Association	It is the national association for the energy storage industry in the country, which represents the complete supply chain of energy storage.

TABLE 12 EUROPE: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	SHORT DESCRIPTION
UK	Department for Business, Energy, and Industrial Strategy	Government Agency	It is the government agency of the UK that is responsible for energy and clean growth, climate change, and businesses.
UK	Office of Gas and Electricity Markets (Ofgem)	Government Agency	It is the non-ministerial government department and an independent National Regulatory Authority. It focuses on delivering a greener energy system.
France	International Energy Agency (IEA)	Government Agency	It is an intergovernmental organization that works with countries worldwide to shape energy policies for a sustainable and secure future.

Belgium	European Association for Storage of Energy	Non-profit Organization	The European Association for Storage of Energy (EASE) is the leading association representing organizations active across the entire energy storage value chain. It supports the deployment of energy storage to support the cost-effective transition to a climate-neutral and secure energy system.
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TABLE 13 ASIA PACIFIC: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	SHORT DESCRIPTION
India	Bureau of Energy Efficiency	Government Regulatory	The mission of the Bureau of Energy Efficiency is to assist in developing policies and strategies with a thrust on self-regulation and market principles within the overall framework of the Energy Conservation Act.
India	India Energy Storage Alliance (IESA)	Association	It is the membership-driven industry alliance focused on green hydrogen, advanced energy storage, and e-mobility technologies in the country.
China	National Energy Administration	Government Agency	It is the state administration under the National Development and Reform Commission. It drafts China's national energy strategy, regulates the energy sector, and implements energy policies.
Japan	Agency for Natural Resources and Energy	Government Agency	It is the government agency of natural resources and energy in Japan and part of the Ministry of Economy, Trade, and Industry. It is responsible for the country's policies pertaining to natural resources and energy.
Australia	Australian Renewable Energy Agency	Agency	It is an independent agency of the Australian federal that manages the renewable energy programs of the country.

TABLE 14 REST OF THE WORLD: LIST OF REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

COUNTRY	ORGANIZATION NAME	ORGANIZATION TYPE	SHORT DESCRIPTION
South Africa	South African Energy Storage Association	Association	It is an energy storage association set up to guide members involved in business development in South Africa and assist in an advisory role within the Southern African Development.
UAE	International Renewable Energy Agency	Non-profit Organization	It is an intergovernmental non-profit organization supporting countries in their transition to a future with sustainable energy.

6 BATTERY ENERGY STORAGE SYSTEM MARKET, BY STORAGE SYSTEM

6.1 INTRODUCTION

Battery energy storage systems (BESS) are integrated with the grid to stabilize the distribution of energy and improve the power quality. In this chapter, storage systems are segmented into two types: front-of-the-meter (FTM) and behind-the-meter (BTM). Any power equipment or utility, which is offsite, away from the property or home is referred to as the front-of-the-meter (FTM) storage system. Electricity users such as industrial; micro, small & medium enterprises (MSME); commercial, hospitals, schools, or residential users have meters installed on their premises, which calculate the amount of energy taken from the grid and consequently, the amount of energy owed to the utility provider. Whereas behind-the-meter refers to the storage system and energy production utility, which is used on-site without passing through the meter. The behind-the-meter energy storage system directly supplies electricity to homes and buildings.

FIGURE 31 BATTERY ENERGY STORAGE SYSTEM MARKET, BY STORAGE SYSTEM



Source: Company Websites, Expert Interviews, White Papers, and MarketsandMarkets Analysis

6.2 FRONT-OF-THE-METER

6.2.1 FRONT-OF-THE-METER STORAGE SYSTEMS ARE WIDELY USED BY UTILITIES

A front-of-the-meter (FTM) battery energy storage system delivers electricity through power lines owned by the utility company, which distributes the energy to the consumers. Front-of-the-meter battery storage systems are also referred to as utility-scale or grid-scale battery storage systems, which can be connected to distribution/transmission networks or power generation assets. This type of storage system is mainly used by power-generating utilities or large-scale power distributors.

. The surplus energy stored in this storage system is referred to as the grid-connected load, which indicates that the storage system is charging. At times of peak demand for electricity, this stored energy is sent to the transmission lines/distribution lines. The stored energy is fed directly into the grid and transmitted to remote end users instead of being used on site. These storage systems supply energy to off-site or remote locations such as commercial buildings and offices, where the energy needs to pass through a meter before it can be used. For instance, the energy received by residential users needs to be passed through a meter before usage, as the storage system and energy generation system are positioned as front-of-the-meter systems. Similarly, the energy fed in transmission or distribution lines in grids or microgrids by the storage system is liable to pass through a meter in a front-of-the-meter system.

6.3 BEHIND-THE-METER

6.3.1 BEHIND-THE-METER (BTM) STORAGE SYSTEMS HELP IN SAVING ELECTRICITY BILLS FOR CUSTOMERS

The behind-the-meter storage system refers to an energy system that provides power, which is utilized on-site without passing through the meter. This system provides energy directly to the end users without interacting with the electric grid. Systems such as home solar panel systems, gas-powered generators, and small wind turbines generate energy, which can be used on the premises. The energy produced and stored by the commercial and residential solar panels is separated from the grid and is not counted by the meter, as, these solar panels are positioned behind-the-meter. Behind-the-meter is also referred to as Distributed Energy Resources (DER), as the electricity is generated or managed behind the electricity meter in a business or house. DERs are renewable energy systems, which provide power to homes and businesses.

Behind-the-meter energy storage systems are often paired with distribution grids to reduce energy bills and/or enhance compensation. Bill reduction is primarily achieved through demand charge reduction (lowering the maximum power consumed, typically per month) and energy arbitrage (shifting electricity consumption from high to low energy cost periods). For instance, the Massachusetts Department of Energy Resources has established an incentive program, to promote the development of renewable battery storage systems in Massachusetts. The incentive program requires information on the installation of the additional meter to measure the amount of electricity produced by the solar panels. According to Solar Massachusetts Renewable Target (SMART) incentive program, customers combining their storage systems with solar energy in three different utility regions are eligible to receive an incentive payment for the users of behind-the-meter systems at a fixed rate per kilowatt-hour of energy produced by the solar panels.

7 BATTERY ENERGY STORAGE SYSTEM MARKET, BY ELEMENT

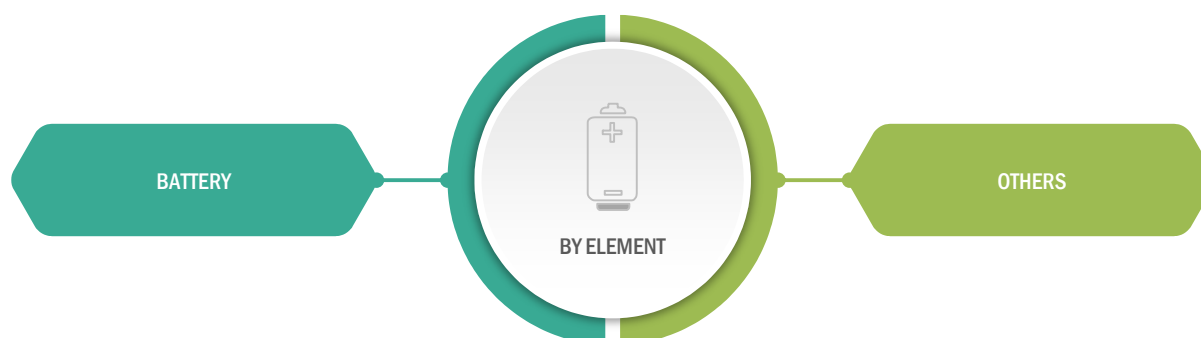
KEY FINDINGS

- The market for battery elements was worth USD 2,157.2 million in 2021 and is projected to reach USD 9,688.3 million by 2027; growing at a CAGR of 28.9% between 2022 and 2027.
- Several key technical features are considered to select the battery for storage systems. A few of the important criteria include performance requirement, battery cell and module cost, installation, infrastructure, ambient condition, application, and safety. The battery element alone constitutes 60% of the total battery energy storage system. It is the major element, where the functioning is regularized by hardware and software, which increases its efficiency.
- The others segment was worth USD 1,377.7 million in 2021 and is projected to reach USD 5,445.9 million by 2027, growing at a CAGR of 26.2% between 2022 and 2027.
- The other segment includes engineering, procurement and construction (EPC), and development services. These services are essential for regulating and controlling systems. The regulating and controlling systems ensure the safety of critical components and the proper functioning of the whole system.
- The hardware used in the battery energy storage system facilitates scheduled energy storage and ensures that communication and functioning are as per the requirement. It involves supervisory control and data acquisition (SCADA), control system, OEM testing, and site preparation for the monitoring of storage and related processes.

7.1 INTRODUCTION

The elements of the battery energy storage system include battery and others. The other elements include hardware and others. The hardware includes power conversion systems (PCS), containers, monitoring and control systems, communication devices, HVAC, meters, AC main panels, and others. Other elements include engineering, procurement, and construction (EPC); and development services, which comprise OEM testing, control systems, lifting and hoisting, labor, EPC overhead, and others.

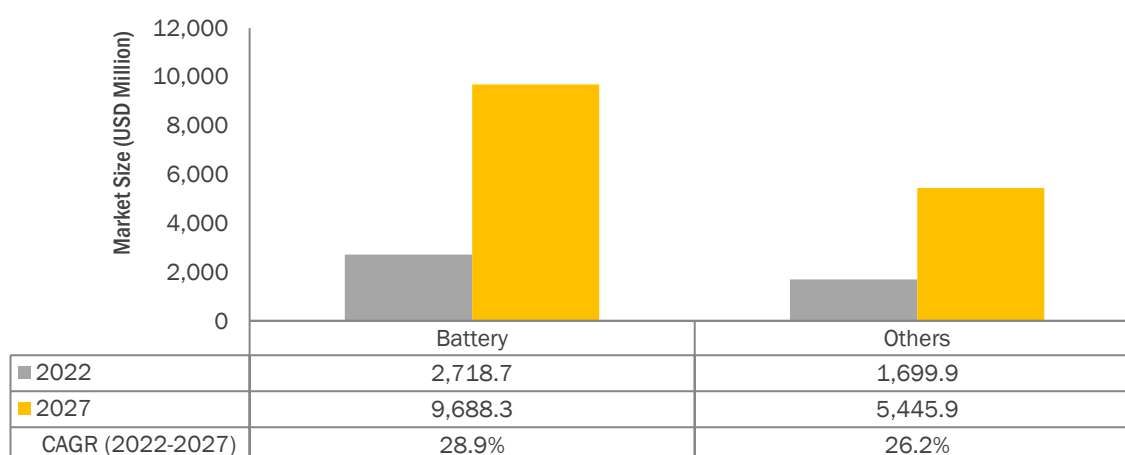
FIGURE 32 BATTERY ENERGY STORAGE SYSTEM MARKET, BY ELEMENT



Note: Others include engineering, procurement, and construction (EPC) and development services.

Source: Company Websites, Expert Interviews, White Papers, and MarketsandMarkets Analysis

FIGURE 33 BATTERY ELEMENT SEGMENT TO HOLD LARGER SHARE OF BATTERY ENERGY STORAGE SYSTEM MARKET DURING FORECAST PERIOD



Note: Other elements include engineering, procurement, and construction (EPC) and development services.

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Batter Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 15 BATTERY ENERGY STORAGE SYSTEM MARKET, BY ELEMENT, 2018–2021 (USD MILLION)

Element	2018	2019	2020	2021
Battery	1,380.4	2,164.0	1,767.9	2,157.2
Others	939.0	1,442.7	1,153.5	1,377.7
Total	2,319.4	3,606.7	2,921.4	3,534.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Note: Others include engineering, procurement, and construction (EPC) and development services.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 16 BATTERY ENERGY STORAGE SYSTEM MARKET, BY ELEMENT, 2022–2027 (USD MILLION)

Element	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Battery	2,718.7	3,451.3	4,414.0	5,688.8	7,394.4	9,688.3	28.9%
Others	1,699.9	2,113.0	2,646.6	3,341.0	4,247.3	5,445.9	26.2%
Total	4,418.6	5,564.3	7,060.6	9,029.8	11,641.7	15,134.2	27.9%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Others include engineering, procurement, and construction (EPC) and development services.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The battery element segment held a larger size of the battery energy storage system market in 2021. The battery energy storage system constitutes components such as a battery, battery management system, energy management system, monitoring and control system, and power conversion system. The battery elements used in the battery energy storage system are in the form of modules and racks. The battery element alone constitutes 60% of the total battery energy storage system. It is the major element, where the functioning is regularized by hardware and software, which increases its efficiency. Mostly used battery types in battery energy storage systems are lithium-ion and advanced lead-acid batteries. In the recent years, flow batteries have also gained momentum and are expected to be adopted widely during the forecast period owing to the rising demand for these batteries from utility and residential applications.

Other element segment includes hardware components that comprise regulating and controlling systems. The regulating and controlling systems ensure the safety of critical components and the proper functioning of the whole system. The hardware used in the battery energy storage system facilitates scheduled energy storage and ensures that communication and functioning are as per the requirement. Besides, it also comprises engineering, procurement, and construction (EPC), and development services, which account for about 20% of the total cost of battery energy storage systems.

7.2 BATTERY

The battery is a major part of the battery energy storage system. It includes 60% of the total cost of the system. Cell-based batteries consist of individual cells connected to the module and then into packs. The different types of batteries used in battery energy storage systems are lithium-ion, advanced lead-acid batteries, sodium-sulfur, and flow batteries, among others. The usage of these batteries depends on their characteristics and intended application. Additionally, the use of these batteries also depends on their capacity need.

7.2.1 BATTERY SELECTION IS CRUCIAL STEP FOR DEVELOPMENT OF BATTERY ENERGY STORAGE SYSTEMS

A number of key technical features are used to compare different energy storage solutions. A few of the most significant considerations of battery selection are represented in the below diagram. A few of these considerations relate to the life and performance of the battery, such as temperature, DoD, calendar and cycle life, as well as efficiency and performance requirements. The other considerations are related to the specific location in which the battery is required, such as installation infrastructure, space limitations, and ambient conditions. Other considerations are economic such as maintenance costs, power component costs, and battery cell & module costs. Safety is also an important criterion when selecting the battery. Additionally, these batteries are widely adopted in applications such as utility, commercial, and residential, hence the capacity of these batteries plays a major role during the selection of the type of battery to be set up in the storage system.

FIGURE 34 CRITERIA FOR BATTERY SELECTION

BATTERY SELECTION	
- Company Warranty/Performance Guarantee - Performance Requirements - Maintenance Cost - Power Component Availability & Cost - Battery Cell and Module Cost - Grid/Utility Requirements - Depth/Length of Discharge	
- Calendar & Cycle Life - Efficiency - Application - Space Limitation - Installation Infrastructure - Ambient Condition/Temperature - Safety	

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

7.3 OTHER ELEMENTS

7.3.1 HARDWARE

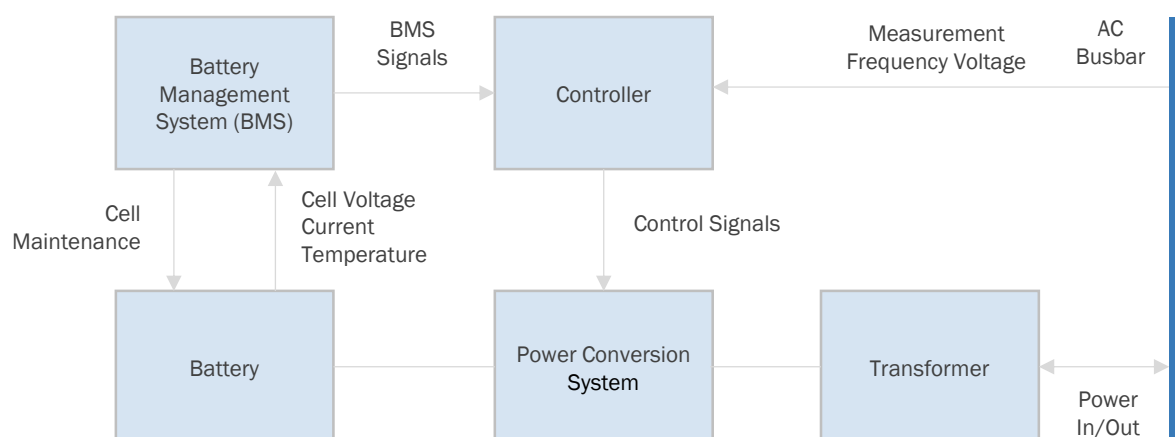
7.3.1.1 Power conversion systems and battery management systems are essential components for high performance of battery energy storage systems

The hardware used in the battery energy storage system facilitates scheduled energy storage and ensures that communication and functioning are as per the requirement. Hardware elements include power conversion systems, containers, monitoring and control systems, AC main panels, electrical conduits, HVAC, fire detection systems, isolation transformers, and Aux power. These equipment help in fixing the hardware part of the system and facilitate the communication of the system with users. Also, the outer infrastructure of the battery energy storage system or enclosure/container contains the metering and monitoring hardware components as well as communication infrastructure. The enclosure protects the battery energy storage system from the outer environment. Thus, batteries are installed in the battery energy storage system as hardware components to support grid flexibility.

The batteries often perform in several critical conditions including weather, high thermal heat, combustion, among others, which may damage the system, and can also cause hazardous situations. To protect these batteries, BMS is installed to monitor critical information of the battery racks. The critical information includes important electrical parameters such as cell temperature, electrolyte levels, and battery enclosure temperature. All this information is saved for periodic and proper maintenance and to ensure safety and maximize performance. The battery management system also prevents individual cells from overcharging and controls the charge and discharge of the battery. The BMS is a very important component of any storage system. A cluster of battery modules is combined to form a tray, which is packed with its battery management system (BMS). For certain model types, these trays are stacked together to form a battery rack.

In addition to batteries, a typical battery energy storage system has the following main components: power conversion systems, monitoring and control systems, switchgear, and controllers. A battery energy storage system can have a power conversion system that comprises multiple power converters depending on the size and design of the battery energy storage system. The batteries used in the battery energy storage system are inherently direct current (DC) equipment, on the other hand, the power grid operates on alternating current (AC). Thus, the power conversion system works as an inverter, which converts AC to DC and rectifies DC to AC. The power conversion system is also responsible for charging or discharging a battery. To address the issue of the fluctuation of electricity voltage, the transformer is used, which converts DC to AC to the desired level between a power converter and a battery energy storage system. The high-level and low-level controllers are used for the smart management of the electric energy generated.

FIGURE 35 LAYOUT OF BATTERY ENERGY STORAGE SYSTEM



Source: MarketsandMarkets Analysis

7.3.2 OTHERS

7.3.2.1 Before commissioning of BESS, EPC and development services are essential

Others comprise engineering, procurement, construction (EPC), and development services. It accounts for about 20% of the total cost of battery energy storage systems. EPC includes modules for various functions and systems, which help in the proper functioning of the battery energy storage system. It involves supervisory control and data acquisition (SCADA), control system, OEM testing, and site preparation, among others. These functions and systems enable the timely monitoring of the storage and related processes. The SCADA system communicates with the battery management system (BMS) and controls the devices throughout the plant, which maximizes the performance of the storage system. These elements are necessary right from the first phase to the commissioning of the project. Each calculation related to the installation of the system and construction of the project is based on this element. It also helps understand the costing required for the construction of the system with favorable and unfavorable conditions.

The integration of battery energy storage systems (BESS) with the on-grid or off-grid connection, is used to supply electricity during the peak demand, improve the stability and power quality of the distributed energy, and operate as a backup during source as intermittency requires a high level of control. To ensure safe operation with high reliability, efficient control is required. Thus, before the commissioning of the system, EPC and development services are essential. The development services here refer to the work related to the handling of the energy storage system efficiently by customers and enabling the interconnection between the system and users.

8 BATTERY ENERGY STORAGE SYSTEM MARKET, BY BATTERY TYPE

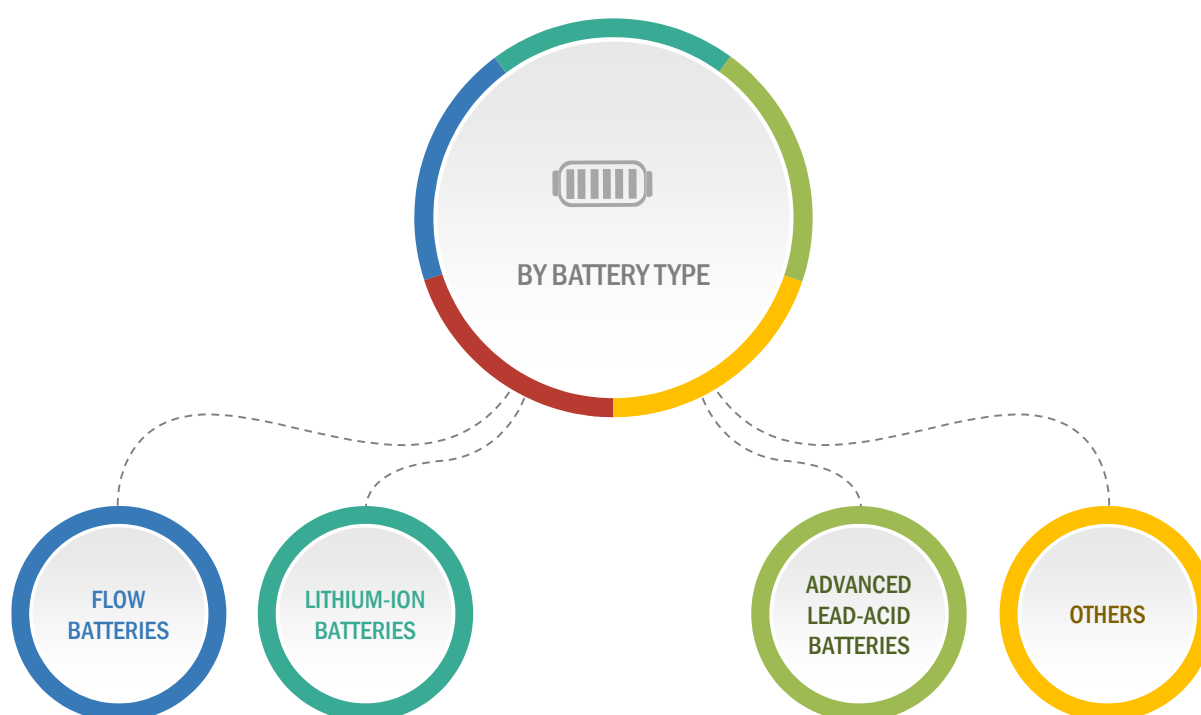
KEY FINDINGS

- Lithium-ion batteries held the largest size of the battery energy storage system (BESS) market in 2021. The market for lithium-ion batteries was worth USD 1,838.1 million in 2021 and is projected to reach USD 7,245.1 million by 2027, growing at a CAGR of 26.1% between 2022 and 2027.
- The growth of this segment is attributed to the advantages of the battery such as high energy and power density, which leads to low standby losses and a high life expectancy of about 5–15 years with 98% efficiency.
- The market for flow batteries is expected to register the highest CAGR of 32.8% between 2022 and 2027, to reach USD 4,525.9 million by 2027 from USD 842.8 million in 2021.
- The rapidly growing demand for efficient, flexible, and durable energy storage systems in utilities, as well as commercial and industrial applications, is responsible for the growth of flow batteries.
- The lead-acid battery is a mature technology for energy storage and is suitable for off-grid storage applications. The advanced lead-acid batteries are specially designed to provide a steady current over a long period, and they can be repeatedly discharged by as much as 80% of their capacities. This ability of lead-acid batteries makes them ideal for use in battery energy storage systems based on off-grid connection.

8.1 INTRODUCTION

The battery energy storage system (BESS) is a device embedded with various rechargeable batteries. The battery energy storage system market based on battery type has been classified into lithium-ion batteries, flow batteries, advanced lead-acid batteries, and others. Other batteries include sodium-sulfur batteries, nickel-cadmium batteries, nickel-metal hydride batteries, nickel-iron batteries, and flywheel batteries. Lithium-ion and advanced lead-acid are the oldest types of batteries and are widely used in utilities. Lithium-ion batteries are mostly used in battery energy storage systems due to their higher energy density and better performance than other batteries. Flow batteries are also gaining traction in the market owing to their technical advantages over conventional batteries such as long-life span, easy maintenance, and large storage capacity. These batteries are witnessing penetration in applications such as residential and electric vehicle charging stations. Currently, these batteries are highly utilized by utility and commercial applications. Large-scale companies are highly involved in the manufacturing of these batteries. For instance, Panasonic Corporation, Samsung SDI Co., Ltd., BYD Company Limited, among others provide a wide variety of lithium-ion batteries, lead-acid batteries, and other types of batteries, which are used in electronics, automobiles, and industrial applications.

FIGURE 36 BATTERY ENERGY STORAGE SYSTEM MARKET, BY BATTERY TYPE



Note: Other battery types include sodium-sulfur batteries, nickel-cadmium, nickel-metal hydride, nickel-iron batteries, and flywheel batteries.

Source: MarketsandMarkets Analysis

The selection of a battery for battery energy storage systems is based on battery capacity (kWh), output power (kW), C-rating of the battery, the number of cycles per period, expected lifetime (years), discharge depth (%), self-discharge (% per month), operation and maintenance costs, the dimension of battery installation, and safety (electrical, mechanical, and operator safety).

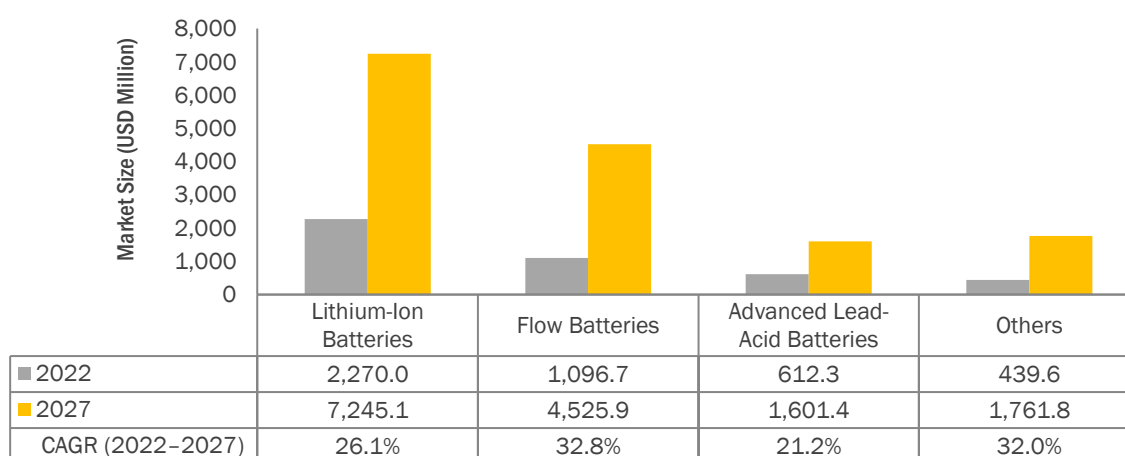
Among all conventional battery types, lithium-ion batteries have a long lifespan of 5–15 years, and up to 98% efficiency (i.e., only 2% of electrical charge is lost during use). Lithium-ion batteries are further categorized into lithium-iron-phosphate, lithium cobalt oxide, and lithium manganese oxide batteries. The table below shows the comparison between lithium-ion and lead-acid batteries used in battery energy storage systems, with respect to their lifespan, cycle time, depth of discharge (DoD), roundtrip efficiency, and operating temperature.

TABLE 17 COMPARISON BETWEEN LITHIUM-ION AND LEAD-ACID BATTERIES

BATTERY	SUITABLE APPLICATION [ENERGY (KWH) OR POWER (KW)]	LIFETIME (YEARS)	CYCLE LIFE (NUMBER OF CHARGE-DISCHARGE CYCLES)	DEPTH OF DISCHARGE (DOD)	ROUND TRIP EFFICIENCY	OPERATING TEMPERATURE
Lithium-ion Phosphate	Both	5–10	3000–6000	50–90%	89%	0–45 °C
Lithium-cobalt Oxide	Both	5–7	800–1200	50%	97%	–10–45 °C
Lithium-manganese Oxide	Both	10–30	5000–10000	90%	75–95%	–10–45 °C
Lead-acid	Energy	3–15	2000–3000	50%	85%	–10–45 °C

Source: Queensland Government and MarketsandMarkets Analysis

FIGURE 37 LITHIUM-ION BATTERIES TO HOLD LARGEST SIZE OF BATTERY ENERGY STORAGE SYSTEM MARKET IN 2022



Note: Other battery types include sodium–sulfur batteries, nickel–cadmium, nickel-metal hydride, nickel–iron batteries, and flywheel batteries.

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 18 BATTERY ENERGY STORAGE SYSTEM MARKET, BY BATTERY TYPE, 2018–2021 (USD MILLION)

Battery Type	2018	2019	2020	2021
Lithium-Ion Batteries	1,249.6	1,911.5	1,534.9	1,838.1
Advanced Lead-Acid Batteries	384.7	577.1	445.5	513.9
Flow Batteries	484.8	793.5	669.1	842.8
Others	200.4	324.6	271.9	340.1
Total	2,319.4	3,606.7	2,921.4	3,534.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 19 BATTERY ENERGY STORAGE SYSTEM MARKET, BY BATTERY TYPE, 2022–2027 (USD MILLION)

Battery Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Lithium-Ion Batteries	2,270.0	2,819.2	3,521.5	4,424.6	5,650.9	7,245.1	26.1%
Advanced-Lead Acid Batteries	612.3	734.9	888.9	1,083.6	1,309.8	1,601.4	21.2%
Flow Batteries	1,096.7	1,437.7	1,899.1	2,528.3	3,364.7	4,525.9	32.8%
Others	439.6	572.5	751.1	993.3	1,316.3	1,761.8	32.0%
Total	4,418.6	5,564.3	7,060.6	9,029.8	11,641.7	15,134.2	27.9%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The lithium-ion batteries segment held the largest size of the battery energy storage system market in 2021. The lithium-ion battery has very high energy and power density, which leads to low standby losses and high life expectancy. Due to these characteristics, lithium-ion batteries are likely to account for a major share of the battery energy storage system market by 2027. Lithium-ion sulfur battery energy storage systems are used in utility applications to address the demand during peak time as these systems offer high response time and can store excess energy. Also, lithium-ion-sulfur battery energy storage systems support frequency regulation, demand charge reduction, grid-buffering, and renewable integration, which are suitable for on-grid connections.

The lead-acid battery is a mature technology for energy storage and is suitable for off-grid storage applications. The advanced lead-acid batteries are specially designed to provide a steady current over a long period, and they can be repeatedly discharged by as much as 80% of their capacities. This ability of lead-acid batteries makes them ideal for use in battery energy storage systems based on off-grid connection. These batteries are being increasingly used for residential energy storage and behind-the-meter (BTM) energy storage purposes. Lead-acid batteries are widely used in vehicles, mainly in autonomous vehicles owing to their characteristics of long battery life and slow discharge rate. According to the International Energy Agency (IEA), worldwide EV sales have surpassed the 10 million mark in 2020. In the coming years, the adoption of EVs is expected to increase, which will also pave the way for increased battery storage usage in the automotive sector. The increase in the installation of battery storage systems at the charging stations of electric vehicles is estimated to fuel the growth of the market during the forecast period.

8.2 LITHIUM-ION BATTERIES

8.2.1 LI-ION BATTERIES ARE WIDELY USED IN BATTERY ENERGY STORAGE SYSTEMS DUE TO THEIR HIGH ENERGY DENSITY

Lithium-ion (Li-ion) batteries were introduced commercially by Sony in 1991 for use primarily in consumer products. Since then, they have become the most widely used battery technology for grid-scale energy storage. Lithium-ion batteries have the versatility to handle small-scale applications, such as powering electric vehicles, as well as grid-scale applications requiring megawatts of power for hours at a time. The major factor for the rapid increase in the use of BESS Li-ion technology has been a 50% decline in the costs of energy storage over the past two years.

Li-ion batteries have been deployed in a wide range of energy storage applications, ranging from energy-type batteries of a few kilowatt-hours in residential systems with rooftop photovoltaic arrays to multi-megawatt containerized batteries for the provision of grid ancillary services. The lithium-ion battery energy storage systems dominate the residential energy-storage applications with small-scale storage batteries (small kWh range). Battery energy storage systems enable self-consumption in the residential application, secure grid stability in the power supply system (including generation, transmission, and distribution) for utility, and can be used to charge EVs.

The lithium-ion batteries have reliable cycle life as well as high energy and power density in terms of volume, high charge/discharge efficiency, and low maintenance needs. Additionally, these batteries are more lightweight than nickel-cadmium and nickel-metal-hydride batteries. Lithium-ion batteries are costlier than other batteries and are also dangerous in a few conditions due to the presence of flammable electrolytes. BMS monitors the function of lithium-ion batteries, which helps extend the storage range of the system in various applications. For energy storage durations of 30 minutes to 3 hours, lithium-ion batteries are currently the most cost-effective solution and have the best energy density compared to other battery technologies. For longer durations, lithium may or may not be the most cost-effective choice depending on the application, particularly when considering lifetime costs. Lithium batteries are also highly configurable into a variety of string sizes and battery racks to create a wide range of voltages, energy increments, or power ratings. This allows for application-specific designs that can range from a few kilowatts with a few minutes of storage, up to multi-megawatt solutions with hours of storage that may be used at a utility substation or a wind farm. Several manufacturers, including Tesla (US) and Panasonic (Japan), are highly investing in R&D for cost reduction, attaining stability in terms of the extension of operational life, and the development of safety mechanisms.

Lithium-ion batteries need a controlled charging/discharging process during storage. The lithium-ion battery modules used in the battery energy storage systems can be easily replaced when required, which helps in saving the replacement cost of the whole battery energy storage system. The storage without a battery management system (BMS) causes a total discharge of the battery modules, which can lead to a defect to battery cells. BMS ensures that battery modules have the optimum temperature during the charging/discharging cycles. Lithium-ion batteries perform the task of frequency regulation in the battery energy storage system. The applications of battery energy storage systems comprising lithium-ion batteries are self-consumption in residential applications, utilities/renewable energy plants, offices and schools, and commercial buildings.

One of the key market opportunities is the decline in prices of lithium-ion batteries in recent years. The cost of lithium-ion batteries has reduced as compared to the previous years and is estimated to reduce further due to the mass production of lithium globally. These batteries are largely used in electronic products. The outbreak of the COVID-19 pandemic has a sudden increase in the sales of electronics products as work-from-home had become the need of the hour. The sales of mobiles and laptops have been increasing steadily. Lithium-ion batteries are widely used in these electronic products, which in turn, is driving the growth of the electronics market, and thus the growth of lithium-ion batteries during the forecast period.

Lithium-ion batteries are also used in medical equipment and electric vehicles, which has created huge opportunities for the adoption of Li-ion batteries to grow rapidly during the forecast period. These batteries are lightweight, portable, and smaller in size as compared to other types of batteries, which has increased their adoption in various applications.

The other key growth opportunities for the market are planned projects and investments in grid-scale battery energy storage projects in North America, Europe, and Asia Pacific from 2020 to 2023. For instance, in the US, the Swiftsure Energy Storage and Gemini Project BESS are the early stages of energy storage projects. Swiftsure Energy Storage will be operational by June 2022, and Gemini Project Bess will be operational in 2023. In Europe, Germany has planned energy storage projects such as grid development plans for 2030, which will create opportunities for the market. In Asia Pacific, GE was awarded the largest grid battery project for generating power by 2021 in Australia. All these developments are expected to fuel the growth of the battery storage industry.

TABLE 20 LITHIUM-ION BATTERY ENERGY STORAGE SYSTEM MARKET, BY CONNECTION TYPE, 2018–2021 (USD MILLION)

Connection Type	2018	2019	2020	2021
On-grid	1,070.7	1,641.5	1,320.7	1,585.0
Off-grid	178.9	270.0	214.2	253.1
Total	1,249.6	1,911.5	1,534.9	1,838.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 21 LITHIUM-ION BATTERY ENERGY STORAGE SYSTEM MARKET, BY CONNECTION TYPE, 2022–2027 (USD MILLION)

Connection Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
On-grid	1,962.8	2,444.5	3,062.4	3,859.1	4,941.8	6,355.6	26.5%
Off-grid	307.3	374.7	459.1	565.5	709.1	889.5	23.7%
Total	2,270.0	2,819.2	3,521.5	4,424.6	5,650.9	7,245.1	26.1%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 22 LITHIUM-ION BATTERY ENERGY STORAGE SYSTEM MARKET, BY ENERGY CAPACITY, 2018–2021 (USD MILLION)

Energy Capacity	2018	2019	2020	2021
Below 100 MWh	141.8	226.2	180.1	214.9
Between 100 and 500 MWh	228.9	336.0	270.6	319.6
Above 500 MWh	878.9	1,349.4	1,084.2	1,303.6
Total	1,249.6	1,911.5	1,534.9	1,838.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 23 LITHIUM-ION BATTERY ENERGY STORAGE SYSTEM MARKET, BY ENERGY CAPACITY, 2022–2027 (USD MILLION)

Energy Capacity	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Below 100 MWh	257.3	308.6	377.3	479.0	605.6	768.8	24.5%
Between 100 and 500 MWh	381.9	459.9	554.9	673.8	847.2	1,055.8	22.6%
Above 500 MWh	1,630.9	2,050.8	2,589.2	3,271.8	4,198.1	5,420.5	27.2%
Total	2,270.0	2,819.2	3,521.5	4,424.6	5,650.9	7,245.1	26.1%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 24 LITHIUM-ION BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Residential	134.3	216.4	171.5	203.7
Commercial	193.8	284.2	231.0	274.8
Utility	921.4	1,410.9	1,132.5	1,359.6
Total	1,249.6	1,911.5	1,534.9	1,838.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 25 LITHIUM-ION BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	245.3	295.6	363.5	464.3	582.9	739.5	24.7%
Commercial	329.1	398.5	484.4	596.2	751.3	943.3	23.4%
Utility	1,695.6	2,125.1	2,673.6	3,364.1	4,316.7	5,562.3	26.8%
Total	2,270.0	2,819.2	3,521.5	4,424.6	5,650.9	7,245.1	26.1%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 26 LITHIUM-ION BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	400.9	606.5	483.5	574.2
Europe	254.0	380.7	297.4	346.4
APAC	526.0	820.9	671.8	820.6
RoW	68.7	103.4	82.2	96.9
Total	1,249.6	1,911.5	1,534.9	1,838.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 27 LITHIUM-ION BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	701.7	861.8	1,063.5	1,319.6	1,671.2	2,119.3	24.7%
Europe	414.5	498.6	604.0	739.0	912.1	1,131.1	22.2%
APAC	1,037.2	1,317.7	1,681.9	2,153.1	2,801.5	3,660.8	28.7%
RoW	116.6	141.1	172.1	212.8	266.2	333.9	23.4%
Total	2,270.0	2,819.2	3,521.5	4,424.6	5,650.9	7,245.1	26.1%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

Lithium-ion batteries are widely used in small-scale energy storage applications and grid-scale applications, which require megawatts of power for hours at a time. A major factor in the rapid increase in the use of BESS Lithium-ion technology has been the 50% decrease in the costs of energy storage over the last two years. Attributes such as low cost, less space required for installation, and low self-discharge rate are a few of the major factors increasing the adoption of lithium-ion batteries in energy storage systems. The lithium-ion battery technology is most compatible with energy storage systems. The technology is being used in power plants to achieve maximum efficiency. These batteries are more resistive to extreme temperature situations, making them suitable for use in remote locations and thermal management.

8.3 LEAD-ACID BATTERIES

8.3.1 INCREASING ADOPTION OF LEAD-ACID BATTERIES IN GRID ENERGY STORAGE

The lead-acid battery is a tested technology that has been used in off-grid energy systems for decades. While the batteries have a relatively short life and lower DoD than other battery types, they are also one of the least expensive options currently available in the home energy storage sector. Lead-acid BESS can be an ideal option for residential applications, which require off-grid systems and many energy storage systems. The lead-acid battery is a mature technology used for grid energy storage, off-grid household electric power systems, and emergency lighting. Advanced lead-acid battery technology, a new generation of affordable, efficient, and environment-friendly lead-acid battery technology, available for use in energy storage programs, was developed after extensive research.

The next technology breakthrough in advanced lead-acid battery technology is the Ultrabattery. The deployment of carbon in a supercapacitor electrode attached to the negative plate resulted in the formation of the Ultrabattery. This Ultrabattery was developed by the Furukawa Battery Co. of Japan and Commonwealth Scientific and Industrial Research Organization (CSIRO) of Australia and tested by the Advanced Lead-Acid Battery Consortium (ALABC). Lead-acid batteries are installed in medium- and large-scale battery energy storage systems (i.e., up to several MW/MWh) that offer different grid services and smoothen renewable power generation. The function of a lead-acid battery energy storage system is to provide and absorb electric power from the grid.

Lead-acid batteries are most popularly used in automobiles due to their lightweight technology. The batteries have a low discharge rate, and fast charging capability. These batteries can also tolerate overcharging, a definite advantage over other batteries. Lead-acid batteries have a longer life and about 95% of the battery can be recycled. The battery can withstand long-term inactivity with or without solvent, because of which these batteries are preferred in automobiles. Due, to their high advantages, and recyclable quality, the market for lead-acid batteries is always increasing.

The outbreak of the pandemic has created several disruptions in the lead-acid battery industry. However, several planned projects and government initiatives pertaining to lead-acid battery technology are increasing the adoption of the technology. In 2020, the Pillar Mountain Wind project in the US used a lead-acid battery and made a battery energy storage system to supplement electricity generated by wind farms. This innovation will increase reliability. The Monash Smart Energy City project in Australia launched BTM battery to achieve zero net emissions by 2030. In Italy, the use of lead-acid batteries is mainly in the utility sector, especially for applications such as peak shaving, grid stability, and, most importantly, renewable integration.

TABLE 28 LEAD-ACID BATTERY ENERGY STORAGE SYSTEM MARKET, BY CONNECTION TYPE, 2018–2021 (USD MILLION)

Connection Type	2018	2019	2020	2021
On-grid	309.9	466.7	361.3	417.9
Off-grid	74.7	110.4	84.3	95.9
Total	384.7	577.1	445.5	513.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 29 LEAD-ACID BATTERY ENERGY STORAGE SYSTEM MARKET, BY CONNECTION TYPE, 2022–2027 (USD MILLION)

Connection Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
On-grid	499.7	601.7	730.1	892.3	1,082.5	1,327.6	21.6%
Off-grid	112.6	133.2	158.8	191.3	227.3	273.9	19.5%
Total	612.3	734.9	888.9	1,083.6	1,309.8	1,601.4	21.2%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 30 LEAD-ACID BATTERY ENERGY STORAGE SYSTEM MARKET, BY ENERGY CAPACITY, 2018–2021 (USD MILLION)

Energy Capacity	2018	2019	2020	2021
Below 100 MWh	55.9	85.7	64.9	73.4
Between 100 and 500 MWh	92.0	133.0	101.6	115.1
Above 500 MWh	236.8	358.4	279.1	325.3
Total	384.7	577.1	445.5	513.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 31 LEAD-ACID BATTERY ENERGY STORAGE SYSTEM MARKET, BY ENERGY CAPACITY, 2022–2027 (USD MILLION)

Energy Capacity	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Below 100 MWh	84.9	98.8	117.0	143.6	168.4	201.9	18.9%
Between 100 and 500 MWh	133.6	156.6	184.7	220.1	257.3	304.1	17.9%
Above 500 MWh	393.7	479.5	587.1	719.9	884.0	1,095.5	22.7%
Total	612.3	734.9	888.9	1,083.6	1,309.8	1,601.4	21.2%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 32 LEAD-ACID BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Residential	43.3	68.4	51.9	59.2
Commercial	57.3	82.5	65.0	75.0
Utility	284.1	426.1	328.6	379.7
Total	384.7	577.1	445.5	513.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 33 LEAD-ACID BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	68.6	79.7	94.7	117.2	138.3	166.4	19.4%
Commercial	87.2	102.7	121.6	146.0	176.5	213.6	19.6%
Utility	456.5	552.5	672.6	820.4	995.1	1,221.4	21.8%
Total	612.3	734.9	888.9	1,083.6	1,309.8	1,601.4	21.2%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 34 LEAD-ACID BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	133.5	199.0	153.0	175.7
Europe	80.5	118.0	88.5	99.1
APAC	147.7	226.1	178.3	210.1
RoW	23.0	33.8	25.7	29.0
Total	384.7	577.1	445.5	513.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 35 LEAD-ACID BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	208.1	248.3	298.5	361.7	434.4	527.4	20.4%
Europe	114.5	133.2	156.4	185.4	216.7	256.8	17.5%
APAC	255.9	313.8	387.3	480.4	592.0	736.9	23.6%
RoW	33.7	39.5	46.8	56.0	66.6	80.3	18.9%
Total	612.3	734.9	888.9	1,083.6	1,309.8	1,601.4	21.2%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The lead-acid battery is a mature technology used for grid energy storage. The lead-acid battery is increasingly used for residential energy storage and behind-the-meter energy storage purposes. Most of the lead-acid batteries are connected on-grid. Residential battery storage is highly dependent on solar PV implementation at homes and commercial buildings.

8.4 FLOW BATTERIES

8.4.1 VARIOUS ADVANTAGES OF FLOW BATTERIES OVER OTHER BATTERIES TO BOOST THEIR DEMAND

A flow battery is a type of rechargeable battery where rechargeability is provided by two chemical components dissolved in liquids contained in the system, separated by a membrane. The main difference between the conventional battery and flow battery is that the energy is stored in the electrode material in a conventional battery, while it is stored in the electrolyte in flow cells. Different classes of flow batteries have been developed, including redox, hybrid, and membrane-less. The most popular flow batteries include vanadium redox flow battery, iron–chromium (ICB) flow battery, and zinc–bromine (ZNBR) flow battery. Compared with lithium-ion batteries, vanadium redox flow batteries (VRB) are non-flammable and environmentally friendly, have a long lifespan, and maintain 90% of their capacity over 20 years, resulting in the reduction of the total cost of ownership. A new generation of redox (reduction–oxidation) flow batteries in battery energy storage systems are much better suited for medium- and large-scale applications where heavy use, long-term cycling, durability, and safety are paramount. Their application areas include campuses, community service centers, commercial and industrial sites, and utility grids.

Flow batteries have several technical advantages over several other batteries, such as lithium-ion, lead-acid, and others. Unlike conventional batteries that store energy in an electrode material, flow batteries store energy as electrolytes in external tanks. As electrolytes can be easily replaced, it is among the major advantages that are increasing the use of flow batteries in battery energy storage systems. Their storage capacity is based on the size of the storage tank, and thus can be scaled to suit a wide variety of energy needs from residential to grid storage applications. In flow batteries, power is determined by the size of the electrochemical cell, whereas the energy is proportional to the size of the storage tanks. In addition, flow batteries are widely adopted as they can provide various benefits such as:

- Scalable design as output power depends on the size of the cell stack and storage capacity of tanks, as well as the concentration of electrolyte reserves,
- High power capacity as compared to other batteries,
- Ability to separate energy and power requirements,
- Long cycle life,
- Quick response time and high electricity-to-electricity equalization requirement,
- Simple state-of-charge indication,
- Tolerance to overcharge and over-discharge,
- Ability to deep discharge without affecting the cycle life,
- Easy maintenance and low operating cost,
- Materials used for flow batteries are safe and free from potential explosion or hazard,
- High tolerance to extreme weather conditions,
- Major impact on stabilization and smooth output of renewable energy,
- Operations can be fully automated.

These factors have helped the flow battery market gain traction in lieu of other battery types in energy storage solutions.

Various companies are involved in the installation of battery energy storage projects that are incorporated with flow batteries. For instance, in December 2021, Invinity Energy Systems, collaborated with developer Pivot Power and has installed a fully energized largest flow battery, located in Oxford, England. This 5 MWh flow battery is combined with a 50 MWh lithium-ion battery energy storage system (BESS) to operate as a single energy storage asset. The project is already in operation and will be supporting the zero-carbon initiative by the city.

TABLE 36 FLOW BATTERY ENERGY STORAGE SYSTEM MARKET, BY CONNECTION TYPE, 2018–2021 (USD MILLION)

Connection Type	2018	2019	2020	2021
On-grid	390.6	641.7	542.5	685.4
Off-grid	94.2	151.7	126.5	157.4
Total	484.8	793.5	669.1	842.8

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 37 FLOW BATTERY ENERGY STORAGE SYSTEM MARKET, BY CONNECTION TYPE, 2022–2027 (USD MILLION)

Connection Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
On-grid	895.0	1,177.1	1,559.8	2,081.9	2,779.0	3,749.9	33.2%
Off-grid	201.7	260.6	339.3	446.4	585.7	776.0	30.9%
Total	1,096.7	1,437.7	1,899.1	2,528.3	3,364.7	4,525.9	32.8%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 38 FLOW BATTERY ENERGY STORAGE SYSTEM MARKET, BY ENERGY CAPACITY, 2018–2021 (USD MILLION)

Energy Capacity	2018	2019	2020	2021
Below 100 MWh	70.4	117.8	97.5	120.4
Between 100 and 500 MWh	115.9	182.9	152.5	188.8
Above 500 MWh	298.4	492.7	419.1	533.5
Total	484.8	793.5	669.1	842.8

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 39 FLOW BATTERY ENERGY STORAGE SYSTEM MARKET, BY ENERGY CAPACITY, 2022–2027 (USD MILLION)

Energy Capacity	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Below 100 MWh	152.1	193.2	250.0	335.0	432.7	570.6	30.3%
Between 100 and 500 MWh	239.3	306.4	394.6	513.5	666.9	874.6	29.6%
Above 500 MWh	705.2	938.0	1,254.4	1,679.8	2,265.1	3,080.6	34.3%
Total	1,096.7	1,437.7	1,899.1	2,528.3	3,364.7	4,525.9	32.8%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 40 FLOW BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Residential	44.9	79.0	66.4	83.9
Commercial	80.0	123.9	105.0	130.5
Utility	359.9	590.6	497.7	628.5
Total	484.8	793.5	669.1	842.8

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

Flow batteries also offer greater flexibility to influence power and energy rating for a given application than other electrochemical means for storing electrical energy. A flow battery, with all its attractive features, has become the most preferred energy storage technology for utility-based storage as large-scale utilities require technologies that can cost-effectively store renewable energy for future grid use at any location. Among all the applications, flow batteries are the most widely commercialized for various applications in the utility sector. The utility sector also holds the highest number of operational flow battery projects to date worldwide. Further, various large-scale projects under construction involve installations of flow batteries with a capacity of over 1,000 MW.

TABLE 41 FLOW BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	107.7	138.6	182.5	250.4	329.5	442.4	32.7%
Commercial	163.3	207.0	263.7	340.7	445.3	582.6	29.0%
Utility	825.7	1,092.1	1,452.8	1,937.3	2,590.0	3,500.9	33.5%
Total	1,096.7	1,437.7	1,899.1	2,528.3	3,364.7	4,525.9	32.8%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 42 FLOW BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	190.9	310.9	260.8	326.9
Europe	101.7	162.3	132.9	162.4
APAC	163.2	273.7	236.9	306.1
RoW	29.0	46.5	38.4	47.4
Total	484.8	793.5	669.1	842.8

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 43 FLOW BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	423.2	551.9	725.4	960.8	1,271.9	1,701.8	32.1%
Europe	205.2	261.1	334.8	432.6	555.6	722.1	28.6%
APAC	407.9	547.1	738.7	1,004.2	1,368.2	1,880.6	35.8%
RoW	60.4	77.5	100.3	130.7	169.1	221.3	29.7%
Total	1,096.7	1,437.7	1,899.1	2,528.3	3,364.7	4,525.9	32.8%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The high number of flow battery installations in North America, mainly in the US and Canada, attributes to the dominating position of the region in the flow battery market. In addition, favorable economic conditions in the US and the increase in the importance of battery energy storage, as well as supportive regulatory policies aiding energy storage in the US and other countries in the region are encouraging many domestic and international manufacturers to deploy flow batteries for efficient energy supply. On the other hand, APAC has several operational flow battery installations with large power ratings. Australia has the largest number of flow battery projects for utilities, residential, industrial, and commercial applications from various players across the world. China holds the highest capacity of flow battery installed to date among other countries in the region. Also, the growth in microgrid projects in India and Japan has encouraged manufacturers to cater to the needs of potential customers in the region.

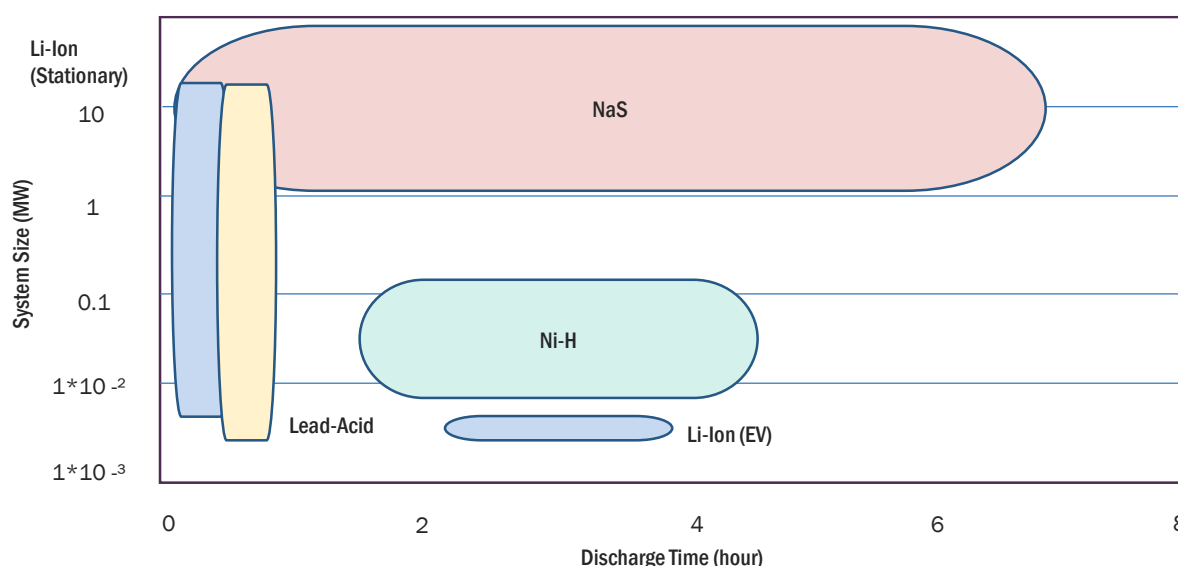
8.5 OTHERS

The other battery technologies include sodium-sulfur batteries, nickel-cadmium batteries, nickel-metal hydride, nickel-iron batteries, and flywheel batteries.

8.5.1 SODIUM-SULFUR BATTERIES

The sodium-sulfur (NaS) batteries were first developed by Ford Motor Company in the 1960s, and subsequently, the technology was sold to NGK (Japan) for stationary applications. A NaS battery system is an energy storage system based on electrochemical charge/discharge reactions that occur between a positive electrode (cathode), which is typically made of molten sulfur (S), and a negative electrode (anode), which is typically made of molten sodium (Na). The sodium-sulfur batteries have a high energy density, high charge/discharge efficiency with long cycle life, and are fabricated from inexpensive materials. Under some market conditions, sodium-sulfur batteries provide value by energy arbitrage (charging the battery when electricity is abundant/cheap and discharging into the grid when electricity is more valuable) and voltage regulation. NaS battery is a possible energy storage technology to support renewable energy generation, specifically solar generation plants and wind farms. The sodium-sulfur batteries are widely used in microgrid applications. The sodium-sulfur battery energy storage system is versatile in function; however, the adoption of these batteries is less than other battery technologies.

FIGURE 38 COMPARISON OF SODIUM-SULFUR BATTERIES WITH OTHER BATTERIES



Source: NGK Company Website and MarketsandMarkets Analysis

As shown in the figure above, sodium-sulfur batteries' discharge time is higher than lithium-ion and lead-acid batteries. The life expectancy of the sodium-sulfur batteries is expected to be 4500 cycles (15 years, 300 cycles per year). They are designed for bulk energy storage and configured for very large grid-scale deployments (10–100 MW).

In October 2021 a long-duration energy storage system using NGK's sodium-sulfur (NAS) batteries has been commissioned by a subsidiary of a German chemicals company, BASF New Business GmbH in Antwerp, Belgium. NGK developed high-temperature rechargeable batteries; the NAS technology used by NGK is said to be suitable for multi-megawatt battery energy storage system (BESS) applications for durations of six to seven hours. The storage system is also said to be designed to last 4500 cycles or 15 years of operational lifetime.

8.5.2 NICKEL-CADMIUM BATTERY

The first nickel-cadmium battery was patented in 1899 by the Swedish inventor and engineer Waldemar Jungner. The nickel-cadmium battery is a type of rechargeable battery that uses nickel oxide-hydroxide and metallic cadmium as electrodes. The nickel-cadmium batteries typically last longer in terms of charge/discharge cycles, offer higher energy density, and are smaller and lightweight compared to lead-acid batteries. Compared to other rechargeable batteries, these batteries have longer cycle life and offer performance at low temperatures. Under rough conditions, these batteries perform well, which makes them adaptable for portable tools. The applications of these batteries include standby power supply devices. Panasonic offers rechargeable nickel-cadmium batteries, which operate at -20°C – 60°C and find applications in telecommunications equipment, emergency lights, power tools, and battery-powered vehicles. The nickel-cadmium battery energy storage systems are used in off-grid renewable energy applications for stabilizing wind-energy systems. They are also used in cordless power tools and camera flash units.

8.5.3 NICKEL-METAL HYDRIDE BATTERY

The nickel-metal hydride (NiMH) batteries have a higher capacity, less toxicity, and more cost-efficiency than the nickel-cadmium batteries. These batteries are designed to meet the increasing requirement for a higher level of energy demanded by electronic products. The features of NiMH batteries are long life, small size, and lightweight, which helps to save space. These batteries provide three times higher capacity than the same size nickel-cadmium batteries. Due to the technical advantages of the nickel-metal hydride batteries such as higher energy density and increased capacity, the longer run time between charges is expected from these batteries. These batteries can be used in a wide range of environments and various applications such as mobile devices, electrically powered devices, and backup equipment.

8.5.4 NICKEL-IRON BATTERY

The nickel-iron (NiFe) battery is a rechargeable battery having nickel (III) oxide-hydroxide on positive plates, and iron on negative plates, with an electrolyte of potassium hydroxide. Thomas Edison commercialized NiFe batteries in the US and patented these batteries in 1901. These batteries were designed as a cost-effective alternative for corrosive lead-acid batteries. The advantages of NiFe batteries over lead-acid batteries are deeper depth of discharge, higher charging rate, longer cycle life, higher efficiency, higher maximum operating temperature, higher safe storage temperature, and no acid content. Nickel-iron batteries are expensive as compared to other batteries, even though they can last 20 years and can withstand overcharge and over-discharge. These batteries are utilized in heavy-weight applications such as wind energy systems and solar energy systems as a backup.

8.5.5 FLYWHEEL BATTERY

A flywheel battery stores electric energy by converting it into kinetic energy, by using a motor to spin a rotor. The motor also works as a generator. The kinetic energy can be converted back to electric energy when needed. The modern high-speed flywheel energy storage system consists of a massive rotating cylinder (a rim attached to a shaft) that is supported on a stator – the stationary part of an electric generator – by magnetically levitated bearings. To maintain efficiency, the flywheel system is operated in a vacuum to reduce drag. The flywheel is connected to a motor generator that interacts with the utility grid through advanced power electronics. Flywheel batteries have a very low charging time. These batteries are generally used in hybrid vehicles.

TABLE 44 OTHER BATTERY ENERGY STORAGE SYSTEM MARKET, BY CONNECTION TYPE, 2018–2021 (USD MILLION)

Connection Type	2018	2019	2020	2021
On-grid	161.5	262.5	220.5	276.6
Off-grid	38.9	62.1	51.4	63.5
Total	200.4	324.6	271.9	340.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 45 OTHER BATTERY ENERGY STORAGE SYSTEM MARKET, BY CONNECTION TYPE, 2022–2027 (USD MILLION)

Connection Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
On-grid	358.8	468.7	616.9	817.9	1,087.5	1,460.1	32.4%
Off-grid	80.8	103.8	134.2	175.4	228.7	301.7	30.1%
Total	439.6	572.5	751.1	993.3	1,316.3	1,761.8	32.0%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 46 OTHER BATTERY ENERGY STORAGE SYSTEM MARKET, BY ENERGY CAPACITY, 2018–2021 (USD MILLION)

Energy Capacity	2018	2019	2020	2021
Below 100 MWh	29.1	48.2	39.6	48.6
Between 100 and 500 MWh	47.9	74.8	62.0	76.2
Above 500 MWh	123.3	201.6	170.3	215.3
Total	200.4	324.6	271.9	340.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 47 OTHER BATTERY ENERGY STORAGE SYSTEM MARKET, BY ENERGY CAPACITY, 2022–2027 (USD MILLION)

Energy Capacity	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Below 100 MWh	61.0	76.9	98.9	131.6	169.3	222.1	29.5%
Between 100 and 500 MWh	95.9	122.0	156.1	201.7	260.9	340.5	28.8%
Above 500 MWh	282.7	373.5	496.1	659.9	886.1	1,199.2	33.5%
Total	439.6	572.5	751.1	993.3	1,316.3	1,761.8	32.0%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 48 OTHER BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Residential	19.0	33.0	27.5	34.4
Commercial	32.8	50.3	42.4	52.4
Utility	148.6	241.3	202.0	253.4
Total	200.4	324.6	271.9	340.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 49 OTHER BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	43.7	55.8	72.8	99.0	129.7	173.0	31.7%
Commercial	65.2	82.2	104.2	133.8	174.3	227.0	28.3%
Utility	330.7	434.4	574.1	760.4	1,012.4	1,361.7	32.7%
Total	439.6	572.5	751.1	993.3	1,316.3	1,761.8	32.0%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 50 OTHER BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	69.6	112.0	93.4	116.3
Europe	41.9	66.4	54.0	65.6
APAC	76.9	127.2	108.8	139.1
RoW	12.0	19.0	15.7	19.2
Total	200.4	324.6	271.9	340.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 51 OTHER BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	149.4	193.4	252.2	331.6	437.2	581.9	31.2%
Europe	82.2	103.8	132.1	170.0	217.2	280.9	27.9%
APAC	183.8	244.5	327.2	440.4	595.5	812.3	34.6%
RoW	24.2	30.8	39.5	51.4	66.3	86.7	29.1%
Total	439.6	572.5	751.1	993.3	1,316.3	1,761.8	32.0%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

Among the batteries included in the others segment, sodium-sulfur batteries are most suitable and widely used for the energy storage requirements in applications such as load leveling, energy shifting of renewables from off-peak to peak time, transmission and distribution (T&D) network management, emergency power supplies, and uninterruptible power supply. These batteries can be deployed to support the electric grid or in off-grid renewable power applications.

9 BATTERY ENERGY STORAGE SYSTEM MARKET, BY CONNECTION TYPE

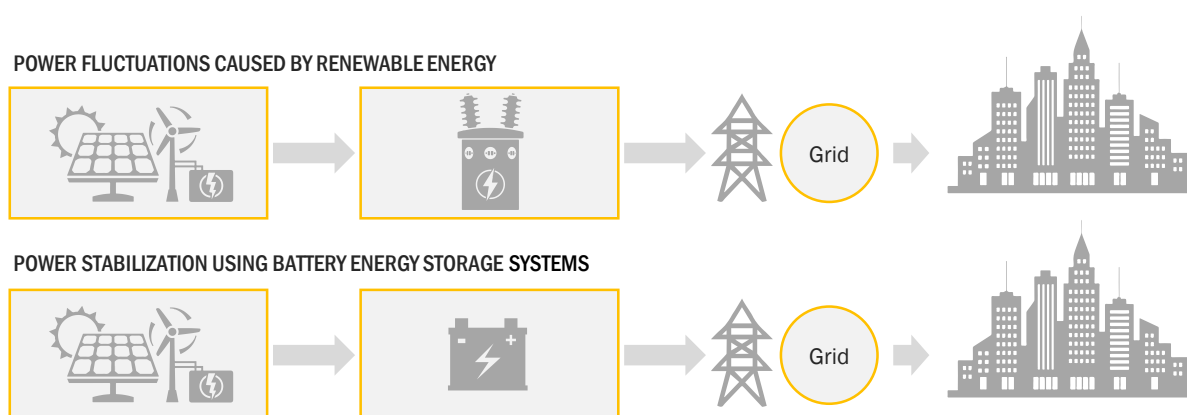
KEY FINDINGS

- The on-grid connection segment held a larger size of the battery energy storage system market in 2021. The market for on-grid connection was worth USD 2,965.0 million in 2021 and is projected to reach USD 12,892.8 million by 2027, growing at a higher CAGR of 28.2% between 2022 to 2027.
- The on-grid connections exhibit lower upfront costs than the off-grid system, as the surplus energy is sent to the grid, which can be used later on a metered basis. The reduction in the energy bills for the customers relying on the utility grid for electricity is expected to drive the growth of the segment. The demand for grid modernization from renewable energy suppliers and new regulatory policies for the energy sector are driving the growth of the on-grid battery energy storage system market for utilities.
- The off-grid battery energy storage system market was worth USD 569.9 million in 2021 and is projected to reach USD 2,241.4 million by 2027, growing at the highest CAGR of 26.1% between 2022 and 2027.
- Off-grid connections are proving to be effective for remote places where grid connections are not available. The installation might be costly but is more efficient and reliable than the on-grid connection as they are self-sufficient (which means having fixed energy costs without receiving a bill).
- The off-grid battery energy storage system market is estimated to grow rapidly in the coming years owing to the increasing installation of solar panels from residential applications.

9.1 INTRODUCTION

The battery energy storage system (BESS) market based on connection type has been segmented into on-grid connection and off-grid connection. On-grid battery energy storage systems are connected to the systems of the local utility providers. When a battery energy storage system is not connected to the main utility grid, it is termed an off-grid connection. The on-grid connections are mainly used by utility providers and independent power producers (IPPs). The utility providers and IPPs use a battery energy storage system for the transmission and distribution (T&D) upgrade deferral, frequency regulation, renewables integration, and peak shifting. The off-grid connections are suitable for remote locations where utility providers are not available. Residential consumers nowadays prefer battery energy storage systems for off-grid connection as this connection enables the users to dissociate themselves from monthly bills and enables the self-consumption of energy storage with a focus on eco-friendly operations by nullifying the dependence on diesel and generators.

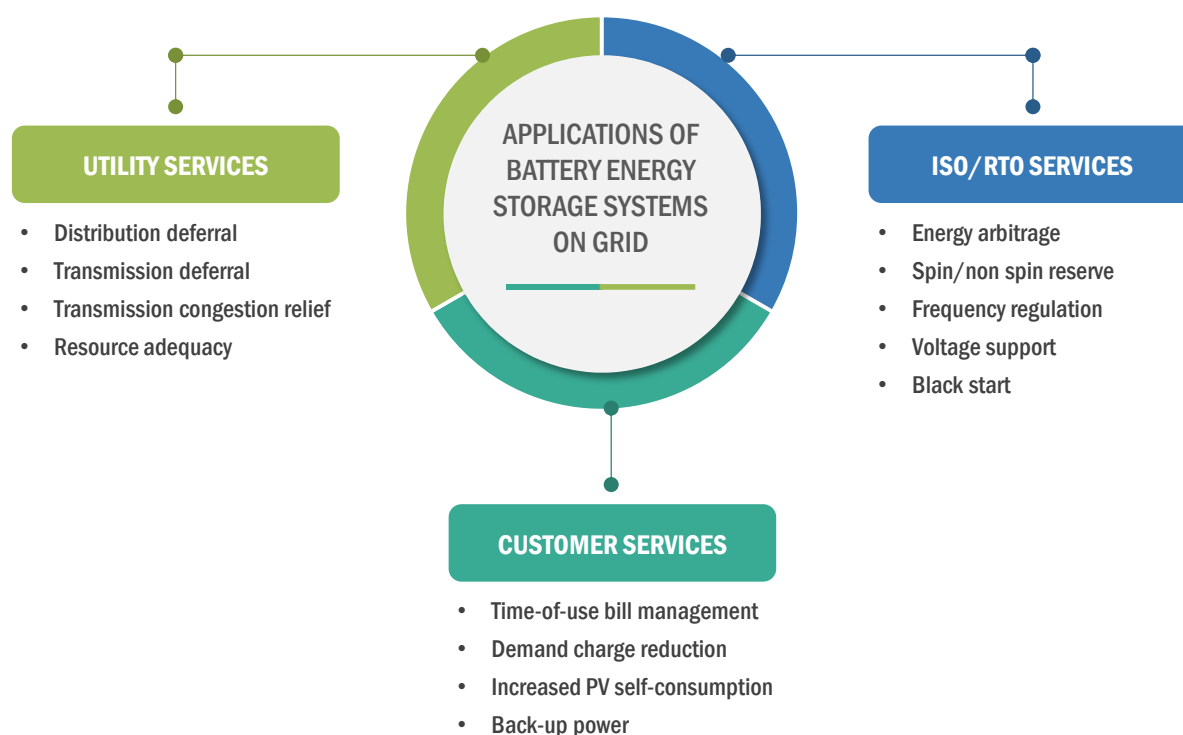
FIGURE 39 IMPORTANCE OF BATTERY ENERGY STORAGE SYSTEMS IN GRID CONNECTION



Source: MarketsandMarkets Analysis

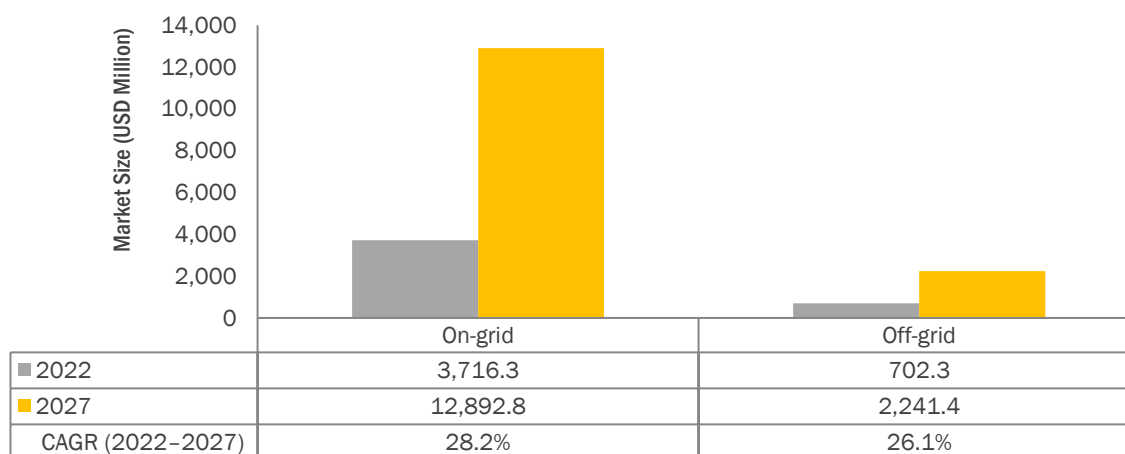
Renewable energy sources are intermittent, and it is not possible always to rely on them to produce energy when required. The on-grid energy storage systems store the electrical energy in the electrical power grids, whereas the off-grid systems do not need the connection of the battery energy storage systems with electrical grids. Thus, the battery energy storage systems used in on- and off-grid connections offer power stabilization and enable to store a large amount of energy, depending on battery capacity. Various companies, such as Invinity Energy System (UK) and TrinaBESS (Australia), are offering battery energy storage systems for on- and off-grid connections for reducing the use of fossil fuel.

National Energy Renewable Laboratory (NERL) conducts research activities on the application of on-grid battery energy storage systems. NERL has categorized the application of battery energy storage systems on the grid through three ways—utility services, ISO/RTO services, and customer services. The battery energy storage systems used in grids help the customer by reducing their bills and increasing self-consumption through the off-grid connection.

FIGURE 40 APPLICATIONS OF BATTERY ENERGY STORAGE SYSTEMS ON GRID

Note: ISO = Independent System Operator (ISO) and RTO = Regional Transmission Organizations

Source: MarketsandMarkets Analysis

FIGURE 41 BATTERY ENERGY STORAGE SYSTEM MARKET FOR ON-GRID CONNECTION TO REGISTER HIGHER CAGR DURING FORECAST PERIOD

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 52 BATTERY ENERGY STORAGE SYSTEM MARKET, BY CONNECTION TYPE, 2018–2021 (USD MILLION)

Connection Type	2018	2019	2020	2021
On-grid	1,932.7	3,012.5	2,445.0	2,965.0
Off-grid	386.7	594.1	476.4	569.9
Total	2,319.4	3,606.7	2,921.4	3,534.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 53 BATTERY ENERGY STORAGE SYSTEM MARKET, BY CONNECTION TYPE, 2022–2027 (USD MILLION)

Connection Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
On-grid	3,716.3	4,692.1	5,969.2	7,651.2	9,891.1	12,892.8	28.2%
Off-grid	702.3	872.2	1,091.3	1,378.6	1,750.6	2,241.4	26.1%
Total	4,418.6	5,564.3	7,060.6	9,029.8	11,641.7	15,134.2	27.9%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The on-grid connection segment held a larger size of the battery energy storage system market in 2021. The on-grid connections have lower upfront costs than off-grid systems, as the surplus energy is sent to the grid, which can be used later on a meter basis. The reduction in the energy bills for the customers relying on the utility grid for electricity is estimated to drive the growth of the segment. The demand for grid modernization from renewable energy suppliers and new regulatory policies for the energy sector are driving the growth of the on-grid battery energy storage system market for utilities.

The off-grid connection utilizes solar energy and is mainly suitable for residential purposes as there is no requirement of the grid, which saves space. Neither the off-grid systems are subject to the terms of policies of the local utility, nor are the system owners subject to the increasing rates, blackouts, and brownouts. Tesla built a massive manufacturing plant Gigafactory, which is based on lithium-ion batteries through a combination of various sources such as solar panels covering the entire rooftop and wind turbines on slopes, which would essentially be suitable for the off-grid type connection. These developments are fueling the adoption of battery energy storage systems with an off-grid connection and driving the growth of the battery energy storage system market.

9.2 ON-GRID CONNECTION

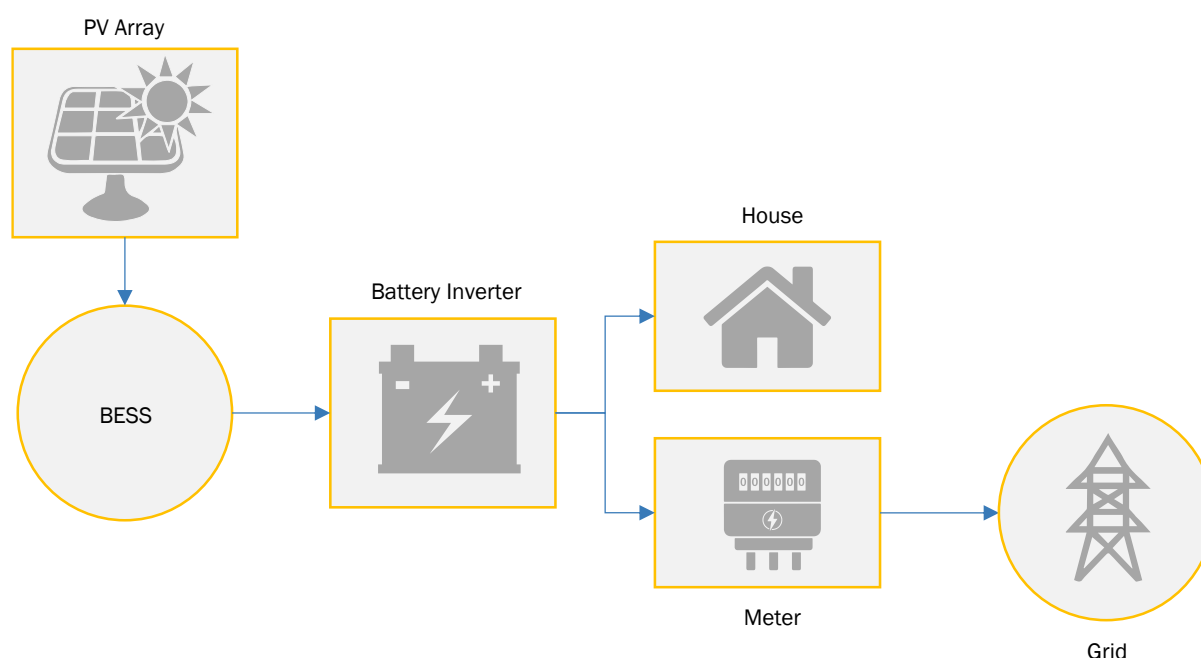
9.2.1 INCREASING ELECTRICITY DEMAND WORLDWIDE TO PROPEL DEMAND FOR ON-GRID STORAGE SYSTEMS

The on-grid battery energy storage systems help in energy storage on a large scale. Electrical energy is stored during times when electricity is available in plenty and is inexpensive or when the demand is low. This electrical energy is returned to the electrical grid when the demands are high and, the electricity price tends to be higher. Thus, on-grid battery energy storage systems use utility grids for supplying electricity to consumers, which also helps reduce energy bills. Load leveling, peak shaving, and power demand management are the main applications of any on-grid connected battery energy storage system installed with electrical grids.

Electricity generated from these renewable sources (wind and solar) can be stored and used later, despite transmitting it for sale elsewhere. Transmission capacity or peak generating can be reduced by the total potential of all storage plus deferrable loads, which results in saving the expense of this capacity. Sometimes, battery energy storage systems are placed at substations for local voltage support that also serve additional applications such as transmission and delivery (T&D) deferral, power quality assurance, voltage regulation, spinning reserve, and increased system reliability. Nowadays, battery energy storage systems serve as the preferred technology for improving the efficiency of the grid infrastructure, and these systems also lead to higher charge–discharge efficiencies along with more efficient production of electricity. Lastly, the grid system can aid in emergency preparedness, in which the vital needs can be met reliably even with no transmission or generation of electricity going on while non-essential needs are deferred. Hence, the grid systems are valuable assets for the battery energy storage system and vice versa.

In October 2021, the ministry of India, revealed its plan to install 1,000 MWh battery energy storage systems, as an on-grid connection to maximize the use of the transmission systems and strengthen grid stability for flexible operation. In addition, the Government of India is also working on the large-scale integration of renewable energy generation with the country's power system, which will help to balance the national electricity grid. Additionally, Northern Europe has underway three projects for battery energy storage systems. The first project is for Finland, where BESS of 30 MW power capacity would be installed near the city of Lappeenranta in the Southeast of the country, which will provide rapid and safe energy storage, stabilize the national grid, and will eliminate the peak demand problems in the country. The second and third projects are undertaken in Sweden, where BESS of 9 MW and 6 MW power capacity would be installed in the Lowen and Edsele sites, respectively. Battery energy storage systems are being set up to avoid grid congestion, store the surplus generated energy, and supply it back to the grid at peak demands, and stabilize the grid in the country. Thus, the increasing power needs across the world are expected to boost the demand for battery energy storage systems in the coming years.

FIGURE 42 ON-GRID CONNECTION



Source: MarketsandMarkets Analysis

As the figure shows, the PV arrays capture solar energy, as it is the complete power-generating unit consisting of a number of PV modules and panels. The battery energy storage system connected to the PV array stores the energy and the battery converter converts the energy from DC to AC. The loads connected to the battery inverter utilize the energy stored by the battery energy storage system. The meter connected to the grid is capable of measuring power going in both directions, from the grid to loads. The battery energy storage system responds to instant voltage spikes and outages, thus impacting the overall efficiency of the grid.

TABLE 54 BATTERY ENERGY STORAGE SYSTEM MARKET FOR ON-GRID CONNECTION TYPE, BY ENERGY CAPACITY, 2018–2021 (USD MILLION)

Energy Capacity	2018	2019	2020	2021
Below 100 MWh	239.5	386.5	309.9	371.9
Between 100 and 500 MWh	390.6	587.8	475.7	569.2
Above 500 MWh	1,302.7	2,038.3	1,659.4	2,023.9
Total	1,932.7	3,012.5	2,445.0	2,965.0

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 55 BATTERY ENERGY STORAGE SYSTEM MARKET FOR ON-GRID CONNECTION TYPE, BY ENERGY CAPACITY, 2022–2027 (USD MILLION)

Energy Capacity	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Below 100 MWh	453.2	554.7	692.6	896.8	1,142.8	1,474.2	26.6%
Between 100 and 500 MWh	694.3	855.6	1,059.8	1,325.0	1,678.1	2,132.2	25.2%
Above 500 MWh	2,568.8	3,281.8	4,216.8	5,429.3	7,070.2	9,286.4	29.3%
Total	3,716.3	4,692.1	5,969.2	7,651.2	9,891.1	12,892.8	28.2%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 56 BATTERY ENERGY STORAGE SYSTEM MARKET FOR ON-GRID CONNECTION TYPE, BY BATTERY TYPE, 2018–2021 (USD MILLION)

Battery Type	2018	2019	2020	2021
Lithium-Ion Batteries	1,070.7	1,641.5	1,320.7	1,585.0
Advanced Lead-Acid Batteries	309.9	466.7	361.3	417.9
Flow Batteries	390.6	641.7	542.5	685.4
Others	161.5	262.5	220.5	276.6
Total	1,932.7	3,012.5	2,445.0	2,965.0

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 57 BATTERY ENERGY STORAGE SYSTEM MARKET FOR ON-GRID CONNECTION TYPE, BY BATTERY TYPE, 2022–2027 (USD MILLION)

Battery Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Lithium-Ion Batteries	1,962.8	2,444.5	3,062.4	3,859.1	4,941.8	6,355.6	26.5%
Advanced Lead-Acid Batteries	499.7	601.7	730.1	892.3	1,082.5	1,327.6	21.6%
Flow Batteries	895.0	1,177.1	1,559.8	2,081.9	2,779.0	3,749.9	33.2%
Others	358.8	468.7	616.9	817.9	1,087.5	1,460.1	32.4%
Total	3,716.3	4,692.1	5,969.2	7,651.2	9,891.1	12,892.8	28.2%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 58 BATTERY ENERGY STORAGE SYSTEM MARKET FOR ON-GRID CONNECTION TYPE, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Residential	203.3	332.8	265.1	317.4
Commercial	277.8	411.2	335.6	401.4
Utility	1,451.6	2,268.6	1,844.3	2,246.2
Total	1,932.7	3,012.5	2,445.0	2,965.0

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 59 BATTERY ENERGY STORAGE SYSTEM MARKET FOR ON-GRID CONNECTION TYPE, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	386.0	470.8	587.5	763.4	964.4	1,238.2	26.3%
Commercial	483.8	590.4	724.0	900.4	1,140.0	1,442.4	24.4%
Utility	2,846.5	3,630.9	4,657.7	5,987.4	7,786.7	10,212.2	29.1%
Total	3,716.3	4,692.1	5,969.2	7,651.2	9,891.1	12,892.8	28.2%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

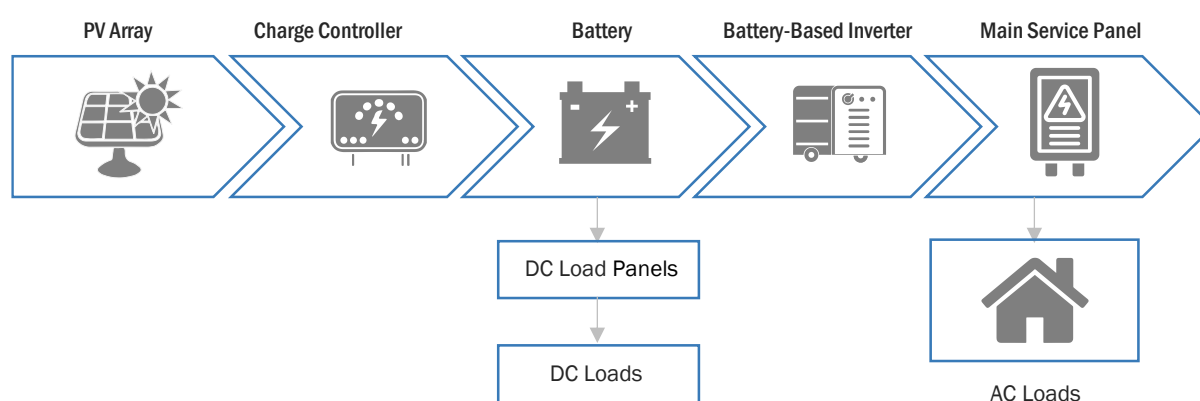
Lithium-ion is the most prevalent battery type used in utility-scale energy storage systems. The high growth of the Li-ion batteries segment, by storage capacity, is driven by the declining costs of the Li-ion technology. About 90% of the large-scale battery energy storage systems deployed are equipped with lithium-ion batteries. The battery energy storage systems with grid systems provide key functions such as peak shaving & load leveling, voltage & frequency regulation, and emergency energy storage. For system operators, battery storage systems with on-grid connections provide grid services such as frequency response, regulation reserves, and ramp rate control for utility applications. Battery storage systems can also defer investments in peak generation and grid reinforcements. Battery energy storage systems for utility applications can enable high penetration of variable renewable energy into the grid by storing the excess generation and by firming the renewable energy output.

9.3 OFF-GRID CONNECTION

9.3.1 OFF-GRID CONNECTIONS TO BE EFFECTIVE ALTERNATIVES TO SUPPLY ENERGY AT REMOTE AREAS

The off-grid connection type is better for users who are unable to connect to the grid due to the geographic location or high cost of bringing in the power supply. The main advantage of an off-grid system is that a user can have fixed energy costs and will not be getting a bill and can power the remote places away from the grid. Along with it, off-grid connections are self-sustainable and do not depend on the grid, which makes them an independent source of energy. They are capable of avoiding power outages, especially in the interior locations of the countries. Off-grid connections help to keep the environment clean and safe as they are not burning fossil fuels and are connected to renewable energy sources such as solar or wind. Off-grid systems require more components during installation, which cost more than a standard grid-tied system. The off-grid installation of the battery energy storage system maintains the life cycle at a charge/discharge time and also reduces the installation space. For instance, Tesla offers Powerwall, which is a solar self-consumption product designed to drag in the excess solar energy back during the night hours instead of sending it back to the grid. The Tesla Powerwall mainly serves the residential application, which minimizes the reliance on utilities.

FIGURE 43 BASIC OFF-GRID SYSTEMS



Source: MarketsandMarkets Analysis

The figure above shows a basic off-grid system in which all loads are being supplied by the batteries, and there is no grid supply. The off-grid system is used in residential, commercial, and industrial applications when located in remote areas having an irregular supply of electricity or even no supply at all.

TABLE 60 BATTERY ENERGY STORAGE SYSTEM MARKET FOR OFF-GRID CONNECTION TYPE, BY ENERGY CAPACITY, 2018–2021 (USD MILLION)

Energy Capacity	2018	2019	2020	2021
Below 100 MWh	57.7	91.4	72.3	85.4
Between 100 and 500 MWh	94.2	139.0	110.9	130.7
Above 500 MWh	234.8	363.8	293.2	353.8
Total	386.7	594.1	476.4	569.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 61 BATTERY ENERGY STORAGE SYSTEM MARKET FOR OFF-GRID CONNECTION TYPE, BY ENERGY CAPACITY, 2022–2027 (USD MILLION)

Energy Capacity	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Below 100 MWh	102.1	122.8	150.7	192.3	233.2	289.3	23.2%
Between 100 and 500 MWh	156.4	189.4	230.5	284.1	354.2	442.7	23.1%
Above 500 MWh	443.8	560.0	710.2	902.2	1,163.2	1,509.4	27.7%
Total	702.3	872.2	1,091.3	1,378.6	1,750.6	2,241.4	26.1%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 62 BATTERY ENERGY STORAGE SYSTEM MARKET FOR OFF-GRID CONNECTION TYPE, BY BATTERY TYPE, 2018–2021 (USD MILLION)

Battery Type	2018	2019	2020	2021
Lithium-Ion Batteries	178.9	270.0	214.2	253.1
Advanced Lead-Acid Batteries	74.7	110.4	84.3	95.9
Flow Batteries	94.2	151.7	126.5	157.4
Others	38.9	62.1	51.4	63.5
Total	386.7	594.1	476.4	569.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 63 BATTERY ENERGY STORAGE SYSTEM MARKET FOR OFF-GRID CONNECTION TYPE, BY BATTERY TYPE, 2022–2027 (USD MILLION)

Battery Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Lithium-Ion Batteries	307.3	374.7	459.1	565.5	709.1	889.5	23.7%
Advanced Lead-Acid Batteries	112.6	133.2	158.8	191.3	227.3	273.9	19.5%
Flow Batteries	201.7	260.6	339.3	446.4	585.7	776.0	30.9%
Others	80.8	103.8	134.2	175.4	228.7	301.7	30.1%
Total	702.3	872.2	1,091.3	1,378.6	1,750.6	2,241.4	26.1%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 64 BATTERY ENERGY STORAGE SYSTEM MARKET FOR OFF-GRID CONNECTION TYPE, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Residential	38.1	64.0	52.1	63.7
Commercial	86.1	129.8	107.8	131.3
Utility	262.5	400.3	316.5	374.8
Total	386.7	594.1	476.4	569.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 65 BATTERY ENERGY STORAGE SYSTEM MARKET FOR OFF-GRID CONNECTION TYPE, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	79.3	98.8	126.1	167.6	216.0	283.1	29.0%
Commercial	161.1	200.1	249.9	316.4	407.3	524.2	26.6%
Utility	462.0	573.2	715.4	894.7	1,127.4	1,434.1	25.4%
Total	702.3	872.2	1,091.3	1,378.6	1,750.6	2,241.4	26.1%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

Battery energy storage systems with off-grid connection types are ideal for islands and off-grid communities, where they can help to save high fuel costs and reduce dependency on fossil fuel. Off-grid systems are most suitable in remote locations devoid of utility service. An off-grid battery energy storage system is a perfect choice for residential applications since it is independent of the electricity grid and requires solar panels to produce enough electricity to cover the whole energy needs of the homeowners. After lithium-ion batteries, flow batteries are considered as an efficient option for off-grid energy storage systems, however, the current challenge is to reduce the costs of these systems for widespread commercial penetration. Companies engaged in the manufacturing of flow batteries are receiving funds, which can further help the companies to deploy utility-scale flow battery storage projects.

10 BATTERY ENERGY STORAGE SYSTEM MARKET, BY OWNERSHIP

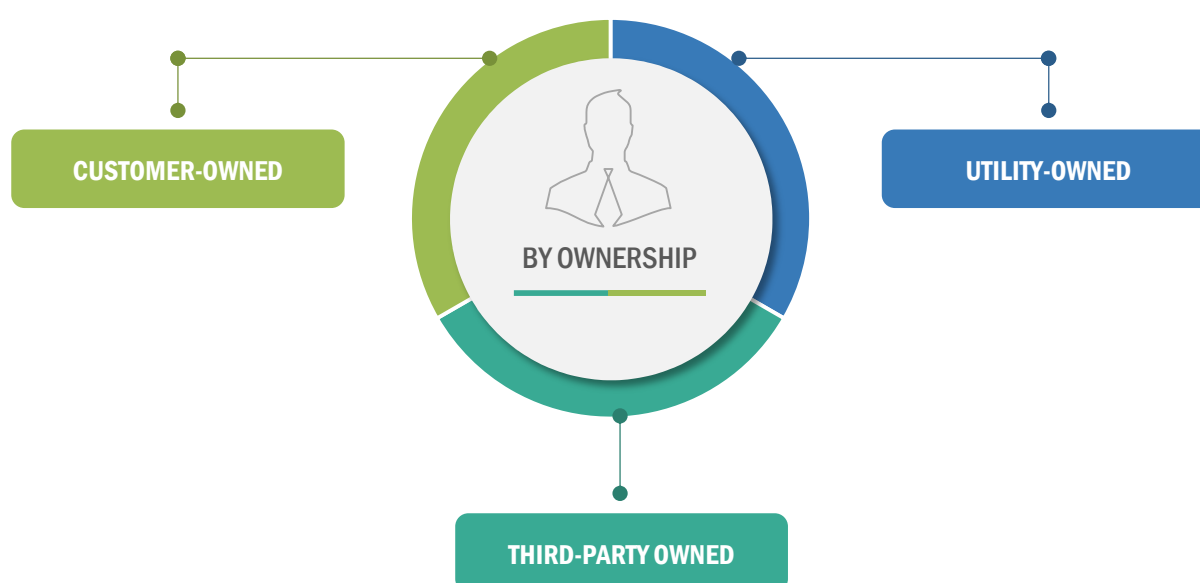
KEY FINDINGS

- The customer-owned battery energy storage system market was worth USD 237.0 million in 2021 and is projected to reach USD 862.8 million by 2027, growing at a CAGR of 24.7% between 2022 and 2027.
- The customer-owned battery energy storage systems reduce the communication burden of grid operators and avoid the need for external utility. Thus, the customer pays the full cost of the system with no utility control.
- The market for the utility-owned segment was valued at USD 2,621.1 million in 2021 and is projected to reach USD 11,646.4 million by 2027, growing at a CAGR of 28.6% between 2022 and 2027.
- The ability of the utility-owned battery energy storage systems to manage large energy requirements during peak hours is increasing their adoption. According to US Energy Information Administration, the large-scale battery capacity of the US grew by 35% in 2020 and the capacity is estimated to continue to grow rapidly during the forecast period. It is also estimated that utilities have planned to install over 10,000 MW of large-scale battery capacity from 2021 to 2023, in the US.
- The increasing investments in renewable energy are also supporting the growth of the utility-owned segment.

10.1 INTRODUCTION

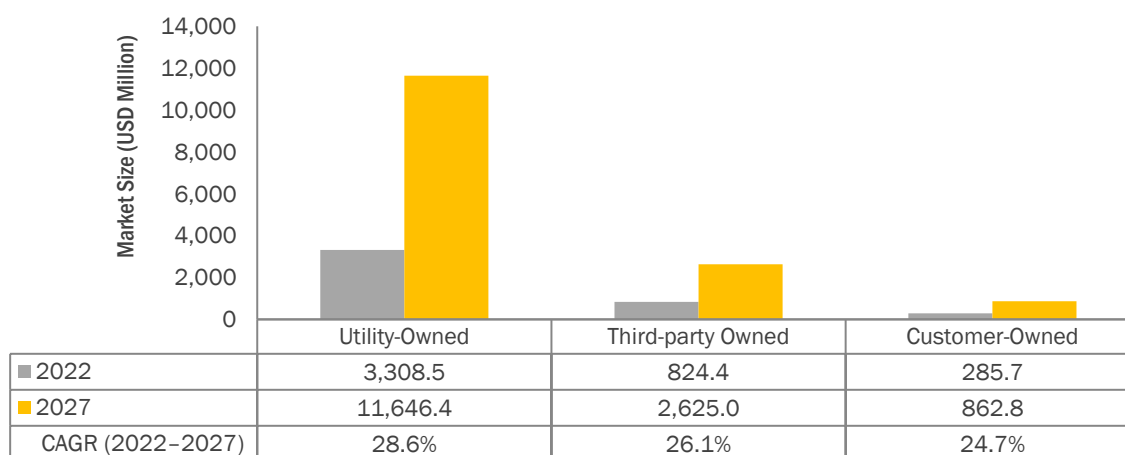
In this chapter, the battery energy storage system market (BESS) has been segmented based on ownership into customer-owned, third-party-owned, and utility-owned. The customer-owned battery energy storage systems are those owned by end users for the energy storage of their applications. The customer-owned battery storage system is installed at the customer premises, which helps the customer reduce electricity bills, and get power supply without interruptions even at the time of outages. The use of renewable energy and battery energy storage systems helps users to store electricity for a longer period. In third-party-owned ownership, the third party owns the energy storage systems, and it buys and sells electricity with the help of arbitrage through end customers and utilities. Furthermore, the utility-owned battery energy storage system involves grid operations to provide grid infrastructure for the transmission and distribution (T&D) of electricity. The utilities set up high-capacity battery storage systems, to store excess generated power at the time of low-peak periods and supply the stored power at the time of high-peak demand for electricity.

FIGURE 44 BATTERY ENERGY STORAGE SYSTEM MARKET, BY OWNERSHIP



Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 45 UTILITY-OWNED BATTERY ENERGY STORAGE SYSTEMS SEGMENT TO HOLD LARGEST MARKET SIZE BY 2027



Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 66 BATTERY ENERGY STORAGE SYSTEM MARKET, BY OWNERSHIP, 2018–2021 (USD MILLION)

Ownership	2018	2019	2020	2021
Customer-Owned	160.4	247.7	199.0	237.0
Third-party-Owned	445.0	690.0	561.7	676.8
Utility-Owned	1,714.0	2,668.9	2,160.8	2,621.1
Total	2,319.4	3,606.7	2,921.4	3,534.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 67 BATTERY ENERGY STORAGE SYSTEM MARKET, BY OWNERSHIP, 2022–2027 (USD MILLION)

Ownership	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Customer-Owned	285.7	347.1	427.4	540.5	680.3	862.8	24.7%
Third-party-Owned	824.4	1,013.0	1,260.1	1,607.2	2,047.4	2,625.0	26.1%
Utility-Owned	3,308.5	4,204.1	5,373.1	6,882.1	8,914.1	11,646.4	28.6%
Total	4,418.6	5,564.3	7,060.6	9,029.8	11,641.7	15,134.2	27.9%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

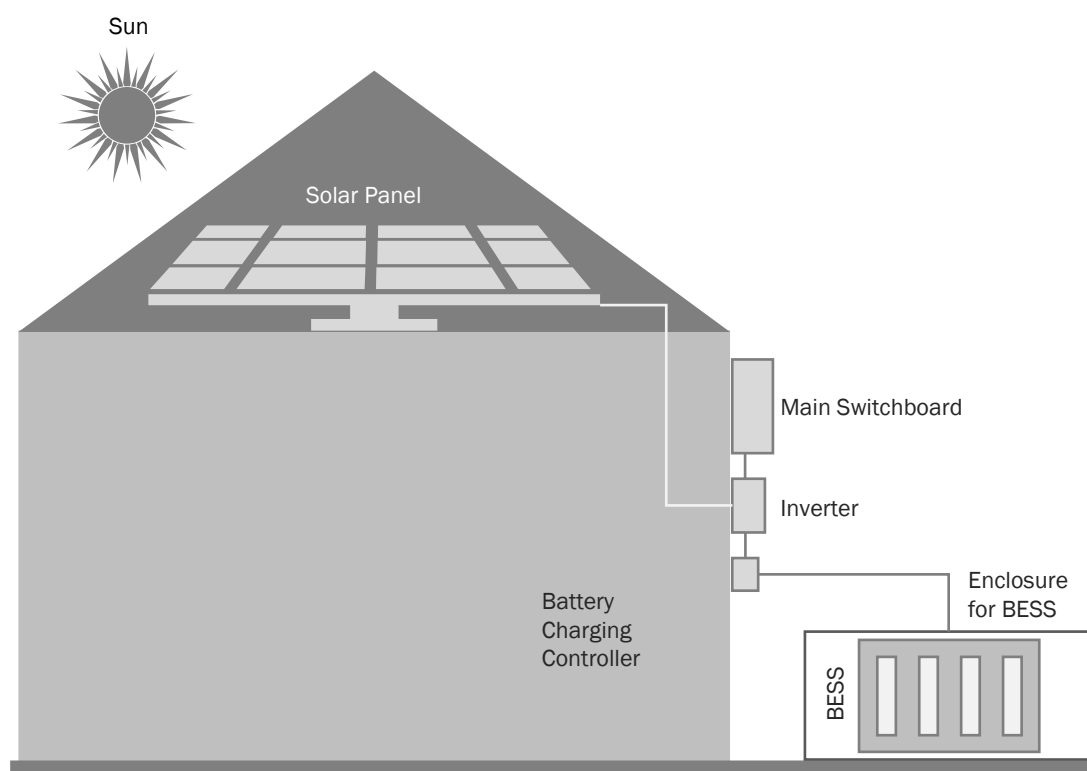
The ability of the utility-owned battery energy storage systems to manage large energy requirements during peak hours is increasing their adoption. The operators of the utility-owned storage systems are investing in energy storage to defer the cost of electrical transmission & distribution upgrades to meet the rising electricity demand.

10.2 CUSTOMER-OWNED

10.2.1 INCREASING ADOPTION OF ENERGY STORAGE SYSTEMS FROM RESIDENTIAL OWNERS TO REDUCE COMMUNICATION LOADS FROM GRIDS

The customer-owned battery energy storage system is used for storing solar energy for self-consumption in homes or residential areas. These customer-owned battery energy storage systems are also utilized in commercial and industrial applications. The customer-owned battery energy storage systems help reduce the electricity cost, as electricity is charged under the time-of-use (TOU) tariff. There are three periods for TOU pricing: peak, shoulder, and off-peak. Commonly, prices are highest in peak times and lowest in off-peak times. Under the TOU tariff, a customer can easily reduce the energy cost (electricity cost and annual investment cost) by using battery energy storage systems. Additionally, the customer-owned battery storage systems are used to manage the electricity cost and reduce the cost associated with lower and unreliable quality power. Battery energy storage systems also help solve the over-voltage issues by controlling the fluctuation of voltage and frequency. Energy storage allows the customer to reduce the time-of-use energy cost by storing the energy during the off-peak time and using this energy during the higher on-peak time, which helps the end users to manage the electricity bill. The customer-owned battery energy storage systems reduce the communication burden from grid operators and avoid the need for an external utility. Thus, customers pay the full cost of the system with no utility control.

FIGURE 46 LAYOUT OF BATTERY ENERGY STORAGE SYSTEM INSTALLATION IN RESIDENTIAL AREA



Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

The figure above depicts the example of a solar photovoltaic (PV) installation with battery storage for domestic use—the layout changes as per solar availability. For instance, in Western Australia, the solar panel should be placed facing north and west to consume high solar radiation for the desirable installation of battery energy storage systems. Whereas, in India, it is recommended that the solar panel should be facing the South to receive maximum radiation. A high amount of radiation generates a high amount of energy. Hence, according to the position of the countries, it is recommended to install the solar panels in the direction in which they will receive the highest radiation.

TABLE 68 BATTERY ENERGY STORAGE SYSTEM MARKET FOR CUSTOMER-OWNED SEGMENT, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Residential	72.4	117.9	93.3	111.0
Commercial	87.9	129.8	105.7	126.0
Utility	-	-	-	-
Total	160.4	247.7	199.0	237.0

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 69 BATTERY ENERGY STORAGE SYSTEM MARKET FOR CUSTOMER-OWNED SEGMENT, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	134.2	162.7	201.8	260.7	327.0	417.1	25.5%
Commercial	151.5	184.4	225.6	279.9	353.2	445.7	24.1%
Utility	-	-	-	-	-	-	-
Total	285.7	347.1	427.4	540.5	680.3	862.8	24.7%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The increasing adoption of wind and solar power has led to an increase in the use of customer-owned battery energy storage technologies to store electricity. Also, the need to reduce carbon emissions has increased the transformation of electrical systems and fueled the demand for intermittent sources of power such as wind and solar energy.

10.3 THIRD-PARTY-OWNED

10.3.1 THIRD-PARTY OWNERSHIP OF BATTERY ENERGY STORAGE SYSTEMS BENEFITS UTILITY APPLICATION

The third-party owner or independent power producer (IPP) acts as an aggregator and makes its rate structure when offering electricity to consumers or end users. In third-party ownership, the battery energy storage system at the customer premise serves both end customers and utilities. The energy generated and stored by the energy storage system can be directly utilized by the end consumer, thus, giving them the privilege of power supply in outages. The third-party-owned battery energy storage system can also be linked to the grid and can supply energy to the utilities' distribution network in high-peak demand periods.

Third-party ownership involves a wide range of agreements between vendors and their customers. For instance, in off-grid connections, the customers make their energy storage and, if interested, can also sell the stored energy in the form of electricity. Sunverge Energy (US), a leading provider of intelligent energy storage systems for residences and small businesses, operates through third-party-owned battery energy storage systems providing its Sunverge Infinity (an energy management platform). This platform deals with a range of battery storage and inverter products, and it is utilized by various vendors, including utilities, electric retailers, and solar companies. In May 2020, Pacific Gas and Electric Company (PG&E) (US) has been awarded contracts for battery energy storage projects of more than 1,000 MWs of capacity to be deployed throughout its service area through 2023. Third-party-owned projects include a transmission-connected 300 MW BESS (also to be located in Moss Landing), 2 MW BESS at PG&E's Gonzales substation in the Salinas Valley, and a transmission-connected 75 MW BESS located near the city of Morgan Hill, California, US.

TABLE 70 BATTERY ENERGY STORAGE SYSTEM MARKET FOR THIRD-PARTY-OWNED SEGMENT, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Residential	169.0	278.9	223.9	270.1
Commercial	275.9	411.2	337.7	406.6
Utility	-	-	-	-
Total	445.0	690.0	561.7	676.8

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 71 BATTERY ENERGY STORAGE SYSTEM MARKET FOR THIRD-PARTY-OWNED, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	331.1	407.0	511.8	670.3	853.3	1,104.2	27.2%
Commercial	493.3	606.1	748.3	936.9	1,194.0	1,520.9	25.3%
Utility	-	-	-	-	-	-	-
Total	824.4	1,013.0	1,260.1	1,607.2	2,047.4	2,625.0	26.1%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The growth of the third-party-owned battery energy storage system market is attributed to the affordability to install solar energy systems and a reduction in electricity bills. A major share of the residential sector in the US has access to third-party-owned battery energy storage systems. According to the US Energy Information Administration, the average battery energy storage capital costs fell by 72% between 2015 and 2019. Thus, the lower costs support larger capacity storage system installations. This shows the high rate of adoption of a third-party-owned battery energy storage system. Moreover, the third party gets complete information about the plant control systems for monitoring it remotely.

10.4 UTILITY-OWNED

10.4.1 GLOBAL INITIATIVES FOR RENEWABLE ENERGY GENERATION TO BOOST DEMAND FOR UTILITY-OWNED ENERGY STORAGE SYSTEMS

The operators of the utility-owned battery energy storage systems use renewable energy sources to a greater extent for delivering a large number of units of electricity. The customers of utility-owned battery energy storage systems need to pay monthly fees based on the power usage. The utility-owned energy storage companies invest in energy storage to overcome the high cost of electrical T&D by fulfilling the growing electricity demand. The battery energy storage systems help improve grid reliability, manage T&D congestion while improving T&D performance, and help avoid the purchase of additional equipment for energy storage as well as increase the lifespan of assets.

The utilities in the US are planning to set a 10 GW of new grid-connected capacity online in the coming years. These battery storages include standalone battery storage, solar battery storage, wind battery storage, and fossil fuel battery storage. The deployment of battery energy storage systems in the country is rapidly increasing, which has enabled the power grids to operate more flexibly and resiliently.

TABLE 72 BATTERY ENERGY STORAGE SYSTEM MARKET FOR UTILITY-OWNED SEGMENT, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Residential	-	-	-	-
Commercial	-	-	-	-
Utility	1,714.0	2,668.9	2,160.8	2,621.1
Total	1,714.0	2,668.9	2,160.8	2,621.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 73 BATTERY ENERGY STORAGE SYSTEM MARKET FOR UTILITY-OWNED SEGMENT, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	-	-	-	-	-	-	-
Commercial	-	-	-	-	-	-	-
Utility	3,308.5	4,204.1	5,373.1	6,882.1	8,914.1	11,646.4	28.6%
Total	3,308.5	4,204.1	5,373.1	6,882.1	8,914.1	11,646.4	28.6%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The demand for utility-owned energy storage systems was largely affected by the COVID-19 pandemic. Most of the projects in China, the US, Germany, and South Korea are operational and are used for integrating renewable energy power generation and grid balancing operations. However, a few of the companies faced short-term operational issues due to the limited workforce. The outbreak of COVID-19 has impacted the sale of energy storage systems used in utilities due to delays in production, especially in countries such as China and South Korea. However, it is expected that the demand for the energy storage systems would be to a greater extent in the coming years owing to the various global initiatives including the zero carbon emissions and RE100, a renewable energy initiative that brings together large businesses, which are committed to 100% renewable electricity. Furthermore, power generation is also moving toward renewable, and cleaner modes, which is also estimated to increase the demand for battery energy storage systems.

11 BATTERY ENERGY STORAGE SYSTEM MARKET, BY ENERGY CAPACITY

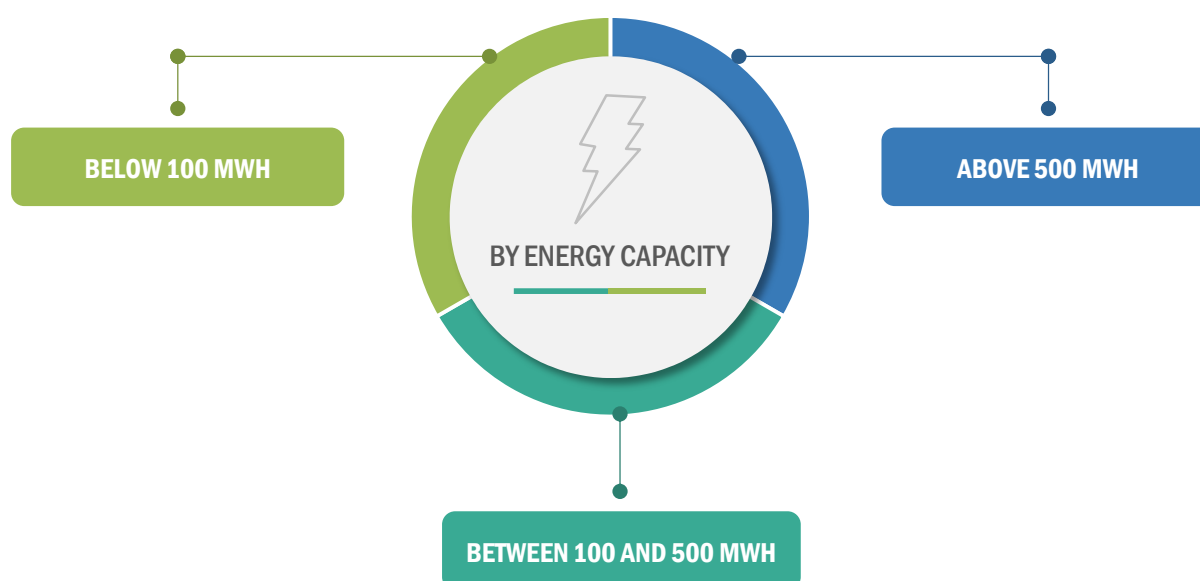
KEY FINDINGS

- The battery energy storage system market for above 500 MWh energy capacity was valued at USD 2,377.8 million in 2021 and is projected to reach USD 10,795.8 million by 2027; it is expected to grow at the highest CAGR of 29.1% from 2022 to 2027.
- These battery energy storage systems help in supplying a surplus of energy during times of high electricity demand. The utilities are among the applications that require high-capacity energy storage systems. In recent years, due to the growing need for electrification in rural and urban areas, government and private bodies are striving to improve the electric grid systems to cater to energy needs. These systems help the utilities meet local capacity requirements and mitigate energy procurement costs.
- The battery energy storage systems market for energy capacity between 100 and 500 MWh accounted for a 20% share in 2021. The market for battery energy storage systems between 100 and 500 MWh was valued at USD 699.8 million in 2021 and is projected to reach USD 2,575.0 million by 2027; it is expected to grow at a CAGR of 24.8% from 2022 to 2027.
- The battery energy storage systems ranging between 100 and 500 MWh increase the need for frequency control services to manage increased variability and uncertainty in the power system due to high energy density.
- The market for lithium-ion battery energy storage systems above 500 MWh energy capacity was valued at USD 1,303.6 million in 2021. It is the most widely used battery by all the applications. Lithium-ion batteries are feasible and can be manufactured according to the need of capacity. Hence, these energy storage batteries are used in residential, commercial as well as utility sectors.

11.1 INTRODUCTION

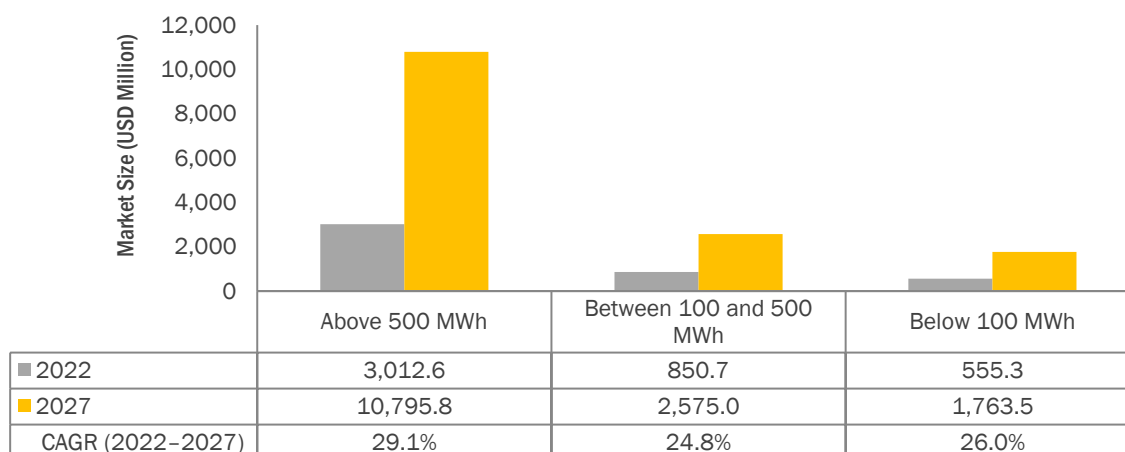
This chapter describes the energy capacity of battery energy storage systems. Energy capacity is the maximum amount of stored energy and is expressed in the form of megawatt-hours (MWh) or kilowatt-hours (KWh). Battery energy storage systems are rechargeable battery systems that store energy from solar arrays or the electric grid and provide that energy to a home or business. Energy is discharged from the battery storage system during times of high usage, reducing or eliminating costly demand charges. In this chapter, the battery energy storage system market has been classified into low, medium, and high ranges. The battery range below 100 MWh is considered in the low range, the medium range includes battery capacity between 100 and 500 MWh, and the high range includes more than 500 MWh energy capacity. The batteries of below 100 MWh range are more commonly used in residential applications, the batteries of medium-range are used at commercial and industrial sites. The high range energy capacity batteries are dedicatedly used for on-grid connection by utility applications.

FIGURE 47 ENERGY CAPACITY RANGES OF BATTERY ENERGY STORAGE SYSTEMS



Source: Secondary Research, Company Websites, Expert Interview, and MarketsandMarkets Analysis

FIGURE 48 BATTERY ENERGY STORAGE SYSTEMS FOR ABOVE 500 MWH ENERGY CAPACITY TO REGISTER HIGHEST CAGR FROM 2022 TO 2027



Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 74 BATTERY ENERGY STORAGE SYSTEM MARKET, BY ENERGY CAPACITY, 2018–2021 (USD MILLION)

Energy Capacity	2018	2019	2020	2021
Below 100 MWh	297.2	477.9	382.2	457.3
Between 100 and 500 MWh	484.7	726.7	586.6	699.8
Above 500 MWh	1,537.5	2,402.0	1,952.6	2,377.8
Total	2,319.4	3,606.7	2,921.4	3,534.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 75 BATTERY ENERGY STORAGE SYSTEM MARKET, BY ENERGY CAPACITY, 2022–2027 (USD MILLION)

Energy Capacity	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Below 100 MWh	555.3	677.4	843.3	1,089.1	1,376.0	1,763.5	26.0%
Between 100 and 500 MWh	850.7	1,045.0	1,290.4	1,609.2	2,032.3	2,575.0	24.8%
Above 500 MWh	3,012.6	3,841.8	4,926.9	6,331.5	8,233.4	10,795.8	29.1%
Total	4,418.6	5,564.3	7,060.6	9,029.8	11,641.7	15,134.2	27.9%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

Based on energy capacity, the market for battery energy storage systems for above 500 MWh capacity accounted for the largest share in 2021 and is expected to register the highest CAGR of 29.1% from 2022 to 2027. A battery energy storage system, having high capacity and low power delivers a low amount of electricity for a long time and is sufficient to run a few crucial appliances. Such battery storage systems help in operating power plants during high energy demand times. The utilities are among the applications that require high-capacity energy storage systems. To date, most of the high-capacity energy storage system installation projects across the world cater to the utility application. In recent years, due to the growing need for electrification in rural and urban areas, government and private bodies are striving to improve the electric grid systems to cater to energy needs. Also, renewable energy generation and their integration in grids are one of the major factors contributing to the growing need for energy storage systems in utilities.

Additionally, the storage of excessive wind and solar energy and the utilization of the energy by supplying it to the grids during the highest demand of electricity across the world is also anticipated to drive the growth of the battery energy storage system market. These systems help the utility meet local capacity requirements and mitigate energy procurement costs. For instance, large lithium-ion battery storage systems of 182.5 MW/730 MWh were provided by Tesla and owned and operated by PG&E. The Tesla system began its commercial operation by the end of 2019. In September 2021, Tesla's Powerpack BESS equipment was installed at the Mohammed Bin Rashid Al Maktoum Solar Park in Dubai, UAE. This pilot project utilizes 1.21 MW / 8.61 MWh of Tesla's Powerpack product.

11.2 BELOW 100 MWH

11.2.1 GROWING DEMAND FOR BATTERY ENERGY STORAGE SYSTEMS WITH LOW ENERGY CAPACITY IN RESIDENTIAL APPLICATIONS

Battery energy storage systems with low energy capacity below 100 MWh are mostly suitable for residential- or community-based storage systems. The batteries used in these systems generally have long life cycles, which makes them well-suited for supplying continuous power. In September 2019, General Electric was selected by Convergent to supply battery energy storage systems in California with a total energy capacity of 100 MWh. This project supported local area reliability and provided fast response resources to help integrate growing renewable energy generation in California. On the other hand, Panasonic's smart energy storage system with an energy capacity of 1 MWh is used for various applications such as residential energy storage, community energy storage, and utility-scale ancillary services.

Similarly, there was a short-term drop-in network frequency in the UK, in August 2019, which resulted in a large-scale power outage. To cope with the situation, and to ensure this kind of emergency does not repeat, the UK government installed storage systems. The 150 MW Minety power plant was developed in 2020 in the UK, which is the largest battery storage development in the European region. The project comprises three 50 MW adjacently located battery units, which are using ternary lithium battery technology/ lithium-iron-phosphate (LiFePO₄) for the storing of electricity. China Huaneng Group is responsible for the operation and construction of the facility.

TABLE 76 BATTERY ENERGY STORAGE SYSTEM MARKET FOR BELOW 100 MWH ENERGY CAPACITY, BY BATTERY TYPE, 2018–2021 (USD MILLION)

Battery Type	2018	2019	2020	2021
Lithium-Ion Batteries	141.8	226.2	180.1	214.9
Advanced Lead-Acid Batteries	55.9	85.7	64.9	73.4
Flow Batteries	70.4	117.8	97.5	120.4
Others	29.1	48.2	39.6	48.6
Total	297.2	477.9	382.2	457.3

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 77 BATTERY ENERGY STORAGE SYSTEM MARKET FOR BELOW 100 MWH ENERGY CAPACITY, BY BATTERY TYPE, 2022–2027 (USD MILLION)

Battery Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Lithium-Ion Batteries	257.3	308.6	377.3	479.0	605.6	768.8	24.5%
Advanced Lead-Acid Batteries	84.9	98.8	117.0	143.6	168.4	201.9	18.9%
Flow Batteries	152.1	193.2	250.0	335.0	432.7	570.6	30.3%
Others	61.0	76.9	98.9	131.6	169.3	222.1	29.5%
Total	555.3	677.4	843.3	1,089.1	1,376.0	1,763.5	26.0%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 78 BATTERY ENERGY STORAGE SYSTEM MARKET FOR BELOW 100 MWH ENERGY CAPACITY, BY CONNECTION TYPE, 2018–2021 (USD MILLION)

Connection Type	2018	2019	2020	2021
On-grid	239.5	386.5	309.9	371.9
Off-grid	57.7	91.4	72.3	85.4
Total	297.2	477.9	382.2	457.3

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 79 BATTERY ENERGY STORAGE SYSTEM MARKET FOR BELOW 100 MWH ENERGY CAPACITY, BY CONNECTION TYPE, 2022–2027 (USD MILLION)

Connection Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
On-grid	453.2	554.7	692.6	896.8	1,142.8	1,474.2	26.6%
Off-grid	102.1	122.8	150.7	192.3	233.2	289.3	23.2%
Total	555.3	677.4	843.3	1,089.1	1,376.0	1,763.5	26.0%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 80 BATTERY ENERGY STORAGE SYSTEM MARKET FOR BELOW 100 MWH ENERGY CAPACITY, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Residential	241.5	396.7	317.2	381.1
Commercial	55.7	81.1	64.9	76.2
Utility	-	-	-	-
Total	297.2	477.9	382.2	457.3

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 81 BATTERY ENERGY STORAGE SYSTEM MARKET FOR BELOW 100 MWH ENERGY CAPACITY, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	465.3	569.7	713.6	930.9	1,180.3	1,521.3	26.7%
Commercial	90.0	107.8	129.7	158.2	195.7	242.2	21.9%
Utility	-	-	-	-	-	-	-
Total	555.3	677.4	843.3	1,089.1	1,376.0	1,763.5	26.0%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The home energy storage technologies have a low capacity with off-grid connections, which are used for powering the home appliances during an outage or as an emergency backup. In the forecast period, the BESS market for residential applications is estimated to grow rapidly owing to the technological advancements in solar-powered energy generation at home and the energy bill savings in the backdrop. For instance, according to the Renewable Energy World, about 65,000 new residential battery solutions were installed in 2019 an increase of 44% in Germany. Lithium-ion batteries are mostly adopted since they are a lighter and more compact option than lead-acid batteries for home energy storage systems. They also have a higher DoD and longer lifespan when compared to lead-acid batteries. However, the lithium-ion battery is more expensive than its lead-acid counterpart.

11.3 BETWEEN 100 AND 500 MWH

11.3.1 TO MANAGE HIGH VARIABILITY AND UNCERTAINTY IN POWER SYSTEMS, BATTERIES OF ENERGY CAPACITY BETWEEN 100 AND 500 MWH HAVE INCREASED NEED FOR FREQUENCY CONTROL SERVICES

The battery energy storage systems ranging between 100 and 500 MWh increase the need for frequency control services to manage increased variability and uncertainty in the power system due to high energy density. In November 2019, the Salt River Project announced to set up the Storey Energy Center, a solar and energy storage system in Coolidge, Arizona, with 88 MW of solar and energy storage capacity, which will be installed by June 2023. The project comprises 88 MW solar generation capacity, with the storage system of a 4-hour battery—totaling the energy capacity up to 352 MWh. The project is set to be on board by June 2023 and will be owned and operated by subsidiaries of NextEra Energy Resources (US). Similarly, in December 2020, Vistra's Moss Landing Phase 1, Energy Storage Facility in California, came into operation, which is connected to power grids. The company has begun operating a 300 MW/1,200 MWh lithium-ion battery storage system on its 1,020-MW combined cycle gas turbine in Phase 1. Furthermore, the second phase of the project comprises an additional 100 MW/400 MWh storage system, which has resulted in a total capacity of 400 MW/1,600 MWh Moss Landing energy storage facility.

TABLE 82 BATTERY ENERGY STORAGE SYSTEM MARKET FOR BETWEEN 100 MWH AND 500 MWH ENERGY CAPACITY, BY BATTERY TYPE, 2018–2021 (USD MILLION)

Battery Type	2018	2019	2020	2021
Lithium-Ion Batteries	228.9	336.0	270.6	319.6
Advanced Lead-Acid Batteries	92.0	133.0	101.6	115.1
Flow Batteries	115.9	182.9	152.5	188.8
Others	47.9	74.8	62.0	76.2
Total	484.7	726.7	586.6	699.8

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 83 BATTERY ENERGY STORAGE SYSTEM MARKET FOR BETWEEN 100 MWH AND 500 MWH ENERGY CAPACITY, BY BATTERY TYPE, 2022–2027 (USD MILLION)

Battery Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Lithium-Ion Batteries	381.9	459.9	554.9	673.8	847.2	1,055.8	22.6%
Advanced Lead-Acid Batteries	133.6	156.6	184.7	220.1	257.3	304.1	17.9%
Flow Batteries	239.3	306.4	394.6	513.5	666.9	874.6	29.6%
Others	95.9	122.0	156.1	201.7	260.9	340.5	28.8%
Total	850.7	1,045.0	1,290.4	1,609.2	2,032.3	2,575.0	24.8%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 84 BATTERY ENERGY STORAGE SYSTEM MARKET FOR BETWEEN 100 MWH AND 500 MWH ENERGY CAPACITY, BY CONNECTION TYPE, 2018–2021 (USD MILLION)

Connection Type	2018	2019	2020	2021
On-grid	390.6	587.8	475.7	569.2
Off-grid	94.2	139.0	110.9	130.7
Total	484.7	726.7	586.6	699.8

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 85 BATTERY ENERGY STORAGE SYSTEM MARKET FOR BETWEEN 100 MWH AND 500 MWH ENERGY CAPACITY, BY CONNECTION TYPE, 2022–2027 (USD MILLION)

Connection Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
On-grid	694.3	855.6	1,059.8	1,325.0	1,678.1	2,132.2	25.2%
Off-grid	156.4	189.4	230.5	284.1	354.2	442.7	23.1%
Total	850.7	1,045.0	1,290.4	1,609.2	2,032.3	2,575.0	24.8%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 86 BATTERY ENERGY STORAGE SYSTEM MARKET FOR BETWEEN 100 MWH AND 500 MWH ENERGY CAPACITY, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Residential	-	-	-	-
Commercial	308.1	459.8	378.5	456.5
Utility	176.6	266.9	208.2	243.3
Total	484.7	726.7	586.6	699.8

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 87 BATTERY ENERGY STORAGE SYSTEM MARKET FOR BETWEEN 100 MWH AND 500 MWH ENERGY CAPACITY, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	-	-	-	-	-	-	-
Commercial	554.8	682.7	844.2	1,058.6	1,351.6	1,724.4	25.5%
Utility	295.9	362.3	446.1	550.6	680.7	850.5	23.5%
Total	850.7	1,045.0	1,290.4	1,609.2	2,032.3	2,575.0	24.8%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

Power outages can hamper businesses to a large extent due to interrupted production, delayed deliveries, and damage to production lines and products. Unexpected power dropouts can last from minutes or hours to days or weeks in some locations. In commercial and industrial companies, battery energy storage systems with medium capacity help in reducing utility costs, improving power quality and reliability. Battery energy storage systems integrated with renewable energy generation like solar panels at the business premises can ensure uninterrupted power supply with reduced grid dependency, which also helps in minimizing the electricity bills. Through the implementation of battery storage systems, critical production lines run unhindered through power blackouts and without penal charges from peak demand tariffs. In Germany, the growth in individual network charges in recent years is encouraging larger commercial and industrial segments to adopt a flow battery energy storage system. Flow battery-based energy storage systems between 100 and 500MWh capacity can deliver emergency backup power for sites ranging from large- and small-scale manufacturing factories to commercial offices, shopping centers, hospitals, farms, and data centers to rural pumping stations and mines.

11.4 ABOVE 500 MWH

11.4.1 LARGE-SCALE BATTERY STORAGE SYSTEMS HELP CONTINUITY OF POWER DISTRIBUTION IN LARGE GRIDS

These battery energy storage systems exhibit extended service life. The battery energy storage systems with high energy capacity can support peak shaving and valley filling in the future global energy interconnection. These systems help in serving the energy demands, which can be used for peak load regulation in support of a large grid. High energy storage systems can be used to smoothen fluctuations of large-scale clean energy generation. The large-scale battery storage system aligns the renewable energy production and rapidly responds to the output of wind and photovoltaic power, smoothen fluctuations of renewable energy generation, and ensures the safety of real-time grid operations. The demand switching off (a sudden drop of load), on energy generation, can hamper the whole system, thus, in times of low demand, the surplus energy generated can be stored in the batteries; and at the peak times, this stored energy can be integrated with the grid for further distribution. This, in turn, results in decreasing the load on traditional power generation plants and making the distribution network more flexible and reliable.

In November 2019, the Salt River Project had announced the setting up of a Sonoran Energy Center of 250 MW, 1,000 MWh battery project in Little Rainbow Valley, Arizona, which will be set up by June 2023. The installation is said to be the largest in the state and one among the largest in the US. Additionally, in July 2020 China's National Development and Reform Commission (NDRC) stated that it will be installing more than 30 gigawatts (GW) of new energy storage capacity by 2025 in the country. This energy storage system will include storage processes that use electrochemical, compressed air, flywheel, and supercapacitor systems.

TABLE 88 BATTERY ENERGY STORAGE SYSTEM MARKET FOR ABOVE 500 MWH ENERGY CAPACITY, BY BATTERY TYPE, 2018–2021 (USD MILLION)

Battery Type	2018	2019	2020	2021
Lithium-Ion Batteries	878.9	1,349.4	1,084.2	1,303.6
Advanced Lead-Acid Batteries	236.8	358.4	279.1	325.3
Flow Batteries	298.4	492.7	419.1	533.5
Others	123.3	201.6	170.3	215.3
Total	1,537.5	2,402.0	1,952.6	2,377.8

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 89 BATTERY ENERGY STORAGE SYSTEM MARKET FOR ABOVE 500 MWH ENERGY CAPACITY, BY BATTERY TYPE, 2022–2027 (USD MILLION)

Battery Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Lithium-Ion Batteries	1,630.9	2,050.8	2,589.2	3,271.8	4,198.1	5,420.5	27.2%
Advanced Lead-Acid Batteries	393.7	479.5	587.1	719.9	884.0	1,095.5	22.7%
Flow Batteries	705.2	938.0	1,254.4	1,679.8	2,265.1	3,080.6	34.3%
Others	282.7	373.5	496.1	659.9	886.1	1,199.2	33.5%
Total	3,012.6	3,841.8	4,926.9	6,331.5	8,233.4	10,795.8	29.1%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 90 BATTERY ENERGY STORAGE SYSTEM MARKET FOR ABOVE 500 MWH ENERGY CAPACITY, BY CONNECTION TYPE, 2018–2021 (USD MILLION)

Connection Type	2018	2019	2020	2021
On-grid	1,302.7	2,038.3	1,659.4	2,023.9
Off-grid	234.8	363.8	293.2	353.8
Total	1,537.5	2,402.0	1,952.6	2,377.8

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 91 BATTERY ENERGY STORAGE SYSTEM MARKET FOR ABOVE 500 MWH ENERGY CAPACITY, BY CONNECTION TYPE, 2022–2027 (USD MILLION)

Connection Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
On-grid	2,568.8	3,281.8	4,216.8	5,429.3	7,070.2	9,286.4	29.3%
Off-grid	443.8	560.0	710.2	902.2	1,163.2	1,509.4	27.7%
Total	3,012.6	3,841.8	4,926.9	6,331.5	8,233.4	10,795.8	29.1%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 92 BATTERY ENERGY STORAGE SYSTEM MARKET FOR ABOVE 500 MWH ENERGY CAPACITY, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Residential	-	-	-	-
Commercial	-	-	-	-
Utility	1,537.5	2,402.0	1,952.6	2,377.8
Total	1,537.5	2,402.0	1,952.6	2,377.8

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 93 BATTERY ENERGY STORAGE SYSTEM MARKET FOR ABOVE 500 MWH ENERGY CAPACITY, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	-	-	-	-	-	-	-
Commercial	-	-	-	-	-	-	-
Utility	3,012.6	3,841.8	4,926.9	6,331.5	8,233.4	10,795.8	29.1%
Total	3,012.6	3,841.8	4,926.9	6,331.5	8,233.4	10,795.8	29.1%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

Utilities require resiliency, long-operating life, and the capability to deal with an increasing amount of renewable energy penetration across the grid. The electric grid has limited ability to store excess energy, therefore, electricity must be constantly generated in excess to assure reliable supply. This induces the need for battery energy storage systems with high energy capacity. These energy storage systems ensure uninterrupted power supply and balance temperatures and fluctuations because of weather conditions, day lengths, and unstable networks. The lithium-ion battery is the most adopted technology in energy storage systems for utility applications. However, large-scale flow batteries are also gaining momentum, allowing electrical systems to run more efficiently at lower prices, fewer emissions, and reliable power.

12 BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION

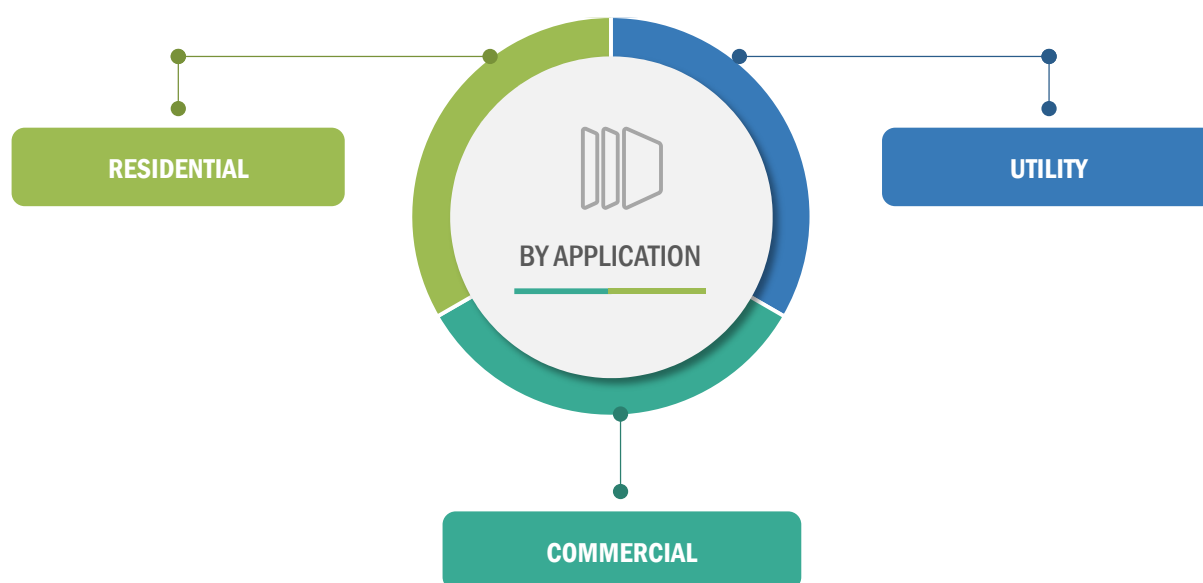
KEY FINDINGS

- The utility application held the largest size of the battery energy storage system market in 2021. The market was worth USD 2,621.1 million in 2021 and is projected to reach USD 11,646.4 million by 2027, growing at a CAGR of 28.6% between 2022 and 2027.
- The need to fulfill the requirement of peak electricity demands is the major factor contributing to the growth of the market for utility. The utility BESS provides power quality, frequency, and voltage control services in case of sudden fluctuation in the energy supply. The battery energy storage systems enable grid operators to save electricity when there is a surplus of renewable energy.
- The market for the residential application is expected to grow at the highest CAGR of 26.7% between 2022 and 2027, to reach USD 1,521.3 million by 2027 from USD 381.1 million in 2021.
- The use of battery energy storage systems in residential applications helps in electricity bill saving, self-consumption, and emergency power backup.
- The increasing demand for a reliable and efficient supply of electricity in most regions and the growing need for renewable energy are responsible for the high demand for battery energy storage systems in the utility sector.

12.1 INTRODUCTION

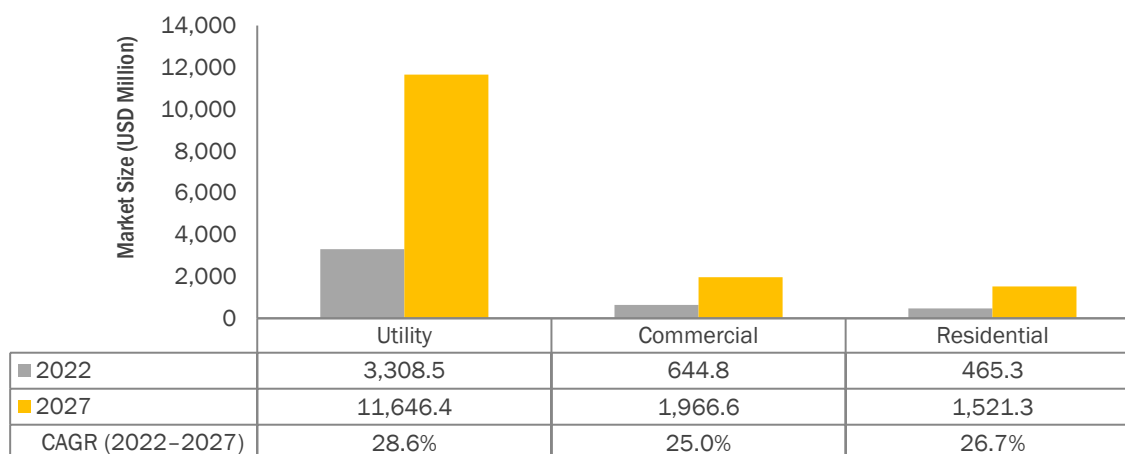
The battery energy storage system (BESS) is a device used to store energy and control the electricity fluctuations during supply to connected components and equipment. The battery energy storage system market based on application has been segmented into residential, commercial, and utility. Battery energy storage systems are most commonly used to store the generated renewable energy at customer sites in the residential sector. Battery energy storage using solar power generation is a more common form of residential BESS. For instance, Tesla Inc. provides solar panels with battery storage systems, which convert solar radiation into electrical energy, which can be used anywhere in the household. Commercial facilities such as malls, multiplexes, offices, and hospitals install battery energy storage systems with a large storage capacity compared to the residential sector. Whereas utilities have the highest storage capacity ranging from several megawatts to gigawatt, to save the surplus energy, and distribute it into the grids when in need.

FIGURE 49 BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION



Source: Expert Interviews and MarketsandMarkets Analysis

FIGURE 50 BATTERY ENERGY STORAGE SYSTEM MARKET FOR UTILITY APPLICATION TO EXHIBIT HIGHEST CAGR DURING FORECAST PERIOD



Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 94 BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

Application	2018	2019	2020	2021
Residential	241.5	396.7	317.2	381.1
Commercial	363.9	541.0	443.4	532.7
Utility	1,714.0	2,668.9	2,160.8	2,621.1
Total	2,319.4	3,606.7	2,921.4	3,534.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 95 BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	465.3	569.7	713.6	930.9	1,180.3	1,521.3	26.7%
Commercial	644.8	790.5	973.9	1,216.8	1,547.3	1,966.6	25.0%
Utility	3,308.5	4,204.1	5,373.1	6,882.1	8,914.1	11,646.4	28.6%
Total	4,418.6	5,564.3	7,060.6	9,029.8	11,641.7	15,134.2	27.9%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The utility segment held the largest size of the battery energy storage system market in 2021 and is expected to register the highest growth during the forecast period. The need to fulfill the requirement of peak electricity demands is the major factor contributing to the growth of the market for utilities. The flexibility offered by these systems to the grid energy storage to deal with the fluctuations in renewable energy, such as solar and wind energy, is increasing the adoption of battery energy storage systems. They provide frequency and voltage control services in case of sudden fluctuation in the energy supply. Battery energy storage systems enable grid operators to save electricity when there is a surplus of renewable energy.

The COVID-19 outbreak has impacted the supply chain of Li-ion batteries and components. The generation of electricity by utility operators was not discontinued, however, they faced tremendous issues such as sudden decrease of power demand from the huge power consumers such as factories, commercial buildings, railways, etc. as they were shut down. Lower economic conditions across the world have affected the utility sector, as electricity bills were not paid on a larger scale, thus impacting the utilities economically. The ease of lockdown in various countries has substantially increased the electricity demand, which, in turn, is responsible for the growth of the market for the utility application. This, in turn, is estimated to fuel the growth of the battery energy storage market during the forecast period.

The commercial application segment accounted for the second-largest market share in 2021. Commercial & industrial customers pay a major portion of their electricity bills based on demand charges, and battery energy storage systems help them save on these bills through peak demand reduction. This is expected to drive the growth of the on-grid connection battery energy storage system market for commercial applications.

12.2 IMPACT OF COVID-19 ON APPLICATION SEGMENTS OF BATTERY ENERGY STORAGE SYSTEM MARKET

Over the past few months, the COVID-19 pandemic has caused an unprecedented global economic and social crisis. This highly impacted the growth of the battery energy storage system market during the initial phase of the pandemic. The residential sector was significantly affected by the pandemic, which caused an economic slowdown, gradually affecting the residential battery storage market. A change in consumer buying behavior was observed due to the economic crisis, as the customers were highly focused on buying essentials, which also made the energy storage market stagnant.

Furthermore, the supply and demand of batteries were also restricted in countries such as the US, Germany, China, and Australia during the first two quarters of 2020, which has also impacted the growth of the energy storage market. The key impacts on the energy storage battery operators in the residential sector are economic slowdown and reduced consumer demand due to lockdowns. The delayed implementation of residential energy storage projects has resulted in a decline in power quality. The pandemic has delayed the implementation of DER in Germany, the US, and Australia, which are among the highly impacted countries. Another key impact is the battery equipment supply constraints due to a decline in dollar value and subsidies for residential sector energy storage projects. The shortage of supply is expected to increase prices in countries that are highly dependent on Chinese players for operations such as the US, Australia, and Germany.

The COVID-19 outbreak has also impacted the supply chain of Li-ion batteries and the components, which has resulted in project delays in the US, Germany, and Australia. Due to these delays, project developers are not able to meet the deadlines for claiming tax credits and renewable incentives. This current uncertainty about the ability to qualify for tax incentives is expected to have substantial negative impacts on the energy storage sector. The key countries to be hit the most by the COVID-19 impact include the US, Germany, Italy, Australia, China, and South Korea. However, with the improvement in the situation, it is expected that the energy storage industry will gradually return to normal within the next couple of months.

12.3 RESIDENTIAL

12.3.1 GOVERNMENT INITIATIVES TO BOOST NUMBER OF RESIDENTIAL ENERGY STORAGE PROJECTS

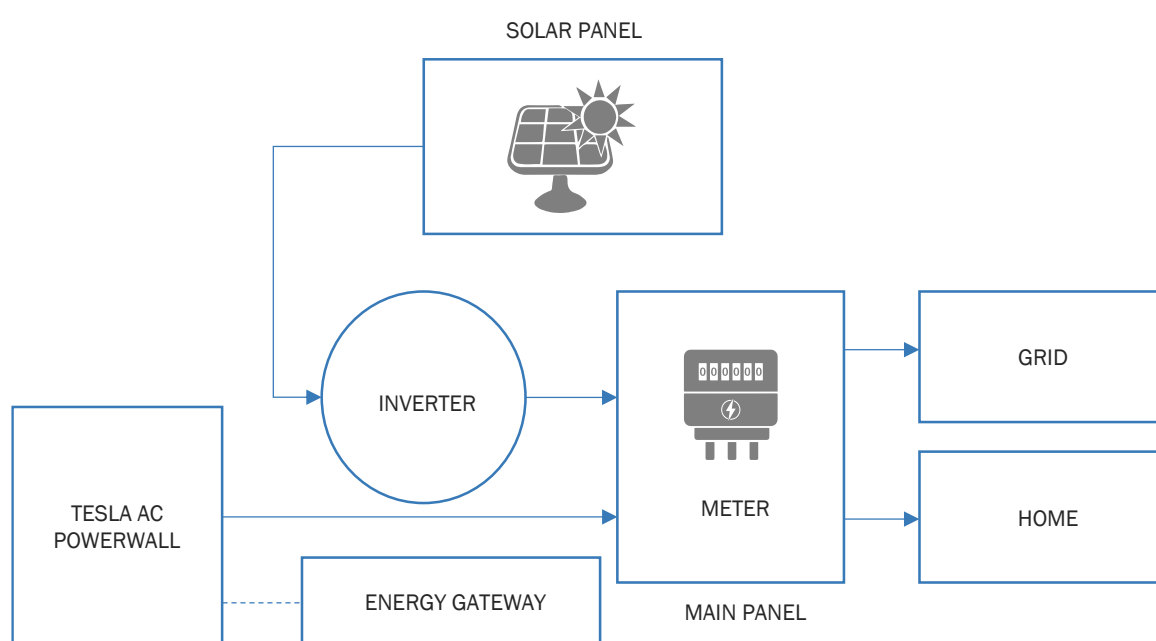
The battery energy storage systems used in residential applications provide power to residential buildings such as condominiums, co-operative societies, and single-family homes, and are also used as an emergency power source in residential areas. Renewable energy sources, such as wind and solar, generate a large quantity of energy. Battery energy storage systems can store these energies during peak production with the help of batteries embedded in them. Thus, this ability to store the energy during the peak production period, leading to less wastage of energy, is driving their adoption in residential applications, as the residential sector worldwide is shifting toward the adoption of renewable energy sources for accomplishing routine tasks. The use of battery energy storage systems in residential applications helps in electricity bill saving, self-consumption, and emergency power backup.

Governments of various countries worldwide have also taken initiatives for the adoption and installation of residential battery energy storage systems. For instance, the UK government has increased the adoption of battery energy storage to meet local, regional, and international goals for the reduction in carbon emission. The demand for customer-owned battery energy storage systems is driven by the need for self-sufficiency and the backup of power during natural disasters. In Australia, recent bushfires had resulted in multiple blackouts in the country, which consequently has increased the demand for residential battery storage systems.

The adoption of electric vehicles globally in the coming years will also create positive demand for battery energy storage systems in residential applications. The increase in sales of electric vehicles would increase the electricity consumption, which, in turn, would increase energy bills, as a huge amount of electricity is used in charging vehicles. Here, residential renewable energy generation will act as a perfect solution. The generation of electricity at customer sites through solar panels would help in reducing the electricity bills and store the excess energy in the energy storage systems.

A few of the companies offering battery energy storage systems for residential applications are Samsung SDI (South Korea), LG Energy Solutions Co., Ltd. (South Korea), and Tesla (US), among others. Tesla offers a residential battery energy storage system, named Powerwall 2. Tesla Powerwall 2 is a fully integrated battery system that powers homes by storing daytime energy and producing electricity during the unavailability of solar energy. The typical system layout for the connection of the Tesla Powerwall with other systems is shown in the figure below.

FIGURE 51 TESLA POWERWALL INSTALLED FOR RESIDENTIAL APPLICATIONS



Source: Company Website and MarketsandMarkets Analysis

The solar panels convert sunlight to DC, which charges the Tesla Powerwall. The inverter converts the DC to AC power for use in the home. Also, the inverter converts AC to DC to charge the Powerwall using cheap off-peak mains power. Tesla Powerwall 2 has a 14-kWh capacity (13.2 kWh usable) and 7 kW peak/5 kW continuous output capacities. The use of Powerwall 2 benefits by ensuring 89% round-trip efficiency, single-phase and three-phase utility grid compatibility, solar self-consumption, and continuous power output capability of 7 kW peak and 5 kW.

TABLE 96 BATTERY ENERGY STORAGE SYSTEM MARKET FOR RESIDENTIAL APPLICATION, BY BATTERY TYPE, 2018–2021 (USD MILLION)

Battery Type	2018	2019	2020	2021
Lithium-Ion Batteries	134.3	216.4	171.5	203.7
Advanced Lead-Acid Batteries	43.3	68.4	51.9	59.2
Flow Batteries	44.9	79.0	66.4	83.9
Others	19.0	33.0	27.5	34.4
Total	241.5	396.7	317.2	381.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 97 BATTERY ENERGY STORAGE SYSTEM MARKET FOR RESIDENTIAL APPLICATION, BY BATTERY TYPE, 2022–2027 (USD MILLION)

Battery Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Lithium-Ion Batteries	245.3	295.6	363.5	464.3	582.9	739.5	24.7%
Advanced-Lead Acid Batteries	68.6	79.7	94.7	117.2	138.3	166.4	19.4%
Flow Batteries	107.7	138.6	182.5	250.4	329.5	442.4	32.7%
Others	43.7	55.8	72.8	99.0	129.7	173.0	31.7%
Total	465.3	569.7	713.6	930.9	1,180.3	1,521.3	26.7%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 98 BATTERY ENERGY STORAGE SYSTEM MARKET FOR RESIDENTIAL APPLICATION, BY CONNECTION TYPE, 2018–2021 (USD MILLION)

Connection Type	2018	2019	2020	2021
On-grid	203.3	332.8	265.1	317.4
Off-grid	38.1	64.0	52.1	63.7
Total	241.5	396.7	317.2	381.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 99 BATTERY ENERGY STORAGE SYSTEM MARKET FOR RESIDENTIAL APPLICATION, BY CONNECTION TYPE, 2022–2027 (USD MILLION)

Connection Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
On-grid	386.0	470.8	587.5	763.4	964.4	1,238.2	26.3%
Off-grid	79.3	98.8	126.1	167.6	216.0	283.1	29.0%
Total	465.3	569.7	713.6	930.9	1,180.3	1,521.3	26.7%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The residential battery energy storage system utilizes the green energy that is available 24×7 in association with the photovoltaic (PV) system. Companies such as H2 Inc., Volterion, VoltStorage, RedFlow, and SCHMID Group provide flow batteries for residential applications. The battery energy storage system acts as a small power plant in residential applications, as it stores the daytime energy with the assistance of PV cells and makes it available at any time in the day, even in the unavailability of sunlight. This minimizes the electricity purchase from the grid, also results in increased energy self-consumption at home.

TABLE 100 BATTERY ENERGY STORAGE SYSTEM MARKET FOR RESIDENTIAL APPLICATION, BY ENERGY CAPACITY, 2018–2021 (USD MILLION)

Energy Capacity	2018	2019	2020	2021
Below 100 MWh	241.5	396.7	317.2	381.1
Between 100 and 500 MWh	-	-	-	-
Above 500 MWh	-	-	-	-
Total	241.5	396.7	317.2	381.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 101 BATTERY ENERGY STORAGE SYSTEM MARKET FOR RESIDENTIAL APPLICATION, BY ENERGY CAPACITY, 2022–2027 (USD MILLION)

Energy Capacity	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Below 100 MWh	465.3	569.7	713.6	930.9	1,180.3	1,521.3	26.7%
Between 100 and 500 MWh	-	-	-	-	-	-	-
Above 500 MWh	-	-	-	-	-	-	-
Total	465.3	569.7	713.6	930.9	1,180.3	1,521.3	26.7%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 102 BATTERY ENERGY STORAGE SYSTEM MARKET FOR RESIDENTIAL APPLICATION, BY OWNERSHIP, 2018–2021 (USD MILLION)

Ownership	2018	2019	2020	2021
Customer-Owned	72.4	117.9	93.3	111.0
Third-Party Owned	169.0	278.9	223.9	270.1
Utility-Owned	-	-	-	-
Total	241.5	396.7	317.2	381.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 103 BATTERY ENERGY STORAGE SYSTEM MARKET FOR RESIDENTIAL APPLICATION, BY OWNERSHIP, 2022–2027 (USD MILLION)

Ownership	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Customer-Owned	134.2	162.7	201.8	260.7	327.0	417.1	25.5%
Third-Party Owned	331.1	407.0	511.8	670.3	853.3	1,104.2	27.2%
Utility-Owned	-	-	-	-	-	-	-
Total	465.3	569.7	713.6	930.9	1,180.3	1,521.3	26.7%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 104 BATTERY ENERGY STORAGE SYSTEM MARKET FOR RESIDENTIAL APPLICATION, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	87.3	142.1	113.2	135.4
Europe	76.1	122.7	95.9	112.6
APAC	55.9	95.4	78.7	97.6
RoW	22.1	36.5	29.4	35.5
Total	241.5	396.7	317.2	381.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 105 BATTERY ENERGY STORAGE SYSTEM MARKET FOR RESIDENTIAL APPLICATION, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	164.4	200.0	248.5	321.2	405.8	519.7	25.9%
Europe	134.4	160.9	197.1	251.4	310.6	390.3	23.8%
APAC	122.9	155.2	200.6	270.0	351.3	465.3	30.5%
RoW	43.5	53.6	67.4	88.4	112.7	146.0	27.4%
Total	465.3	569.7	713.6	930.9	1,180.3	1,521.3	26.7%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The residential battery energy storage system market in APAC is expected to register the highest growth rate during the forecast period. Government schemes to encourage the adoption of residential energy storage systems in Australia and the upcoming deployments of battery energy storage systems in other countries are the major factors responsible for the growth of the market in the region. Also, the rising population across the world has increased the number of construction facilities, which in turn, would increase the installation of solar panels to self-power the homes. Thus, the growing residential sector will contribute to the growth of the market in the region.

12.4 COMMERCIAL

12.4.1 INCREASED USAGE OF UPS IN COMMERCIAL APPLICATIONS IS DRIVING DEMAND FOR BATTERY ENERGY STORAGE SYSTEMS

In commercial applications, battery energy storage systems are used in commercial buildings, municipalities, agriculture and water district schools, enterprises, affordable housing arrangements, government buildings, solar & wind farms, telecoms, small- and large-scale organizations, and others. In non-residential applications, battery energy storage systems situated at the customer premise serve both end customers and the utility. The ownership involves a wide range of agreements between vendors and their customers. As the installation and initial capital costs are extremely high for the battery energy storage, third-party vendors buy the systems and supply them to non-residential consumers. The two leading third-party-owned business models are—power purchase and power efficiency agreement plans. These models have allowed a large number of commercial and industrial customers to benefit from energy storage without incurring significant upfront expenses.

In the commercial sector, another prominent application of battery energy storage systems is UPS, which is used in data centers, factories, and critical infrastructures such as hospitals. The utilization of UPS in the commercial sector has widened in the past years owing to its high performance and efficiency. In production facilities, machine downtime leads to numerous losses. The interruption during the production process due to power outages, even for a shorter period may affect the processes or can even damage the heavy machinery. Here, UPS plays a critical role and acts as a power backup in industries, thus saving them from huge losses. For UPS applications, lithium-ion battery-based energy storage systems are preferred due to their benefits such as saving battery space, longer life, and fast charge/discharge rate.

TABLE 106 BATTERY ENERGY STORAGE SYSTEM MARKET FOR COMMERCIAL APPLICATION, BY BATTERY TYPE, 2018–2021 (USD MILLION)

Battery Type	2018	2019	2020	2021
Lithium-Ion Batteries	193.8	284.2	231.0	274.8
Advanced Lead-Acid Batteries	57.3	82.5	65.0	75.0
Flow Batteries	80.0	123.9	105.0	130.5
Others	32.8	50.3	42.4	52.4
Total	363.9	541.0	443.4	532.7

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 107 BATTERY ENERGY STORAGE SYSTEM MARKET FOR COMMERCIAL APPLICATION, BY BATTERY TYPE, 2022–2027 (USD MILLION)

Battery Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Lithium-Ion Batteries	329.1	398.5	484.4	596.2	751.3	943.3	23.4%
Advanced Lead-Acid Batteries	87.2	102.7	121.6	146.0	176.5	213.6	19.6%
Flow Batteries	163.3	207.0	263.7	340.7	445.3	582.6	29.0%
Others	65.2	82.2	104.2	133.8	174.3	227.0	28.3%
Total	644.8	790.5	973.9	1,216.8	1,547.3	1,966.6	25.0%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 108 BATTERY ENERGY STORAGE SYSTEM MARKET FOR COMMERCIAL APPLICATION, BY CONNECTION TYPE, 2018–2021 (USD MILLION)

Connection Type	2018	2019	2020	2021
On-grid	277.8	411.2	335.6	401.4
Off-grid	86.1	129.8	107.8	131.3
Total	363.9	541.0	443.4	532.7

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 109 BATTERY ENERGY STORAGE SYSTEM MARKET FOR COMMERCIAL APPLICATION, BY CONNECTION TYPE, 2022–2027 (USD MILLION)

Connection Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
On-grid	483.8	590.4	724.0	900.4	1,140.0	1,442.4	24.4%
Off-grid	161.1	200.1	249.9	316.4	407.3	524.2	26.6%
Total	644.8	790.5	973.9	1,216.8	1,547.3	1,966.6	25.0%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

Redox vanadium flow battery is the most efficient utilities-scale battery storage technology to emerge in the commercial market. These batteries enable commercial or industrial enterprises to lower costs while independently producing and storing energy. It reduces the risk of production loss during the shortage of power supply or blackouts. Moreover, the long operational life and safe operation of flow batteries have attracted several owners of commercial and industrial enterprises to adopt these batteries in lieu of conventional batteries such as lithium-ion and lead-acid. The on-grid battery energy storage systems help in energy storage on a large scale, which helps to power crucial commercial & industrial applications.

TABLE 110 BATTERY ENERGY STORAGE SYSTEM MARKET FOR COMMERCIAL APPLICATION, BY ENERGY CAPACITY, 2018–2021 (USD MILLION)

Energy Capacity	2018	2019	2020	2021
Below 100 MWh	55.7	81.1	64.9	76.2
Between 100 and 500 MWh	308.1	459.8	378.5	456.5
Above 500 MWh	-	-	-	-
Total	363.9	541.0	443.4	532.7

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 111 BATTERY ENERGY STORAGE SYSTEM MARKET FOR COMMERCIAL APPLICATION, BY ENERGY CAPACITY, 2022–2027 (USD MILLION)

Energy Capacity	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Below 100 MWh	90.0	107.8	129.7	158.2	195.7	242.2	21.9%
Between 100 and 500 MWh	554.8	682.7	844.2	1,058.6	1,351.6	1,724.4	25.5%
Above 500 MWh	-	-	-	-	-	-	-
Total	644.8	790.5	973.9	1,216.8	1,547.3	1,966.6	25.0%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 112 BATTERY ENERGY STORAGE SYSTEM MARKET FOR COMMERCIAL APPLICATION, BY OWNERSHIP, 2018–2021 (USD MILLION)

Ownership	2018	2019	2020	2021
Customer-Owned	87.9	129.8	105.7	126.0
Third-Party Owned	275.9	411.2	337.7	406.6
Utility-Owned	-	-	-	-
Total	363.9	541.0	443.4	532.7

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 113 BATTERY ENERGY STORAGE SYSTEM MARKET FOR COMMERCIAL APPLICATION, BY OWNERSHIP, 2022–2027 (USD MILLION)

Ownership	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Customer-Owned	151.5	184.4	225.6	279.9	353.2	445.7	24.1%
Third-Party Owned	493.3	606.1	748.3	936.9	1,194.0	1,520.9	25.3%
Utility-Owned	-	-	-	-	-	-	-
Total	644.8	790.5	973.9	1,216.8	1,547.3	1,966.6	25.0%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 114 BATTERY ENERGY STORAGE SYSTEM MARKET FOR COMMERCIAL APPLICATION, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	139.2	205.6	168.2	201.4
Europe	85.4	124.4	99.6	116.9
APAC	98.6	151.5	127.6	157.5
RoW	40.6	59.5	48.0	56.8
Total	363.9	541.0	443.4	532.7

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 115 BATTERY ENERGY STORAGE SYSTEM MARKET FOR COMMERCIAL APPLICATION, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	242.9	296.3	362.9	450.2	571.4	722.9	24.4%
Europe	138.3	165.6	199.4	243.4	301.3	373.1	22.0%
APAC	196.0	246.9	312.7	401.5	522.6	680.6	28.3%
RoW	67.6	81.6	98.9	121.7	152.0	190.0	22.9%
Total	644.8	790.5	973.9	1,216.8	1,547.3	1,966.6	25.0%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The battery energy storage system market in North America for commercial applications held the largest market share in 2021. Increasing deployment of large-scale battery energy storage systems for industrial purposes in the US and growing focus on the decarbonization of the economy are a few of the factors that are fueling the growth of the market in the region.

12.5 UTILITY

12.5.1 UTILITY APPLICATION TO DOMINATE BATTERY ENERGY STORAGE SYSTEM MARKET DURING FORECAST PERIOD

Public utility applications or utilities involve the electric grid, a system where the electricity generated at power plants is immediately used by the loads that are connected to it. The generation and consumption of electricity need to be always matched, as mismatch leads to voltage and frequency deviations and causes power outages and damage to the equipment. To overcome this problem, a battery energy storage system is installed with the electric grids, which controls the voltage and frequency deviations. The battery energy storage system used in utilities is a cost-effective alternative to replace a conventional infrastructure, especially in helping substations and transmission and distribution (T&D) lines to meet growing peak demand.

Avoiding the wastage of the generated renewable energy is a key opportunity for long-duration, utility-scale battery energy storage systems. Battery energy storage systems enable better management of renewable energy by storing energy generated from solar and wind sources, as the amount of energy generated through these sources is variable, and this energy is stored in the electrical grids. The installed battery energy storage systems control the variable energies. The battery energy storage system is well-suited for smoothing the variable output of renewables and controlling the rapid ramping up and down of solar as well as wind generation. The grid operators and regulators are aware of the importance of battery energy storage systems and their ability to cater through multiple services. The utility operators are using these systems in their resource planning processes, which reduces system costs and increases storage capacity; this proves these systems to be an attractive alternative to certain infrastructure investments.

Lithium-ion batteries dominate the current market for utility-scale battery storage in the world. The grid operators are witnessing an upsurge in the usage of power across the world. Besides, the resumption of the BESS projects is also fueling the growth of the market. For instance, in March 2021, India installed the country's first grid-connected community energy storage system in Delhi. The new 0.52 MWh grid-connected system will charge during off-peak times and discharge to the grid during peak times. The newly installed battery-based energy storage (BESS) will provide the agility to better integrate intermittent solar and wind energy resources into India's electric grid and ensure high-quality power for consumers.

TABLE 116 BATTERY ENERGY STORAGE SYSTEM MARKET FOR UTILITY APPLICATION, BY BATTERY TYPE, 2018–2021 (USD MILLION)

Battery Type	2018	2019	2020	2021
Lithium-Ion Batteries	921.4	1,410.9	1,132.5	1,359.6
Advanced Lead-Acid Batteries	284.1	426.1	328.6	379.7
Flow Batteries	359.9	590.6	497.7	628.5
Others	148.6	241.3	202.0	253.4
Total	1,714.0	2,668.9	2,160.8	2,621.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 117 BATTERY ENERGY STORAGE SYSTEM MARKET FOR UTILITY APPLICATION, BY BATTERY TYPE, 2022–2027 (USD MILLION)

Battery Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Lithium-Ion Batteries	1,695.6	2,125.1	2,673.6	3,364.1	4,316.7	5,562.3	26.8%
Advanced Lead-Acid Batteries	456.5	552.5	672.6	820.4	995.1	1,221.4	21.8%
Flow Batteries	825.7	1,092.1	1,452.8	1,937.3	2,590.0	3,500.9	33.5%
Others	330.7	434.4	574.1	760.4	1,012.4	1,361.7	32.7%
Total	3,308.5	4,204.1	5,373.1	6,882.1	8,914.1	11,646.4	28.6%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 118 BATTERY ENERGY STORAGE SYSTEM MARKET FOR UTILITY APPLICATION, BY CONNECTION TYPE, 2018–2021 (USD MILLION)

Connection Type	2018	2019	2020	2021
On-grid	1,451.6	2,268.6	1,844.3	2,246.2
Off-grid	262.5	400.3	316.5	374.8
Total	1,714.0	2,668.9	2,160.8	2,621.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 119 BATTERY ENERGY STORAGE SYSTEM MARKET FOR UTILITY APPLICATION, BY CONNECTION TYPE, 2022–2027 (USD MILLION)

Connection Type	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
On-grid	2,846.5	3,630.9	4,657.7	5,987.4	7,786.7	10,212.2	29.1%
Off-grid	462.0	573.2	715.4	894.7	1,127.4	1,434.1	25.4%
Total	3,308.5	4,204.1	5,373.1	6,882.1	8,914.1	11,646.4	28.6%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The battery energy storage system used in utilities is a cost-effective alternative to replace a conventional infrastructure, especially in helping substations and transmission and distribution (T&D) lines to meet the growing peak demand. According to the US EIA and NREL, utility-owned battery energy storage systems play a crucial role in the operations of the current and future energy generation mix of a country. In addition to the system flexibility they provide, a battery energy storage system also offers reliable interconnections in neighboring power systems. The growth of the utility-owned segment is mainly driven by increasing financial incentives and regulatory support from governments globally.

TABLE 120 BATTERY ENERGY STORAGE SYSTEM MARKET FOR UTILITY APPLICATION, BY ENERGY CAPACITY, 2018–2021 (USD MILLION)

Energy Capacity	2018	2019	2020	2021
Below 100 MWh	-	-	-	-
Between 100 and 500 MWh	176.6	266.9	208.2	243.3
Above 500 MWh	1,537.5	2,402.0	1,952.6	2,377.8
Total	1,714.0	2,668.9	2,160.8	2,621.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 121 BATTERY ENERGY STORAGE SYSTEM MARKET FOR UTILITY APPLICATION, BY ENERGY CAPACITY, 2022–2027 (USD MILLION)

Energy Capacity	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Below 100 MWh	-	-	-	-	-	-	-
Between 100 and 500 MWh	295.9	362.3	446.1	550.6	680.7	850.5	23.5%
Above 500 MWh	3,012.6	3,841.8	4,926.9	6,331.5	8,233.4	10,795.8	29.1%
Total	3,308.5	4,204.1	5,373.1	6,882.1	8,914.1	11,646.4	28.6%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 122 BATTERY ENERGY STORAGE SYSTEM MARKET FOR UTILITY APPLICATION, BY OWNERSHIP, 2018–2021 (USD MILLION)

Ownership	2018	2019	2020	2021
Customer-Owned	160.4	247.7	199.0	237.0
Third-Party Owned	445.0	690.0	561.7	676.8
Utility-Owned	1,714.0	2,668.9	2,160.8	2,621.1
Total	2,319.4	3,606.7	2,921.4	3,534.9

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 123 BATTERY ENERGY STORAGE SYSTEM MARKET FOR UTILITY APPLICATION, BY OWNERSHIP, 2022–2027 (USD MILLION)

Ownership	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Customer-Owned	285.7	347.1	427.4	540.5	680.3	862.8	24.7%
Third-Party Owned	824.4	1,013.0	1,260.1	1,607.2	2,047.4	2,625.0	26.1%
Utility-Owned	3,308.5	4,204.1	5,373.1	6,882.1	8,914.1	11,646.4	28.6%
Total	4,418.6	5,564.3	7,060.6	9,029.8	11,641.7	15,134.2	27.9%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 124 BATTERY ENERGY STORAGE SYSTEM MARKET FOR UTILITY APPLICATION, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	568.5	880.7	709.4	856.1
Europe	316.4	480.4	377.3	444.0
APAC	759.2	1,201.0	989.5	1,220.7
RoW	69.9	106.8	84.5	100.3
Total	1,714.0	2,668.9	2,160.8	2,621.1

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

TABLE 125 BATTERY ENERGY STORAGE SYSTEM MARKET FOR UTILITY APPLICATION, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	1,075.1	1,359.2	1,728.2	2,202.3	2,837.5	3,687.8	28.0%
Europe	543.6	670.1	830.8	1,032.3	1,289.2	1,626.2	24.5%
APAC	1,565.9	2,021.0	2,621.7	3,406.6	4,483.3	5,944.8	30.6%
RoW	123.8	153.8	192.3	240.9	304.2	387.7	25.6%
Total	3,308.5	4,204.1	5,373.1	6,882.1	8,914.1	11,646.4	28.6%

Note: Values are provided up to 1 decimal to exhibit minute differences.

Source: Secondary Research, Journals, Expert Interviews, and MarketsandMarkets Analysis

The battery energy storage system market for utility applications in APAC is expected to register with the highest growth rate due to the increasing number of large-scale energy storage projects in developing countries for the distribution of electricity. According to the US Energy Information Administration's (EIA) Annual Energy Outlook 2021 (AEO2021), it is estimated that the installation of battery energy storage systems in the US power grid is expected to increase in the coming years owing to the reduced battery costs, growth in non-dispatchable renewables, and the application of the Investment Tax Credit to co-located storage systems. It is also estimated that 59 gigawatts (GW) of battery storage will serve the power grid in 2050.

13 GEOGRAPHIC ANALYSIS

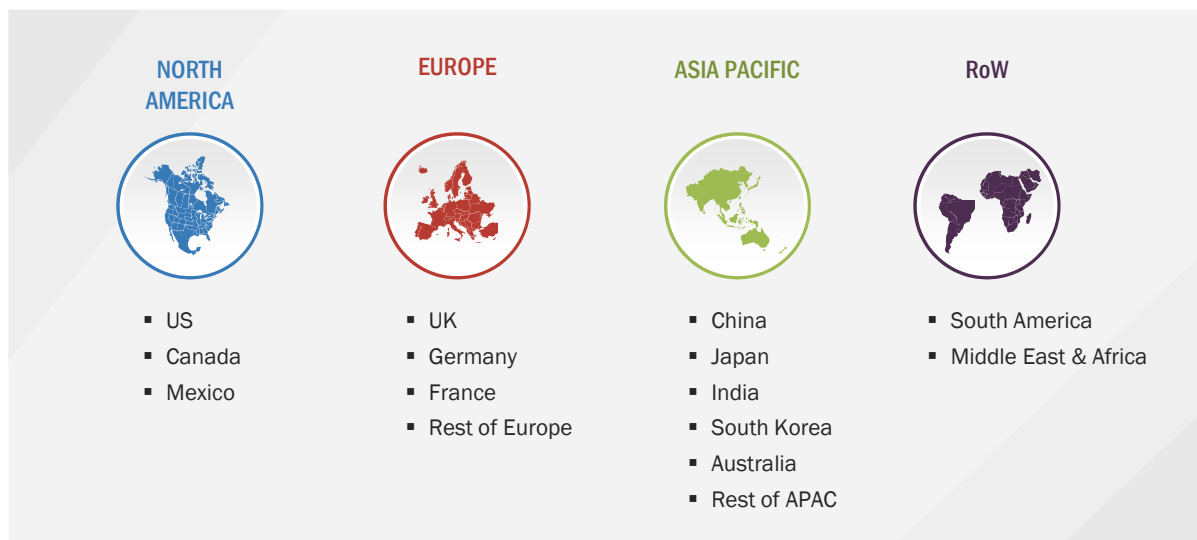
KEY FINDINGS

- APAC led the battery energy storage system market in 2021, with a market size of USD 1,475.8 in 2021, and is projected to reach USD 7,090.6 by 2027, growing at a CAGR of 30.3% between 2022 and 2027.
- APAC is expected to witness plans for electrification in remote areas, which are mostly off-grid in several countries. The use of residential energy storage technology will enable to increase access to electricity for these remote and island communities in the region. Energy storage investments are expected to increase substantially within the APAC region as the governments in growing economies are making new policies to improve the reliability and quality of the power distribution facilities to the residential customers.
- The battery energy storage system market in North America was valued at USD 1,193.0 in 2021 and is projected to reach USD 4,930.4 by 2027, growing at the highest CAGR of 27.2% between 2022 and 2027.
- Due to the increasing cost of building and maintaining plants, increasing concern over burning fossil fuels, and government policies, the US government has adopted battery energy storage systems to eliminate the dependency on fossil fuels. By the end of 2020, the US accounted for about 1.65 GW of battery energy storage systems installed. There is a pipeline of giant lithium-ion projects scheduled for completion by 2021 in the US.
- The presence of prominent market players, increasing importance of battery energy storage, and supportive regulatory policies are fuelling the demand for energy storage systems in the US.

13.1 INTRODUCTION

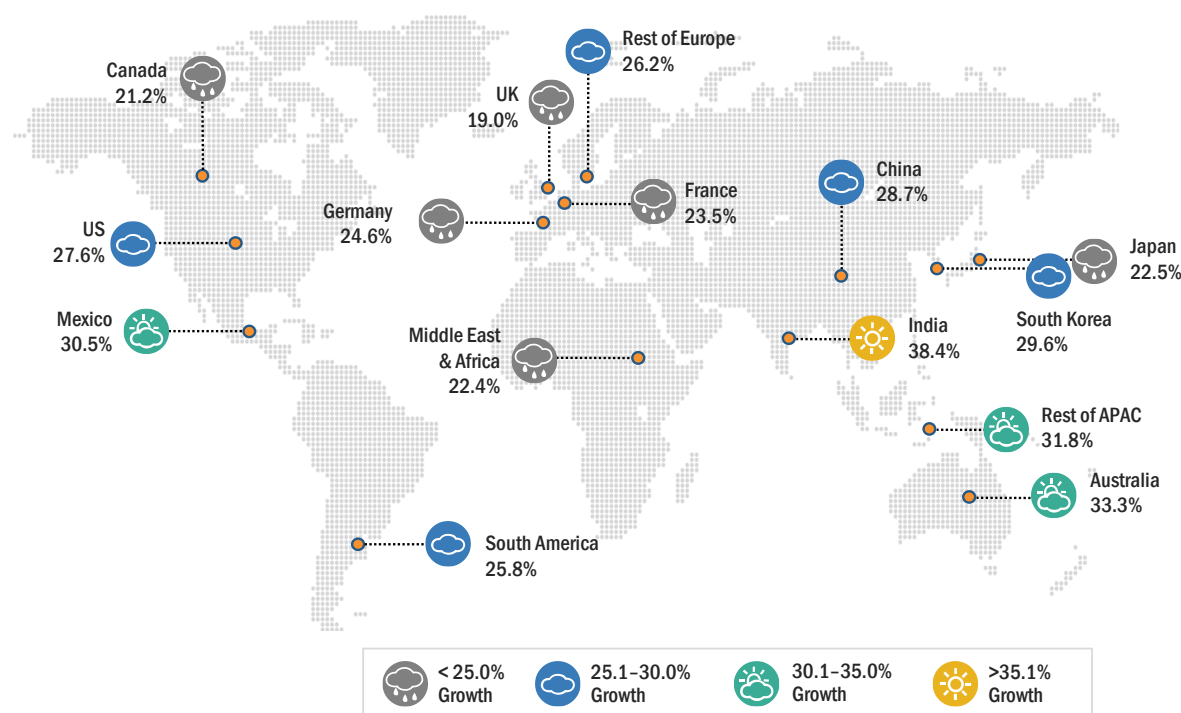
The battery energy storage system (BESS) market has been segmented based on region into North America, Europe, Asia Pacific (APAC), and the Rest of the World (RoW). The battery energy storage system market in the US, Mexico, France, Germany, Australia, and South America is expected to exhibit high growth in the coming years due to the increasing adoption of renewable energy. Moreover, the governments of various countries, such as the US, the UK, Germany, France, and India, are investing in the development and adoption of battery energy storage systems to reduce carbon emissions. In recent years, the installation of battery energy storage systems for various projects has increased across the world.

FIGURE 52 BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION



Note: The Rest of Europe (RoE) includes Italy, Spain, Norway, Sweden, and Switzerland. The Rest of APAC includes New Zealand, Thailand, Singapore, Taiwan, Malaysia, and Indonesia.

Source: Expert Interviews, White Papers, and MarketsandMarkets Analysis

FIGURE 53 GEOGRAPHIC SNAPSHOT: BATTERY ENERGY STORAGE SYSTEM MARKET

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 126 BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
North America	794.9	1,228.4	990.8	1,193.0
Europe	478.0	727.5	572.8	673.5
APAC	913.8	1,447.9	1,195.9	1,475.8
RoW	132.6	202.8	161.9	192.5
Total	2,319.4	3,606.7	2,921.4	3,534.9

Note: RoW includes South America and the Middle East & Africa.

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 127 BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
North America	1,482.5	1,855.4	2,339.6	2,973.6	3,814.7	4,930.4	27.2%
Europe	816.4	996.7	1,227.3	1,527.0	1,901.0	2,389.6	24.0%
APAC	1,884.8	2,423.1	3,135.0	4,078.1	5,357.1	7,090.6	30.3%
RoW	234.9	289.0	358.7	451.0	568.9	723.6	25.2%
Total	4,418.6	5,564.3	7,060.6	9,029.8	11,641.7	15,134.2	27.9%

Note: RoW includes South America and the Middle East & Africa.

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

APAC is a hub for the battery energy storage system industry. The region accounted for the largest share of the battery energy storage system market in 2021 and is expected to grow at the highest CAGR during the forecast period. South Korea and China, both added to their energy storage installation capacities, i.e., around 0.7 GW and 1.6 GW, respectively, in 2020. APAC is expected to witness plans for electrification in remote areas, which are mainly off-grid in several countries. The use of residential energy storage technology will enable to increase access to electricity for these remote and island communities in the region. Energy storage investments are expected to increase substantially within the region as the governments in growing economies are making new policies to improve the reliability and quality of the power distribution facilities to the residential customers. The governments of various countries in this region are focusing on minimizing the adverse effects of the energy sector on the environment. These factors lead to the growth in the deployment of battery energy storage systems in residential and public utility applications. A few of the major battery energy storage system manufacturers based in Asia Pacific are Samsung SDI (South Korea), LG Chem (South Korea), Hitachi (Japan), Panasonic (Japan), and NEC Corporation (Japan).

The growth of the North American market is dependent on the increasing demand for renewable energy storage systems in the residential, non-residential, and utility sectors. By the end of 2020, the US accounted for about 1.65 GW of battery energy storage systems installed. Companies such as Tesla, ABB, General Electric, and Honeywell are the key players in the region, which are focusing on installing and providing battery energy storage systems. There is a pipeline of giant lithium-ion projects scheduled for completion by 2021 in the US, for instance, 100 MW storage projects (Clean Power Alliance Lancaster, AES Alamos, and Arizona, Strata Oxnard) as well as the 300 MW Vistra Moss Landing project and the massive 409 MW Florida Power and Light Manatee project. Besides, North America held the largest share of the flow battery market worldwide. The region is expected to hold a significant market size in the near future as it is home to several active players as well as has the highest number of flow battery projects for various applications such as utilities, microgrids, industrial, commercial, and military.

13.2 NORTH AMERICA

The US, Canada, and Mexico are the major countries contributing to the growth of the North American battery energy storage system market. The growth of the battery energy storage system market in North America is attributed to the increasing demand for renewable energy storage in the residential, non-residential, and utility sectors. A few of the players involved in the battery energy storage system market in North America are The AES Corporation, Tesla, Inc., BYD Company Limited, Primus Power, Johnson Controls, and General Electric. The market players are expected to create immense growth opportunities in the coming years. The companies are focusing on strategic partnerships and new product launches and developments to increase their product portfolios and fulfill the requirements of various applications. Besides, the US has ample lithium deposits that can be derived in the future to produce lithium-ion batteries. This will drive the demand for lithium-ion batteries in the manufacturing of energy storage systems in the region.

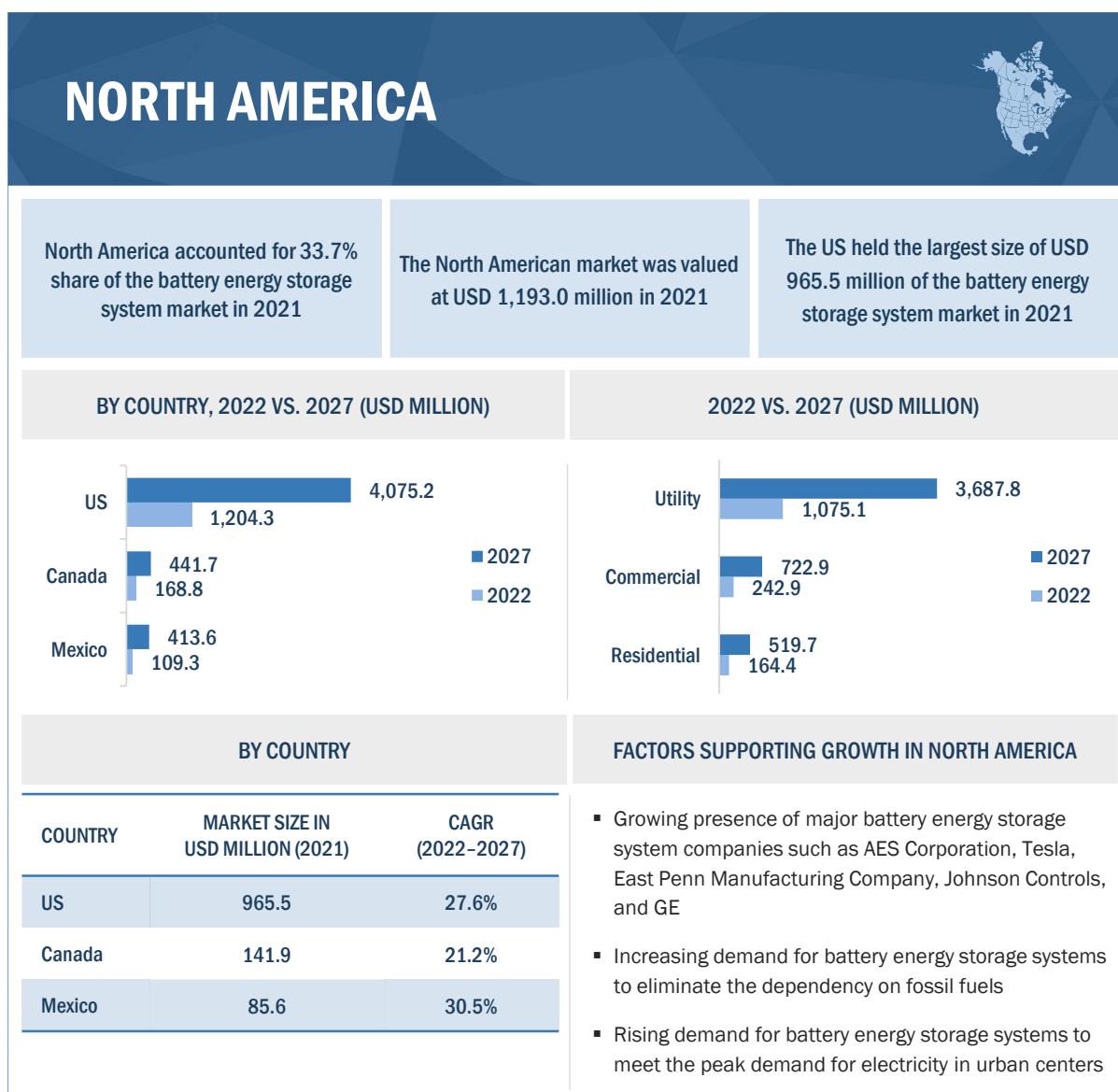
13.2.1 IMPACT OF COVID-19 ON NORTH AMERICAN MARKET

Before the outbreak of COVID-19, there was a constant increase in demand for battery energy storage systems in the region owing to a large inflow of investments in the renewable energy sector. However, COVID-19 transformed into a global pandemic from a regional crisis in just a matter of a few weeks. The pandemic has severely hit North America, especially countries such as the US and Mexico.

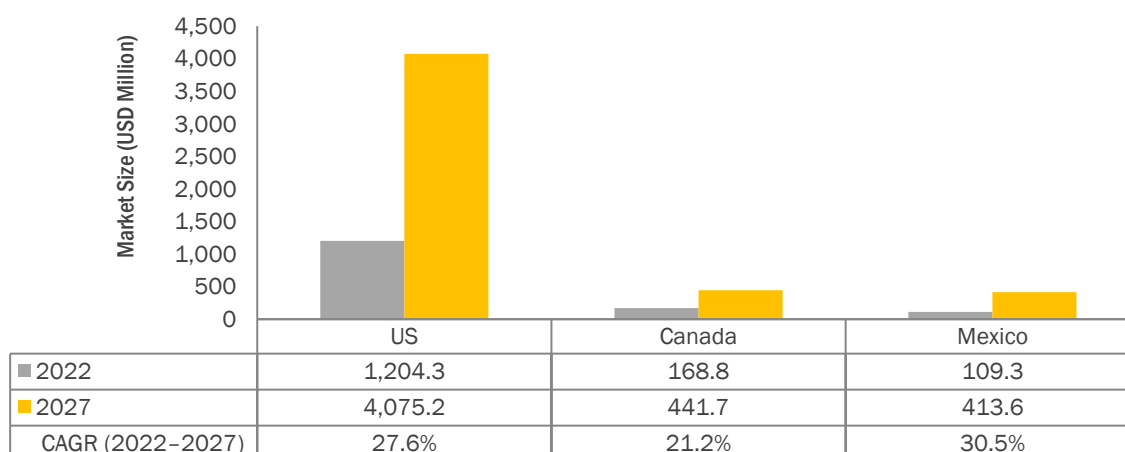
The sudden outbreak of the COVID-19 pandemic and its effects rippled through several lives and had numerous implications across sectors, the energy storage market is no exception. In the initial months of the spread of the pandemic, the North American countries suffered from business shutdowns, especially in the US. The total shutdown in the initial months of the pandemic impacted the high power-consuming sectors such as industrial, commercial, manufacturing, and import & export operations in the country. Thus, the reduced load in the grid reduced the requirement for distribution infrastructure upgrades, which further reduced the demand for battery energy storage. Production facilities such as that of battery and automobiles etc. suffered acute labor and raw material shortage owing to the stringent quarantine measures and the disruption of global supply chains leading to the freezing of shipments worldwide. Due to production shutdowns in the initial months of the pandemic, the economic slowdown was observed in countries across North America, which, in turn, led to a decline in the installation of battery energy storage systems owing to the restriction in the supply of batteries and other components.

The COVID-19 pandemic significantly affected the supply chain pertaining to the energy industry in North America, especially renewable energy technologies and battery energy storage systems. Moreover, the fatal impact of overdependence on countries in APAC, especially China for business inputs affected the business of the energy storage systems in the US. Several countries, including a few North American countries, had imposed restrictions on material flow in and out of China, thereby impacting the supply of essential components. According to the US ESA, the impact of the pandemic resulted in delays in several energy infrastructure projects.

However, post-pandemic, a few improvements have been observed in the battery energy storage system market. The increased adoption of automation in production facilities across North America has ensured reduced human intervention, thus curtailing the effects of labor shortages up to a certain extent. Additionally, the surge in demand for personal electronics and electric vehicles is helping stabilize the demand in the energy storage market. An increase in remote electrification and grid modernization projects and other government measures across North American countries through R&D efforts are adding resilience to the energy industry supply chain and thereby helping to mitigate future interruptions.

FIGURE 54 MARKET IN NORTH AMERICA: SNAPSHOT

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 55 MEXICO TO REGISTER HIGHEST CAGR DURING FORECAST PERIOD

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

13.2.2 US

13.2.2.1 Increasing adoption of large-scale battery storage systems in power grids to drive market growth

The US is facing issues related to the use of fossil fuels such as natural gas and coal for electricity generation. Due to the increasing cost of building and maintaining a plant, growing concern over burning fossil fuels, and government policies, the US government has adopted battery energy storage systems to eliminate the dependence on fossil fuels. According to the US Energy Information Administration, large-scale battery storage systems are increasingly being adopted across power grids in the US. Over 90% of large-scale battery storage power capacity in the US was provided by batteries based on lithium-ion technology. In August 2021, the US-based energy company, Vistra announced the completion of the construction of Phase II of the Moss Landing battery energy storage facility located in California. It is claimed to be the largest of its kind in the world and is intended to store power and release it to California's grid. The Phase II addition of 100 MW storage capacity to the facility has increased its total capacity to 400 MW/1,600 MWh, thereby making it the largest facility globally. As of June 2021, California was the leading state in the nation with the highest utility-scale battery energy storage capacity of 1,438 MW. These planned additions put California in line to meet its energy storage mandate (Assembly Bill 2514), which requires its investor-owned utilities to install 1,325 MW of energy storage across the transmission, distribution, and customer levels by 2024. As of May 2020, five states besides California have set battery energy storage mandates or targets. The governments of Massachusetts and New York also have implemented mandates for energy storage and have planned battery storage projects in the coming years. In February 2020, Virginia passed House Bill 1526, which has established a goal of setting 3,100 MW of energy storage by 2035.

As of February 2020, more than 90% of the operating capacity from the renewable generation sites and the co-located battery have been installed in nine states. By December 2023, it is planned that 47% (2.3 GW of the 4.9 GW) of the operating battery storage capacity would be paired onsite with renewable generation. As of December 2020, most of the large-scale battery energy storage systems in the US were built as standalone facilities, i.e., they were not located at sites that generated power from renewable sources. Only 38% of the capacity of the large-scale battery storage systems was co-located with other generators, out of which 30% were co-located with facilities for the generation of power from renewable sources such as wind or solar PV. Furthermore, in 2021, the US-based firm Strata Clean Energy announced the commencement of a USD 450 million battery energy storage system project to augment the APS grid by 2024. The Strata Clean Energy project is projected to have a capacity of 400 MW/1600 MWh of battery

storage and therefore would be enough to power about 400,000 homes for an hour. As per the Annual Energy Outlook 2020, the large-scale battery storage capacity in the US would increase from 1 GW in 2019 to 17 GW in 2050.

A few non-profit organizations organize various conferences and exhibitions to support the implementation of battery energy storage systems. EES North America is the ultimate hotspot for energy storage solutions in the US, which covers the entire value chain of innovative battery and energy storage technologies. The EES North America exhibition takes place in California, which is an epicenter of the US storage market, and Intersolar North America is another exhibition held in California, which focuses on the areas of PV systems, smart renewable energy, and solar heating and cooling technologies. EES North America and Intersolar North America support the battery energy storage system market by providing solutions through exhibitions and international conferences. For instance, an event named EES North America 2021, Electrical Energy Storage was held in San Diego, California on 12 January 2021.

13.2.3 CANADA

13.2.3.1 Growing focus on advancements in battery energy storage technology to meet peak demand for electricity in urban centers

Canada is focusing on advancements in battery energy storage technology to meet the peak demand for electricity in urban centers. The manufacturing and transportation sectors are playing a vital role in the growth of the battery energy storage market in Canada. Natural Resources Canada provided funding of USD 2 billion to the Ontario Power Generation (OPG) to finance the Giizis Energy Micro-Grid Project. The project integrates a solar PV-installed plant and battery power system, which will supply renewable power to the existing electricity grid. This project also received financial support from OPG, Gull Bay First Nation, and other partners. Indigenous Services Canada is expected to provide First Nation Inc. with about USD 1.2 billion monetary funding in support of the project. Additionally, in May 2021, NRStor and the Six Nations' Grand River Development Corporation, under the partnership with Oneida Energy Storage LP, announced a deal with the Canada Infrastructure Bank to create a 250 MW energy storage facility, which is claimed to be the largest of its kind in Canada. The Canada Infrastructure Bank would invest USD 170 million in the first billion-dollar energy storage project, which would be in Southwestern Ontario. This will allow Ontario to efficiently manage its peak electricity demand and will help manage future demands. The project will also enable a reduction in greenhouse gas emissions by 4.1 million tons. Thus, all these projects are expected to increase the installation of battery energy storage systems in the country.

13.2.4 MEXICO

13.2.4.1 Rising demand for electricity to propel market growth

Mexico was one of the nascent energy storage markets in North America in 2021. Most of the battery energy storage systems are installed in automobiles in the country. A few grid stabilization units, along with the mining industries, have incorporated battery energy storage systems. According to the Mexico Ministry of Energy, the government has identified energy storage as a pillar of its transition to cleaner energy. Mexico has set goals for reducing carbon emissions, targeting 35% of energy from clean energy sources by 2035 and 50% by 2050. Falling capital costs for wind and solar PV generation and commitments to clean generation and emission reductions are leading to a large uptake in the wind and solar PV generation in Mexico, which will further drive the growth of the battery energy storage system market. While battery storage does not currently provide several services to the Mexican electric grid, there is considerable potential for battery storage to offer valuable economic and reliability services soon. For instance, in 2021, the Powin-Monterrey Microgrid battery energy storage system, which uses lithium-ion technology and has a capacity of 12,000 KW became operational in Mexico. Similarly, in September 2019, Developer ON Energy Storage installed a 30 kW/60kWh battery energy storage system in Puebla, Mexico. It is claimed to be the first such system to deliver peak shaving and frequency regulation services in the country.

Mexico is home to the world's largest lithium deposit and the nation receives more solar irradiation than most other countries. Additionally, in 2021, the Mexican government announced plans for the development of a USD 2 billion PV project as a part of Mexico's modernization plan and will be built in Puerto Penasco, in the state of Sonora. Therefore, Mexico is ideally positioned to make the transition to renewable energy generation, thereby creating opportunities to create new industries, high value-added chains, and thousands of jobs in the areas of energy storage. In 2020, the installed solar PV energy generation capacity of Mexico surpassed 5.6 GW, which marked an increase of over 27% as compared to 2019.

TABLE 128 NORTH AMERICA: BATTERY ENERGY STORAGE SYSTEM MARKET, BY COUNTRY, 2018–2021 (USD MILLION)

Country	2018	2019	2020	2021
US	636.0	985.6	798.6	965.5
Canada	106.6	159.3	123.1	141.9
Mexico	52.3	83.4	69.2	85.6
Total	794.9	1,228.4	990.8	1,193.0

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 129 NORTH AMERICA: BATTERY ENERGY STORAGE SYSTEM MARKET, BY COUNTRY, 2022–2027 (USD MILLION)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
US	1,204.3	1,512.5	1,913.1	2,438.4	3,141.8	4,075.2	27.6%
Canada	168.8	202.3	244.3	297.4	360.6	441.7	21.2%
Mexico	109.3	140.6	182.1	237.9	312.3	413.6	30.5%
Total	1,482.5	1,855.4	2,339.6	2,973.6	3,814.7	4,930.4	27.2%

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 130 NORTH AMERICA: BATTERY ENERGY STORAGE SYSTEM MARKET, BY BATTERY TYPE, 2018–2021 (USD MILLION)

BATTERY TYPE	2018	2019	2020	2021
Lithium-ion Batteries	400.9	606.5	483.5	574.2
Lead-acid Batteries	133.5	199.0	153.0	175.7
Flow Batteries	190.9	310.9	260.8	326.9
Others	69.6	112.0	93.4	116.3
Total	794.9	1,228.4	990.8	1,193.0

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 131 NORTH AMERICA: BATTERY ENERGY STORAGE SYSTEM MARKET, BY BATTERY TYPE, 2022–2027 (USD MILLION)

BATTERY TYPE	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Lithium-Ion Batteries	701.7	861.8	1,063.5	1,319.6	1,671.2	2,119.3	24.7%
Lead-acid Batteries	208.1	248.3	298.5	361.7	434.4	527.4	20.4%
Flow Batteries	423.2	551.9	725.4	960.8	1,271.9	1,701.8	32.1%
Others	149.4	193.4	252.2	331.6	437.2	581.9	31.2%
Total	1,482.5	1,855.4	2,339.6	2,973.6	3,814.7	4,930.4	27.2%

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 132 NORTH AMERICA: BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

APPLICATION	2018	2019	2020	2021
Residential	87.3	142.1	113.2	135.4
Commercial	139.2	205.6	168.2	201.4
Utility	568.5	880.7	709.4	856.1
Total	794.9	1,228.4	990.8	1,193.0

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 133 NORTH AMERICA: BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

APPLICATION	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	164.4	200.0	248.5	321.2	405.8	519.7	25.9%
Commercial	242.9	296.3	362.9	450.2	571.4	722.9	24.4%
Utility	1,075.1	1,359.2	1,728.2	2,202.3	2,837.5	3,687.8	28.0%
Total	1,482.5	1,855.4	2,339.6	2,973.6	3,814.7	4,930.4	27.2%

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

The US has the largest number of battery energy storage systems installed in North America. The growth of the battery energy storage market depends on government policies. Due to the increasing cost of building and maintaining the plants and rising concern over burning fossil fuels, the US adopted battery energy storage to eliminate the dependence on fossil fuels. The country also has a presence of major providers of battery energy storage systems, and several ongoing and under construction MW capacity lithium-ion battery energy storage projects, as of 2021.

13.3 EUROPE

The growth of the residential sector in Europe is supporting the growth of the battery energy storage system market. The European battery energy storage system market is sub-segmented into the UK, Germany, France, and the Rest of Europe (RoE). This region is witnessing developments in the residential sector such as rural electrification in which lithium-ion battery is used in battery energy storage systems. A few of the major companies such as ABB, AEG Power Solutions, and Siemens AG that provide battery energy storage systems worldwide are based in Europe.

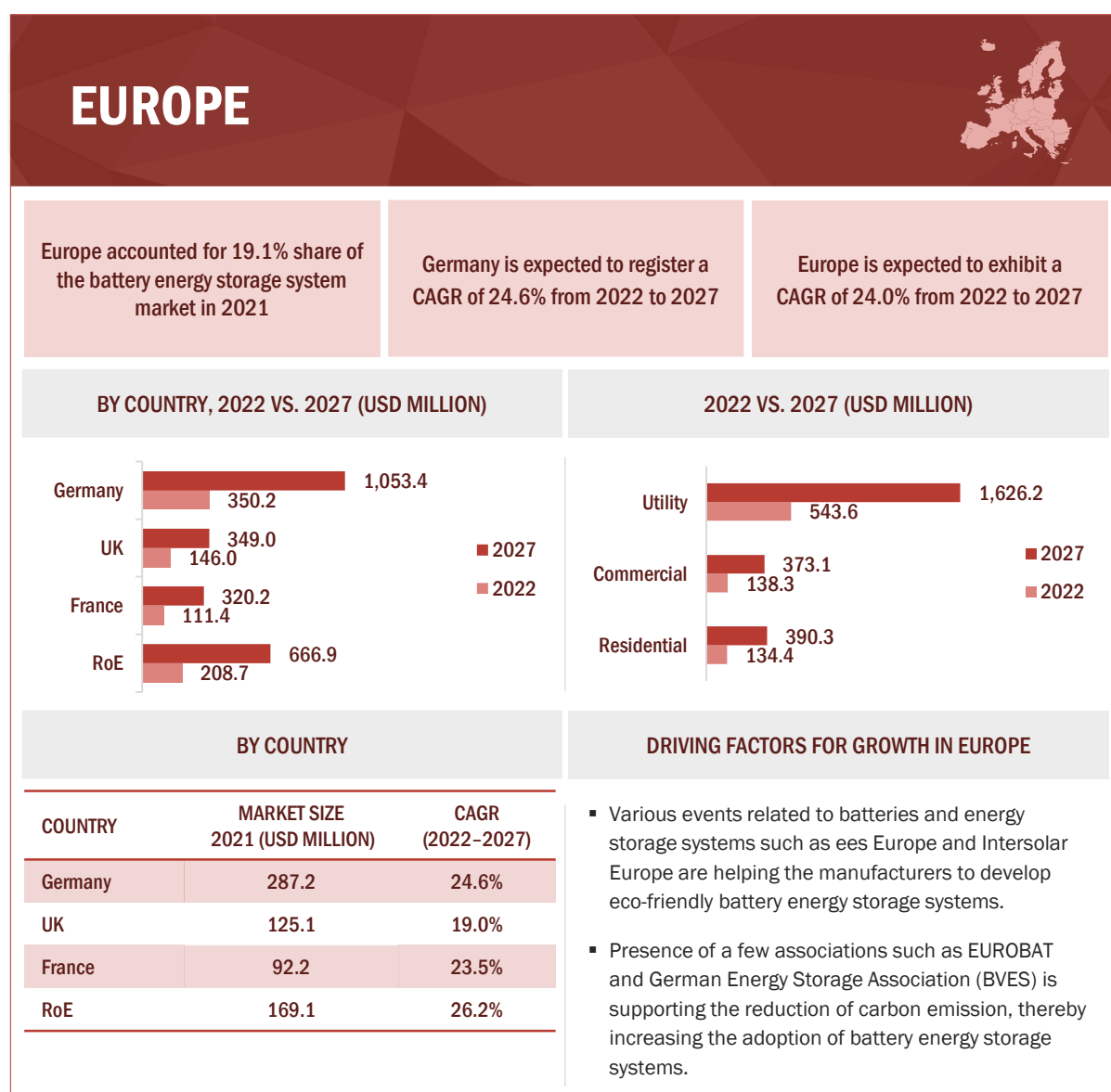
The associations such as EUROBAT and German Energy Storage Association (BVES) are promoting the adoption of battery energy storage systems, which efficiently and economically use renewable energies for energy storage. The EU's 20-20-20 strategy is planned for a 20% drop in CO₂ emissions, a 20% improvement in energy efficiency, and an increase in the share of renewable energy to 20% by 2020. As of July 2021, certain changes have been incorporated in the directive, which has established a new binding renewable energy target for the EU to achieve at least 32% renewable energy of the final energy consumption by 2030. The new directive also includes a clause for a possible upward revision in the current renewable energy generation target by 2023. Furthermore, it comprises a second clause entailing another 14% increase in the target set for the share of renewable energy in transportation by 2030. These plans are targeting the smart, sustainable, and inclusive energy sector, which is further promoting the growth of the battery energy storage system market in Europe.

The EES Europe is the largest exhibition of batteries and energy storage systems. This exhibition is the industry hotspot for suppliers, manufacturers, distributors, and users of stationary and mobile electrical energy storage solutions. This exhibition is organized annually in conjunction with Intersolar Europe, the world's largest exhibition for the solar industry, in Munich, Germany. The scope of the exhibition involves energy storage technologies, energy storage system components and equipment, battery production technologies/machine and plant construction energy storage systems, and grid infrastructure and solutions for the integration of renewable energy in electricity generation.

13.3.1 IMPACT OF COVID-19 ON EUROPE

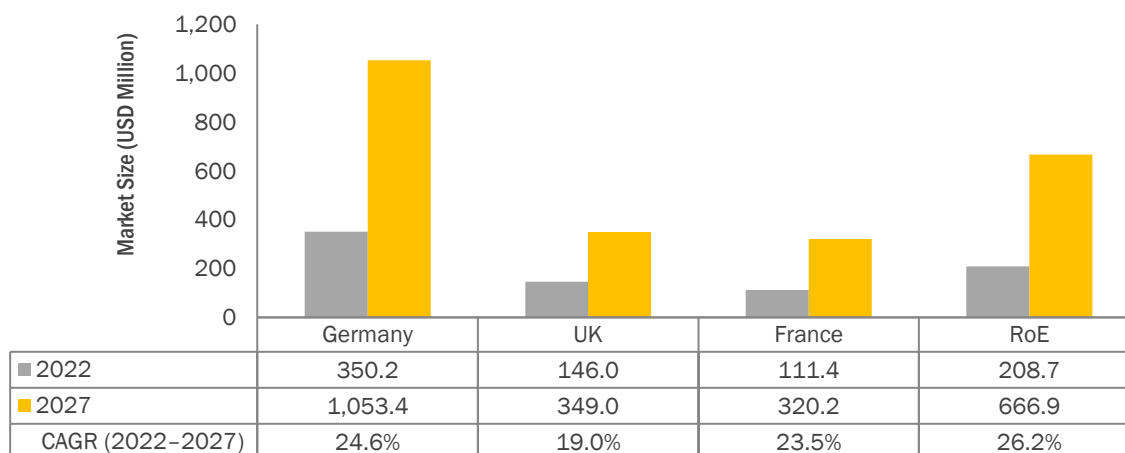
The advent of the COVID-19 pandemic and its rapidly spreading effects unhinged the social, political, and economic landscape across countries, the European countries being no exception. The pandemic led to diminishing revenues across businesses owing to workforce restrictions and delaying or cancellation of planned projects by customers. There was a reduction in demand for energy from grids during the initial phase of the pandemic due to the shutdowns in the countries. Several electrifications and grid modernization projects were halted due to a lack of raw materials, batteries, and other essential components. Additionally, the decarbonization trends, which needed rapid storage deployment to continue the renewable energy revolution were affected due to the economic slowdown. This led to allied business segments shifting to other modes of energy storage systems other than batteries to sustain and avoid project delays. Overdependence on essential inputs from countries in APAC, such as China combined with disruptions in the global supply chain also affected energy storage system manufacturers in the European region, especially in Germany.

However, post-pandemic, the implementation of much-needed energy infrastructure flexibility and system services are helping the energy storage system market to cope with the changing trends. Energy storage deployments have accelerated post-pandemic providing the much-needed flexibility and resilience to the energy storage system market, thereby helping in serving a larger customer base. Several initiatives are being undertaken by the government to prioritize the energy storage sector, which has resulted in positive commitments toward economic recovery and an increase in demand for energy storage systems across segments.

FIGURE 56 EUROPE MARKET: SNAPSHOT

Source: Press Releases, Investor Relation Presentations, National Renewable Energy Laboratory (NREL), Expert Interviews, and MarketsandMarkets Analysis

FIGURE 57 GERMANY TO DOMINATE BATTERY ENERGY STORAGE SYSTEM MARKET, IN TERMS OF SIZE, DURING FORECAST PERIOD



Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

13.3.2 GERMANY

13.3.2.1 Growing expansion of residential storage sector to boost market growth

Germany is a well-developed country that focuses on innovation and more advanced solutions in the energy storage sector. In 2020, renewable energy sources in Germany accounted for the majority share in the electricity supply and provided 45.3% of the total electricity demand. Germany has pledged to transform its electricity requirement to 100% renewable energy by 2050. According to the German Renewable Energy Federation (BEE), Germany's renewable energy industry has been growing steadily due to the Renewable Energy Act (REG). The country aims to phase out its nuclear plants by 2025. According to the goals set by Energiewende, the share of power generated from renewable sources is set to increase to 40–45% by 2025 and to more than 80% by 2050.

A growing number of homeowners in Germany are installing battery energy storage systems. One out of every two rooftop solar panels is now sold with a battery storage system. German retail electricity prices have surged in recent years. Meanwhile, the nation has chosen to shut down its existing nuclear capacity through the Energiewende (energy transition) policy, replacing it with intermittent renewables. This has resulted in the shifting of homeowners toward home batteries, which has further propelled the use of home storage systems in Germany. As of 2021, there are more than 300,000 battery storage systems installed in German households with the average installation representing about 8.5 KWh capacity. Most of the stationary battery storage systems in Germany are purchased as a part of photovoltaic solar systems for homes. The increasing popularity of solar panels and battery systems, low-cost production of solar panels from China, and drop in lithium-ion battery prices are encouraging the use of home energy storage systems in Germany. A few of the key companies providing home storage systems in Germany are Enerix, Sonnen, and SOLARWATT. For instance, the sonnenBatterie is a high-tech storage system that has proven itself in thousands of households in Germany. The combination of a PV system and a sonnenBatterie covers about 75% of yearly energy requirements with self-produced and clean energy. Germany is expected to show potential growth during the forecast period. Various large-scale energy storage projects are being installed in Germany. In July 2021, Siemens Smart Infrastructure and Zukunftsenergie Nordostbayern GmbH signed a letter of intent for the establishment of a 100 MW battery energy storage facility in Wunsiedel, Germany. The major objective of the project would be to provide support to the power grid and manage peak electricity demands. The project will have an energy storage capacity of 200 MWh and would cover an estimated area of 5000 square meters. Also, in September 2021, the German energy group RWE AG

announced the development of two solar-plus-energy storage projects, the contracts of which it acquired through the latest innovation tender in Germany. The project will enable the installment of 31.5 MWp of solar power capacity along with a battery energy storage capacity of 10.5 MW in the state of North Rhine-Westphalia. The two projects, namely, the Garzweiler project and Jackerath project will involve the setting up of a solar plant with a battery energy storage capacity of 6.5 MW and 4.1 MW, respectively. In 2019, Bordesholm battery energy storage system was set up in Germany, which is a 10,000 KW energy storage project located in Bordesholm. The system uses Lithium-ion as its storage technology and the project is owned by Versorgungsbetriebe Bordesholm. The battery energy storage system provides grid stabilization and backup power, which is sourced from 100% renewables to Bordesholm.

13.3.3 UK

13.3.3.1 Increasing investments and support by UK government in battery storage to boost market growth

The UK government is providing funding to boost the energy storage segment to reduce carbon emission levels. The funding in R&D pertaining to energy storage has led to the development of the next-generation storage technologies of battery energy storage systems. In 2021, the UK government's Department for Business, Energy, and Industrial Strategy (BEIS) supervised an energy storage competition as a part of the UK's Net Zero Innovation Portfolio backed by almost USD 92 million capital funding to support new energy storage technology projects to support renewable power generation. The Faraday Challenge is a 4-year investment as a major part of the industrial strategy of the current government. Such challenges lead to competitions that boost innovation and R&D in the battery sector. An electric vehicle charging infrastructure company Pivot Power is aiming to develop 2 GW of grid-scale energy storage arrays in the UK by 2030. The batteries will be used to facilitate rapid EV charging, both in the form of co-located EV charging points and supporting local grids to cope with increasing electricity demand. Besides, in 2020, the government announced a relaxation in the planning legislation to facilitate the construction of large batteries to enhance the storing of renewable energy from solar and wind farms across the UK. Following this, the UK government decided to remove barriers for storage projects above 50 MW in England and 350 MW in Wales, allowing for more clean energy to be stored and used all year round. Hence, these kinds of projects are increasing the adoption of battery energy storage systems for different applications in the UK.

The investments and support from the UK government in battery storage have minimized the country's dependence on large fossil generation. As of June 2021, more than 16.1 GW of battery energy storage system capacity is either operating, under construction, or in the pipeline across 729 projects in the UK. 2021 was a record-breaking year for energy storage in the UK in terms of submitted capacity. Also, the UK government is investing more than GBP 3 billion in low-carbon innovation, as the UK aims to encourage the use of clean energy.

13.3.4 IRELAND

13.3.4.1 Rising adoption of renewable energy and enhanced need for grid modernization to accommodate reliable energy storage systems are driving market growth in Ireland

The Government of Ireland, to achieve its 70X30 ambitions to source 70% of its electricity from renewable sources by 2030, is setting up safe and reliable energy storage systems, which will be the key to delivering cleaner, more resilient, and more efficient power system infrastructures. According to the Irish Energy Storage Association, although energy storage in Ireland is being regarded as a relatively new concept, the country has been able to successfully meet its targets for 2020 of achieving 40% renewable electricity enabling energy storage project developers to win contracts in the DS3 (Delivering a Secure, Sustainable Electricity System) program, a multi-year program began by the grid operator EirGrid's. Moreover, with the commencement of a fixed tariff regime under the DS3, the need for fast-responding grid service assets such as energy storage is growing and there is an increased demand for DS3-like grid services from

industries across Ireland. In 2020, when the frequency of the electricity grid of Ireland dropped below normal operating levels, two large-scale battery energy storage projects stepped in to help stabilize the network. The Kilathmoy 11 MW and the 26 MW Kelwin-2 battery energy storage system by Statkraft were the ones, which responded to the under-frequency event, which proved the importance of these systems to support the electricity network on the island of Ireland, as the Republic of Ireland and Northern Ireland share a single grid. This has also catapulted a growing need for the transition of energy systems from fossil fuels to renewable sources in Ireland. Statkraft has recently announced an additional 68 MW battery energy storage system to be set up in the Republic of Ireland, which would further strengthen the grid services. These projects are in line with the 70X30 goal because to hit the average of 70% renewable energy capacity, Eirgrid is looking at a target of over 95% capacity from renewables as the Irish grid at times would require significantly more renewable penetration than 70%.

In Ireland, there is an increasing demand for a transition toward intermittent renewables and distributed generation technologies. New technologies such as EV, Storage, Digital Platforms, etc. are being adopted to enhance the adoption of increasingly complex grid management systems to ensure bi-directional power and data flow, thus giving rise to smarter energy networks in the country. Keeping this in mind, several projects are being undertaken such as the Gore Street battery energy storage projects, which will provide rapid frequency response and reserves to the national electricity grid within milli-seconds when there is a sudden drop-off in generation or supply. Furthermore, the Fluence projects, which is a joint venture of Siemens and AES corporation and is ESB's first major battery project, is an initiative to undertake close to 100 MWh portfolio of projects, which upon completion will have the capacity to power approximately 100,000 Irish homes and businesses for several hours. More than 2.5 GW of planned utility-scale project capacity has been set up in Ireland as of 2020, which comprises large-scale (>20 MW) capacity sites.

13.3.5 FRANCE

13.3.5.1 Surging penetration of renewable energy is boosting market growth in France

The Government of France aims to achieve diversified electricity production and reduce the share of nuclear power to 50% by 2025, while also reducing the emission of greenhouse gases by 40% by 2030. The country has stated its vision of deploying 7 million electric vehicle charging outlets by 2030. Such developments would enhance the overall demand for electric motors, eventually impacting the market for battery energy storage positively. In 2021, France-headquartered multinational energy company, TotalEnergies set up the biggest battery energy storage system in Mainland France. The contract for the project was supervised by the RTE and has an output capacity of 25 MW and a storage capacity of 25 MWh. It was built on the site of a former oil refinery in Dunkirk, which was owned by TotalEnergies. The lithium-ion battery storage technology, which was used for the project was provided by Saft, an energy storage provider owned by TotalEnergies. The system is based on Saft's Intensium Max 20 High Energy solution and comprises 11 integrated 2.3 MWh containers. The project is set to provide fast reserve services for grid stability and backs the government's policy to support the development of electrical capacity through capacity mechanisms.

Additionally, in 2020, the Brittany battery energy storage system, a 7,000 KW energy storage project was announced, which uses lithium-ion technology and would be set up in Brittany, France. The project is developed by Paris-based energy giant, Neoen. Neoen has won contracts for a total capacity of 13 MW of energy storage, which would be split into two phases, one of which is underway and is expected to be completed by 2027 and the other phase having 6 MW capacity of projects is expected to commence in 2022 and continue until 2028. Developments in the power sector have further contributed to the decarbonization of various sectors for the country, which further drives the battery energy storage market. According to a study by EUPD Research, French installers have started providing energy storage offerings in their respective portfolios. The number of such installers went up by 71% in 2021. The efforts of France to bring energy storage projects are expected to boost the growth of the regional market.

13.3.6 REST OF EUROPE

The Rest of Europe includes Italy, Spain, Norway, Sweden, and Switzerland. The European Union is focusing on cutting its greenhouse gas emissions by 40% by 2030, according to the European Commission. The European Union (EU) countries (Hungary, Latvia, Slovenia, Estonia, Lithuania, Romania, Bulgaria, the Czech Republic, and Slovakia), and Southern European countries (Croatia, Serbia, and Georgia) are bound to the EU laws regarding the deregulation of the electricity market and the reduction of greenhouse gas emissions. The need to reduce carbon emissions in these countries has led to the high adoption of battery energy storage systems in the region. The growth of the battery storage market in these countries is driven by government incentive plans to meet climate change ambitions. Countries in this region are adopting battery energy storage systems to modernize the grid network and stabilize the power grid.

In 2020, the Aragon Solar PV Phase III battery energy storage system was announced to be set up in Spain and is expected to be commissioned by 2026. It is a 105,000 KW energy storage project, which is in Andorra, Aragon, and has the objective of promoting renewable energy integration. The project is being developed by a Spanish multinational electric utility company, Endesa. Furthermore, Spain's Integrated National Energy and Climate Plan (PNIEC) 2021–2030 aims to maximize the capacity of non-dispatchable renewable technologies and thereby plans to increase the battery energy storage installation capacity to approximately 2.5 GW by 2030. Similarly, in 2021, the Agnes Floating Solar PV battery energy storage system was announced in Ravenna, Emilia-Romagna, Italy, and is expected to be commissioned by 2025. It is a 50,000 KW energy storage project and the rated storage capacity of the project is 100,000 KWh. The project is developed by AGNES, Qintx Holding, and Saipem and has an objective to balance the grid. The project uses lithium-ion as the battery storage technology.

TABLE 134 EUROPE: BATTERY ENERGY STORAGE SYSTEM MARKET, BY COUNTRY, 2018–2021 (USD MILLION)

Country	2018	2019	2020	2021
Germany	200.0	305.5	242.5	287.2
UK	98.5	145.5	110.4	125.1
France	66.2	100.4	78.7	92.2
RoE	113.4	176.1	141.2	169.1
Total	478.0	727.5	572.8	673.5

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 135 EUROPE: BATTERY ENERGY STORAGE SYSTEM MARKET, BY COUNTRY, 2022–2027 (USD MILLION)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Germany	350.2	429.9	531.8	664.3	833.3	1,053.4	24.6%
UK	146.0	171.8	203.8	244.3	290.3	349.0	19.0%
France	111.4	135.5	166.3	206.1	255.7	320.2	23.5%
RoE	208.7	259.5	325.4	412.3	521.7	666.9	26.2%
Total	816.4	996.7	1,227.3	1,527.0	1,901.0	2,389.6	24.0%

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 136 EUROPE: BATTERY ENERGY STORAGE SYSTEM MARKET, BY BATTERY TYPE, 2018–2021 (USD MILLION)

BATTERY TYPE	2018	2019	2020	2021
Lithium-ion Batteries	254.0	380.7	297.4	346.4
Lead-acid Batteries	80.5	118.0	88.5	99.1
Flow Batteries	101.7	162.3	132.9	162.4
Others	41.9	66.4	54.0	65.6
Total	478.0	727.5	572.8	673.5

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 137 EUROPE: BATTERY ENERGY STORAGE SYSTEM MARKET, BY BATTERY TYPE, 2022–2027 (USD MILLION)

BATTERY TYPE	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Lithium-ion Batteries	414.5	498.6	604.0	739.0	912.1	1,131.1	22.2%
Lead-acid Batteries	114.5	133.2	156.4	185.4	216.7	256.8	17.5%
Flow Batteries	205.2	261.1	334.8	432.6	555.6	722.1	28.6%
Others	82.2	103.8	132.1	170.0	217.2	280.9	27.9%
Total	816.4	996.7	1,227.3	1,527.0	1,901.0	2,389.6	24.0%

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 138 EUROPE: BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

APPLICATION	2018	2019	2020	2021
Residential	76.1	122.7	95.9	112.6
Commercial	85.4	124.4	99.6	116.9
Utility	316.4	480.4	377.3	444.0
Total	478.0	727.5	572.8	673.5

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 139 **EUROPE: BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2022–2027 (USD MILLION)**

APPLICATION	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	134.4	160.9	197.1	251.4	310.6	390.3	23.8%
Commercial	138.3	165.6	199.4	243.4	301.3	373.1	22.0%
Utility	543.6	670.1	830.8	1,032.3	1,289.2	1,626.2	24.5%
Total	816.4	996.7	1,227.3	1,527.0	1,901.0	2,389.6	24.0%

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

Germany is a well-developed country that focuses on innovation and the development of advanced solutions in the energy storage sector. Large battery energy storage system projects in Germany would help this region to retain its major share in the European market in the future. The COVID-19 outbreak in the European economy affected both the services and manufacturing sectors alike. However, the resumption of the projects in the various countries of the region has highly fueled the growth of the market.

13.4 APAC

The market in APAC is further segmented into China, Japan, India, Australia, and the Rest of APAC (RoAPAC). The region is the largest market for battery energy storage and is home to some of the fastest-growing economies in the world. APAC also plans for electrification in remote areas that are mainly off-grid in several countries. The residential energy storage technology would further enable increased access to electricity for these remote and island communities in the region.

Energy storage investments are expected to increase substantially within the region as the governments in growing economies are taking new policies to improve the reliability and quality of the power distribution facilities to the residential customers. The governments of various countries in this region focus on minimizing the adverse effects of the energy sector on the environment. These factors lead to the growth in the deployment of battery energy storage systems in residential and public utility applications. A few of the major battery energy storage system manufacturers based in APAC are Samsung SDI Co., Ltd., LG Energy Solutions Co., Ltd., Hitachi Ltd, Toshiba Corporation, TrinaBESS, NGK INSULATORS, LTD., Delta Electronics, Inc., Panasonic Corporation, and NEC Corporation.

13.4.1 IMPACT OF COVID-19 ON MARKET IN APAC

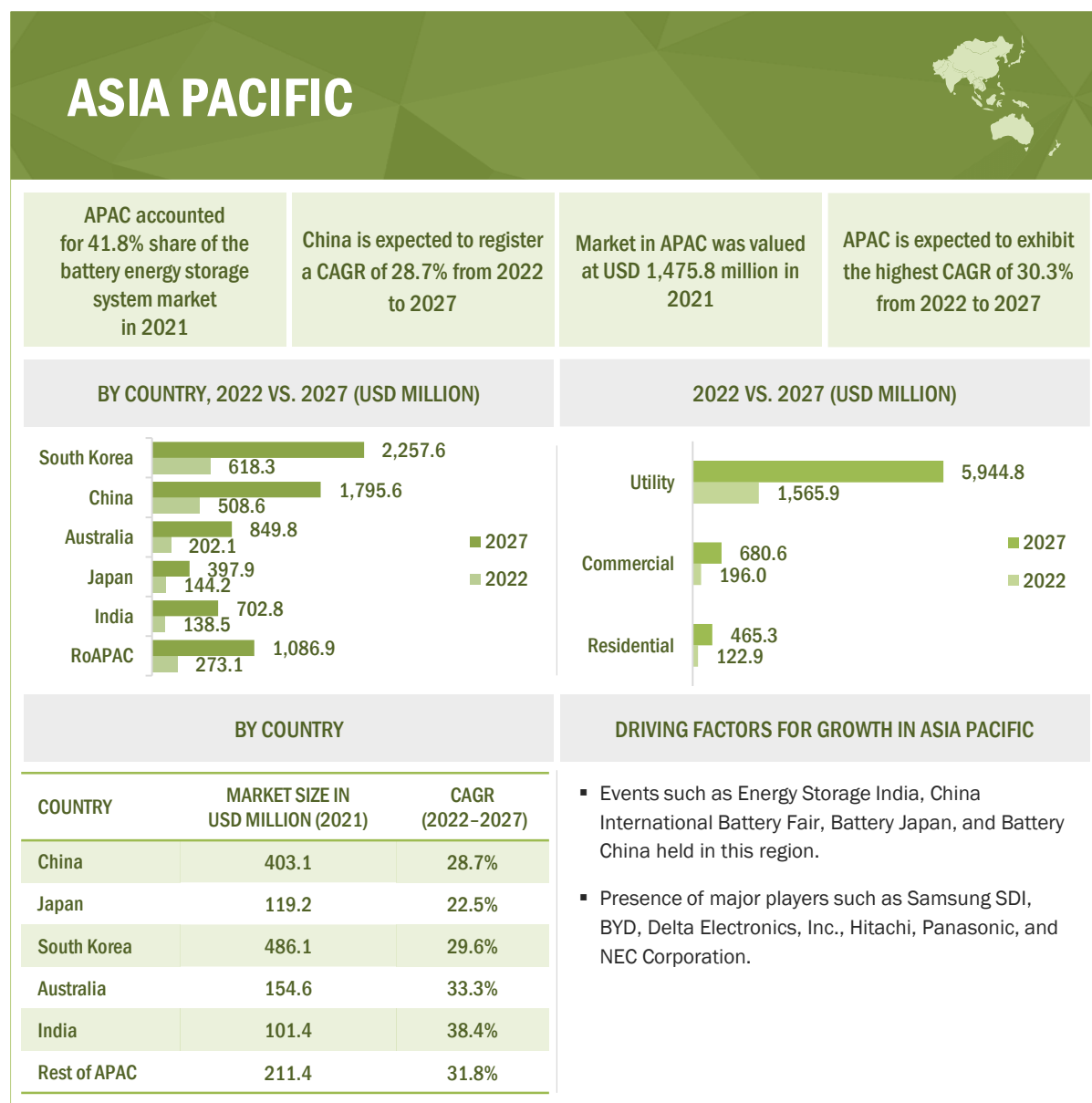
The impact of the COVID-19 pandemic was felt across the countries in the APAC region, especially China. The stringent measures undertaken by the government to curb the spread of the pandemic led to reduced workforce engagements and numerous facilities were shut down. According to a World Bank report by World Economic and Financial Surveys, growth momentum in the largest economies in Asia Pacific was impacted due to COVID-19, reflected in government policies and projects over concern in China and Japan, which, in turn, harmed other economies in Asia. The worldwide battery manufacturing and energy storage business was affected due to global overdependence on China for essential components as China suffered through complete shutdowns in the initial months of the pandemic.

The business of the energy storage system manufacturers in Australia has been adversely affected by the shutdown of operations in China and other countries of Asia Pacific. It has also resulted in the limited production of energy storage system units resulting in project delays and cancellations across countries. In addition, the per capita income of individuals has also been affected due to the pandemic, which has consequently declined the demand for residential battery energy storage systems. Moreover, curtailments

across segments such as mobility, construction, and manufacturing, etc. have affected the demand in the energy industry, adversely affecting the energy storage system market.

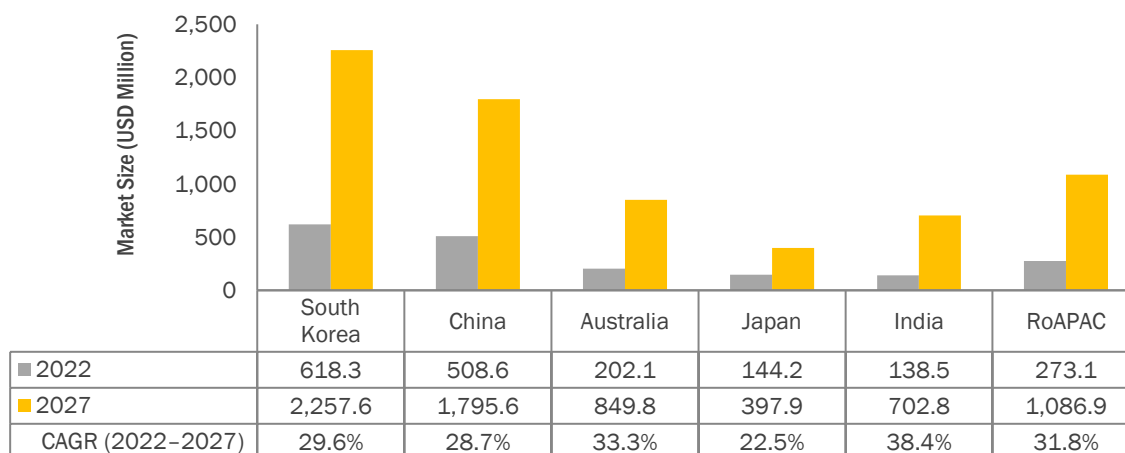
However, post-pandemic, certain improvements in demand have been seen across the countries in the APAC region. An increase in demand for electronics, enhanced adoption of remote electrification projects under government schemes, and increase in adoption of power backup energy sources have helped stabilize the demand for battery energy storage systems in various countries of the region.

FIGURE 58 APAC MARKET: SNAPSHOT



Source: Press Releases, Investor Relation Presentations, National Renewable Energy Laboratory (NREL), Expert Interviews, and MarketsandMarkets Analysis

FIGURE 59 BATTERY ENERGY STORAGE SYSTEM MARKET IN INDIA TO REGISTER HIGHEST CAGR DURING FORECAST PERIOD



Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

13.4.2 CHINA

13.4.2.1 Increasing focus on setting up large flow vanadium battery energy storage projects

The demand for electricity in China is on the rise due to the rapid population growth in the country. To fulfill the increasing demand, the state-owned State Grid Corporation of China, the world's largest utility, has deployed battery energy storage systems to provide ancillary services throughout its grid. Moreover, China allows the non-state-owned power providers to enter the energy market-opening opportunities for independent power producers (IPPs) to provide ancillary and capacity services, along with battery energy storage systems.

China has extensive natural reserves of graphite. It is making all the efforts to achieve a threefold increase in its overall graphite processing capacity by 2020 to meet the ever-increasing demand for lithium-ion batteries. It is expected that China's sales of lithium-ion batteries would rise to 51.5 GWh by 2023 from 37.8 GWh in 2020. Shanshan Technology, BTR New Energy Materials, and LuiMao Graphite, the firms in China, have collaborated with BAIC Automotive Group. These three companies are building lithium-ion battery-graphite anode megafactories with a total processing capacity of 260,000 metric tons per year. Such projects involving the graphite megafactories are likely to complement and support the growth of the battery energy storage system market in China soon. China's flow vanadium energy storage market is also projected to grow, as the ability to offer virtually unlimited storage capacity, rapid response time, long-scale duration, and negligible self-discharge are expected to be a few of the key features that make the technology suitable for large-scale renewable integration applications over its available alternatives.

China is focusing on building massive flow battery projects. For instance, in 2021, the world's first 1 MWh Na-ion battery energy storage system started operating in Taiyuan, which is in North China's Shanxi province. The system is based on Na-ion battery storage technology and is combined with municipal electricity, photovoltaic, and charging facilities, which would create a micro-grid framework, which can interact smartly with public networks. Additionally, in 2021, the Dalian-UET/Rongke Power battery energy power system was commissioned, which is a 200,000 KW energy storage project set up in Dalian, Liaoning, China. The system uses large vanadium redox flow batteries as its storage technology and has a rated battery storage capacity of 800,000 KWh. They allow charging and discharging at the same time and can be charged more than 20,000 times without reducing their storage capacity.

The project is owned by Dalian-UET/Rongke Power and the National Energy Administration of China and a few of the key applications of the project include electric energy time shift, renewable energy time shift, renewable capacity farming, and resiliency.

By 2030, China is planning to install more than 30 GW of new battery energy storage systems. As of 2020, China has approximately 3.3 GW of energy storage capacity, mostly consisting of battery energy storage systems and excluding pumped hydro systems. Lithium-ion batteries comprise the largest portion of non-pumped hydro energy storage in China. These factors are bound to drive the market for battery energy storage in the coming years.

NEC's Energy Solutions division, which has been responsible for NEC Corporation's activities in the battery energy storage industry, suffered a decline in business due to COVID-19. The company's energy solutions segment has battery maintenance contracts lasting up to March 2030 and will be operational until then, although the company's energy storage solutions, such as GSS (Grid Storage Solution) and Distributed Storage Solution (DSS), will not be actively seeking new projects. CATL and BYD, two of the largest battery manufacturers in China, are widening their production capacities abroad, backed by the Chinese Government. China's major Li-ion manufacturing players, BYD and CATL, are faced with a high probability of additional production delays.

13.4.3 JAPAN

13.4.3.1 Growing installations of solar and wind farms combined with battery energy storage to meet requirements of local grid operators and utilities

Japan has an active policy approach to energy storage, and it develops the battery storage industry policies for the adoption of storage technology. Japan has been a world leader in the production of sodium-sulfur batteries and is installing sodium-sulfur battery energy storage system of 50 MW/300 MWh to date. Toshiba and Panasonic, the Japanese giants, are competing with Samsung SDI and LG Energy Solutions Co., Ltd., the South Korean conglomerates, for the leadership in the fast-growing battery energy storage market. A few of the leading battery manufacturers providing batteries for the battery energy storage systems are GS Yuasa, Hitachi Ltd., Mitsubishi, NEC Corporation, Panasonic Corporation, and Toshiba Corporation for lithium-ion; GS Yuasa and Shin-Kobe Electric Machinery for lead-acid batteries; NGK Insulators for sodium-sulfur batteries; and Sumitomo Electric for redox flow batteries.

In July 2020, Sumitomo Electric Industries received a supply contract from transmission and distribution network operator Hokkaido Electric Power to provide a grid-scale flow battery energy storage system for a wind farm in northern Japan. The project will be completed by the end of March 2022. In 2021, the Shiriuchi Solar PV Park battery energy storage system became operational in Shiriuchi, Hokkaido, Japan. It is a 12,500 KW energy storage project and uses lithium-ion storage technology with a rated storage capacity of 7,200 KWh. The key applications of the project include reactive power support, frequency regulation, and voltage control and are being developed by ORIX and Solar Frontier. Additionally, in 2021, Japan has started the development of a marine battery energy storage vessel, which would be charged from offshore wind at the sea and would carry the power back to land. Also, Tesla Inc., a US auto major is partnering with Japanese power retailer and aggregator Global Engineering and engineering firm Ene-Vision to develop an energy storage facility in Northern Japan, which would be connected to the grid with 6.095 KWh capacity to help in stabilizing the power system, which will use the Megapack battery technology of Tesla. The energy storage system would be able to power about 500 homes.

The government of Japan is focusing on installing solar and wind farms to meet the requirements of local grid operators and utilities. A large-scale solar farm has been completed on the island of Hokkaido, Japan. Telecoms and tech company Softbank's clean energy subsidiary SB Energy and financing entity Mitsubishi commissioned Tomatoh Abira Solar Park 2, a 64.6-MW solar farm that is combined with 19 MWh of onsite battery energy storage. The country also issued a new international standard for the safety of large-scale

stationary energy storage systems. On April 16, 2020, the International Electrotechnical Commission (IEC) issued a new international standard for the safety of storage battery-based large-scale stationary electrical energy storage (EES) systems. This issuance of the first international standard for the safety of EES systems is expected to contribute to the development of environments for assessing such safety at home and abroad and to encourage industries to make use of renewable energy in the future.

13.4.4 INDIA

13.4.4.1 Government investments and upcoming battery energy storage projects to revolutionize India's power systems

India is an upcoming country in the battery energy storage market in APAC in 2021. According to the India Energy Storage Alliance (IESA), the industry has crossed 1,000 MWh of lithium-ion battery deployment for distributed applications, including telecom towers and bank ATMs. There are initiatives undertaken for achieving the ambitious target of the installation of 225 GW of renewable energy by 2022. Besides, according to the recent study by Bloomberg New Energy Finance (BNEF), India will supersede China to become the world's largest producer of renewable energy by 2050 and consequently, renewables will account for 75% of India's electricity demands. This will pose a USD 1.6 trillion opportunity for India by 2050 in the power generation segment. The increasing urban population in India is driving the need for expanding both the electricity generation capability and T&D infrastructure across the country. Moreover, the country is facing frequent power outages due to changing weather, insufficient generation capacity, and fragile infrastructure. Such challenges generate the requirement of heavy investments to improve the grid's resilience and reliability. The adoption of a battery energy storage system is helping the country to overcome such challenges. India has revolutionized its power systems and augmented grid efficiency with its recent battery energy storage projects. In 2021, NTPC Renewable Energy Limited (NREL) floated a global tender for the setting up of a 3 GW renewable energy project with battery energy power systems, which would be worth USD 2 billion. Similarly, in August 2020, Bharat Heavy Electricals Limited (BHEL) issued a tender for 410 kWh/415 kW of battery energy storage systems (BESS) to be installed at three sites in Delhi. Hence, these upcoming projects have boosted the growth of the battery energy storage system market in India. All this is a part of the government's initiative of inviting global bids for a total of 4,000 MWh of battery energy storage projects.

According to Central Electricity Authority (CEA), as of 2021, India's renewable energy capacity is 104 GW, excluding large hydro plants. The renewable energy capacity of India comprises 49 GW solar and 40 GW of wind energy. The Central Electricity Authority (CEA) also estimates that the peak electricity demand in India will grow at the rate of 6.9% CAGR and will reach 340 GW by 2030. According to Power System Operation Corporation Limited (POSOCO), in July 2021, the country witnessed the highest ever demand for power, which crossed 200 GW and was 17.6% higher compared to the demand in July 2020. Hence, an alternative battery energy storage system (BESS) based on low-cost lithium-ion batteries may enable India to meet the morning and evening peak demands. According to the AES, in the next 4–5 years, India will be the potential market for energy storage to invest about USD 50 billion in battery storage facilities that could help integrate renewable energy into the grid, replace polluting diesel-fueled power plants, and boost electric mobility. This investment will drive the growth of the battery energy storage market in India in the next few years. For instance, the government has planned to set up solar energy power plants with a cumulative capacity of about 14 MW with a battery storage capacity of 42 MWh in Leh and Kargil. The government, in September 2021, also announced plans for setting up a 12 GWh battery energy storage project in Ladakh. In September 2021, the Government of India (GOI) announced the setting up of a 14 GWh grid-scale battery storage system in Khavda, Gujarat. Additionally, in October 2021, the Solar Energy Corporation of India (SECI), under the Ministry of New and Renewable Energy has announced the installation of a pilot project for the setting up of a battery energy storage system. The project would be aimed toward setting up a 500 MW/1000 MWh standalone battery energy storage system in India under the Global Competitive Bidding (ESS-I).

13.4.5 AUSTRALIA

13.4.5.1 Government schemes and deployment of large-scale battery energy storage projects

The energy storage sector in Australia is starting to pick up momentum, with the 23,796 small-scale batteries installed in 2020 taking Australia's household storage capacity to 1.2 GWh in 2020. State government schemes are continuing to drive this uptake, with South Australia and Victoria both offering generous subsidies to help households install batteries to reduce their energy costs. According to Australia's Energy and Environment Agency, the state government subsidies worth USD 104 million, coupled with low-interest loans and demand response schemes for residential energy storage, would result in an increased storage uptake in the country during the forecast period. In December 2021, the Government of Australia announced a funding of USD 72.3 million to be provided through the Australian Renewable Energy Agency (ARENA) aimed toward the construction of new large-scale batteries, which would be targeted at providing affordable and reliable electricity for industries and households in the region. In 2018, Australia launched the largest battery subsidy program, AUD 100 million Home Battery Scheme, which will subsidize the cost of buying a home energy storage system for up to 40,000 households in the country. This initiative has already attracted a few major players in the residential energy storage market, such as Sonnen Group and Eguana Technologies, to set up local assembly plants in Australia to further reduce the cost. Similarly, the State of Victoria has pledged an AUD 40 million to subsidize the installation of battery storage in 10,000 homes. The Australian Energy Storage Council is at the forefront in terms of implementing industrial voluntary standards regime for safety and quality. A few Australian Standards applicable to battery technology are AS 3011-1992: electrical installations-secondary batteries installed in buildings; AS 2676.1-1992: a guide to the installation, maintenance, testing, and replacement of secondary batteries in buildings-vented cells; AS/NZS 4509.1:2009, AS/NZS 4509.2:2010: stand-alone power systems-system design; and AS 4086.2—1997.

Sixteen large-scale batteries were also under construction at the end of 2020, several of which are co-located with solar and wind farms. This is an increasingly common trend as developers look to maximize returns from their wind and solar projects. In 2021, CEP. Energy, a specialist renewable energy fund company in Australia, announced plans for setting up four battery storage plants with a total capacity of over 2,000 MW to be installed at different locations around Australia. Similarly, in 2021, another project by Origin Energy, an integrated energy company is underway. This would establish a 700 MW/2,800 MWh battery capacity project in New South Wales. In January 2020, Australian utility company AGL had signed a 15-year contract for “operational dispatch rights” to a 100 MW battery storage system with Vena Energy Australia. Vena Energy constructed a 100 MW/150 MWh battery at its Wandoan South Solar Farm in Queensland, which can power up more than 57,000 average homes annually. Vena Energy also announced plans for a USD 650 million solar project, which would be a part of the Wandoan South Project. French renewable energy developer, Neoen has plans to develop a 500 MW/1,000 MWh massive battery storage system near the Australian city of Geelong. Furthermore, another massive battery energy storage system, which is powered by Tesla's utility-scale Megapack batteries became operational in the Australian state of Victoria in 2021. This facility is also developed by Neoen in partnership with Tesla energy and AusNet.

13.4.6 SOUTH KOREA

13.4.6.1 South Korea to hold largest size of battery energy storage system market in 2027

South Korea dedicatedly focuses on the adoption of battery energy storage systems for electrical grids. The country lacks domestic energy reserves and is one of the major global energy importers. The Ministry of Trade, Industry, and Energy of South Korea launched a new program to increase grid reliability by investing USD 391.6 million in energy storage systems by 2020. It has also targeted to reduce carbon emission by 37% by 2030. The government of South Korea hopes to achieve 10% renewable electricity by 2023 and 20% by 2030. The battery energy storage installation in the country witnessed a significant spike in 2020.

In January 2022, Singapore-Norwegian company, G8 Subsea, a specialist in offshore renewable energy projects has entered into an agreement with South Korean industrial business group Holim tech to develop a 1.5 GW offshore wind power plant, which will be paired with energy storage systems provided by lithium-ion batteries. In November 2021, energy storage solutions provider, VFlowTech has announced a collaboration with Seoul National University of Science & Technology (SeoulTech) and Korean-based organization CompanyWE Inc to deploy a 500 KWh vanadium redox flow energy storage system for renewable energy. The major application of the storage system would be to enhance the electric-car fast-charging infrastructure in South Korea. The project will use VFlowTech's modular PowerCube batteries, which will be installed as a part of the pilot project to support the electrification needs of the rapidly expanding South Korean EV market. These developments will support the adoption of battery energy storage systems in the country.

13.4.7 REST OF APAC

The Rest of APAC comprises New Zealand, Thailand, Singapore, Taiwan, Malaysia, and Indonesia, among others. Singapore has set up the Energy Market Authority (EMA), an energy storage program to develop technologies that would enhance the overall stability and resilience of Singapore's power system. The country has identified energy storage systems to play a crucial role in harnessing solar energy more effectively and maintaining a stable and reliable supply of electricity. These countries are island nations, and land constraint is a major factor for any new power generation project. The development of energy storage in these countries will help in countering the intermittency of renewables and enable the use of clean power even at times of non-availability of renewable power. The technological innovations, growing public awareness, and government initiatives toward energy storage are likely to boost the growth of the battery energy storage system market in this region.

In 2021, power distribution company WEL Networks in partnership with renewables developer, Infratec announced the setting up of a project for the establishment of a 35 MW battery energy storage system to be constructed in the Waikato district of New Zealand. The project will cost about USD 17.66 million and is expected to be commissioned by October 2022. Similarly, in 2021, New Horizons Ahead (NHOA) would be working along with TCC, a pre-eminent Asian industrial group, to develop a technology for two landmark energy storage projects in Taiwan. The two energy storage projects would have an aggregate capacity of 160 MW/420 MWh and would be located at TCC SuAo cement plant and HePing Industrial Park to provide ancillary services to the Taiwanese grid.

TABLE 140 APAC: BATTERY ENERGY STORAGE SYSTEM MARKET, BY COUNTRY, 2018–2021 (USD MILLION)

Country	2018	2019	2020	2021
China	258.8	405.4	330.7	403.1
Japan	85.5	130.3	101.9	119.2
South Korea	304.6	477.8	394.7	486.1
Australia	89.0	144.8	122.4	154.6
India	49.5	86.9	76.8	101.4
RoAPAC	126.4	202.7	169.4	211.4
Total	913.8	1,447.9	1,195.9	1,475.8

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 141 APAC: BATTERY ENERGY STORAGE SYSTEM MARKET, BY COUNTRY, 2022–2027 (USD MILLION)

Country	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
China	508.6	645.8	825.2	1,060.3	1,374.4	1,795.6	28.7%
Japan	144.2	175.6	215.1	265.1	323.0	397.9	22.5%
South Korea	618.3	790.1	1,013.8	1,305.0	1,714.6	2,257.6	29.6%
Australia	202.1	266.0	352.2	469.0	628.7	849.8	33.3%
India	138.5	190.5	263.7	367.0	504.3	702.8	38.4%
RoAPAC	273.1	355.2	464.9	611.7	812.2	1,086.9	31.8%
Total	1,884.8	2,423.1	3,135.0	4,078.1	5,357.1	7,090.6	30.3%

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 142 APAC: BATTERY ENERGY STORAGE SYSTEM MARKET, BY BATTERY TYPE, 2018–2021 (USD MILLION)

BATTERY TYPE	2018	2019	2020	2021
Lithium-ion Batteries	526.0	820.9	671.8	820.6
Lead-acid Batteries	147.7	226.1	178.3	210.1
Flow Batteries	163.2	273.7	236.9	306.1
Others	76.9	127.2	108.8	139.1
Total	913.8	1,447.9	1,195.9	1,475.8

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 143 APAC: BATTERY ENERGY STORAGE SYSTEM MARKET, BY BATTERY TYPE, 2022–2027 (USD MILLION)

BATTERY TYPE	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Lithium-ion Batteries	1,037.2	1,317.7	1,681.9	2,153.1	2,801.5	3,660.8	28.7%
Lead-acid Batteries	255.9	313.8	387.3	480.4	592.0	736.9	23.6%
Flow Batteries	407.9	547.1	738.7	1,004.2	1,368.2	1,880.6	35.8%
Others	183.8	244.5	327.2	440.4	595.5	812.3	34.6%
Total	1,884.8	2,423.1	3,135.0	4,078.1	5,357.1	7,090.6	30.3%

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 144 APAC: BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

APPLICATION	2018	2019	2020	2021
Residential	55.9	95.4	78.7	97.6
Commercial	98.6	151.5	127.6	157.5
Utility	759.2	1,201.0	989.5	1,220.7
Total	913.8	1,447.9	1,195.9	1,475.8

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 145 APAC: BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

Application	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	122.9	155.2	200.6	270.0	351.3	465.3	30.5%
Commercial	196.0	246.9	312.7	401.5	522.6	680.6	28.3%
Utility	1,565.9	2,021.0	2,621.7	3,406.6	4,483.3	5,944.8	30.6%
Total	1,884.8	2,423.1	3,135.0	4,078.1	5,357.1	7,090.6	30.3%

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

South Korea and China witnessed the highest annual installations globally in 2021. Besides, Australia has several utility and residential lithium-ion and flow battery installations for energy storage. Tesla announced the development of the world's largest lithium-ion large-scale battery energy storage system in South Australia for utility applications. These developments attribute to the high growth of the region through the forecast period.

13.5 REST OF THE WORLD (ROW)

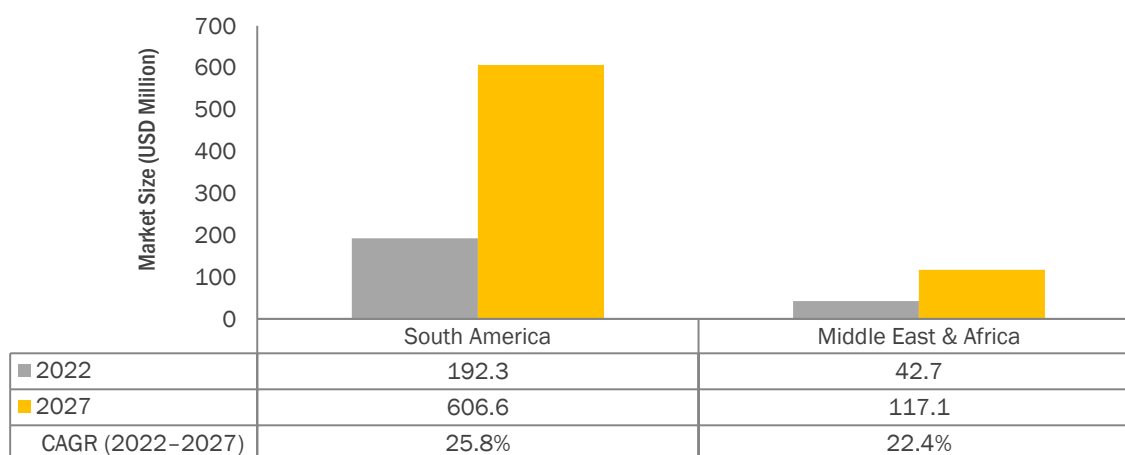
The scope of the market in RoW comprises South America and the Middle East & Africa. The rising adoption of battery energy storage systems in utilities in this region is contributing to the growth of the battery energy storage system market. Due to the harsh weather conditions, the application of a battery energy storage system is important to ensure the storage of energy along with continuous power supply in the residential and commercial sectors.

13.5.1 IMPACT OF COVID-19 ON MARKET IN ROW

Like other countries worldwide, the countries across the Middle East and Africa were equally affected by the COVID-19 pandemic. There were lockdowns and shutdowns across countries in the initial phases of the pandemic. This catapulted the issues pertaining to labor shortages and production facility shutdowns, leading to reduced revenue across businesses. Major application segments across the Middle East such as manufacturing, mining, and oil & gas, etc. were affected by workforce restrictions and disruptions in equipment supply, which resulted in the reduction of energy demand, adversely affecting the energy storage industry. There were increasing delays in projects, which led to businesses shifting toward other modes of energy storage systems such as PHS and CAES, etc. affecting the demand for battery energy storage systems. Economic uncertainty and mass layoffs curtailed consumer spending, thereby adversely affecting the energy demand.

However, post-pandemic, the demand increased, and the market recovered owing to the deployment of battery energy storage in numerous projects in the utility sector, which, in turn, supported the recovery of the market.

FIGURE 60 SOUTH AMERICA TO DOMINATE BATTERY ENERGY STORAGE SYSTEM MARKET DURING FORECAST PERIOD



Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

13.5.2 SOUTH AMERICA

13.5.2.1 Increasing government investments in diversifying power generation mix

South America comprises countries such as Brazil, Argentina, Venezuela, and Colombia. The South America (i.e., electrical energy storage South America) and Intersolar South America are the special exhibitions held in South America for energy storage technologies. These exhibitions cover the entire value chain of innovative battery and energy storage technologies, from components and production to specific user applications. The exhibition includes investors, utilities, installers, manufacturers, and project developers from all over the world related to energy storage. These exhibitions fuel the innovations and advancements of solutions in the battery energy storage system market in South America.

Brazil is the largest and the most developed country in the region. It is one of the fastest-growing economies, and its chemical, iron and steel, cement, petroleum processing, automobiles, and paper & pulp industries contribute to a significant portion of its GDP. According to the Ministry of Mines and Energy, Brazil's installed power generation capacity is expected to reach 212.5 GW by 2024, with an increase of 174.74 GW from 2020. The Ministry of Mines and Energy also stated that an investment of USD 531 billion is required by 2024 for the modernization of power infrastructure in the country. The government is investing in thermal power generation, solar, biomass, and offshore wind power generation to mitigate the risk of power shortages during severe droughts. It has set a target to deploy 3.5 GW of solar PV by 2023 and 7 GW by 2024. Brazil also has some of the largest manufacturing plants for electronics in South America. The government's investments in diversifying the power generation mix of the country are driving the Brazilian battery energy storage market.

Similarly, in June 2021, Venezuela announced the installation of the first photovoltaic energy storage system in the country, which is built in the Sombrero II substation in Guarico state to support the broadband transport network and thus guarantee uninterrupted energy in the event of any failure of the grid. Additionally, in September 2021, Stem, Inc., a global leader in Artificial Intelligence (AI) technology for energy storage, in partnership with Copec, which is one of South America's largest energy companies, announced the completion of their first energy storage system project in South America. Furthermore, in July 2021, Canadian Solar Inc., one of the world's largest solar technology and renewable energy companies, announced the acquisition of the contract for the first utility-scale battery energy storage project in Colombia, which would have an energy storage capacity of 45 MW/45 MWh.

13.5.3 MIDDLE EAST & AFRICA

13.5.3.1 Growing development and deployment of battery energy storage systems

The Middle East comprises Saudi Arabia, the UAE, and Kuwait. These countries are known for harsh weather conditions. The storing of energy is important in this region, and the battery energy storage system plays an important role in energy storage. The UAE aims to generate 25% of its electricity from renewables by 2030, and Kuwait is determined to produce 15% of power demand from renewable energy sources by 2030, which will further increase the installation of energy storage systems. In December 2021, the Middle East Energy Transition highlights showed that no oil-powered or gas-fueled power station projects were commissioned in the first half of 2021, however, during the same period USD 2.8 billion worth of renewable energy project contracts were awarded in the region. For instance, the Al Dhafra Solar Project is a 2 GW PV independent project, which will be in the UAE. 60% of the project will be owned by two Abu Dhabi-based government-held electricity companies, Abu Dhabi National Energy Company (TAQA) and Masdar, and the rest 40% of the project will be owned by a Chinese company Jinko Power in collaboration with the French electricity company EDF Renewables. In 2021, the technology company Huawei announced a partnership with Chinese engineering and construction company SEPCO111 to develop the energy storage system of 1300 MWh capacity as a part of the Red Sea Project, which will be integrated

with a 400 MW solar PV plant that is being developed by ACWA Power in Saudi Arabia. Also, in 2021, Dubai Electricity and Water Authority (DEWA), as a part of its revolution to diversify and enhance the energy storage technologies, inaugurated a pilot project for energy storage at the Mohammed bin Rashid Al Maktoum Solar Park. The project has a power capacity of 1.21 MW and an energy storage capacity of 8.61 MWh. The project uses Tesla's lithium-ion battery storage solution and is projected to have a life span of 10 years. The deployment of battery energy storage systems in the Middle East is growing with the manufacturing facilities of the systems in this region. For instance, in May 2020, German manufacturing equipment provider Schmid Group and Nusaned Investment, a subsidiary of the Kingdom's oil giant Saudi Aramco, have joined forces to develop a 3 GWh redox flow battery production facility in Saudi Arabia. The production facility started the manufacturing activities in 2021 and manufactured redox flow storage systems for utility-scale renewables projects, mining sites, telecom towers, remote cities, and off-grid locations.

Africa experiences the hottest climate and largest sunlight duration that provides excess solar energy. The battery energy storage system stores the excess PV energy and produces electricity at night. According to the report "Energy Storage Trends and Opportunities in Emerging Markets" by International Finance Corporation (IFC) and Energy Sector Management Assistance Program (ESMAP) (organized by World Bank Group), the growth of the Middle East and North Africa markets is driven mainly by growing urban population as well as expected additions to renewable energy capacity. According to the International Energy Agency's Africa Energy Outlook, Africa has 13% of the world's population, but accounts for only 4% of the world's energy demand, leaving a gap that corresponds to around 600 million people with no access to electricity. To overcome this issue, the deployment of battery energy storage system projects is growing with increased funding. For instance, in November 2021, the African Development Bank granted loans worth USD 57.67 million to South Africa's public electricity utility, Eskom, to be used for the development cost large-scale battery energy storage system projects. The loan has been approved by the bank to assist in the establishment of 200 MW of four-hour duration battery storage (800 MWh) at seven different sites in South Africa. The African Development Bank will be a concessional loan, which is a part of the USD 5.4 billion multi-donor trust fund known as the Clean Technology Fund. Additionally, in June 2021, JinkoSolar, a solar PV module manufacturer, announced that it will deliver a 1.2 MWh battery energy storage system for an undisclosed customer in West Africa. In 2020, Eskom's integrated report prioritized the distributed battery energy storage system projects, which involved the development of storage systems with 360 MW capacity at various Eskom distribution sites in several provinces of Africa. Also, in 2021, the Umoyilanga Energy battery energy storage system project was announced to be set up in South Africa. The project is due to be commissioned in 2022 and will have an energy storage capacity of 75,000 KW. Another project due to be commissioned in 2022 is the ACWA Power DAO battery energy storage system with an energy storage capacity of 150,000 KW. Furthermore, in July 2020, the state-owned electricity utility of South Africa passed a tender for the deployment of a battery energy storage system of minimum size and capacity of 80 MW/320 MWh. The utility has received financial backing from the African Development Bank, World Bank, and New Development Bank. The funds received will be used directly for the deployment of BESS.

TABLE 146 ROW: BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION, 2018–2021 (USD MILLION)

Region	2018	2019	2020	2021
Middle East & Africa	26.2	39.3	30.7	35.7
South America	106.4	163.5	131.2	156.8
Total	132.6	202.8	161.9	192.5

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 147 ROW: BATTERY ENERGY STORAGE SYSTEM MARKET, BY REGION, 2022–2027 (USD MILLION)

Region	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Middle East & Africa	42.7	51.3	62.3	76.7	94.3	117.1	22.4%
South America	192.3	237.7	296.4	374.3	474.6	606.6	25.8%
Total	234.9	289.0	358.7	451.0	568.9	723.6	25.2%

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 148 ROW: BATTERY ENERGY STORAGE SYSTEM MARKET, BY BATTERY TYPE, 2018–2021 (USD MILLION)

BATTERY TYPE	2018	2019	2020	2021
Lithium-ion Batteries	68.7	103.4	82.2	96.9
Lead-acid Batteries	23.0	33.8	25.7	29.0
Flow Batteries	29.0	46.5	38.4	47.4
Others	12.0	19.0	15.7	19.2
Total	132.6	202.8	161.9	192.5

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 149 ROW: BATTERY ENERGY STORAGE SYSTEM MARKET, BY BATTERY TYPE, 2022–2027 (USD MILLION)

BATTERY TYPE	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Lithium-ion Batteries	116.6	141.1	172.1	212.8	266.2	333.9	23.4%
Lead-acid Batteries	33.7	39.5	46.8	56.0	66.6	80.3	18.9%
Flow Batteries	60.4	77.5	100.3	130.7	169.1	221.3	29.7%
Others	24.2	30.8	39.5	51.4	66.3	86.7	29.1%
Total	234.9	289.0	358.7	451.0	568.9	723.6	25.2%

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 150 ROW: BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2018–2021 (USD MILLION)

APPLICATION	2018	2019	2020	2021
Residential	22.1	36.5	29.4	35.5
Commercial	40.6	59.5	48.0	56.8
Utility	69.9	106.8	84.5	100.3
Total	132.6	202.8	161.9	192.5

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

TABLE 151 ROW: BATTERY ENERGY STORAGE SYSTEM MARKET, BY APPLICATION, 2022–2027 (USD MILLION)

APPLICATION	2022	2023	2024	2025	2026	2027	CAGR (2022–2027)
Residential	43.5	53.6	67.4	88.4	112.7	146.0	27.4%
Commercial	67.6	81.6	98.9	121.7	152.0	190.0	22.9%
Utility	123.8	153.8	192.3	240.9	304.2	387.7	25.6%
Total	234.9	289.0	358.7	451.0	568.9	723.6	25.2%

Source: Press Releases, International Battery Forum, Energy Storage and Stationary Battery Journals and Magazines, Investor Relation Presentations, Expert Interviews, and MarketsandMarkets Analysis

South America has emerged as a manufacturing hub and foreign investment destination. The availability of skilled laborers at an economical cost has made South America a preferred destination for manufacturing companies to establish their manufacturing units. In 2020, the Brazilian energy company, Cemig undertook the implementation of a project involving the establishment of three energy storage systems with a combined capacity of 1.58 MWh in the Uberlandia municipality in the Triangulo Mineiro area in the west of Minas Gerais state in Brazil, thereby creating opportunities for the growth of the battery energy storage system market.

The Middle East & Africa have witnessed a significant economic turnaround due to the rapid growth in GDP, an increase in trade and investments, and strong demand for domestic goods worldwide. The manufacturers see Africa as a prominent market for battery energy storage systems. South Africa has attracted several flow battery providers to install flow battery systems for several applications such as telecom, industrial, and utilities. Invinity Energy Systems, Redflow Limited, Primus Power, and UniEnergy Technologies are the prominent providers of flow batteries.

14 COMPETITIVE LANDSCAPE

14.1 INTRODUCTION

This section of the report provides an overview of the prevailing competitive scenario in the battery energy storage system market. The market share/ranking analysis was carried out based on the revenues generated by key players from various business segments serving the said market. A benchmarking of the strategies adopted by key players in the aforementioned market, involving an analysis of strategic developments such as product launches and developments, partnerships, contracts, and agreements from February 2020 to January 2022, was also carried out.

TABLE 152 KEY DEVELOPMENTS IN BATTERY ENERGY STORAGE SYSTEM MARKET (2020–2022)

KEY PLAYERS	PRODUCT TYPE	STRATEGIC DEVELOPMENTS/PARTNER SHIPS	MERGERS & ACQUISITIONS/ PRODUCT LAUNCHES	REGION
SAMSUNG SDI Co., Ltd.	- Utility-Commercial Solution - UPS-UES Solution - Residential Solution - Base Transceiver Station Solution - Prismatic Lithium-Ion Battery Cell - Prismatic Cell - Polymer Cell - Mini Cell - Cylindrical Cell	The company launched a new battery brand, PRiMX, which gives consumers quality, performance, and proven advantage; the products allow consumers to conveniently use the company's independently developed technology. The brand has been trademarked in Korea and Europe and will soon be registered in the US.	-	Europe contributed 34% of its total revenue.
LG Energy Solutions Co., Ltd.	- Automotive Battery - Electric Vehicle Battery Solution - IT & New Application Battery - RESU 3.3 (48V) - RESU 6.5 (48V) - RESU 10 (48V) - RESU 13 (48V) - RESU 7H (400V) - RESU 10H (400V) - High-Energy Module - High-Energy Rack	-	LG Energy Solutions Co., Ltd. has signed a Memorandum of Understanding (MoU) with Siemens AG for collaboration in the field of battery manufacturing, particularly the digitization of the production process. Through this strategic collaboration, LG Energy Solutions Co., Ltd. implemented smart battery manufacturing processes at its factories with advanced techniques.	Europe contributed 56% of its total revenue.

	▪ M4863P3S ▪ M48126P3S ▪ M7830P7B ▪ M4864P6B ▪ M48128P6B		
Panasonic Corporation	▪ Smart Storage Battery Systems ▪ EverVolt Battery Storage	Panasonic signed a new agreement with Span.IO, Inc. (US) to create a residential energy storage system. This system is likely to be integrated into Panasonic's EverVolt, paired with Span smart panel, creating a new level of control over their battery backup power with intuitive energy management for homes.	The company announced a robust solar energy portfolio of Total Home Energy Solution offerings. This system offers a complete solution of solar energy storage and conversion. The system includes solar panels, built-in micro-inverters, and battery storage systems. Japan contributed 46% of its total revenue.
Siemens AG	▪ Siestorage ▪ BlueVault ▪ Fluence's Cube Designed Battery Energy Storage System ▪ Advancion 5 ▪ SunFlex	-	Fluence (US), a firm formed with the joint venture of Siemens and AEG Corporation, unveiled its sixth-generation energy storage technology stack combining factory-built hardware, advanced software, and data-driven intelligence. Fluence has been selected for launching 2,300 MWh of projects from Enel, Siemens, and LS Power. The technology stack creates the foundation for three purpose-built systems configured for renewable, grid and commercial, and industrial applications. Europe, CIS, Africa, and the Middle East contributed 50% of its total revenue.
Tesla, Inc.	▪ PowerWall ▪ PowerPack ▪ Megapack	-	Tesla launched the new lowest-cost solar panels for residential applications in the US. Such product launches can boost the demand for energy storage products. The US contributed 45% of its total revenue.

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

In this chapter, strategies of the various battery energy storage system manufacturers have been assessed along with the choices they have made:

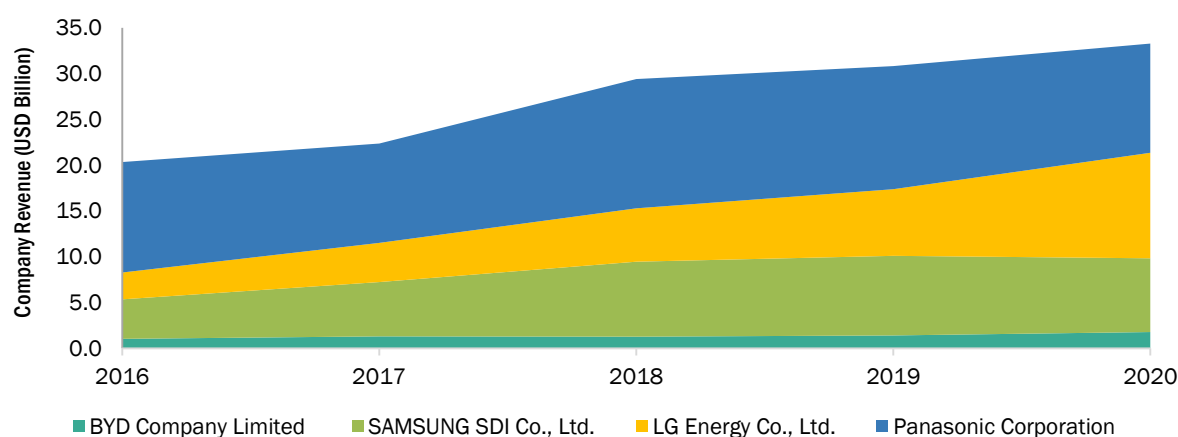
- **Product portfolio:** is their portfolio comprehensive, covering most solutions, or are they focused on a handful of solutions?
- **Regional focus:** Are they well-diversified globally or focused on a few key regions? This has been assessed based on where they source their revenues from and where their major client base exists.
- **Manufacturing footprint:** Are they centralized in terms of manufacturing footprint, having the bulk of their plants in one or two countries/regions? Or are they decentralized in terms of manufacturing locations with plants closer to demand centers?
- **Inorganic play/partnerships:** Are they primarily deploying organic means to grow, or are they leveraging partnerships (whether technology-based or go-to-market-based) to grow their revenues? Have they been active on the M&A front or investing in startups, etc.?

Considering the above factors, we have concluded that various companies are deploying different strategies to win in the market under discussion. Panasonic Corporation, SAMSUNG SDI Co., Ltd., LG Energy Solutions Co., Ltd. Siemens AG, and Tesla Inc. have a more diversified regional strategy to gain significant revenues from North America, Europe, and APAC. In general, the five companies are all strong in their home region and are exploring geographic diversification alternatives to grow their businesses.

14.1.1 REVENUE ANALYSIS OF TOP COMPANIES

The market has several major players, with many of them offering their products at competitive prices. The revenue analysis of the top companies in the market has been given below.

FIGURE 61 BATTERY ENERGY STORAGE SYSTEM MARKET: REVENUE ANALYSIS (2020)



Note: For BYD Company Limited, SAMSUNG SDI Co., Ltd., and LG Energy Co., Ltd., the revenues of the segments providing battery energy storage systems were considered. For Panasonic Corporation, the segment providing lithium-ion batteries was considered.

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

14.2 MARKET SHARE ANALYSIS OF TOP PLAYERS, 2021

This section provides a detailed market share analysis of the top players who provide batteries in the battery energy storage system market. These players account for a significant market share and have been relying on their strong R&D capabilities, a balanced portfolio of products and services, as well as their presence in mature and emerging markets to increase their market shares.

Players have been evaluated based on their geographic presence, product and service offerings, current market shares, consumer perception index, product launches and developments, and service agreements and contracts in the market. The following table depicts the top players who provide batteries in the battery energy storage system market in 2021.

TABLE 153 BATTERY ENERGY STORAGE SYSTEM MARKET: DEGREE OF COMPETITION

DEGREE OF COMPETITION	COMPETITIVE
Total Market Share of Top Players	25–35%
BYD Company Limited	08–11%
SAMSUNG SDI Co., Ltd.	07–09%
LG Energy Solutions Co., Ltd.	06–08%
Panasonic Corporation	04–07%

DEGREE OF COMPETITION IS DEFINED AS BELOW:

Fragmented: When top 5 players have a total market share of <25%

Competitive: When top 5 players have 25–50% market share

Consolidated: When top 5 players have a total market share of >50%

Source: Annual Reports, SEC Filings, Press Releases, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

The major providers of batteries to the battery energy storage system market are BYD Company Limited, SAMSUNG SDI Co., Ltd., LG Energy Solution Co., Ltd., and Panasonic Corporation. The battery energy storage system market is competitive, as the major players cumulatively accounted for 25–35% market share in 2021. The market share analysis for the major players has been calculated based on the number of battery energy storage system projects as well as inputs from the primary interviews. There might be instances when a company can get several major projects to provide their batteries to battery energy storage system providers.

In the market mentioned above, a few other companies are working toward strengthening their market position by developing innovative technologies. These companies include Panasonic Corporation, Hitachi Ltd, General Electric, Siemens AG, Tesla, Inc., and ABB.

BYD Company Limited

BYD Company Limited (BYD) is a top high-tech enterprise in China and a pioneer in battery technology. The company offers its products under three main segments: Rechargeable Batteries and Photovoltaic; Mobile Handset Components, Assembly Service and Other Products; and Automobiles and Related Products, and Other Products. Under its Rechargeable Batteries and Photovoltaic business, the company provides iron, nickel, and lithium-ion battery products, including energy storage stations and iron battery packs, which are widely used in electronic tools, mobile phones, and other portable electronic devices, energy storage products, photovoltaic products, and electric vehicles. BYD also offers large-scale energy storage systems, distributed energy storage systems, and micro-grid systems.

SAMSUNG SDI Co., Ltd.

Samsung SDI Co., Ltd. is a leading manufacturer of rechargeable batteries used in the IT industry, automobiles, and ESS. The company leads technologically in the battery manufacturing market on the foundation of change and innovation. It operates through two business segments—Electronic Materials and Energy Solutions. The major product lines under the Energy Solutions business segment include automotive batteries, small-sized li-ion batteries, and ESS (Energy Storage Systems). In 2010, Samsung SDI started the lithium battery ESS business based on small-sized lithium-ion rechargeable battery technology. ESS integrated with these batteries are used for the latest mobile devices and electric cars. The company offers ESS from kWh to MWh for the residential, utilities, commercial, UPS, and base transceiver station applications. Samsung ESS has a long-lasting service life. They are safe to use and provide excellent performance.

LG Energy Solutions Co., Ltd.

LG Chem is a chemical company that provides advanced materials and business and energy solutions. The company operates through segments such as Petrochemicals, Life Sciences, Energy Solutions, Advanced Materials, and Common and Others. In December 2020, the company split off its Energy Solution business and formed LG Energy Solutions Co., Ltd. Energy Solutions Co., Ltd. is involved mainly in the manufacture and supply of batteries ranging from IT and new application batteries for laptops, mobile phones, and computers to automotive batteries for vehicles and energy storage systems (ESS). The company offers ESS to improve energy efficiency by enhancing the quality of renewable energy that stabilizes power supply systems. An ESS integrates a battery cell, a battery pack, a battery module, and a battery rack system. ESS is used in electric grids, homes, and UPS. Various ESS available in the market for residential, commercial, and UPS applications include RESU10H, RESU 7H, RESU 3.3, M48126P3S, M48128P6B, and others.

Siemens AG

Siemens, formerly known as Siemens & Halske, is a Germany-based electronics and electrical engineering company that operates in electrification, automation, and digitalization. The company is a leading conglomerate pioneer in electrical engineering. With its vast product portfolio, solutions, and services, Siemens covers almost the entire energy value chain—from power generation and transmission to storage. Siemens operates through four reportable segments, namely, Digital Industries, Smart Infrastructure, Mobility, Siemens Healthineers, Siemens Financial Services, and Portfolio Companies. In the battery energy storage system market, the company offers Siemens Energy Storage, also known as SIESTORAGE, a modular energy storage system with state-of-the-art lithium-ion battery technology. SIESTORAGE helps optimize energy performance during various stages, ranging from generation to consumption, by stabilizing frequency and voltage and balancing supply and demand. It is an intelligent grid solution that provides flexible energy with unmatched versatility and optimizes grid connection.

Tesla, Inc.

Tesla, Inc. (formerly Tesla Motors) designs, develops, manufactures, and sells high-performance EVs and energy products. The company operates through wholly owned subsidiaries in North America, Europe, and Asia. The key operating segments of the company are Automotive and Energy Generation and Storage. The Energy Generation and Storage segment offers batteries, power electronics, and energy storage products. Energy storage products provided by the company serve a wide range of applications, from homes to large grid-scale projects. Powerpack and Powerwall are two devices offered by Tesla in the Energy Generation and Storage segment. Powerwall is a 14-kWh rechargeable lithium-ion battery designed to store energy in homes or small commercial facilities. Residential and commercial users use the energy stored in the system for self-consumption during solar power generation, and the backup power is stored. The Powerwall battery integrated into an inverter and battery packs monitors energy level in real time, alerts the operator in severe weather conditions, and functions with almost no noise.

14.3 COMPANY EVALUATION QUADRANT, 2021

This section maps 15 players in the battery energy storage system market. These players were rated based on two major factors—market share/rank and strength of product portfolio.

Factors such as the geographic footprint, effectiveness of growth strategies, and vision alignment of the companies were considered for rating companies on the “market share/rank” parameter. Similarly, factors such as the focus on product innovations and the breadth of product offerings were considered for rating companies on the strength of their product portfolio.

14.3.1 STAR

Star players generally receive high scores for most of the evaluation criteria. They have a strong product portfolio, a robust market presence, and effective business strategies. These are the leading players in terms of developments such as product launches, innovative technologies, and the adoption of strategic growth plans. These players have a broad portfolio of products for customers worldwide. Such players primarily focus on acquiring the leading market position through strong financial capabilities and well-established brand equity.

14.3.2 EMERGING LEADER

Emerging leaders demonstrate substantial product innovations than their competitors. They have a highly focused product portfolio. However, they do not have powerful growth strategies for their overall business. Emerging leaders can make profits even if there is an increased demand for innovative battery energy storage systems offered by established players. These players hold a small share of the market; however, they have innovative product and service portfolios.

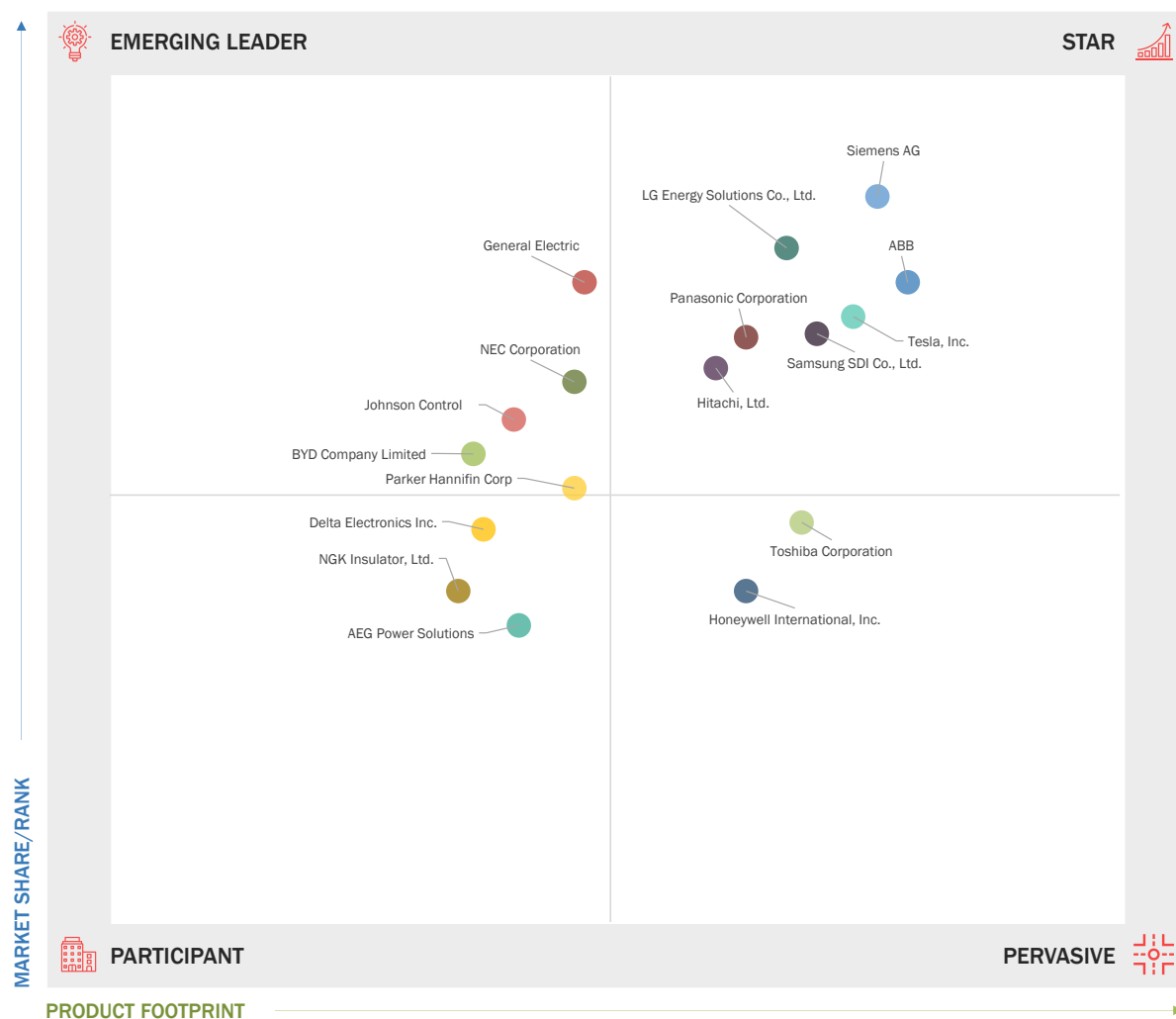
14.3.3 PERVASIVE

Pervasive players are established vendors with robust business strategies. However, they have a limited product portfolio, generally focusing on a specific type of technology related to a particular product. These companies largely depend on their competitive R&D activities. Such players have established business channels and necessary resources but less competitive products than star players.

14.3.4 PARTICIPANT

Participant companies have niche product offerings and have started gaining a position in the market. Such players have weaker business strategies than other established vendors. Participants can either be new entrants or old establishments in the market and may require some extra time to gain significant traction in the market.

FIGURE 62 BATTERY ENERGY STORAGE SYSTEM MARKET (GLOBAL): COMPANY EVALUATION QUADRANT, 2021



Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

14.4 SMALL AND MEDIUM-SIZED ENTERPRISES (SME) EVALUATION QUADRANT, 2021

The SME Evaluation Quadrant provides information about the players that offer battery energy storage systems on a reasonably smaller scale compared with the large market players. Vendor evaluations are based on two broad categories: strength of product portfolio and market share/rank.

14.4.1 PROGRESSIVE COMPANY

Progressive companies offer strong product portfolios in the battery energy storage system market. These vendors have been marking their presence in the market by providing customized storage systems as per customer requirements and adopting effective growth strategies to deliver advanced product offerings.

14.4.2 RESPONSIVE COMPANY

Responsive companies offer innovative solutions for several applications requiring battery energy storage systems. These companies mainly focus on strengthening their product portfolio and have a robust potential to build strong business strategies to expand their businesses and stay at par with progressive companies. These vendors offer customized battery energy storage systems to fulfill the specific requirements of their customers.

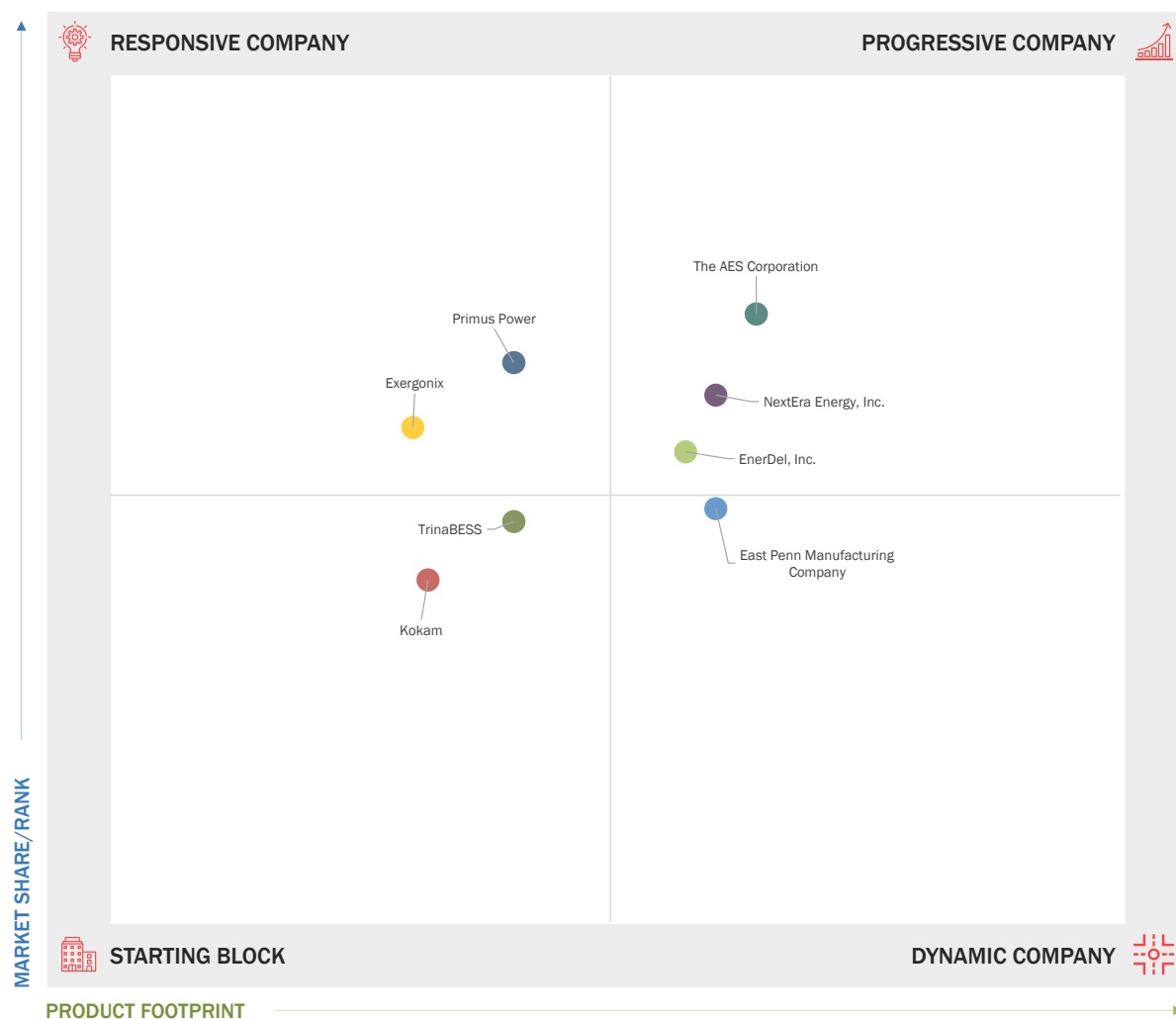
14.4.3 DYNAMIC COMPANY

Dynamic companies offer innovative solutions and have an extensive network of channel partners and resellers to increase the deployment of their solutions across a multitude of vertical markets. These companies have been generating consistent revenue growth in the battery energy storage system market and implementing organic and inorganic strategies to prosper.

14.4.4 STARTING BLOCK

Compared with other established vendors, starting blocks do not have strong business strategies. These are new entrants in the market and, hence, would require some time to gain significant traction in the market.

FIGURE 63 BATTERY ENERGY STORAGE SYSTEM MARKET (GLOBAL): SME EVALUATION QUADRANT, 2021



Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

14.4.5 BATTERY ENERGY STORAGE SYSTEM MARKET: STARTUP MATRIX

TABLE 154 BATTERY ENERGY STORAGE SYSTEM MARKET: DETAILED LIST OF KEY STARTUPS/SMES

COMPANY NAME	CATEGORY	OWNERSHIP STATUS	HQ LOCATION	YEAR FOUNDED	EMPLOYEES	FINANCING STATUS	LATEST FUNDING ROUND	TOTAL FUNDING (USD MN)
Ambari	Battery Provider	Private	Marlborough, Massachusetts, US	2010	11-50	Venture Capital	Series E	211.4
ESS, Inc.	Manufacturer	Private	Wilsonville, Oregon, US	2011	201-500	-	Post-IPO Equity	297.0
Fluence	Manufacturer	Private	Arlington, Virginia, US	2018	501-1000	Venture Capital	Private Equity	680.8
Sonnen GmbH	Manufacturer	Private	Wildpoldsried, Bavaria, Germany	2010	501-1000	-	Series E	169.2
Eos Energy Enterprises, Inc	Battery Provider	Public	Edison, New Jersey, US	2008	201-500	-	Post-IPO Debt	432.0
Highview Power	Manufacturer	Private	London, England, UK	2005	11-50	-	Corporate Round	61.7
Powervault	Manufacturer	Private	London, England, UK	2012	11-50	Venture Capital	Equity Crowdfunding	5.4
Solid Power Inc.	Battery Provider	Public	Louisville, Colorado, US	2011	101-250	-	Post-IPO Equity	381.5
Redflow Limited	Battery Provider	Public	Brisbane, Queensland, Australia	2005	11-50	Corporation	Post-IPO Equity	2.0
Ion Storage Systems	Battery Provider	Private	Beltsville, Maryland, US	2015	11-50	Venture Capital	Series A	52.5
Form Energy, Inc.	Manufacturer	Private	Somerville, Massachusetts	2017	51-200	Venture Capital	Series D	368.8
Zenobe Energy	Manufacturer	Private	London, England, UK	2017	51-200	-	-	687.4
Electriq Power, Inc.	Manufacturer	Private	San Leandro, California, US	2014	11-50	Venture Capital	-	12.0
Hydrostor Inc.	Battery Provider	Private	Toronto, Ontario, Canada	2010	11-50	Private Equity	Venture-Series	297.0
Revterra	Manufacturer	Private	Houston, Texas, US	2017	2-10	-	-	223.5

TABLE 155 BATTERY ENERGY STORAGE SYSTEM MARKET: COMPETITIVE BENCHMARKING OF KEY STARTUPS/SMES

COMPANY NAME	BATTERY TYPE				APPLICATION			GEOGRAPHY				
	LITHIUM-ION BATTERIES	LEAD-ACID BATTERIES	FLOW BATTERIES	OTHERS	RESIDENTIAL	COMMERCIAL	UTILITIES	NORTH AMERICA	EUROPE	APAC	MIDDLE EAST & AFRICA	SOUTH AMERICA
Ambri				✓				✓				
ESS Inc.			✓			✓	✓	✓				
Fluence	✓						✓	✓	✓	✓		
Sonnen GmbH	✓				✓				✓			
Eos Energy Enterprises, Inc	✓				✓	✓	✓	✓				
Highview Power	✓						✓	✓	✓	✓		
Powervault	✓								✓			
Solid Power Inc.				✓				✓				
Redflow Limited			✓		✓	✓				✓		
Ion Storage Systems				✓			✓	✓				
Form Energy, Inc.	✓			✓		✓		✓				
Zenobe Energy									✓			
Electriq Power, Inc.						✓						
Hydrostor Inc.	✓							✓				
Revterra	✓							✓				

14.4.6 BATTERY ENERGY STORAGE SYSTEM MARKET: COMPANY FOOTPRINT

TABLE 156 COMPANY FOOTPRINT: BY APPLICATION

COMPANY	RESIDENTIAL	COMMERCIAL	UTILITY	TOTAL SCORE
ABB	Y	Y	Y	4.00
AEG Power Solutions	N	N	Y	1.33
BYD Company Limited	Y	Y	Y	4.00
Delta Electronics Inc.	Y	Y	Y	4.00
General Electric	Y	Y	N	2.67
Hitachi, Ltd.	Y	Y	Y	4.00
Honeywell International, Inc.	N	Y	N	1.33
LG Energy Solutions Co., Ltd.	Y	Y	Y	4.00
NEC Corporation	Y	Y	Y	4.00
NGK Insulator, Ltd.	Y	Y	Y	4.00
Panasonic Corporation	Y	N	N	1.33
Samsung SDI Co., Ltd.	Y	Y	Y	4.00
Siemens AB	N	N	Y	1.33
Tesla, Inc.	Y	Y	Y	4.00
Toshiba Corporation	Y	N	N	1.33
East Penn Manufacturing Company	Y	Y	N	2.67
EnerDel, Inc.	Y	Y	N	2.67
Exergonix	Y	Y	Y	4.00
Johnson Controls	N	Y	N	1.33
Kokam	N	N	N	0.00
NextEra Energy, Inc.	Y	N	Y	2.67
Parker Hannifin Corp	N	N	N	0.00
Primus Power	N	Y	Y	2.67
The AES Corporation	N	Y	Y	2.67
TrinaBESS	Y	Y	Y	4.00

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

TABLE 157 COMPANY FOOTPRINT: BY REGION

COMPANY	NORTH AMERICA	EUROPE	APAC	ROW	TOTAL SCORE
ABB	Y	Y	Y	Y	4.00
AEG Power Solutions	N	Y	Y	Y	3.00
BYD Company Limited	Y	Y	Y	N	3.00
Delta Electronics Inc.	Y	Y	Y	Y	4.00
General Electric	Y	Y	Y	Y	4.00
Hitachi, Ltd.	Y	Y	Y	Y	4.00
Honeywell International, Inc.	Y	Y	Y	Y	4.00
LG Energy Solutions Co., Ltd.	Y	Y	Y	N	3.00
NEC Corporation	Y	Y	Y	Y	4.00
NGK Insulator, Ltd.	Y	Y	Y	N	3.00
Panasonic Corporation	Y	Y	Y	Y	4.00
Samsung SDI Co., Ltd.	Y	Y	Y	N	3.00
Siemens AB	Y	Y	Y	Y	4.00
Tesla, Inc.	Y	Y	Y	N	3.00
Toshiba Corporation	Y	Y	Y	Y	4.00
East Penn Manufacturing Company	Y	N	Y	N	2.00
EnerDel, Inc.	Y	N	N	N	1.00
Exergonix	Y	N	Y	N	2.00
Johnson Controls	Y	Y	Y	Y	4.00
Kokam	N	N	Y	N	1.00
NextEra Energy, Inc.	Y	N	N	N	1.00
Parker Hannifin Corp	Y	Y	Y	Y	4.00
Primus Power	Y	Y	Y	N	3.00
The AES Corporation	Y	Y	Y	Y	4.00
TrinaBESS	Y	Y	Y	N	3.00

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

TABLE 158 COMPANY FOOTPRINT

COMPANY	APPLICATION FOOTPRINT	REGION FOOTPRINT	TOTAL SCORE
ABB	4.00	4.00	4.00
AEG Power Solutions	1.33	3.00	2.17
BYD Company Limited	4.00	3.00	3.50
Delta Electronics Inc.	4.00	4.00	4.00
General Electric	2.67	4.00	3.33
Hitachi, Ltd.	4.00	4.00	4.00
Honeywell International, Inc.	1.33	4.00	2.67
LG Energy Solutions Co., Ltd.	4.00	3.00	3.50
NEC Corporation	4.00	4.00	4.00
NGK Insulator, Ltd.	4.00	3.00	3.50
Panasonic Corporation	1.33	4.00	2.67
Samsung SDI Co., Ltd.	4.00	3.00	3.50
Siemens AB	1.33	4.00	2.67
Tesla, Inc.	4.00	3.00	3.50
Toshiba Corporation	1.33	4.00	2.67
East Penn Manufacturing Company	2.67	2.00	2.33
EnerDel, Inc.	2.67	1.00	1.83
Exergonix	4.00	2.00	3.00
Johnson Controls	1.33	4.00	2.67
Kokam	0.00	1.00	0.50
NextEra Energy, Inc.	2.67	1.00	1.83
Parker Hannifin Corp	0.00	4.00	2.00
Primus Power	2.67	3.00	2.83
The AES Corporation	2.67	4.00	3.33
TrinaBESS	4.00	3.00	3.50

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

14.5 COMPETITIVE SCENARIO AND TRENDS

This segment studies the growth strategies adopted by leading players between February 2020 to January 2022 to expand their global presence and increase their market shares. Product launches and developments, partnerships, and agreements were the major strategies adopted by market players to achieve growth in the battery energy storage system market.

14.5.1 PRODUCT LAUNCHES

TABLE 159 BATTERY ENERGY STORAGE SYSTEM MARKET: PRODUCT LAUNCHES, FEBRUARY 2020–JANUARY 2022

MONTH & YEAR	COMPANY	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
January 2022	Toshiba Corporation	Li-ion Battery	20Ah-HP SCiB Rechargeable Battery	Toshiba Corporation launched and expanded its SCiB product offerings by including the 20AH-HP rechargeable battery, with wide applications ranging from automotive and industrial and in energy storage systems. The product features include long life, excellent compatibility, and efficient heat suppression.
December 2021	Samsung SDI Co., Ltd.	Battery	PRiMX	The company launched a new battery brand, PRiMX, which gives consumers quality, performance, and proven advantage, which means user convenience achieved with the company's independent developed technology. The brand has been trademarked in Korea and Europe and will soon be registered in the US.
November 2021	Panasonic Corporation	Battery Storage	EverVolt 2.0	The company announced a robust solar energy portfolio of Total Home Energy Solution offerings. This system offers a complete solution of solar energy storage and conversion. The system includes solar panels, built-in micro-inverters, and battery storage systems.
May 2021	Panasonic Corporation	Battery Storage	EverVolt Solar Modules series	Panasonic announced an expanded lineup of solar modules. The new modules feature higher efficiency and enhanced performance, even in high temperatures. This series of solar modules produce more power, has high conversion efficiency, and has a low degradation rate.
December 2021	Samsung SDI Co., Ltd.	Battery	PRiMX	The company launched a new battery brand, PRiMX, which gives consumers quality, performance, and proven advantage, which means user convenience achieved with the company's independent developed

				technology. The brand has been trademarked in Korea and Europe and will soon be registered in the US.
June 2020	Tesla, Inc.	Solar roof	Solar Panels	Tesla launched the new lowest-cost solar panels for residential applications in the US. Such product launches can boost the demand for energy storage products.
February 2020	General Electric Company	Energy Storage	7.4 MW battery	GE achieved a black start of a GE-heavy duty gas turbine using a 7.4 MW battery-based energy storage system in Louisiana, US.

Source: Press Releases, Annual Reports, News Releases and Business Articles, and MarketsandMarkets Analysis

14.5.2 DEALS

TABLE 160 BATTERY ENERGY STORAGE SYSTEM MARKET: DEALS, FEBRUARY 2020–DECEMBER 2021

MONTH & YEAR	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION	DEAL SIZE
December 2021	Partnership	Hitachi, Ltd.	SEV	SEV, the power company owned by the municipalities of the Faroe Islands, has partnered with Hitachi, Ltd. The latter would provide a battery energy storage system (BESS) to SEV to enable a transition of up to 100% of the renewable energy produced in the area by 2030. SEV has selected a BESS solution rated at 6MW/7.5MWh for the project integrating the 6.3MW Porkeri wind farm into the local grid of the southernmost island, Suðuroy.	NA
December 2021	Agreement	LG Energy Solutions Co., Ltd.	Siemens AG	LG Energy Solutions Co., Ltd. has signed a Memorandum of Understanding (MoU) with Siemens AG for collaboration in the field of battery manufacturing, particularly the digitization of the production process. Through this strategic collaboration, LG Energy Solutions Co., Ltd. implemented smart battery manufacturing processes at its factories with advanced techniques.	NA
June 2021	Partnership	Honeywell International, Inc.	Alturus	Alturus-Honeywell collaborated to offer an energy storage-as-a-service model where customers save energy and money by reducing their electricity consumption during peak demand periods, increasing the model's operational efficiency, and decreasing emissions. Honeywell provides BESS and software platforms and enables smart dispatch of the batteries' stored energy to capture revenue streams and maximize electricity cost savings.	NA

April 2021	Partnership	ABB	Stadler	ABB received a contract from Stadler, a Switzerland-based rail vehicle manufacturer, to provide energy-efficient traction converters and high-power lithium-ion battery-based onboard energy storage systems. The traction converters and energy storage systems are installed on 55 bi-mode electric multiple units (BEMUs).	NA
January 2021	Partnership	ABB	Shanghai Bailonggang Wastewater Treatment Plant	ABB completed the delivery of medium- and low-voltage digital solutions for the expansion of the Shanghai Bailonggang Wastewater Treatment Plant in China. The company's electrical distribution solutions also provide data-driven energy management and predictive maintenance functions.	NA
July 2020	Acquisition	Hitachi, Ltd.	ABB Ltd.	Hitachi acquired nearly 80% stake of ABB's power grids business since the power grids market has a significant potential for growth with the increasing utilization of renewable energy, such as solar power, for powering BESS.	NA
June 2020	Agreement	Panasonic Corporation	Span.IO, Inc.	Panasonic signed a new agreement with Span.IO, Inc. (US) to create a residential energy storage system. This system is likely to be integrated into Panasonic's EverVolt, paired with Span smart panel, creating a new level of control over their battery backup power with intuitive energy management for the homes.	NA
April 2020	Partnership	Toshiba Corporation	Sumitomo Corp.	Sumitomo Corp., a Japan-based trader, collaborated with Toshungiba to build a battery power storage system in Ohio to balance the electricity grid serving Northeastern US. The company deploys Toshiba lithium-ion batteries in its systems.	NA
March 2020	Divestment	ABB	FIMER SpA	ABB divested its solar inverter business to an Italy-based company called FIMER SpA. The divestment enabled ABB's Electrification segment to improve its focus on priority growth markets and enhance business opportunities for the solar inverter business under FIMER.	NA
February 2020	Contract	AEG Power Solutions	swb Erzeugung AG & Co. KG	AEG PS received a contract from swb Erzeugung AG & Co. KG (swb), a Bremen-based German utility, to provide its Convert SC Flex for deployment in battery energy storage systems designed for Erzeugung's customers under the Automotive sector.	NA

Source: Press Releases, Annual Reports, Business Articles, and MarketsandMarkets Analysis

15 COMPANY PROFILES

15.1 KEY PLAYERS

15.1.1 ABB

15.1.1.1 Business overview

ABB was founded through a merger between ASEA (Sweden) and BBC Brown Boveri (Switzerland). It is fully engaged in power and automation technologies. The company operates through four business segments—Electrification, Motion, Industrial Automation, and Robotics and Discrete Automation.

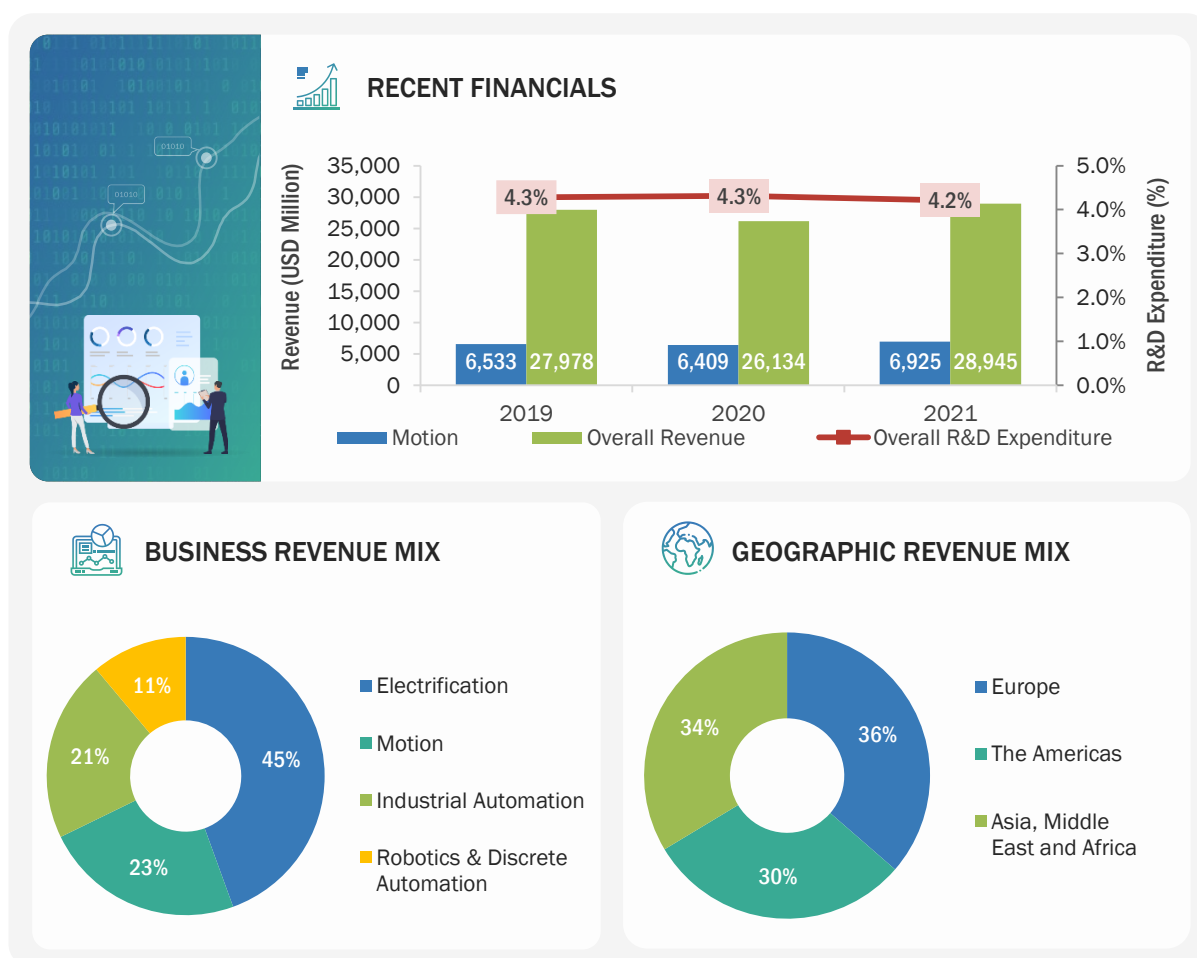
The Motion segment designs and manufactures generators, electric motors, and drives and provides integrated digital powertrain solutions and services. The Motion segment comprises six divisions, including Motors & Generators, System Drives, Drive Products, Mechanical Power Transmission, Traction Systems, and Service. Under the Traction Systems division, the company provides battery energy storage systems, converters and motors, and auxiliary converts. Energy Storage System (ESS) that store electrical energy and transform it into electrochemical energy by using battery systems and then supply it for a longer duration are known as battery energy storage systems. Battery energy storage systems play a vital role in power generation, transmission, distribution, and end-user applications (utilities and residential). The use of battery energy storage systems increases the efficiency of the power grid at all levels, from power generation to its supply to end users. The battery energy storage system is beneficial for frequency regulation, load leveling, and improving power quality.

ABB also offers the essential components of battery energy storage systems, such as power grids, converters for battery energy storage, DC products for battery protection units (BPUs), and power conversion systems (PCS); AC products for power conversion systems (PCS) and grid-connected applications; auxiliary products for battery management systems (BMS); and other products for the installation of battery energy storage systems. ABB also offers the ultra-capacitor technology based ESS, which has ten times higher power density than batteries. Furthermore, ABB offers a renewable energy accumulator and conversion technology (REACT) residential energy storage system. REACT-3.6/4.6-TL is an innovative photovoltaic (PV) inverter integrated with a lithium-ion battery that stores the energy produced during the daytime by the installed PV systems (installed by homeowners and landlords). The extra battery modules used in REACT help increase their energy storage capacity, thereby offering efficiency and flexibility to residential users. Also, in July 2020, ABB completed the divestment of 80.1% of the power grids business to Hitachi Ltd., which includes several energy storage products, such as PQstorl—an energy storage inverter, PQplus—a battery energy storage system, and e-mesh PowerStore—a microgrid and energy storage solution.

TABLE 161 **ABB: BUSINESS OVERVIEW**

Founding Year	1988
Headquarters Country	Switzerland
Headquarters City	Zurich
Ownership Type	Public

Source: Company Website

FIGURE 64 ABB: COMPANY SNAPSHOT

Note: The percentages for the pie chart calculations have been rounded off. Thus, there might be instances where the percentages in the pie chart may not match the total revenue of the company.

In the Recent Financials section, the R&D expenditure has been considered as a percentage of the total revenue.

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

15.1.1.2 Products offered

TABLE 162 ABB: PRODUCT OFFERING

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
Battery Energy Storage Solutions	The battery energy storage solutions increase the operational reliability of the grid. They help reduce operating costs and utility bills.	▪ REACT 2 – Photovoltaic Energy Storage Solutions ▪ EssPro Grid – Battery Energy Storage System ▪ EssPro PCS Energy Storage – Power Conversion System (PCS) ▪ Battery Energy Storage Systems ▪ REACT 2 String Inverter ▪ Power GRIDS ▪ Energy Storage Modules ▪ Community Energy Storage (CES) Modules ▪ Distributed Energy Storage (DES) Modules ▪ Compact Energy Storage Modules (CESM) ▪ Photovoltaic Energy Storage Solutions ▪ Converters for Battery Energy Storage	Residential Commercial Utilities

Source: Company Website

15.1.1.3 Recent developments

15.1.1.3.1 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION	DEAL SIZE
April 2021	Partnership	ABB	Stadler	ABB received a contract from Stadler, a Switzerland-based rail vehicle manufacturer, to provide energy-efficient traction converters and high-power lithium-ion battery-based onboard energy storage systems. The traction converters and energy storage systems are installed on 55 bi-mode electric multiple units (BEMUs).	NA
January 2021	Partnership	ABB	Shanghai Bailonggang Wastewater Treatment Plant	ABB completed the delivery of medium- and low-voltage digital solutions for the expansion of the Shanghai Bailonggang Wastewater Treatment Plant in China. The company's electrical distribution solutions also provide data-driven energy management and predictive maintenance functions.	NA

Source: Company Website

15.1.1.3.2 Others

March 2020	Divestment	ABB	FIMER SpA	ABB divested its solar inverter business to an Italy-based company called FIMER SpA. The divestment enabled ABB's Electrification segment to improve its focus on priority growth markets and enhance business opportunities for the solar inverter business under FIMER.	NA
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Source: Company Website

15.1.2 AEG POWER SOLUTIONS

15.1.2.1 Business overview

AEG Power Solutions is a global provider of power supply systems and solutions for all types of critical and demanding applications such as energy storage in power generation plants and UPS systems in data and IT solutions. AEG Power Solutions (AEG PS) Group is the sole subsidiary of 3W Power S.A. It is one of the leading providers of UPS systems and power electronic solutions for commercial, industrial, renewable, and distributed energy markets. The company ensures continuous power availability through its comprehensive portfolio of power supply systems, AC and DC UPS, rectifier systems, battery chargers, and fully customized UPS systems. The company also offers products such as DC systems, inverter systems, DC modules, converters and renewable integration systems, monitoring solutions, and battery energy storage systems. The company provides battery energy storage systems for on-grid and off-grid solutions.

AEG PS offers different outdoor solutions and engineers complete battery energy storage systems built with certified partners. The company also provides core components of battery energy storage systems. For example, ConvertSC FLEX is a power converter offered by AEG PS, used in battery energy storage systems. The company's battery energy storage solutions range from single-unit housing to metal-sheet iso-container housing up to six units. A complete battery energy storage solution fixed in a container includes batteries, a power conversion system, a medium-voltage transformer, switchgear and all-auxiliary components, as well as the energy management system. The company also provides a hybrid storage solution. In the hybrid energy storage system engineered by AEG PS, energy is stored in a battery system and an electrical heating system; these systems are connected to the power converter. This hybrid storage system significantly reduces the cost of primary-frequency power operation. The company operates in Europe, APAC, and the Middle East and has direct sales and service offices at 14 locations worldwide.

TABLE 163 AEG POWER SOLUTIONS: BUSINESS OVERVIEW

Founding Year	1887
Headquarters Country	Zwanenburg
Headquarters City	Netherlands
Ownership Type	Private

Source: Company Website

15.1.2.2 Products offered

TABLE 164 AEG POWER SOLUTIONS: PRODUCT OFFERING

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
Energy Storage & Renewable Integration	The company offers solutions to connect the energy storage system to any grid for several types of applications or to support entire off-grid power generation. The systems manage peak demand.	- Off-grid Solutions - Grid Voltage Regulation - Battery Energy Storage Systems - Micro-grid Solutions - Solar Inverters - Hybrid Storage System	Power Generation Data and IT Systems

Source: Company Website

15.1.2.3 Recent developments

15.1.2.3.1 Deals

TABLE 165 AEG POWER SOLUTIONS: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION	DEAL SIZE
February 2020	Contract	AEG Power Solutions	swb Erzeugung AG & Co. KG	AEG PS received a contract from swb Erzeugung AG & Co. KG (swb), a Bremen-based German utility, to provide its Convert SC Flex for deployment in battery energy storage systems designed for Erzeugung's customers under the Automotive sector.	NA

Source: Company Website and Press Releases

15.1.3 BYD COMPANY LIMITED

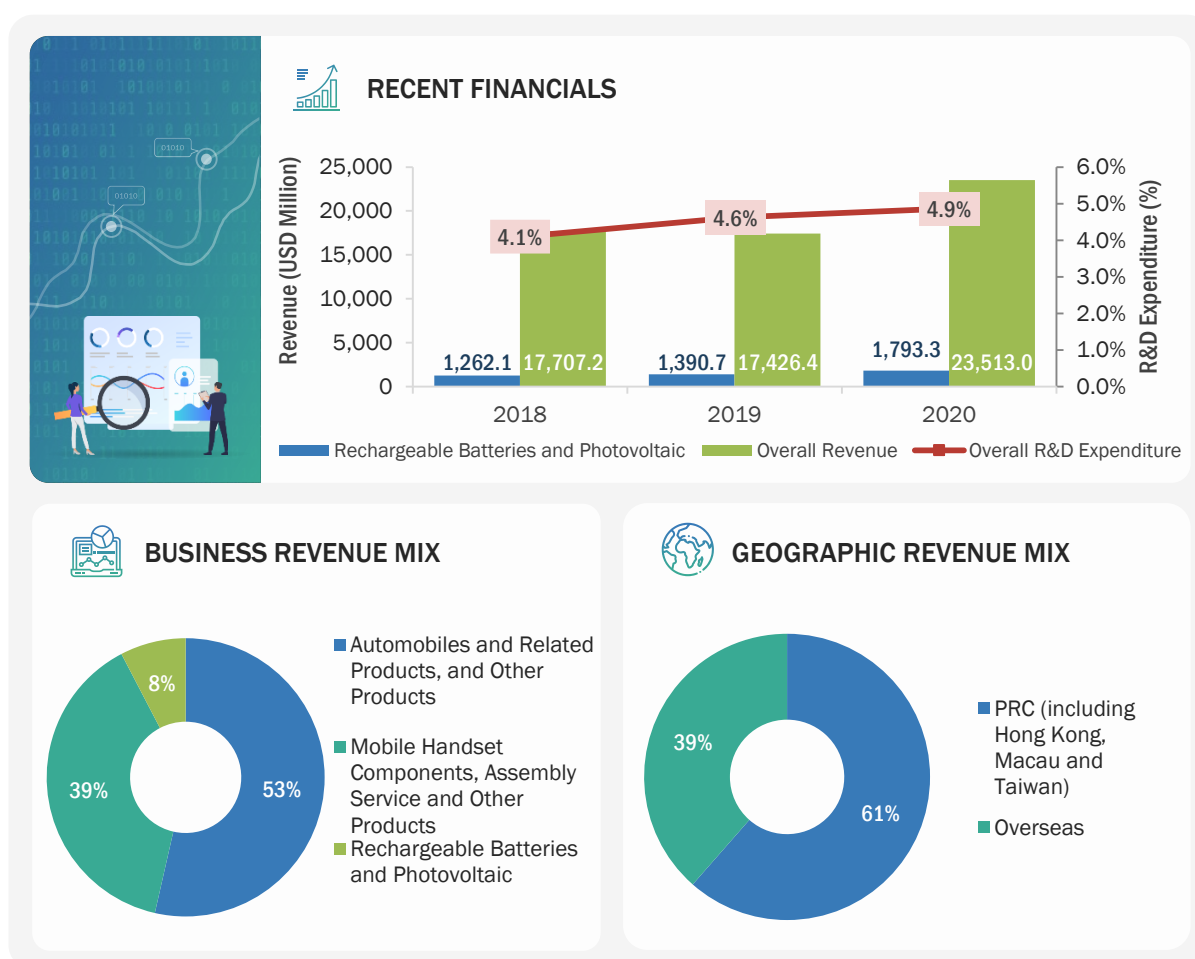
15.1.3.1 Business overview

BYD Company Limited (BYD) is a top high-tech enterprise in China and a pioneer in battery technology. The company offers its products under three main segments: Rechargeable Batteries and Photovoltaic; Mobile Handset Components, Assembly Service and Other Products; and Automobiles and Related Products, and Other Products. Under its Rechargeable Batteries and Photovoltaic business, the company provides iron, nickel, and lithium-ion battery products, including energy storage stations and iron battery packs, which are widely used in electronic tools, mobile phones, and other portable electronic devices, energy storage products, photovoltaic products, and electric vehicles. BYD also offers large-scale energy storage systems, distributed energy storage systems, and micro-grid systems. The company's products are deployed in buses, battery-electric automobiles, and medium- and heavy-duty trucks and forklifts. It also provides the SkyRail monorail system, solar power generation and energy storage systems, and consumer electronics. The company has two principal subsidiaries, BYD Automobile and BYD Electronic. BYD is a public company listed on the Stock Exchange of Hongkong as SEHK: 01211 and Shenzhen Stock Exchange as SZSE: 002594.

TABLE 166 BYD COMPANY LIMITED: BUSINESS OVERVIEW

Founding Year	1995
Headquarters Country	Shenzhen
Headquarters City	China
Ownership Type	Public

Source: Company Website

FIGURE 65 BYD COMPANY LIMITED: COMPANY SNAPSHOT

Note: The percentages for the pie chart calculations have been rounded off. Thus, there might be instances where the percentages in the pie chart may not match the total revenue of the company.

In the Recent Financials section, the R&D expenditure has been considered as a percentage of the total revenue.

The currency conversion rates used for 2018, 2019, and 2020 are RMB 1 = USD 0.14539, RMB 1 = USD 0.14310, and RMB 1 = USD 0.15321, respectively.

Values are represented up to one decimal to show minute differences.

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

15.1.3.2 Products offered

TABLE 167 BYD COMPANY LIMITED: PRODUCT OFFERINGS

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
Energy Storage System	The company offers customized energy storage system solutions for different customer applications. These solutions range from kW to GW sizes.	Utility ESS	Utilities
		Commercial ESS	Residential
		B-Box ESS	Commercial
		Mini ESS	Electronics

Source: Company Website

15.1.3.3 MnM view

15.1.3.3.1 Key strengths/right to win

BYD Company Limited is one of the leading providers of battery energy storage systems; these storage systems are offered under its significant segment, Rechargeable Batteries, and Photovoltaic Products. The company's major clients for rechargeable batteries include leading professional robot brands, such as iRobot and Ecovacs, as well as leading consumer electronics manufacturers, such as Dell and Samsung. The company invests heavily in R&D activities for designing innovative products to match the market needs and drive growth and profitability. In addition to this, the company recorded full-year organic sales growth of above 26% in 2020 compared with 2019.

15.1.3.3.2 Strategic choices made

The company's strategy includes the development and expansion of its product line to meet the rising needs of the customers. For the Rechargeable Batteries segment, the company applies new technologies to expand its customer base and increase the market share in related niches. BYD Company Limited relies on establishing a good reputation and brand image to strengthen the battery market domestically as well as globally.

15.1.3.3.3 Weaknesses and competitive threats

BYD Company Limited lacks a presence in regions such as the Middle East and Africa, which is expected to weaken its position in the battery energy storage system market. The company faces intense competition in the global battery energy storage system market as the market players have a more comprehensive product portfolio, whereas BYD is still expanding its product line. Additionally, the company will have to expand its presence to the untapped regions through strategic developments and differentiating its product line from its competitors to strengthen its position and remain competitive in the market.

15.1.4 DELTA ELECTRONICS INC.

15.1.4.1 Business overview

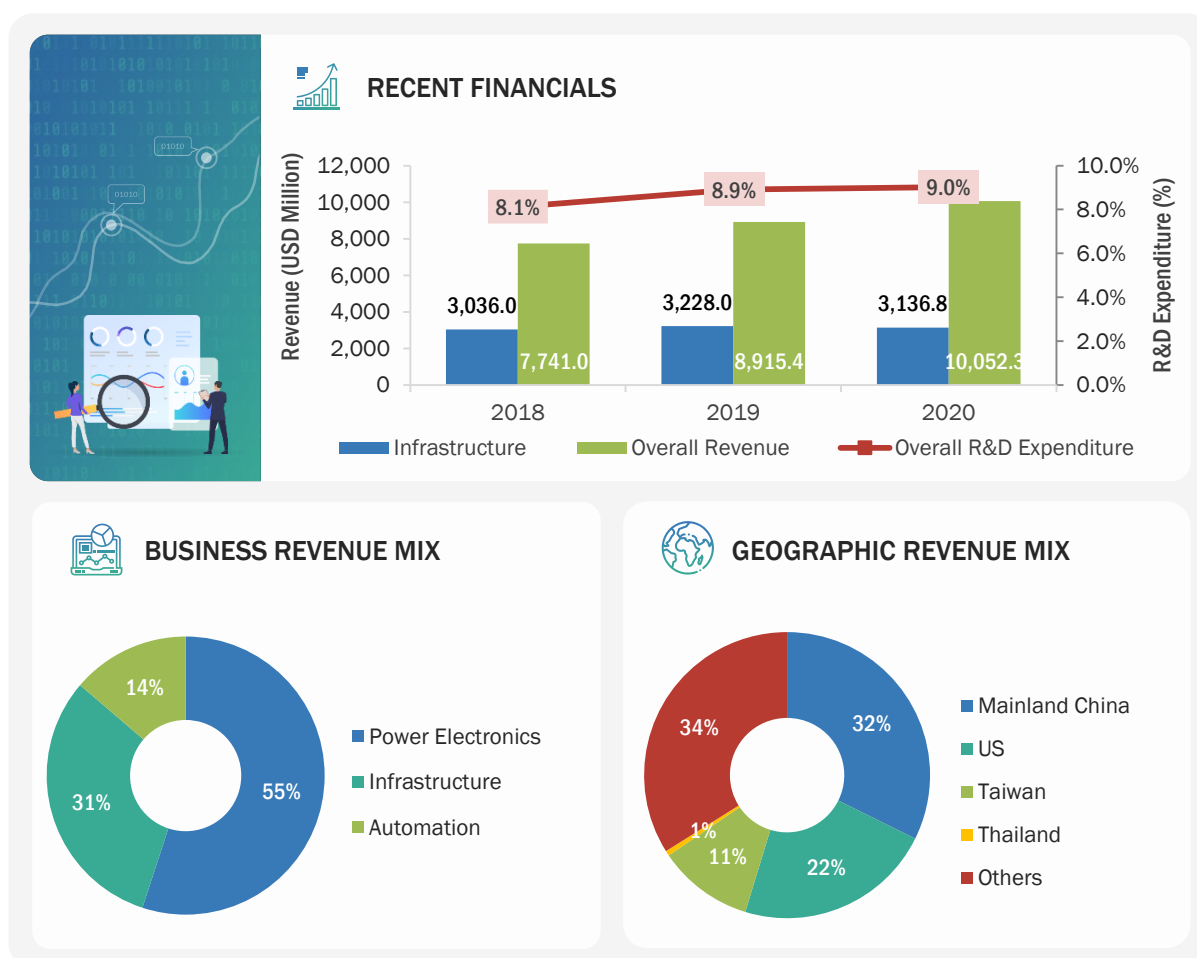
Delta Electronics, Inc. is one of the leading global power and thermal management solutions providers. It offers its products through business segments, namely, Power Electronics, Automation, and Infrastructure. The company provides energy storage solutions under the Infrastructure segment. Delta Electronics provides an energy storage system with high-efficiency power conditioning capabilities for demand management, power dispatch, renewable energy smoothing, etc. The ESS integrates bi-directional power conditioning and battery devices, site controllers, and a cloud management system to provide a comprehensive energy storage system for commercial, residential, and utility applications.

Through its broad product portfolio, the company caters to multiple verticals, such as communications, medical, IT, renewable energy, and power. It has a significant presence in the Americas, Asia, EMEA, and Oceania. As of December 2020, Delta Electronics Inc. had an employee base of 9,451. The company has 54 R&D centers, 105 offices, 31 plants, and subsidiaries worldwide.

TABLE 168 DELTA ELECTRONICS INC.: BUSINESS OVERVIEW

Founding Year	1971
Headquarters Country	Taiwan
Headquarters City	Taipei
Ownership Type	Public

Source: Company Website

FIGURE 66 DELTA ELECTRONICS, INC.: COMPANY SNAPSHOT

Note: The percentages for the pie chart calculations have been rounded off. Thus, there might be instances where the percentages in the pie chart may not match the total revenue of the company.

In the Recent Financials section, the R&D expenditure has been considered as a percentage of the total revenue.

The currency conversion rates used for 2018, 2019, and 2020 were NTD 1 = USD 0.03266, NTD 1 = US 0.03325, and NTD 1 = USD 0.03557, respectively.

Values are represented up to one decimal to show minute differences.

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

15.1.4.2 Products offered

TABLE 169 DELTA ELECTRONICS INC.: PRODUCT OFFERING

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
Energy Storage System	The BESS has high-efficiency power conditioning capabilities for demand management, power dispatch, renewable energy smoothing, etc. The ESS integrates bi-directional power conditioning and battery devices, site controllers, and a cloud management system to provide comprehensive energy storage.	Battery Energy Storage System	- Residential - Commercial - Utilities

Source: Company Website

15.1.5 GENERAL ELECTRIC

15.1.5.1 Business overview

General Electric is one of the world's largest multinational conglomerates and operates as an infrastructure and financial services company. The company is involved in developing and manufacturing products for the generation, transmission, distribution, control, and utilization of electricity. It develops and manufactures power generators, aircraft engines, water processors, medical imaging technology-based solutions, security solutions, business and consumer financing solutions, and industrial products. The company operates via five business segments—Power, Aviation, Healthcare, Renewable Energy, and Capital.

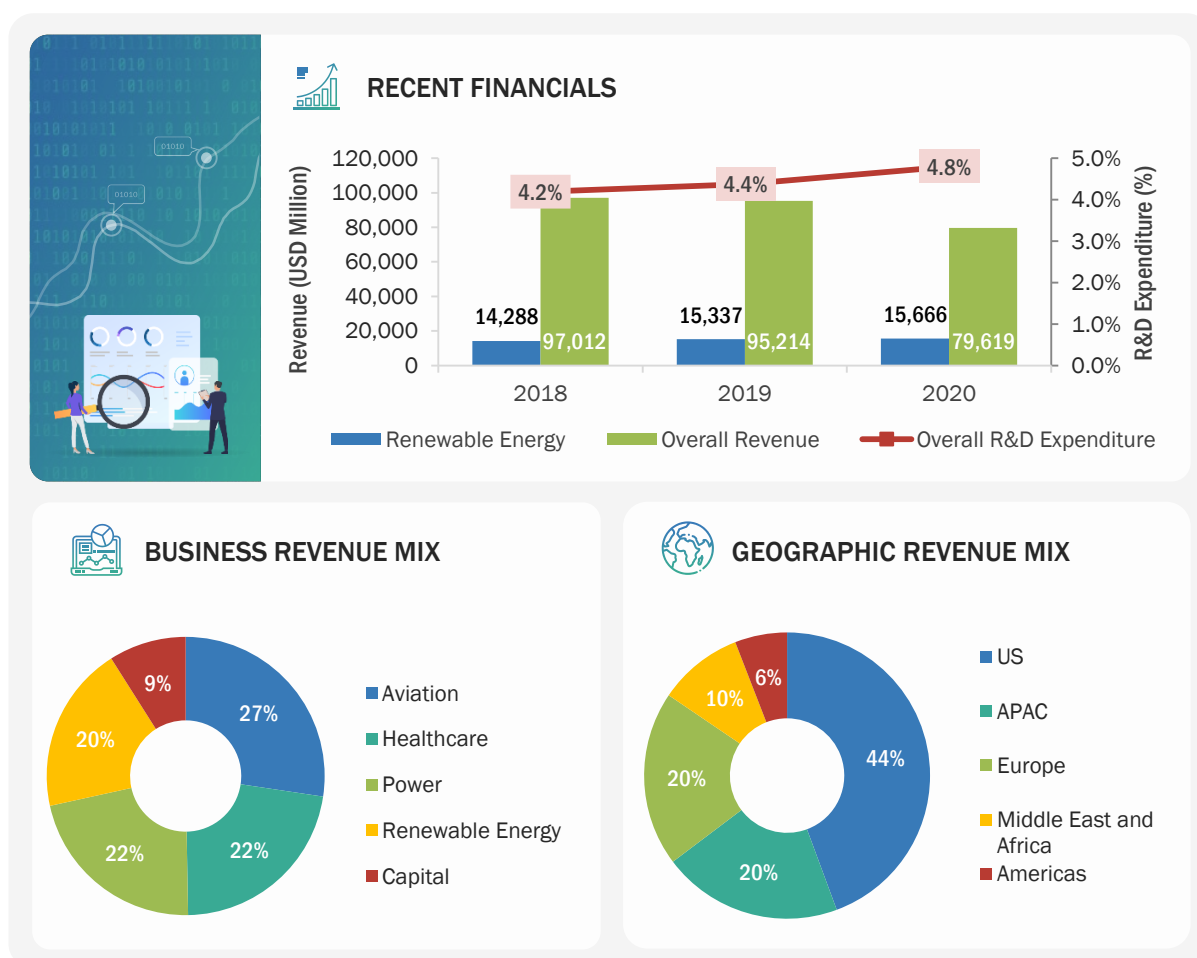
GE provides innovative solutions for renewable integration and grid stability through its battery energy storage systems. The Renewable Energy segment offers battery energy storage systems with one of the two configurations—a power configuration or an energy configuration—depending on the intended application. Battery energy storage systems are integrated with a suitable type of battery according to the intended application (e.g., lithium-ion, with or without hybrid advanced lead-acid batteries), providing advantages such as peak demand reduction, backup, PV self-consumption, and better power quality. The company's battery energy storage systems are available in three standardized, flexible packages—M, L, and XL—and are equipped with a power converter and real-time control software to efficiently match the energy supply to demand. The battery energy storage system by GE is an integrated turnkey product, which comprises a battery system, power electronics, substation control, a grid connection, and network management systems. GE's Reservoir is a flexible, compact energy storage solution that combines GE's advanced technologies and expertise in plant controls, battery management systems, and power electronics. The company also offers battery energy storage solutions for applications such as hybrid grid connection, hybrid renewables, and hybrid electric-gas turbines.

General Electric is listed on the New York Stock Exchange with the ticker symbol GE. The company has a geographical presence across Europe, Asia, the Americas, and the Middle East & Africa. The company serves in over 180 countries. Some of the company's significant subsidiaries are GE Aviation Service Operation LLP (Singapore), GE Energy Europe B.V. (Netherlands), GE Europe Holdings LLC (US), GE Infrastructure, Inc. (US), and GE Pacific Holding Pte. Ltd. (Singapore).

TABLE 170 GENERAL ELECTRIC: BUSINESS OVERVIEW

Founding Year	1878
Headquarters Country	US
Headquarters City	Boston
Ownership Type	Public

Source: Company Website

FIGURE 67 GENERAL ELECTRIC: COMPANY SNAPSHOT

Note: The percentages for the pie chart calculations have been rounded off. Thus, there might be instances where the percentages in the pie chart may not match the total revenue of the company.

In the Recent Financials section, the R&D expenditure has been considered as a percentage of the company's total revenue.

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

15.1.5.2 Products offered

TABLE 171 GENERAL ELECTRIC: PRODUCT OFFERING

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
Battery Energy Storage System	The company offers energy storage reservoirs to all energy sectors, such as wind, solar, hydro, hybrid, and grid solutions.	GE's Reservoir	Commercial Residential

Source: Company Website and Press Releases

15.1.5.3 Recent developments

15.1.5.3.1 Product launches and developments

TABLE 172 GENERAL ELECTRIC COMPANY: PRODUCT LAUNCHES AND DEVELOPMENTS

MONTH & YEAR	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
February 2020	General Electric Company	Energy Storage	7.4 MW battery	GE achieved a black start of a GE-heavy duty gas turbine using a 7.4 MW battery-based energy storage system in Louisiana, US.

Source: Company Website and Press Releases

15.1.6 HITACHI, LTD.

15.1.6.1 Business overview

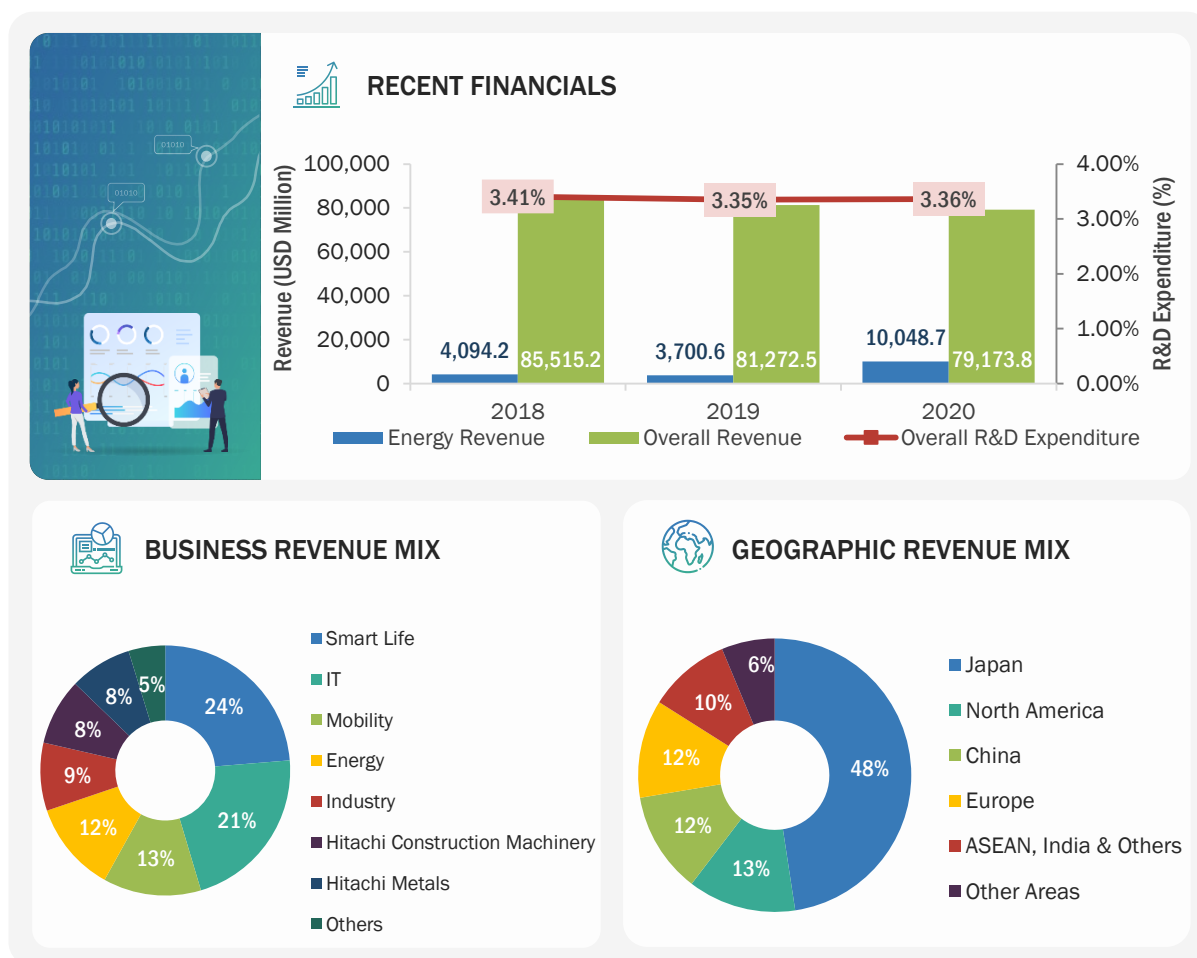
Hitachi, Ltd. is a Japan-based multinational conglomerate and parent company of the Hitachi Group. It has highly diversified operating business segments: IT, Energy, Industry, Mobility, Smart Life, Hitachi High-Tech, Hitachi Construction Machinery, Hitachi Metals, Hitachi Chemicals, and Others. Under the Energy business segment, the company provides power grid solutions, power generation systems, and energy solutions, among others. The company also provides energy storage systems under the Energy business segment. Hitachi developed container-type lithium-ion battery energy storage systems of 1 MW as an optimal solution for frequency regulation in ancillary services. These systems comprise two 500 kW power conditioning systems (PCS) in parallel lithium-ion battery sets with a capacity equivalent to 450 kWh, a controller, a data logger, an air conditioning system, and an optional automatic fire extinguisher. The service life of these storage systems is about 10 years, and they are used for consumer, industrial, and automotive applications. The company also developed a 1.5 MW hybrid energy storage system that combines high input/output, long-life lead-acid batteries with lithium-ion capacitors to suppress short-period voltage and frequency fluctuations.

Hitachi, Ltd. forms a part of the DKB Group of companies. The company has a presence in the Americas, Asia, Europe, the Middle East & Africa, and Oceania. Hitachi, Ltd. also has subsidiaries, namely Hitachi Vantara, Hitachi Metals, Hitachi Rail, Hitachi Astemo, and others. In July 2020, Hitachi, Ltd. acquired an 80.1% stake in ABB's power grid business and formed a new company, Hitachi ABB Power Grids Ltd. The acquisition aimed to expand Hitachi's Energy solution business worldwide.

TABLE 173 HITACHI, LTD.: BUSINESS OVERVIEW

Founding Year	1947
Headquarters City	Tokyo
Headquarters Country	Japan
Ownership Type	Public

Source: Company Website and Press Releases

FIGURE 68 HITACHI, LTD.: COMPANY SNAPSHOT

Note: The percentages for the pie chart calculations have been rounded off. Thus, there might be instances where the percentage in the pie may not match the total revenue of the company.

In the Recent Financials section, the R&D expenditure has been considered as a percentage of the total revenue.

In the financials, Hitachi High-Tech and Hitachi Chemicals data are unavailable in the annual report 2020.

The currency conversion rates used for 2018, 2019, and 2020 are YEN 1 = 0.00902 USD, YEN 1 = USD 0.00927, and YEN 1 = USD 0.00907, respectively.

Values are represented up to one decimal to show minute differences.

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

15.1.6.2 Products offered

TABLE 174 HITACHI, LTD.: PRODUCT OFFERINGS

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
Energy Storage Systems	Energy storage systems are designed to efficiently address the needs of the fast-growing energy storage market for behind-the-meter applications.	- Hybrid Energy Storage System - Container Type Energy Storage System (CrystEna) - and Energy Storage Solution	- Industrial - Commercial - Utilities
ABB Power Grid Products	Energy storage systems are designed to ensure grid reliability, flexibility, and renewable integration.	- Energy Storage Inverter PQstorl - Battery Energy Storage System PQpluS - e-mesh PowerStore-Microgrid and Energy Storage Solution	- Peak Shaving - Backup Power - Power Quality - Utility-scale - Applications such as Load Levelling, Frequency Response, Capacity Firming, and Integration of Renewables

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

15.1.6.3 Recent developments

15.1.6.3.1 Deals

TABLE 175 HITACHI, LTD.: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION	DEAL SIZE
December 2021	Partnership	Hitachi, Ltd.	SEV	SEV, the power company owned by the municipalities of the Faroe Islands, has partnered with Hitachi, Ltd. The latter would provide a battery energy storage system (BESS) to SEV to enable a transition of up to 100% of the renewable energy produced in the area by 2030. SEV has selected a BESS solution rated at 6MW/7.5MWh for the project integrating the 6.3MW Porkeri wind farm into the local grid of the southernmost island, Suðuroy.	NA
July 2020	Acquisition	Hitachi, Ltd.	ABB	Hitachi acquired nearly 80% stake of ABB's power grids business since the power grids market has a significant potential for growth with the increasing utilization of renewable energy, such as solar power, for powering BESS.	NA

Source: Company Website, Press Releases, and MarketsandMarkets Analysis

15.1.7 HONEYWELL INTERNATIONAL, INC.

15.1.7.1 Business overview

Honeywell International, Inc. is an industrial software company that offers industry-specific solutions for aerospace and automotive applications and related services. It operates through four segments: Aerospace, Honeywell Building Technologies, Performance Materials and Technologies, and Safety and Productivity Solutions. The company provides battery energy storage solutions to its customers under the Performance Materials and Technologies segment.

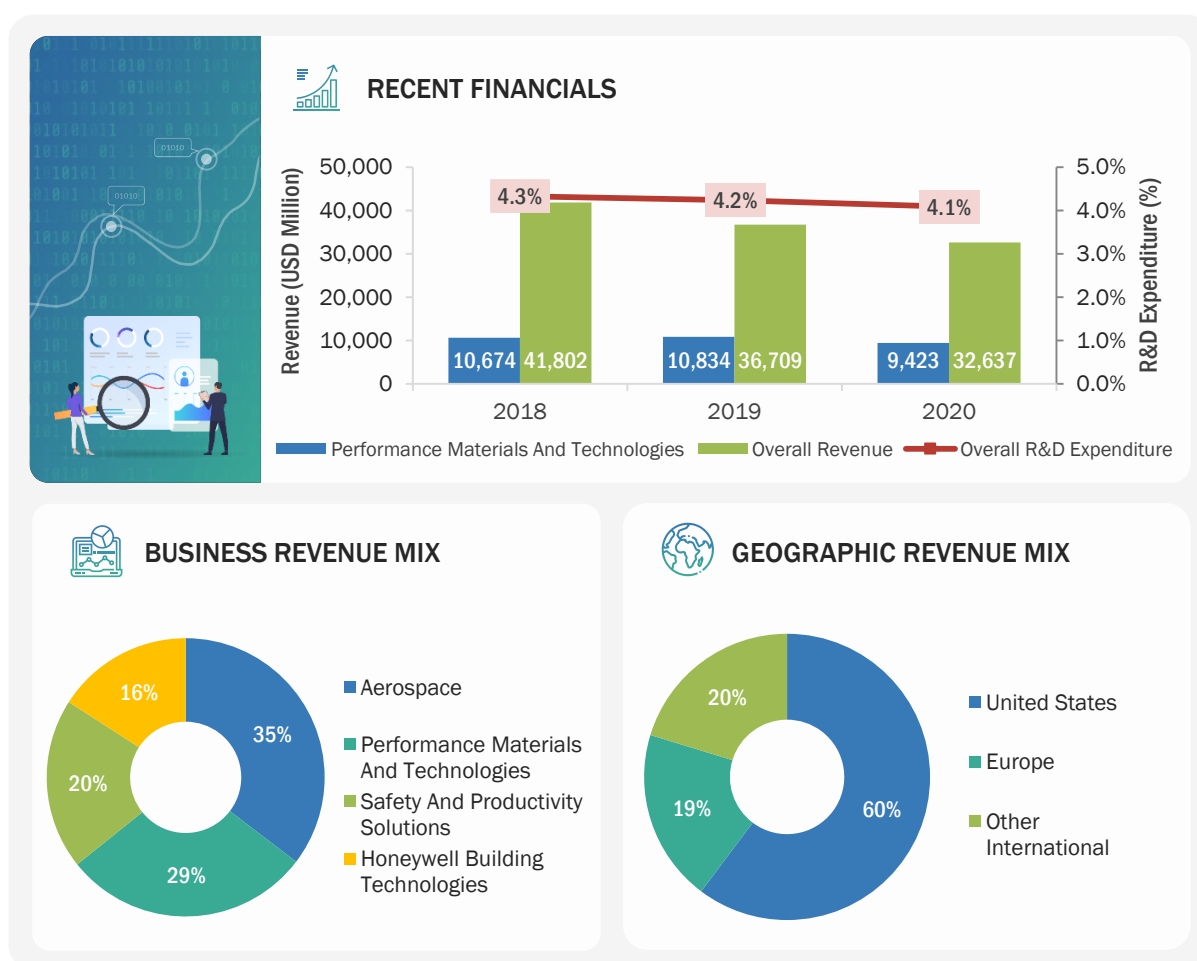
The BESS Platform is aimed at large commercial and industrial (C&I) users, independent power producers (IPPs), and utilities. Honeywell signed an agreement with NRStor Inc. (Canada) in which both will jointly develop and operate 300 megawatts (MW) of BTM battery energy storage systems (BESS) across the US and Canada starting in early 2020. NRStor Inc. develops energy storage projects for utilities, residential, commercial, and industrial consumers.

Honeywell International, Inc. is a publicly listed company on NASDAQ Stock Exchange with the ticker symbol HON. The company operates across various regions, including North America, Latin America, Europe, Asia Pacific, and the Middle East.

TABLE 176 HONEYWELL INTERNATIONAL, INC.: BUSINESS OVERVIEW

Founding Year	1975
Headquarters Country	US
Headquarters City	Washington
Ownership Type	Public

Source: Company Website

FIGURE 69 HONEYWELL INTERNATIONAL INC.: COMPANY SNAPSHOT

Note: The percentages for the pie chart calculations have been rounded off. Thus, there might be instances where the percentage in the pie may not match the total revenue of the company.

In the Recent Financials section, the R&D expenditure has been considered as a percentage of the total revenue

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

15.1.7.2 Products offered

TABLE 177 HONEYWELL INTERNATIONAL, INC: PRODUCT OFFERINGS

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
BTM Battery Energy Storage System	Through BTM battery energy storage systems (BESS), the company is focusing on helping plant owners and operators make the most of new, smarter technologies of energy storage systems and guiding them toward best practices for energy management.	Battery Storage	Commercial

Source: Company Website

15.1.7.3 Recent developments

15.1.7.3.1 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION	DEAL SIZE
June 2021	Partnership	Honeywell International, Inc.	Alturus	Alturus-Honeywell collaborated to offer an energy storage-as-a-service model where customers save energy and money by reducing their electricity consumption during peak demand periods, increasing the model's operational efficiency and decreasing emissions. Honeywell provides BESS and software platforms and enables smart dispatch of the batteries' stored energy to capture revenue streams and maximize electricity cost savings.	NA

Source: Company Website

15.1.8 LG ENERGY SOLUTIONS CO., LTD.

15.1.8.1 Business overview

LG Chem is a chemical company that provides advanced materials and business and energy solutions. The company operates through segments such as Petrochemicals, Life Sciences, Energy Solutions, Advanced Materials, and Common and Others. In December 2020, the company split off its Energy Solution business and formed LG Energy Solutions Co., Ltd. Energy Solutions Co., Ltd. is involved mainly in the manufacture and supply of batteries ranging from IT and new application batteries for laptops, mobile phones, and computers to automotive batteries for vehicles and energy storage systems (ESS). The company offers ESS to improve energy efficiency by enhancing the quality of renewable energy that stabilizes power supply systems. An ESS integrates a battery cell, a battery pack, a battery module, and a battery rack system. ESS is used in electric grids, homes, and UPS. Various ESS available in the market for residential, commercial, and UPS applications include RESU10H, RESU 7H, RESU 3.3, M48126P3S, M48128P6B, and others.

The company also provides mid- and large-sized rechargeable lithium-ion batteries for automotive applications. Electric vehicles (EVs) require automotive batteries with high stability, high output, and high energy density. Automotive batteries are also installed in hybrid vehicles (HEVs) and plug-in hybrid vehicles (PHEVs). The IT and new application batteries manufactured by the company include cylindrical, prismatic, and polymer models, which are provided in cells or battery packs. Automotive lithium-ion batteries are embedded in EVs and the EV charging infrastructure due to their high performance. The battery energy storage system supports the EV charging infrastructure by efficiently storing large capacity energy and handling energy fluctuations.

TABLE 178 LG ENERGY SOLUTIONS CO., LTD.: BUSINESS OVERVIEW

Founding Year	2020
Headquarters Country	South Korea
Headquarters City	Seoul
Ownership Type	Public

Source: Company Website

15.1.8.2 Products offered

TABLE 179 LG ENERGY SOLUTIONS CO., LTD.: PRODUCT OFFERING

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
Energy Solutions	The energy solutions offered by the company provide remote battery monitoring and easy handling.	- Automotive Battery - Electric Vehicle Battery Solution - IT & New Application Battery	- Residential - Commercial - Utilities
		RESU 3.3 (48V)	
		RESU 6.5 (48V)	
		RESU 10 (48V)	
		RESU 13 (48V)	
		RESU 7H (400V)	
		RESU 10H (400V)	
		High-Energy Module	

- High-Energy Rack
- M4863P3S
- M48126P3S
- M7830P7B
- M4864P6B
- M48128P6B

Source: Company Website

15.1.8.3 Recent developments

15.1.8.3.1 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION	DEAL SIZE
December 2021	Agreement	LG Energy Solutions Co., Ltd.	Siemens AG	LG Energy Solutions Co., Ltd. has signed a Memorandum of Understanding (MoU) with Siemens AG for collaboration in the field of battery manufacturing, particularly the digitization of the production process. Through this strategic collaboration, LG Energy Solutions Co., Ltd. implemented smart battery manufacturing processes at its factories with advanced techniques.	NA

Source: Company Website

15.1.8.4 MnM view

15.1.8.4.1 Key strengths/right to win

LG Energy Solutions Co., Ltd. is an established manufacturer of batteries for various applications. The company has a robust portfolio of high-quality energy storage systems for residential, telecom, utilities, and UPS applications, accounting for a significant share of the battery energy storage system market in 2021. The company aims to expand its product portfolio by launching innovative products and investing heavily in R&D activities. For instance, the R&D investment of the company increased from 4.6% in FY 2019 to 4.9% in FY 2020. Its major customers for energy storage systems are Hewlett-Packard Co. and General Motors Corp., among many others.

15.1.8.4.2 Strategic choices made

The company has adopted both organic and inorganic strategies to continue to grow in the market. The company focused on collaborations and product launches. For instance, in December 2021, LG Energy Solutions Co., Ltd. has signed an agreement with Siemens AG for collaboration in the field of battery manufacturing. In addition, the rising focus on adopting inorganic strategies, such as collaborations and partnerships, is expected to increase its market share and provide high-quality battery energy storage systems.

15.1.8.4.3 Weaknesses and competitive threats

The company is mainly focused on Korea and has weak geographical footprints in the Americas, Europe, and the Middle East & Africa. This could hamper its reach and growth in the long run, which is expected to weaken its position in the battery energy storage system market. Additionally, the company competes with various battery providers such as Panasonic (Japan), SAMSUNG SDI (South Korea), and BYD Company Limited (China), as well as with the battery energy storage system providers such as Siemens AG (Germany), Tesla (US), and ABB (Switzerland). The intense competition from established players is expected to impact the company.

15.1.9 NEC CORPORATION

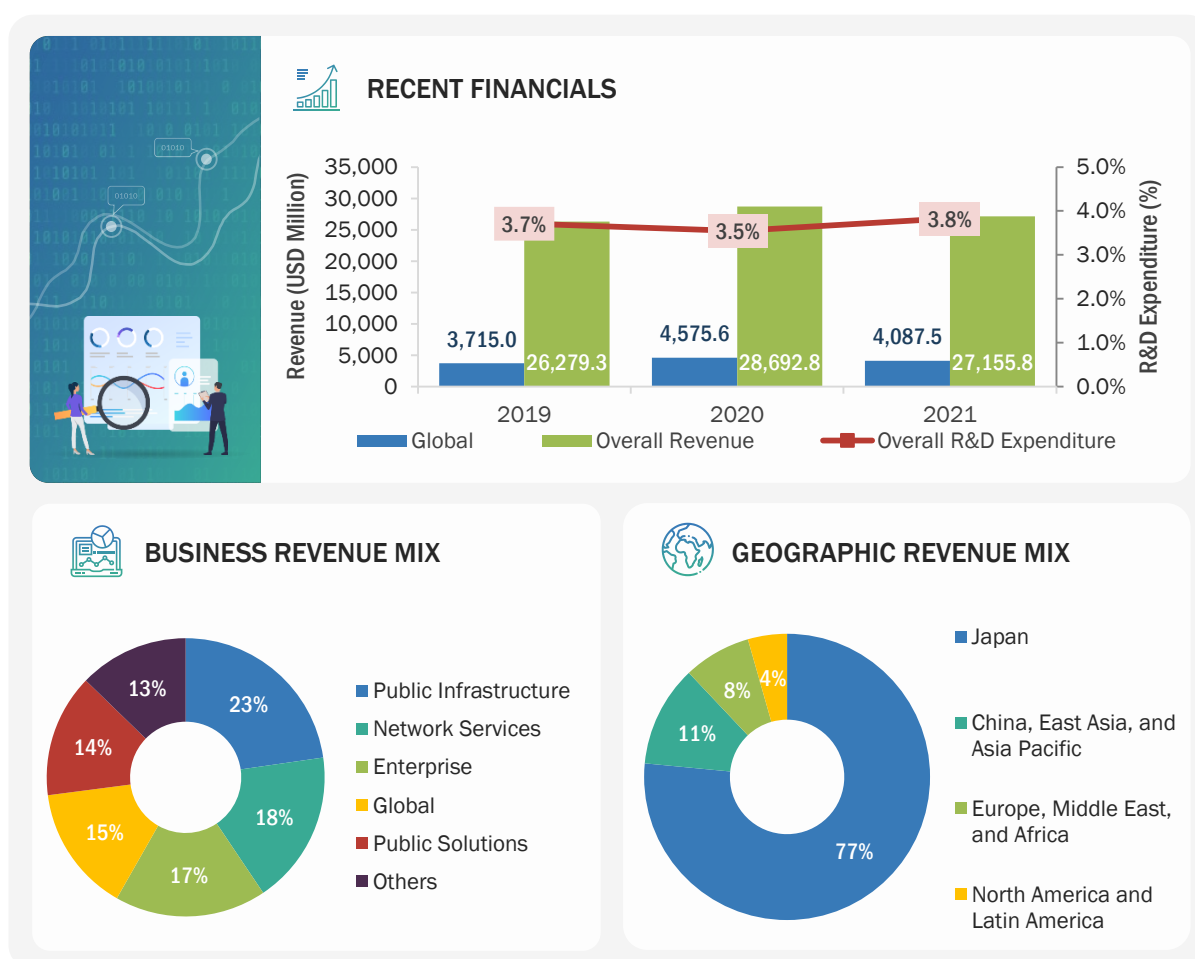
15.1.9.1 Business overview

NEC Corporation is a Japan-based multinational IT service and product provider. The company operates through six business segments, including Public Infrastructure, Network Services, Enterprise, Global, Public Solutions, and Others. Under its Global business segment, the company provides major products and services, including digital government, public safety, and digital finance; software services for service providers, which include OSS/BSS; system devices, including display and projectors and energy storage systems; and network infrastructure, including wireless backhaul and submarine systems. To date, the company has undertaken 141 energy storage projects with an experience of 12 years. The company provides grid storage solutions, distributed solutions, battery systems, and battery components and accessories. It is a public company listed on the Tokyo Stock Exchange. The company has 114,714 employees as of March 31, 2021. It has a wide geographic presence across regions such as North America, Europe, Asia Pacific, Latin America, the Middle East, and Africa.

TABLE 180 NEC CORPORATION: BUSINESS OVERVIEW

Founding Year	1899
Headquarters City	Minato, Tokyo
Headquarters Country	Japan
Ownership Type	Public

Source: Company Website

FIGURE 70 NEC CORPORATION: COMPANY SNAPSHOT

Note: The percentages for the pie chart calculations have been rounded off. Thus, there might be instances where the percentages in the pie chart may not match the total revenue of the company.

In the Recent Financials section, the R&D expenditure has been considered as a percentage of the total revenue.

The currency conversion rates used for 2019, 2020, and 2021 are YEN 1 = 0.00902 USD, YEN 1 = USD 0.00927, and YEN 1 = USD 0.00907, respectively.

Values are represented up to one decimal to show minute differences.

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

15.1.9.2 Products offered

TABLE 181 NEC CORPORATION: PRODUCT OFFERING

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
Energy Storage System	The energy storage system provided by the company supports various applications and enables fine-grained control of the energy storage system according to the customer's conditions. Also, grid battery storage zone subsystems, which are an integration of energy storage systems, controls, and protection equipment, can be deployed in buildings or custom-sized enclosures.	- GSS-Grid Storage Solution - DSS-Distributed Storage Solution - ALM Lithium-ion Batteries - Battery Components and Accessories	- Renewable Integration - Industrial - Commercial - Utilities

Source: Company Website

15.1.10 NGK INSULATORS, LTD.

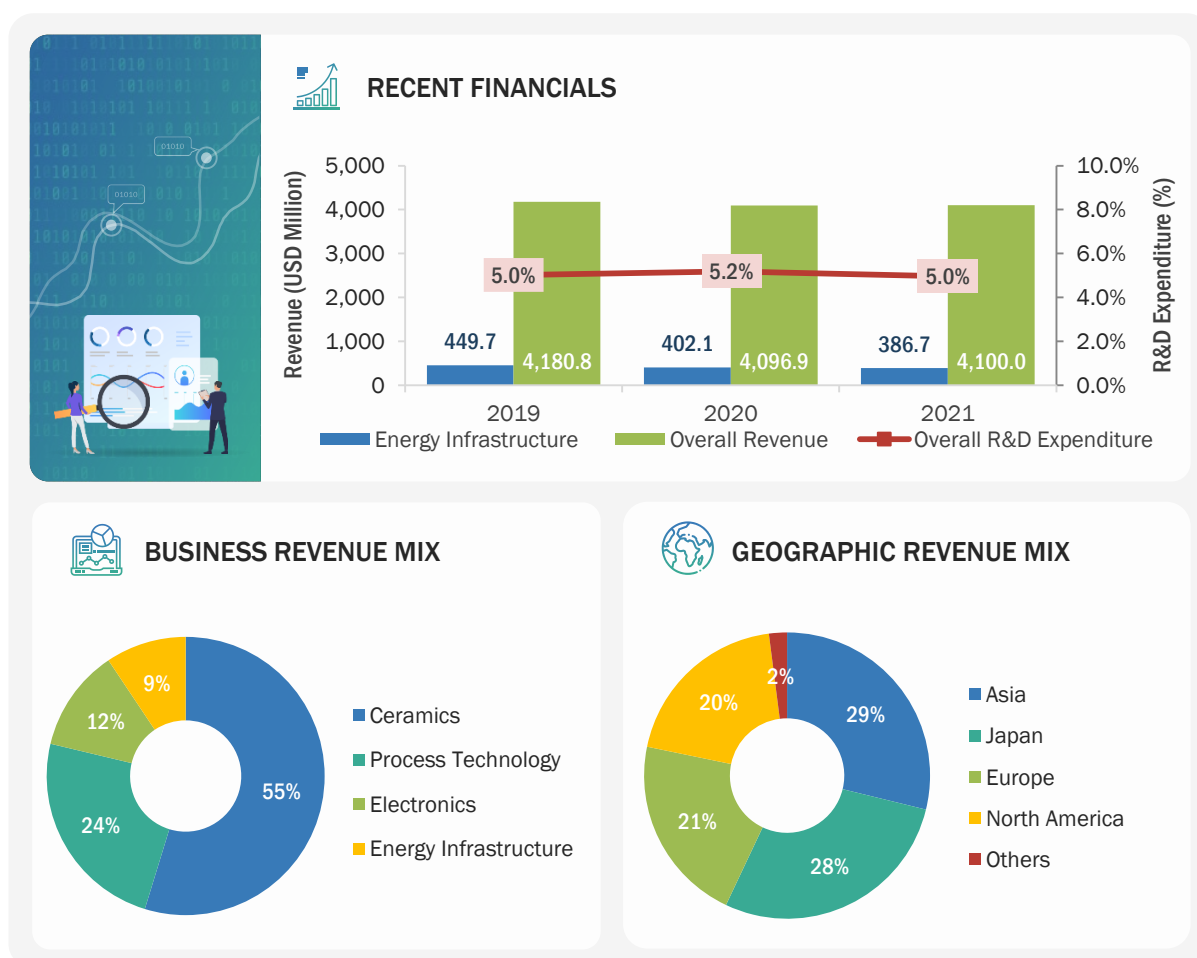
15.1.10.1 Business overview

NGK Insulators, Ltd. is a leading company in the field of ceramics. The company operates through four business segments, namely, Energy Infrastructure, Ceramics, Electronics, and Process Technology. Under its Energy Infrastructure segment, the company provides insulators, NAS (sodium-sulfur) batteries, hardware for insulator assemblies, bushing shells, current limiting arching horn, and line arresters, among others. The company began R&D on sodium-sulfur (NAS) batteries in 1984 and jointly commercialized them in 2002 with TEPCO. NGK's sodium-sulfur ESS is a proven large-scale solution for utilizing variable renewable energy types, which replace the fossil fuels used in energy generation. Sodium-sulfur batteries are used throughout the grids for renewable stabilization, microgrid, generation management, investment deferral, and ancillary services. Sodium-sulfur batteries are used in various applications, such as industrial, commercial, and residential. Additionally, NGK's products include the thin and high-energy-density lithium-ion rechargeable "EnerCera" battery line. Furthermore, the company is active in more than 20 countries worldwide.

TABLE 182 NGK INSULATORS, LTD.: BUSINESS OVERVIEW

Founding Year	1919
Headquarters City	Nagoya, Aichi Prefecture
Headquarters Country	Japan
Ownership Type	Public

Source: Company Website

FIGURE 71 NGK INSULATORS, LTD.: COMPANY SNAPSHOT

Note: The percentages for the pie chart calculations have been rounded off. Thus, there might be instances where the percentages in the pie chart may not match the total revenue of the company.

In the Recent Financials section, the R&D expenditure has been considered as a percentage of the total revenue.

The currency conversion rates used for 2019, 2020, and 2021 are YEN 1 = 0.00902 USD, YEN 1 = USD 0.00927, and YEN 1 = USD 0.00907, respectively.

Values are represented up to one decimal to show minute differences.

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

15.1.10.2 Products offered

TABLE 183 NGK INSULATORS, LTD.: PRODUCT OFFERING

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
Energy Storage System	The company has a large-capacity battery energy storage system, which can provide a stable supply of renewable energy. The NAS energy storage system has features including large capacity, high energy density, and long service life, which gives high performance for a prolonged period. NAS batteries are large-capacity batteries with a compact design, making them flexible for many power grid applications.	- NAS Energy Storage System - NAS Batteries	- Residential - Commercial - Utilities

Source: Company Website

15.1.11 PANASONIC CORPORATION

15.1.11.1 Business overview

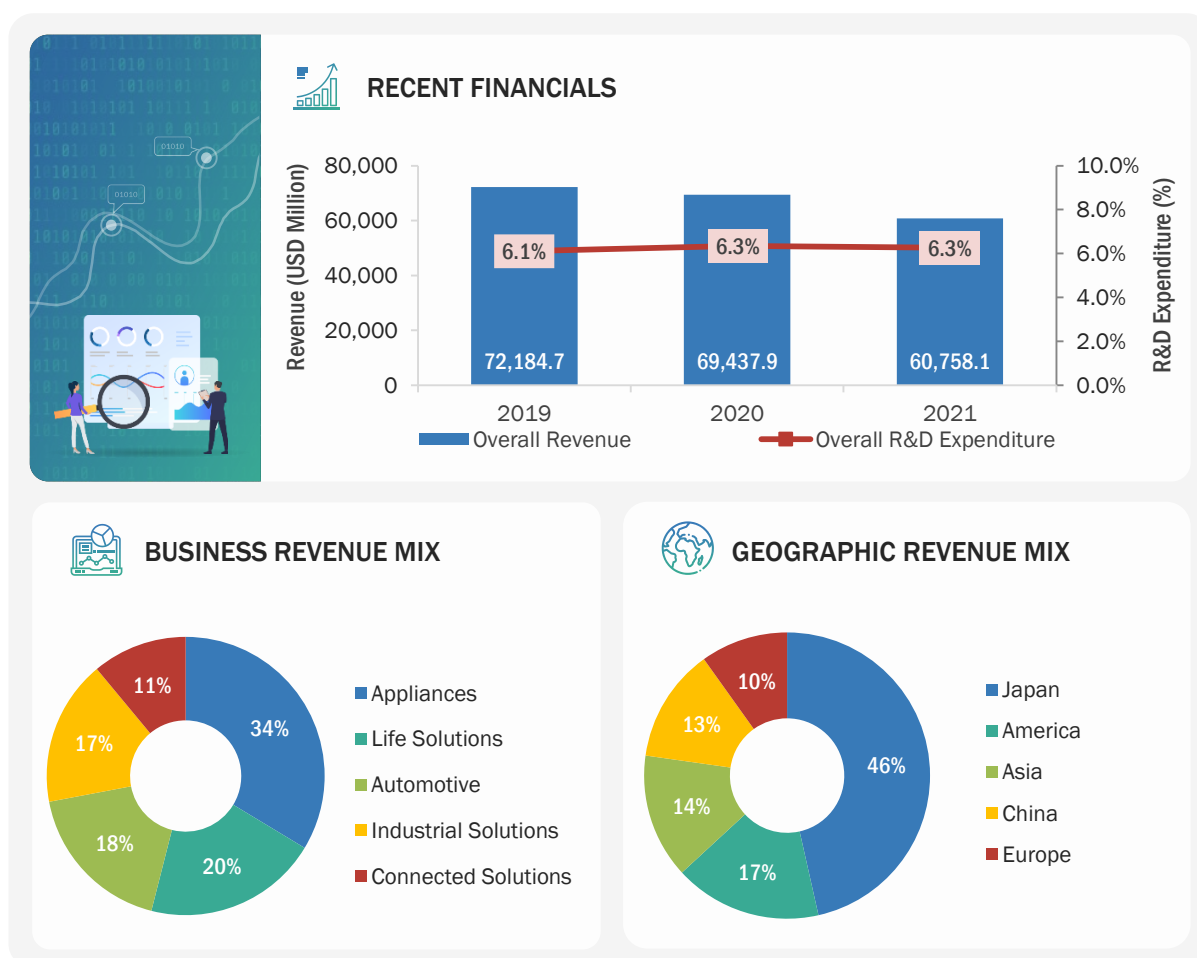
Panasonic Corporation is a Japan-based multinational electronics corporation (formerly known as Matsushita Electric Industrial Co., Ltd.). It is one of the leading companies in the development of innovative technologies and products for the consumer electronics and automotive industries, as well as the housing and industrial sectors. The company operates through the following business segments: Appliances, Life Solutions, Connected Solutions, Automotive, and Industrial Solutions. Automobiles, consumer electronics, home appliances, displays, HVAC, smart homes, industrial automation, and IoT devices are some of the key application areas of Panasonic Corporation's products and services.

The company offers battery energy storage systems under the portfolio of energy solutions. Moreover, the company also offers an extensive lineup of lithium-ion batteries, ranging from small applications, such as consumer batteries for laptops, to more extensive applications, such as automotive batteries. Panasonic Corporation is a leading provider of battery technology and smart storage battery systems used in UPS, residential, utilities, and industrial applications. The smart storage battery systems, which can be used as an emergency power source, provide various benefits, such as reduced power bills through self-consumption of photovoltaic (PV) electricity and reduced CO2 emissions, to residential users. Panasonic Corporation's smart storage battery systems used in the community grid minimize capital investment by utilizing distributed energy storage systems throughout the grids, while in the community off-grid, storage battery systems help reduce the consumption of natural gas and diesel fuel. Recently, in October 2021, Panasonic Corporation launched the 4680 cylindrical battery. Tesla, Inc. will use these batteries in their upcoming car models due to its increased storage. The company has 523 subsidiaries/consolidated companies and 69 associated companies globally. It is a public company listed on the Tokyo Stock Exchange.

TABLE 184 PANASONIC CORPORATION: BUSINESS OVERVIEW

Founding Year	1918
Headquarters Country	Japan
Headquarters City	Osaka
Ownership Type	Public

Source: Company Website

FIGURE 72 PANASONIC CORPORATION: COMPANY SNAPSHOT

Note: The percentages for the pie chart calculations have been rounded off. Thus, there might be instances where the percentages in the pie chart may not match the total revenue of the company.

In the Recent Financials section, the R&D expenditure has been considered as a percentage of the total revenue.

The revenue given in the bar chart is for the "total revenue."

The currency conversion rates used for 2019, 2020, and 2021 are YEN 1 = USD 0.00902, YEN 1 = US 0.00927, and YEN 1 = USD 0.00907, respectively.

Values are represented up to one decimal to show minute differences.

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

15.1.11.2 Products offered

TABLE 185 PANASONIC CORPORATION: PRODUCT OFFERING

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
Storage Battery Systems	The battery systems are designed to collect and store surplus energy. These products are available in AC and DC coupling according to the customers' preferences.	- Smart Storage Battery Systems - EverVolt Battery Storage	- Residential - Automotive - Electronic Devices

Source: Company Website

15.1.11.3 Recent developments

15.1.11.3.1 Product launches and developments

TABLE 186 PANASONIC CORPORATION: PRODUCT LAUNCHES AND DEVELOPMENTS

MONTH & YEAR	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
November 2021	Panasonic Corporation	Battery Storage	EverVolt 2.0	The company announced a robust solar energy portfolio of Total Home Energy Solution offerings. This system offers a complete solution of solar energy storage and conversion. The system includes solar panels, built-in micro-inverters, and battery storage systems.
May 2021	Panasonic Corporation	Battery Storage	EverVolt Solar Modules series	Panasonic announced an expanded lineup of solar modules. The new modules feature higher efficiency and enhanced performance, even in high temperatures. This series of solar modules produces more power, has high conversion efficiency, and has a low degradation rate.

Source: Company Website and Press Releases

15.1.11.3.2 Deals

TABLE 187 PANASONIC CORPORATION: DEALS

MONTH & YEAR	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION	DEAL SIZE
June 2020	Agreement	Panasonic Corporation	Span.IO, Inc.	Panasonic signed a new agreement with Span.IO, Inc. (US) to create a residential energy storage system. This system is likely to be integrated into Panasonic's EverVolt, paired with Span smart panel, creating a new level of control over their battery backup power with intuitive energy management for the homes.	NA

Source: Company Website and Press Releases

15.1.11.4 MnM view

15.1.11.4.1 Key strengths/right to win

Panasonic Corporation is one of the leading battery technologies and energy storage system providers. The company is fostering its growth by extending its innovative and technological leadership into new markets and regions. The company expanded its production facility in North America in 2022 in response to the surging demand for the batteries; this has helped Panasonic increase its revenue by increasing its customer base. The company operates in a dynamic environment driven by innovation.

15.1.11.4.2 Strategic choices made

The company aims to enhance its product and service capabilities by continuously focusing on technological advancements. The company is also focusing on developing battery energy storage systems for residential applications; it has manufactured several products to serve customers from the residential sector with a wide range of products. In November 2021, the company launched the solar energy portfolio of Total Home Energy System to allow the homeowners to generate and store clean energy and make the home more energy independent.

15.1.11.4.3 Weaknesses and competitive threats

Panasonic Corporation has a worldwide presence; however, according to its geographic revenue mix for 2020, the company relied heavily on Japan for revenue generation (approximately 44 to 46% of total sales). The company also competes with various battery manufacturers such as SAMSUNG SDI (South Korea) and LG Energy Solutions Co., Ltd. (South Korea), along with several small-scale domestic and foreign competitors. The company primarily competes with these players based on several parameters, such as quality, brand reputation, reliability, innovative features, pricing, and product availability. Intense competition from companies offering battery energy storage systems with different energy and power capacities at lower prices acts as a competitive threat for the company.

15.1.12 SAMSUNG SDI CO., LTD.

15.1.12.1 Business overview

Samsung SDI Co., Ltd. is a leading manufacturer of rechargeable batteries used in the IT industry, automobiles, and ESS. The company leads technologically in the battery manufacturing market on the foundation of change and innovation. It operates through two business segments—Electronic Materials and Energy Solutions. The major product lines under the Energy Solutions business segment include automotive batteries, small-sized li-ion batteries, and ESS (Energy Storage Systems).

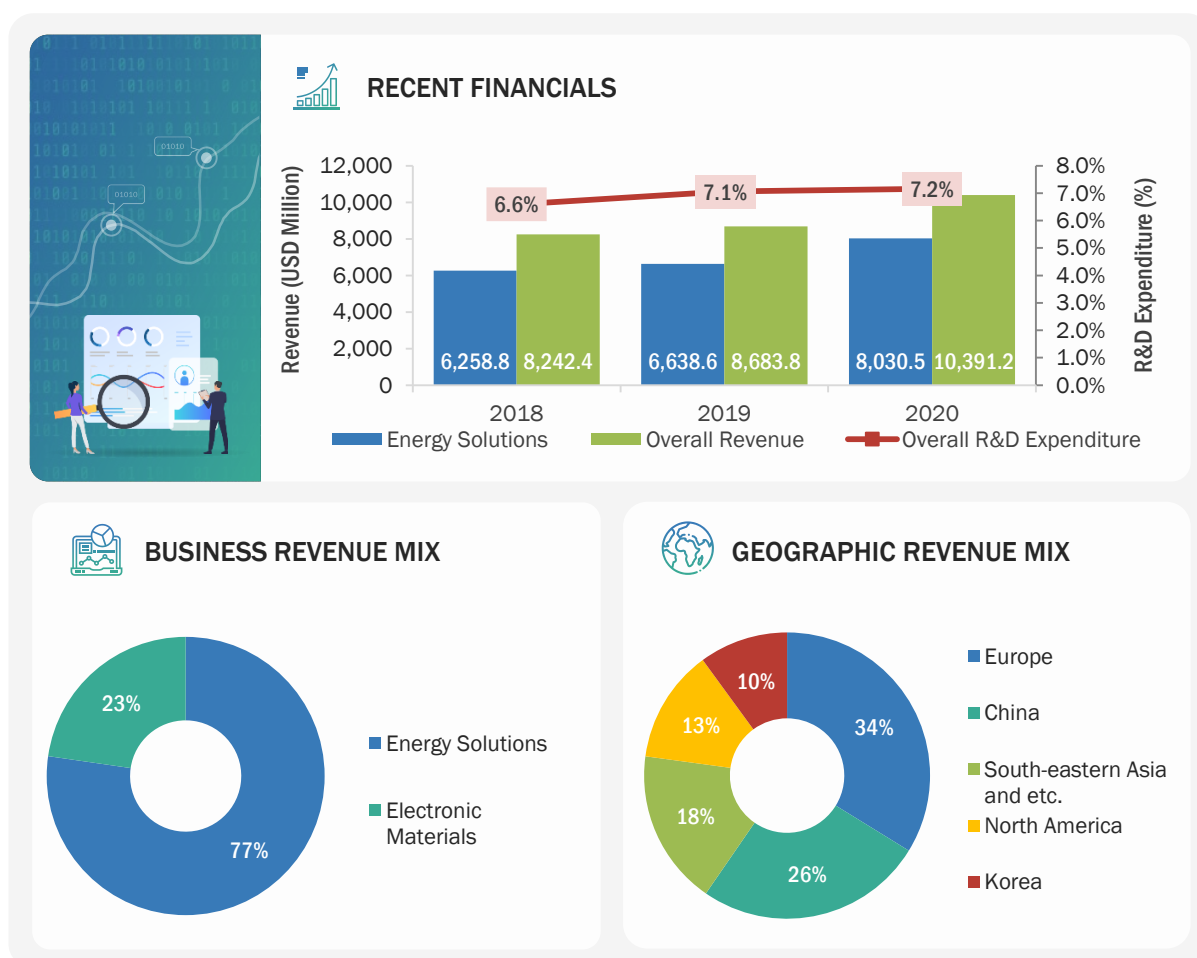
In 2010, Samsung SDI started the lithium battery ESS business based on small-sized lithium-ion rechargeable battery technology. ESS integrated with these batteries are used for the latest mobile devices and electric cars. The company offers ESS from kWh to MWh for the residential, utilities, commercial, UPS, and base transceiver station applications. Samsung ESS has a long-lasting service life. They are safe to use and provide excellent performance. The company aims to provide the means for environment-friendly production of electric power into green energy. Samsung ESS benefit users in various ways: reduce the investment cost, help grid system operators to maintain a constant frequency, minimize output fluctuation with improved power quality, and help users to achieve full utilization of renewable energy sources. In November 2019, the company signed a long-term battery supply contract with BMW until 2031. This deal includes battery cells worth 2.9 billion Euros for the BMW electric vehicles. The company is also a regular battery supplier for Volvo and Volkswagen.

Samsung SDI Co., Ltd. is a public company listed on the Korea Exchange (KRX) market. The company focuses on expanding its global presence and building a strong network worldwide. The company has production factories and sales offices located in North America, Europe, and APAC.

TABLE 188 SAMSUNG SDI CO., LTD.: BUSINESS OVERVIEW

Founding Year	1970
Headquarters Country	South Korea
Headquarters City	Yongin
Ownership Type	Public

Source: Company Website

FIGURE 73 SAMSUNG SDI CO., LTD.: COMPANY SNAPSHOT

Note: The percentages for the pie chart calculations have been rounded off. Thus, there might be instances where the percentages in the pie chart may not match the total revenue of the company.

In the Recent Financials section, the R&D expenditure has been considered as a percentage of the total revenue.

The currency conversion rates used for 2018, 2019, and 2020 are WON 1 = USD 0.00090, WON 1 = US 0.00086, and WON 1 = USD 0.00092, respectively.

Values are represented up to one decimal to show minute differences.

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

15.1.12.2 Products offered

TABLE 189 SAMSUNG SDI CO., LTD.: PRODUCT OFFERING

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
ESS Solutions	ESS stabilizes the output of renewable energy by charging & discharging irregular and intermittent power generated by wind and solar panels. The company offers ESS solutions from small-sized batteries to large MWh energy batteries, expanding the reach of the ESS market to various applications.	- Utility-Commercial Solution - UPS-UES Solution - Residential Solution - Base Transceiver Station Solution	- Commercial - Residential - Power Generation - Utilities

Automotive Battery	The company is developing and supplying secondary batteries for electric motorized vehicles; these batteries are used in fully electric vehicles (EVs), plug-in hybrid vehicles (PHEVs), hybrid electric vehicles (HEV), micro/mild hybrid electric vehicles (Micro-/Mild HEVs)	Prismatic Lithium-Ion Battery Cell	Automotive
Small-Sized Li-ion Battery	The company provides a wide range of batteries according to the need specified with improved quality and high performance	- Prismatic Cell - Polymer Cell - Mini Cell - Cylindrical Cell	- IT Devices - Power Devices

Source: Company Website

15.1.12.3 Recent developments

15.1.12.3.1 Product launches and developments

TABLE 190 SAMSUNG SDI CO., LTD.: PRODUCT LAUNCHES AND DEVELOPMENTS

MONTH & YEAR	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
December 2021	Samsung SDI Co., Ltd.	Battery	PRiMX	The company launched a new battery brand, PRiMX, which gives consumers quality, performance, and proven advantage, which means user convenience achieved with the company's independent developed technology. The brand has been trademarked in Korea and Europe and will soon be registered in the US.

Source: Company Website and Press Releases

15.1.12.4 MnM view

15.1.12.4.1 Key strengths/right to win

Samsung SDI is fostering its growth by providing energy storage solutions for different applications. The company offers energy storage solutions for telecom, residential, commercial, UPS, and utility applications by focusing on sustainable technology. The company provides energy storage systems under the Energy Solutions segment, which held the major share of 77.3% of the total revenue of the company in 2020. The company also focuses on strengthening its brand position in the market. In December 2021, the company launched a battery brand PRiMX, which conveys the company's brand identity. Besides, the company invests heavily in R&D for designing innovative products to match the market needs and explore new opportunities.

15.1.12.4.2 Strategic choices made

Samsung SDI equally focuses on organic and inorganic growth strategies to strengthen its position in the battery energy storage system market. It is focused on launching new battery energy storage systems

having enhanced efficiency. For instance, in August 2019, the company launched various products, including high-capacity energy storage systems, to strengthen its position in the market.

15.1.12.4.3 Weaknesses and competitive threats

Samsung SDI is highly dependent on a single market for revenue generation. According to its geographic revenue mix for 2020, the company's relied over-dependency on Europe for revenue generation can be a weakness. The company's lack of partnerships, mergers, and acquisitions are also expected to decrease its growth rate. Furthermore, it faces intense competition from major players operating in the battery energy storage system market. This has increased the pressure on the company. Developing new products and enhancing its existing products could help the company maintain its position in the market.

15.1.13 SIEMENS AG

15.1.13.1 Business overview

Siemens, formerly known as Siemens & Halske, is a Germany-based electronics and electrical engineering company that operates in electrification, automation, and digitalization. The company is a leading conglomerate pioneer in electrical engineering. With its vast product portfolio, solutions, and services, Siemens covers almost the entire energy value chain—from power generation and transmission to storage. Siemens operates through four reportable segments, namely, Digital Industries, Smart Infrastructure, Mobility, Siemens Healthineers, Siemens Financial Services, and Portfolio Companies.

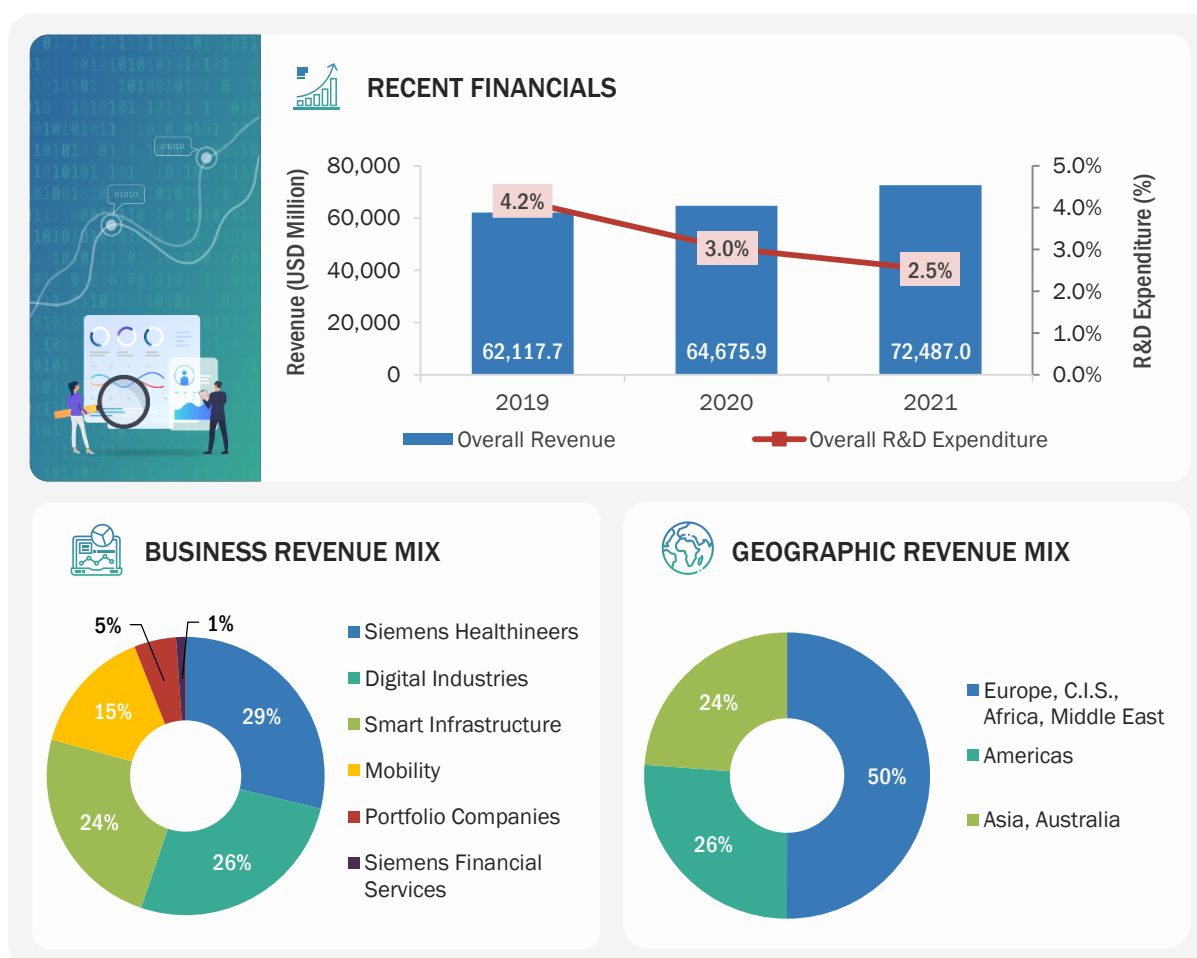
In the battery energy storage system market, the company offers Siemens Energy Storage, also known as SIESTORAGE, a modular energy storage system with state-of-the-art lithium-ion battery technology. SIESTORAGE helps optimize energy performance during various stages, ranging from generation to consumption, by stabilizing frequency and voltage and balancing supply and demand. It is an intelligent grid solution that provides flexible energy with unmatched versatility and optimizes grid connection. The system is commissioned and installed in e-houses, existing buildings, or standard containers. SIESTORAGE helps secure energy even during outages and provides energy for emergency generators when the main supply is unavailable. SIESTORAGE is typically used in renewable generation; conventional generation; grid, off-grid, and microgrid; and industrial applications. BlueVault is another advanced lithium-ion battery-based energy storage solution suitable for all-electric as well as hybrid energy storage applications.

In July 2017, The AES Corporation and Siemens AG formed a joint venture called Fluence to provide battery energy storage systems based on three technologies: Gridstack, Sunstack, and Edgestack. Fluence offers market-leading energy storage products and services and digital applications for renewable energy and storage. The company provides three battery energy storage systems, namely, Advancion 5, Siestorage, and SunFlex. This will further ramp up commercial and grid-scale storage deliveries worldwide.

TABLE 191 SIEMENS AG: BUSINESS OVERVIEW

Founding Year	1847
Headquarters Country	Germany
Headquarters City	Munich
Ownership Type	Public

Source: Company Website

FIGURE 74 SIEMENS AG: COMPANY SNAPSHOT

Note: The percentages for the pie chart calculations have been rounded off. Thus, there might be instances where the percentage in the pie may not match the total revenue of the company.

In the Recent Financials section, the R&D expenditure has been considered as a percentage of the total revenue.

The currency conversion rates used for 2019, 2020, and 2021 are EURO 1 = 1.09368 USD, EURO 1 = USD 1.17052, and EURO 1 = USD 1.16417, respectively.

Values are represented up to one decimal to show minute differences.

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

15.1.13.2 Products offered

TABLE 192 SIEMENS: PRODUCT OFFERINGS

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
Energy Storage Solutions	The energy storage technology from Siemens is designed for the most demanding industrial applications for medium to high voltage needs. The simple, scale-out designs drive efficiencies in project permitting, delivery, and operations of the surplus stored energy, which creates consistency of supply.	- SIESTORAGE - BlueVault - Fluence's Cube Designed Battery Energy Storage System - Advancion 5 - SunFlex	Grid Management

Source: Company Website

15.1.13.3 Recent developments

15.1.13.3.1 Product launches and developments and developments

MONTH & YEAR	COMPANY	PRODUCT TYPE	PRODUCT NAME	DESCRIPTION
June 2020	Siemens AG	Energy storage system	Sixth Generation Large-scale Energy Storage Stack	Fluence (US), a firm formed with the joint venture of Siemens and AEG Corporation, unveiled its sixth-generation energy storage technology stack combining factory-built hardware, advanced software, and data-driven intelligence. Fluence has been selected for launching 2,300 MWh of projects from Enel, Siemens, and LS Power. The technology stack creates the foundation for three purpose-built systems configured for renewable, grid, commercial, and industrial applications.

Source: Company Website

15.1.13.4 MnM view

15.1.13.4.1 Key strengths/right to win

Siemens AG is a globally integrated technology company with core activities in automation, digitalization, and electrification. It also has expert partners spread across the world. The company develops photovoltaic inverters and battery storage and charging solutions. It aims to expand its product portfolio by continuously launching cutting-edge products and investing heavily in R&D for developing high-quality energy storage and distributed energy storage products. Furthermore, the company has been expanding its customer base by winning several contracts. Siemens formed a new company called Fluence by signing a joint venture agreement with The AES Corporation. This has enabled Siemens to expand its grid-scale energy storage system portfolio with the addition of SunFlex and Advancion 5 products.

15.1.13.4.2 Strategic choices made

The company primarily focuses on product developments to offer technically advanced products to its customers. Siemens is also emphasize adopting inorganic growth strategies, such as mergers and acquisitions and partnerships, to target emerging and rapidly growing markets in APAC, such as India, China, and South Korea. Recently in 2020, the firm formed by Siemens and the AEG Corporation, Fluence, unveiled its sixth-generation energy storage technology stack with its digital platform, an operating system that serves utility-based projects up to 1MW at an industrial site.

15.1.13.4.3 Weaknesses and competitive threats

The company only has a strong geographic presence in APAC, which can be considered one of its weaknesses; it needs to expand its geographic presence to stay competitive in the market. Strong technical expertise and brand image help the company gain a competitive edge in major geographic regions. This also helps the company create a vertically integrated business model centered on developing innovative devices. However, the company competes with major players operating in the market.

15.1.14 TESLA, INC.

15.1.14.1 Business overview

Tesla, Inc. (formerly Tesla Motors) designs, develops, manufactures, and sells high-performance EVs and energy products. The company operates through wholly owned subsidiaries in North America, Europe, and Asia. The key operating segments of the company are Automotive and Energy Generation and Storage.

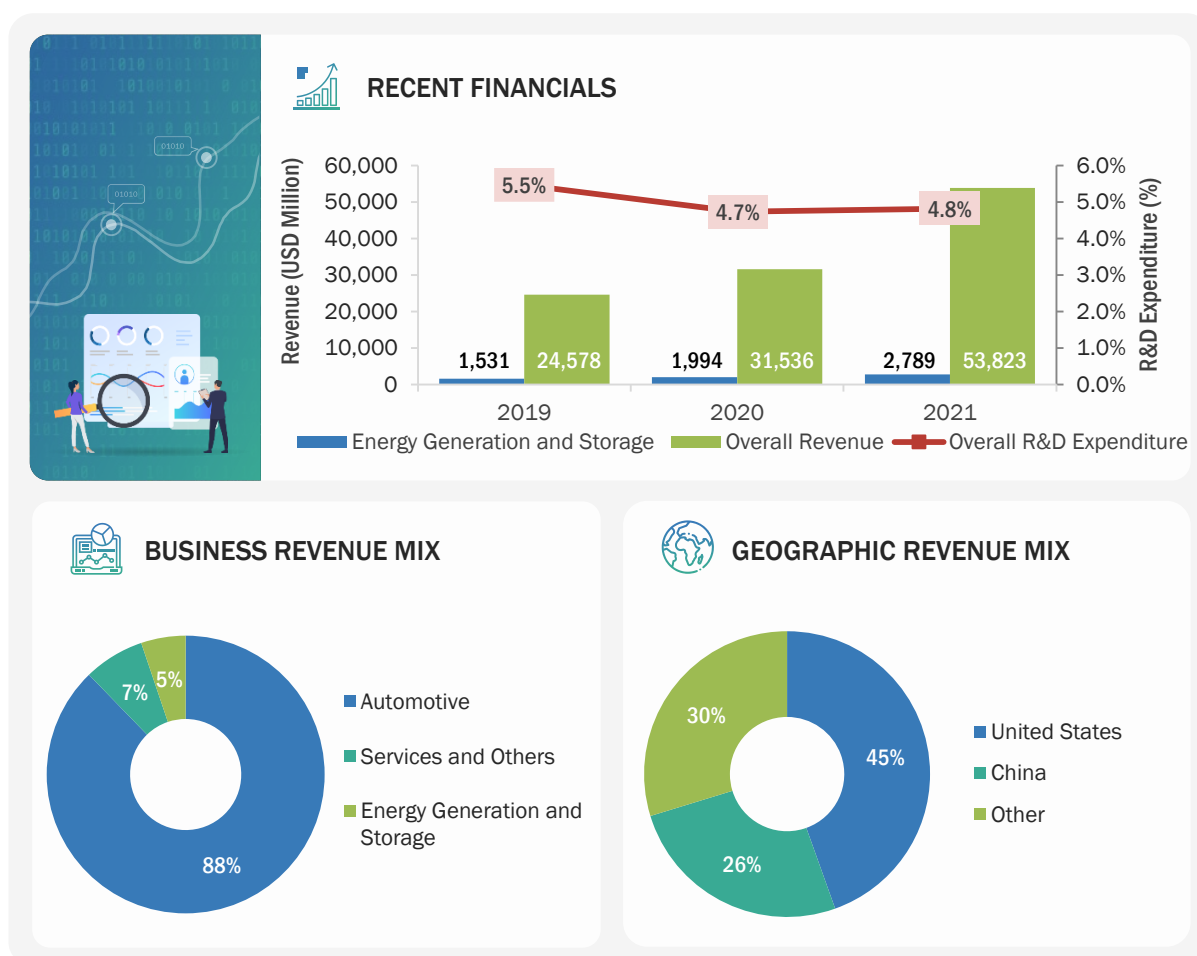
The Energy Generation and Storage segment offers batteries, power electronics, and energy storage products. Energy storage products provided by the company serve a wide range of applications, from homes to large grid-scale projects. Powerpack and Powerwall are two devices offered by Tesla in the Energy Generation and Storage segment. Powerwall is a 14-kWh rechargeable lithium-ion battery designed to store energy in homes or small commercial facilities. Residential and commercial users use the energy stored in the system for self-consumption during solar power generation, and the backup power is stored. The Powerwall battery integrated into an inverter and battery packs monitors energy level in real time, alerts the operator in severe weather conditions, and functions with almost no noise. Additionally, the 200 kWh Powerpack system is used for peak shaving, load shifting, self-consumption of solar power, as well as for responding to the demand raised by the commercial and industrial applications. The company develops energy storage products for applications in homes, commercial facilities, and utility sites. Tesla designed a new battery product, especially for utility-scale projects, in 2019. Megapack, which is easy to install and connect, significantly reduces the complexity of large-scale battery storage systems. In October 2021, Tesla, Inc. announced the 40GWh battery energy storage factory in California that would produce only Megapack systems for utilities.

The company offers integrated solutions for its utility customers, which combines hardware, software, installation, and services. It also offers 24/7 monitoring for a reliable installation of the system. Tesla has its manufacturing facilities: Gigafactory (large-scale battery manufacturing plant in the US), the Fremont factory, and the New York solar panel factory. Gigafactory is designed to achieve economies of scale by minimizing operating and overhead costs by utilizing innovative manufacturing processes, reducing logistics waste, and optimizing co-located processes.

TABLE 193 TESLA, INC.: BUSINESS OVERVIEW

Founding Year	2003
Headquarters Country	US
Headquarters City, State	Austin, Texas
Ownership Type	Public

Source: Company Website

FIGURE 75 TESLA, INC.: COMPANY SNAPSHOT

Note: The percentages for the pie chart calculations have been rounded off. Thus, there might be instances where the percentage in the pie may not match the total revenue of the company.

In the Recent Financials section, the R&D expenditure has been considered as a percentage of the total revenue.

The company operates through two business segments; however, the total revenue comprises the revenue for services & others.

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

15.1.14.2 Products offered

TABLE 194 TESLA, INC.: PRODUCT OFFERINGS

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
Energy Storage Products	Powerwall is a compact home battery that stores the energy that the customer produces with the Solar Roof; so, they can power their home anytime—at night or during an outage. Powerpack is a fully integrated, AC-connected energy storage system with everything needed to connect to a building or utility network. Megapack is designed for utilities and large-scale commercial customers	- PowerWall - PowerPack - Megapack	- Utilities - Residential - Commercial

Source: Company Website

15.1.14.3 Recent developments

15.1.14.3.1 Product launches and developments

MONTH & YEAR	COMPANY	PRODUCT TYPE	PRODUCT NAME	DESCRIPTION
June 2020	Tesla, Inc.	Solar roof	Solar Panels	Tesla launched the new lowest-cost solar panels for residential applications in the US. Such product launches can boost the demand for energy storage products.

Source: Company Website and Press Releases

15.1.14.4 MnM view

15.1.14.4.1 Key strengths/right to win

Tesla, Inc. has proficiency in electrical, mechanical, civil, and software engineering, which permits them to design, customize, and manufacture highly efficient energy storage solutions to cater to different industries and applications. This flexibility and reliability have gained the customer's maximum attention. The company focuses on a modular approach to designing battery systems, which helps maximize the manufacturing capacity of the energy storage products. The company has spent more in 2021 on R&D in comparison with 2020, which indicates the development of the company's product portfolio. In 2021, the company's Energy Generation and Storage business segment revenue rose by 40% compared with 2020, primarily due to the increase in the deployments of Megapacks, Powerwall, solar cash and loan jobs, and solar roofs. The company held a significant share of the battery energy storage system market in 2021.

15.1.14.4.2 Strategic choices made

As part of its growth strategy, the company focuses on product launches. For instance, the company has launched the lowest cost solar panels for residential users. Additionally, the company has also launched Megapack, a utility-scale energy storage product that combines up to 3 MWh of storage capacity and a 1.5 MW inverter. The company's main focus is to develop new products and enhance the existing products to meet the rising needs of the customers for electricity. Tesla is also focused on expanding the production capacity at the existing facilities. Tesla, Inc. believes that the long-term success in the energy storage business is dependent on the increasing margin through large volumes of sales. The demand has increased due to the growing grid stability concerns and better availability.

15.1.14.4.3 Weaknesses and competitive threats

Tesla, Inc. as a company has established itself as a vehicle manufacturer very well, but at the same time, the energy storage solutions offered by the company face high competition in the market by other experienced players. Moreover, the company is still dependent on other market suppliers for lithium-ion battery cells and has limited flexibility to change this situation. Hence, any disruption from the supply side will hamper the production of energy storage solutions of Tesla, Inc., which may further affect the growth of the company in the market. Besides, the company is competing with well-established global players who have a broad product portfolio. Therefore, high-degree competition from market players may lead to lower profit margins for Tesla, Inc.

15.1.15 TOSHIBA CORPORATION

15.1.15.1 Business overview

Toshiba Corporation (Toshiba) manufactures and sells electronic and electrical products. The company operates through the following segments: Infrastructure Systems & Solutions, Energy Systems & Solutions, Building Solutions, Electronic Devices & Storage Solutions, Digital Solutions, Retail & Printing Solutions, and Others. The Energy Systems & Solutions segment's main business is divided into Nuclear Power Systems, Thermal & Hydro Power Systems, Transmission & Distribution Systems, and Others. Under the Transmission & Distribution Systems, the company provides energy storage systems, microgrids, HVDC, system controls, substation equipment, protection relays, and IoT solutions.

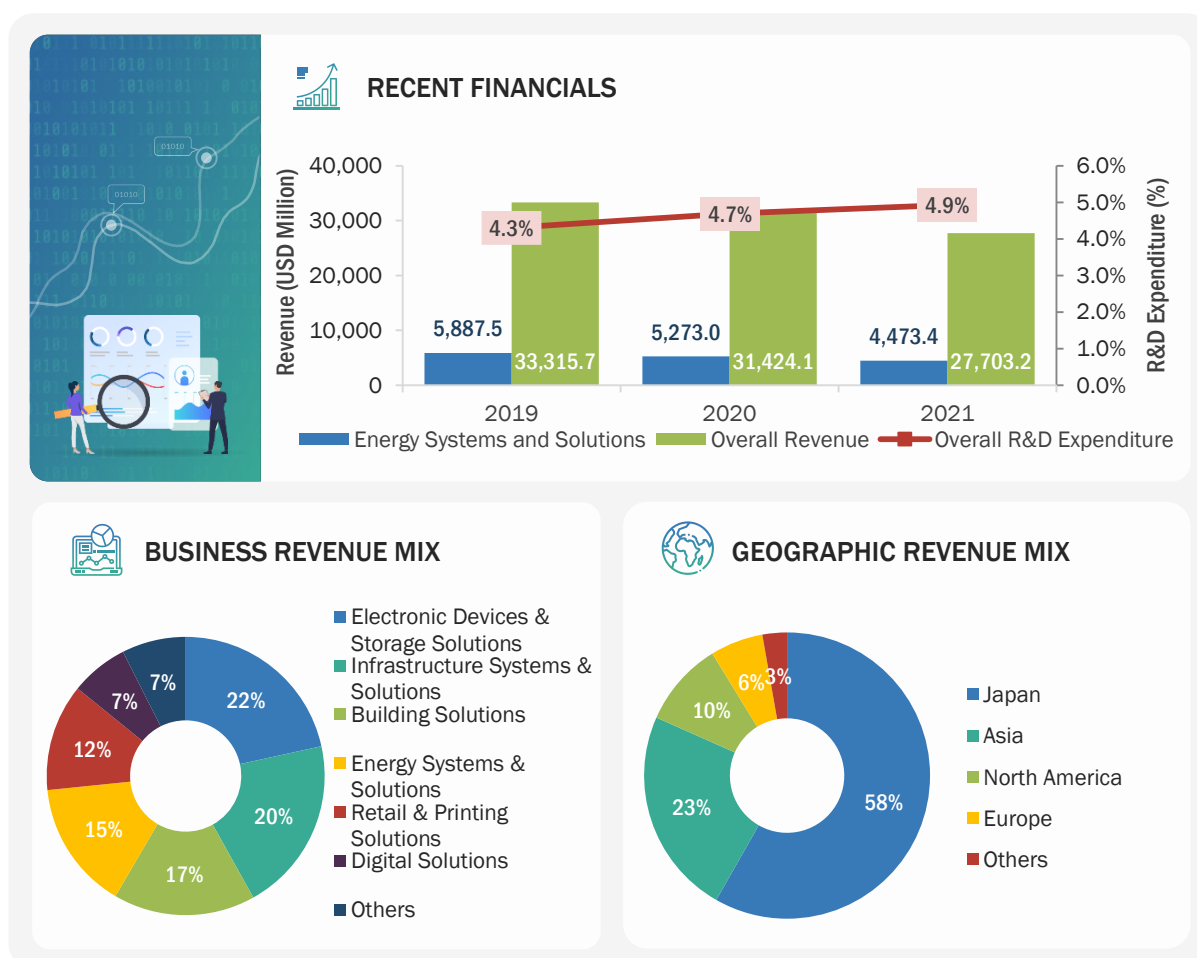
Toshiba offers battery energy storage systems designed to provide a secured and stable electricity supply, serving as a backup power source. SCiB systems can be used for a wide range of residential, public, industrial, and large-scale electrical grid applications. The medium-scale lithium-ion battery energy storage system is designed for use at schools, city halls, train stations, factories, etc., for energy saving and electric load leveling and shifting. The large-scale lithium-ion BESS called Smart Battery makes it possible to reduce grid frequency fluctuations caused by wind and PV generation and provide a secure, stable electricity supply.

The company serves its customers, mainly in Japan, Asia, North America, and Europe. Toshiba Corporation is listed on the Tokyo Stock Exchange with the ticker symbol Toshiba Corp. Toshiba Group has several subsidiaries in different industries, such as Toshiba Plant Systems & Services Corporation and Toshiba Energy Systems & Solutions Corporation in its Energy Systems & Solutions division.

TABLE 195 TOSHIBA CORPORATION: BUSINESS OVERVIEW

Founding Year	1875
Headquarters Country	Japan
Headquarters City	Tokyo
Ownership Type	Public

Source: Company Website

FIGURE 76 TOSHIBA CORPORATION: COMPANY SNAPSHOT

Note: The percentages for the pie chart calculations have been rounded off. Thus, there might be instances where the percentages in the pie chart may not match the total revenue of the company.

In the Recent Financials section, the R&D expenditure has been considered as a percentage of the total revenue.

The currency conversion rates used for 2019, 2020, and 2021 are YEN 1 = 0.00902 USD, YEN 1 = USD 0.00927, and YEN 1 = USD 0.00907, respectively

Others in the business revenue mix include logistics and battery.

Values are represented up to one decimal to show minute differences.

Source: Company Website, Annual Reports, and MarketsandMarkets Analysis

15.1.15.2 Products offered

TABLE 196 TOSHIBA CORPORATION: PRODUCT OFFERINGS

PRODUCT	DESCRIPTION	PRODUCT TYPE	APPLICATION
Battery Energy Storage Systems	SCiB has characteristics such as safety, long life, low-temperature performance, rapid charging, high input/output power, and large effective capacity. This acts as an easy replacement for automotive lead-acid batteries due to its high-capacity power storage and long life.	SCiB Systems	- Utilities - Residential - Commercial - Automotive - Industrial

Source: Company Website

15.1.15.3 Recent developments

15.1.15.3.1 Product launches and developments

MONTH & YEAR	COMPANY	PRODUCT TYPE	PRODUCT NAME	DESCRIPTION
January 2022	Toshiba Corporation	Li-ion Battery	20Ah-HP SCiB Rechargeable Battery	Toshiba Corporation launched and expanded its SCiB product offerings by including the 20AH-HP rechargeable battery, with wide applications ranging from automotive and industrial and in energy storage systems. The product features include long life, excellent compatibility, and efficient heat suppression.

Source: Company Website and Press Releases

15.1.15.3.2 Deals

MONTH & YEAR	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION	DEAL SIZE
April 2020	Partnership	Toshiba Corporation	Sumitomo Corp.	Sumitomo Corp., a Japan-based trader, collaborated with Toshungiba to build a battery power storage system in Ohio to balance the electricity grid serving Northeastern US. The company deploys Toshiba lithium-ion batteries in its systems.	NA

Source: Company Website

15.2 OTHER PLAYERS

15.2.1 EAST PENN MANUFACTURING COMPANY

Founding Year	1946
Headquarters Country	US
Headquarters City/State	Pennsylvania
Ownership Type	Private
Business Overview	East Penn Manufacturing Company is a family-owned company producing high-quality batteries and battery accessories for marine and fleet. The company designs and manufactures lead batteries for transportation applications such as automotive, marine, trucks, buses and equipment, power sports, and recreation vehicles; motive power applications such as industrial lift trucks, AGV, and rail and automotive; reserve power applications such as communication (5G/wireline/cable), uninterruptible power supply, and renewable energy. The company has over 10,500 employees with 515 product designs and a global customer base. It is ISO 9001, ISO 14001, and IATF 16949 certified company. In 2010, East Penn Manufacturing Company acquired Ecoult. Ecoult delivers battery energy storage solutions powered by Ultrabattery (an entirely new class of advanced lead-acid battery technology) that improve energy reliability and efficiency. Ecoult offers various EES with different capacities, which include UltraPod (7 kWh), UltraFlex (28 kWh), and UltraRax, as well as custom and large solutions (up to 840 kWh). These energy storage solutions can be used for frequency regulation and grid services, microgrids, telecommunications and remote diesel, renewable integration and weak grids, and commercial and residential energy storage. Ecoult also provides engineering and support services for energy storage systems.
Products	▪ UltraFlex ▪ UltraRax ▪ DEKA SOLAR BATTERIES ▪ DEKA ULTRABATTERY
Geographic Presence	Americas and Asia

Note: This is a privately held company; hence, no financial information is available in the public domain.

Source: Company Website and Press Releases

15.2.2 ENERDEL, INC.

Founding Year	2004
Headquarters Country	US
Headquarters City	Indianapolis
Ownership Type	Private
Business Overview	EnerDel, Inc. is a leading manufacturer and designer of lithium-ion energy storage solutions and battery systems. The company focuses on on- and off-grid electric, heavy-duty transportation, mass transit, and task-oriented applications. Its products suite delivers energy-dense solutions for high-energy and high-power applications. ESS offered by EnerDel Secure ensures greater reliability, scalability, and efficiency than other battery-based solutions for residential, commercial, and industrial applications. Lithium-ion batteries and BMS-integrated Secure+ Energy Storage System find applications in bulk energy storage solutions, renewable energy storage solutions, substation equipment backup systems, microgrids, community energy storage systems, residential backup, and storage solutions. EnerDel's affiliate companies are EnerTech and EnerZ. The company's distributors include Cullen Diesel Power Ltd., Clarke Power Services, Inc., and Johnson & Towers.
Products	▪ Energy Storage System ▪ SE100-590 Secure+ Grid Energy Storage System ▪ SP90-590 Secure+ Energy Storage System
Geographic Presence	Americas

Note: This is a privately held company; hence, no financial information is available in the public domain.

Source: Company Website and Press Releases

15.2.3 EXERGONIX

Founding Year	2010
Headquarters Country	US
Headquarters City	Kansas City
Ownership Type	Private
Business Overview	Exergonix develops proprietary energy storage technology and supplies integrated ESS. It is unparalleled in the development of renewable energy projects worldwide. The company provides battery energy storage systems for residential, utilities, and commercial applications. The residential-scale distributed energy storage system (RS-DESS) is modular; therefore, it can be accommodated in various homes, based on the size of the house and its application. The company also offers a commercial-scale distributed energy storage system (CS-DESS), which can be used as community energy storage (CES) in smart grid applications, typically used in industrial and retail facilities. The utility-scale distributed energy storage system is designed to house a modular configuration of racks that can form multi-megawatts of storage. The company also offers an advanced battery management system (BMS) hardware platform, CODA CORE Single Tower Power Controller (STPC), Multi Tower Power Controllers (MTPC). The company's management and investment partners include Coda Energy, The Grove at Lee's Summit, Neighbourhood Power Corporation, and Verd2GO.
Products Offered	▪ CODA Core Tower ▪ Residential-scale Distributed Energy Storage System (RS-DESS) ▪ Commercial-scale Distributed Energy Storage System (CS-DESS) ▪ Utility-Scale Distributed Energy Storage Systems
Geographic Presence	North America and Asia

Note: This is a privately held company; hence, no financial information is available in the public domain.

Source: Company Website and Press Releases

15.2.4 JOHNSON CONTROLS

Founding Year	1885
Headquarters Country	US
Headquarters City/State	Milwaukee
Ownership Type	Public
Revenue (2021)	USD 23,668 million
Business Overview	Johnson Controls is a global provider of batteries and building systems. It is engaged in manufacturing, installing, and servicing automatic temperature regulation systems for buildings. The company operates through two business segments, namely Building Efficiency and Automotive Experience. It provides containerized distributed energy storage systems equipped with lithium-ion batteries. L2000 SU50 is a containerized DES system that supplies 50 kW for up to four hours and ranges from 50 kWh to 200 kWh of stored energy. L2000 DES is also designed for HVAC, fire detection, and fire suppression systems that require storage energy from 150 kWh to 5,000 kWh. Services offered by the company include HVAC operations, maintenance, and repair services; fire maintenance and support; security maintenance and support; training services; and managed services. Solutions provided by the company are energy and efficiency solutions, design and construction solutions, and optimization and retrofit solutions. The company serves industries such as healthcare, transportation, sports and entertainment, industrial manufacturing, and marine and navy.
Products	▪ L2000 SU50 Modular Distributed Energy Storage System ▪ L2000 Modular Distributed Energy Storage System
Geographic Presence	North America, Europe, Asia Pacific, Latin America, Middle East, and Africa

Source: Company Website and Press Releases

15.2.5 KOKAM

Founding Year	1989
Headquarters Country	South Korea
Headquarters City	Gyeonggi-do
Ownership Type	Private
Business Overview	Kokam is engaged in the manufacturing of lithium-ion for energy storage systems (ESS), Uninterrupted Power Supply (UPS), electric vehicles (EV), marine, aviation/UAV, defense, and industrial applications. The company is an affiliate of SolarEdge, a leading manufacturer of PV (Photovoltaic) Solar Inverters. The company's containerized energy storage system is used for applications such as frequency regulation, peak shaving, integration with microgrids, renewable energy, and UPS. It can store 250kWh–5.4MWh of energy in transferable 20ft / 40ft / 53ft ISO containers. The company also provides systems such as battery rack (~200 KWh), battery rack system (~MWh), ESS solution (MWh container), and UPS battery.
Products	Battery Energy Storage System
Geographic Presence	Asia

Note: This is a privately held company; hence, no financial information is available in the public domain.

Source: Company Website and Press Releases

15.2.6 NEXTERA ENERGY, INC.

Founding Year	1984
Headquarters Country	US
Headquarters City/State	Florida
Ownership Type	Public
Revenue (2020)	USD 17,997 million
Business Overview	NextEra Energy, Inc. is a leading electric power and energy infrastructure company. The company comprises two principal businesses, namely Florida Power & Light Company (FPL), including Gulf Power, and NextEra Energy Resources (NEER). NEER generates renewable energy from the sun and wind and is a global leader in battery storage. NextEra Energy, Inc. provides lithium-ion battery energy storage solutions for residential and utility applications. Its largest capacity battery energy storage solution in operation is the 30-MW Blue Summit Battery Energy Storage system installed in Texas. NextEra Energy, Inc. is a public company listed on the New York Stock Exchange. The company's subsidiaries are NextEra Energy Resources, Florida Power & Light Company, NextEra Energy Services, and NextEra Energy Transmission.
Products	Battery Energy Storage Systems
Geographic Presence	North America

Source: Company Website and Press Releases

15.2.7 PARKER HANNIFIN CORP

Founding Year	1917
Headquarters Country	US
Headquarters City/State	Cleveland
Ownership Type	Public
Revenue (2021)	USD 14,347.6 million
Business Overview	Parker Hannifin Corp is one of the leading manufacturers of motion and control technologies and systems. The company also provides engineered solutions for industrial, mobile, and aerospace markets. It operates through two business segments, namely Aerospace Systems and Diversified Industrial. The Diversified Industrial business segment is further segregated into Engineered Materials Group, Filtration Group, Fluid Connectors Group, Instrumentation Group, and Motion Systems Group. Under the Fluid Connectors Group, the company provides numerous sizes and configurations for various renewable energy applications. Parker Hannifin developed a battery energy storage system by building a power conversion system that connects energy storage units (large banks of batteries) to the grid. It supplies the power conversion system for the AES Gener Los Andes 12 MW installation at the Norgener power station, which has been delivering improved grid reliability since 2009. The company provides battery energy storage systems mainly for industrial applications. The company serves industries such as aerospace, electronics and semiconductor, healthcare and life sciences, oil & gas, and transportation. It is a public company listed on the New York Stock Exchange as NYSE: PH. The company is located in 50 countries worldwide and supports 100 divisions with 336 manufacturing locations.
Products	Power Conversion System
Geographic Presence	North America, Asia Pacific, Europe, Latin America, Middle East, and Africa

Source: Company Website and Press Releases

15.2.8 PRIMUS POWER

Founding Year	2009
Headquarters Country	US
Headquarters City	Hayward
Ownership Type	Private
Business Overview	Primus Power is one of the leading providers of low-cost, fade-free, grid-scale, and long-duration energy storage solutions. The company provides a zinc bromide flow battery, EnergyPod2, which exhibits a long-duration, full power, multi-decade life without any fire risk and electrode stack replacement. The energy storage solutions by Primus Power serve multiple storage applications for numerous commercial and industrial, data center, microgrid, and utilities customers. The company is gaining demand for flow batteries globally—EnergyPod2 is being shipped worldwide. Primus Power is one of the few companies funded by the US Department of Energy (DOE), the US Government's Advanced Research Projects Agency-Energy (ARPA-E), and the California Energy Commission (CEC). The company has received 35 patents from seven countries and has a subsidiary in Asia.
Products	EnergyPod 2 – Long Duration Energy Storage
Geographic Presence	Americas, Europe, and Asia Pacific

Note: This is a privately held company; hence, no financial information is available in the public domain.

Source: Company Website and Press Releases

15.2.9 THE AES CORPORATION

Founding Year	1981
Headquarters Country	US
Headquarters City	Arlington County
Ownership Type	Public
Revenue (2020)	USD 9,660 million
Employees (2020)	8,200
Business Overview	The AES Corporation is one of the leading companies involved in lithium-ion-based energy storage for commercial applications. It operates through two lines of business, such as Utilities and Generation. In the utility line of business, the company owns or operates utilities to generate or purchase, distribute, transmit and sell electricity to end-user customers in the residential, commercial, industrial, and governmental sectors within a defined service area. In the generation line of business, the company owns or operates power plants to generate and sell power to customers, such as utilities, industrial users, and other intermediaries. The company's product offerings include new clean energy, advanced energy networks, scalable clean energy solutions, and an energy portfolio with cleaner energy. The company provides the Advancion 4 energy storage solution for utilities, power markets, and independent power producers. The company's production of grid battery systems depends on prominent suppliers, including Hitachi, Samsung SDI, Panasonic, BYD, and NEC Corporation. The company has partnered with companies such as DP&L Foundation, United Way, The trust of Americas, and STEM Careers Coalition. The company is organized in four more-oriented SBUs, which include the US and Utilities (United States, Puerto Rico, and El Salvador); MCAC (Mexico, Central America, and the Caribbean); South America (Chile, Colombia, Argentina, and Brazil); and Eurasia (Europe and Asia). In 2008, AES Energy Storage introduced the first grid-scale advanced battery-based energy storage solution for commercial operations. In July 2017, The AES Corporation and Siemens AG joined forces to create a new energy storage technology company called Fluence, which delivers the Advancion energy storage platform.
Products Offered	Advancion Energy Storage
Geographic Presence	North America, Europe, South America, and Asia

Source: Company Website and Press Releases

15.2.10 TRINABESS

Founding Year	2010
Headquarters Country	China
Headquarters City	Changzhou
Ownership Type	Public
Business Overview	TrinaBESS is a battery company working under the global panel manufacturing giant Trina Solar. TrinaBESS is engaged in supplying a large variety of retrofit and hybrid battery solutions on both off-grid and on-grid installations. These systems are designed for various applications in commercial, residential, and utility-scale battery storage applications. The company also provides a higher voltage energy storage system—S & T series. TrinaMega is a custom-made large-capacity energy storage solution used in commercial and utility projects; TrinaCommercial is a perfect business-grade storage solution that provides continuous power supply; and TrinaHome is a smart energy storage solution for green energy management at home.
Products	▪ TrinaBESS - S & T Series ▪ TrinaMega ▪ Portable ESS (Handy Power) ▪ TrinaHome ▪ Trinabess Powercube
Geographic Presence	North America, Europe, Asia, and Oceania

Source: Company Website and Press Releases

16 APPENDIX

16.1 INSIGHTS OF INDUSTRY EXPERTS

“APAC and North America are expected to hold the largest shares of the battery energy storage market during the forecast period.”



“The increasing need for reducing carbon emission, declining prices of lithium-ion batteries, and the use of renewable energy have supported the adoption of battery energy storage systems.”



“The lithium-ion batteries held the largest share of the battery energy storage system in 2021.”



“The utility application held the largest share of the battery energy storage system market in 2021.”



Source: Expert Interviews and MarketsandMarkets Analysis

16.2 DISCUSSION GUIDE

Q. 1. Could you provide a brief introduction to your company and its role in the battery energy storage system market?

Primary's viewpoint _____

Q. 2. What are your views on the growth of the global battery energy storage system market? How is the current scenario, and how is the market expected to grow in the future?

Primary's viewpoint _____

Q. 3. What are the key drivers, restraints, opportunities, and challenges for the battery energy storage system market?

FACTOR	DESCRIPTION
Drivers	
Restraints	
Opportunities	
Challenges	

Primary's viewpoint _____

Q. 4. According to you, what were the percentage contributions of the following elements to the overall size of the battery energy storage system market in 2021?

ELEMENT	MARKET SHARE (%) 2021	CAGR (2022–2027)
Battery		
Others		

Primary's viewpoint _____

Q. 5. According to you, what were the percentage contributions of the following battery types to the battery energy storage system market in 2021?

BATTERY TYPE	MARKET SHARE (%) 2021	CAGR (2022–2027)
Lithium-Ion Batteries		
Advanced Lead-acid Batteries		
Flow Batteries		
Others		

Primary's viewpoint _____

Q. 6. In the context of the connection type of battery energy storage systems, what were the percentage contributions of the following types to the battery energy storage system market in 2021?

CONNECTION TYPE	MARKET SHARE (%) 2021	CAGR (2022–2027)
On-Grid Connection		
Off-Grid Connection		
Primary's viewpoint_____		

Q. 7. In the context of the ownership of battery energy storage systems, what were the percentage contributions of the following ownership types to the battery energy storage system market in 2021?

OWNERSHIP	MARKET SHARE (%) 2021	CAGR (2022–2027)
Customer-Owned		
Third-party-Owned		
Utility-Owned		
Primary's viewpoint_____		

Q. 8. In the context of the energy capacity of battery energy storage systems, what were the percentage contributions of the following energy capacity ranges to the battery energy storage system market in 2021?

ENERGY CAPACITY	MARKET SHARE (%) 2021	CAGR (2022–2027)
Low (Below 100MWh)		
Medium (Between 100–500 MWh)		
High (Above 500 MWh)		
Primary's viewpoint_____		

Q. 9. According to you, what were the percentage contributions of the following applications to the battery energy storage system market in 2021?

APPLICATION	MARKET SHARE (%) 2021	CAGR (2022–2027)
Residential		
Commercial		
Utilities		
Primary's viewpoint_____		

Q. 10. Please list the shares of the regions identified below in the global battery energy storage system market in 2021. What do you expect their CAGRs to be for the next five years?

REGION	MARKET SHARE (%) 2021	CAGR % (2022-2027)
North America		
Europe		
APAC		
RoW		

Primary's viewpoint_____

Q. 11. According to you, what will be the CAGR of the battery energy storage system market from 2022 to 2027? Could you please mention the total size of the battery energy storage system market in 2021 and what it will be by 2022?

Primary's viewpoint_____

Q. 12. How has the COVID- 19 pandemic impacted the growth of the global battery energy storage system market in 2020 and 2021?

Primary's viewpoint_____

Q. 13. How emerging technologies are expected to have an impact on the battery energy storage system market in the near future?

Primary's viewpoint_____

Q. 14. What will be the revenue pockets for the battery energy storage system market in the next five years?

Primary's viewpoint_____

Q. 15. How do you see your company's position change in this market in the next five years?

Primary's viewpoint_____

16.3 KNOWLEDGE STORE: MARKETSandMARKETS' SUBSCRIPTION PORTAL

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  Communication & Connectivity Technology
  Display Technology
  Drones & Robotics
  Electrical System & Components

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  Industrial Automation
  Information System & Analytics
  Security & Surveillance
  Semiconductor Materials & Components
  Sensors and Control

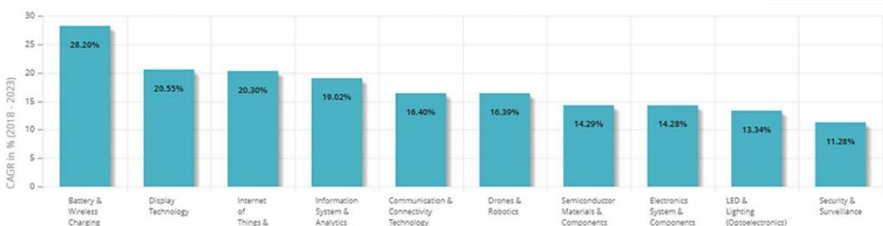
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HIGH GROWTH MARKETS

Industries 

- Aerospace & Defence (285)
- Agriculture (210)
- Automotive & Transportation (270)
- Chemicals & Material (1119)
- Energy & Power (363)
- Food & Beverage (309)
- Healthcare (928)
- Information & Communications Technology (852)
- Packaging, Construction, Mining & Gases (158)
- Semiconductor & Electronics (915)**
 - Battery & Wireless Charging (13)
 - Communication & Connectivity Technology (81)
 - Display Technology (72)
 - Drones & Robotics (22)
 - Electrical System & Components (56)
 - Electronics System & Components (161)
 - Industrial Automation (104)
 - Information System & Analytics (40)
 - Internet of Things & M2M (34)
 - LED & Lighting (Optoelectronics) (50)
 - Security & Surveillance (71)
 - Semiconductor Materials & Components (81)
 - Sensors and Control (89)
 - Testing, Inspection & Certification (41)



Market	CAGR in % (2018 - 2023)
Battery & Wireless Charging	29.20%
Display Technology	20.55%
Internet of Things & M2M	20.30%
Information System & Analytics	19.02%
Communication & Connectivity Technology	16.40%
Drones & Robotics	16.39%
Semiconductor Materials & Components	14.29%
Electronics System & Components	14.28%
LED & Lighting (Optoelectronics)	13.34%
Security & Surveillance	11.28%

Search Reports   

REPORT TITLE	DOMAIN	MARKET SIZE (USD Bn - 2018)	CAGR %	PUBLISH DATE
 LTE Advanced Pro Market by Communication Infrastructure (Small Cell, Macro Cell, RAN Equipment, DAS), Core Network Technology (SDN, NFV), Deployment Location (Urban Areas, Public Spaces, Rural Areas, Residential) and Geography - Global Forecast to 2023	Semiconductor & Electronics	0.15	121.6	Oct 2018
 G.Fast Chipset Market by Deployment Type (DPU and CPE), End User (Residential, and Enterprise/Commercial), Copper-Line Length (Lines Shorter Than 100 Meters, Lines Of 100 Meters-150 Meters, Lines Longer Than 250 Meters) - Global Forecast to 2022	Semiconductor & Electronics	0.19	116.5	Nov 2016
5G Fixed Wireless Access (FWA) Market by Offering (Hardware: Access Units & CPE, Services), Demography (Urban, Semi-Urban, Rural), Application (Residential, Commercial, Industrial, Government), and Region - Global Forecast to 2026	Semiconductor & Electronics	0.2	97.47	Feb 2019
 Free Space Optics (FSO) and Visible Light Communication (VLC)/Light Fidelity (Li-Fi) Market by Component (LED, Photodetector, Microcontroller, and Software), Transmission Type, Application, and Geography - Global Forecast to 2023	Semiconductor & Electronics	2.83	93.34	Sep 2018
 Neuromorphic Computing Market by Offering (Hardware, Software), Application (Image Recognition, Signal Recognition, Data Mining), Industry (Aerospace & Defense, IT & Telecom, Automotive, Medical & Industrial) and Geography - Global Forecast to 2022	Semiconductor & Electronics	0.03	86	Sep 2016
 Micro-LED Market by Application (Display (Smartwatch, NTE Devices (AR, VR), Smartphone, Television, HUD, Digital Signage, Laptop & Monitor) and Lighting (General, Automotive)), Display Panel Size, Vertical, and Geography - Global Forecast to 2025	Semiconductor & Electronics	0.33	80.1	May 2018
 Blockchain in Manufacturing Market by Application (Business Process Optimization, Logistics and Supply Chain Management, Counterfeit Management), End Use (Automotive, Energy & Power, Industrial, Pharmaceuticals), and Region - Global Forecast to 2025	Semiconductor & Electronics	0.01	80	Sep 2018

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16.4 AVAILABLE CUSTOMIZATIONS

With the given market data, MarketsandMarkets offers customizations according to the specific requirements of companies. The following customization options are available for the report:

COMPANY INFORMATION

- Detailed analysis and profiling of additional market players (up to five)

16.5 RELATED REPORTS

SR. NO.	REPORT TITLE	PUBLISHED DATE
1	LITHIUM-ION BATTERY MARKET WITH COVID-19 IMPACT ANALYSIS- GLOBAL FORECAST TO 2030 By Type (Li-NMC, LFP, LCO, LTO, LMO, NCA), Capacity, Voltage, Industry (Consumer Electronics, Automotive, Power, Industrial), & Region (North America, Europe, APAC & RoW) https://www.marketsandmarkets.com/Market-Reports/lithium-ion-battery-market-49714593.html	June 2021
2	FLOW BATTERY MARKET WITH COVID-19 IMPACT- GLOBAL FORECAST TO 2026 By Type (Redox Flow Battery, Hybrid Flow Battery), Material, Storage (Compact and Large scale), Application (Utilities, Commercial & Industrial, EV Charging Station), and Region https://www.marketsandmarkets.com/Market-Reports/flow-battery-market-82953888.html	June 2021
3	BATTERY MANAGEMENT SYSTEM MARKET - GLOBAL FORECAST TO 2024 By Battery Type (Lithium-Ion, Advanced Lead-acid), Topology (Centralized, Modular, Distributed), Type, Application (Automotive, UPS, Telecommunications, Renewable Energy Systems), and Geography https://www.marketsandmarkets.com/Market-Reports/battery-management-bms-market-234498189.html	November 2019

16.6 AUTHOR DETAILS

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